

Studies in Mechanical Engineering I

**Analysis, Synthesis
and Design of Hydraulic
Servosystems
and Pipelines**

Taco J. Viersma

Elsevier

TH 137-43
V 665

STUDIES IN MECHANICAL ENGINEERING, 1

Analysis, Synthesis and Design of Hydraulic Servosystems and Pipelines

TACO J. VIERSMA

*Department of Mechanical Engineering
Delft University of Technology
Delft, The Netherlands*



ELSEVIER SCIENTIFIC PUBLISHING COMPANY
Amsterdam — Oxford — New York 1980

上海图书馆藏书



A541 221 0038 0886B



137-43

ELSEVIER SCIENTIFIC PUBLISHING COMPANY
335 Jan van Galenstraat
P.O. Box 211, 1000 AE Amsterdam, The Netherlands

Distributors for the United States and Canada:

ELSEVIER/NORTH-HOLLAND INC.
52, Vanderbilt Avenue
New York, N.Y. 10017

Library of Congress Cataloging in Publication Data

Viersma, T J
Analysis, synthesis, and design of hydraulic
servosystems and pipelines.

(Studies in mechanical engineering ; v. 1)
Includes bibliographical references and indexes.
1. Hydraulic servomechanisms. 2. Pipe lines.
3. Fluid-film bearings. I. Title. II. Series.
TJ857.V53 629.8'042 80-11302
ISBN 0-444-41869-5

ISBN 0-444-41869-5 (Vol. 1)
ISBN 0-444-41872-5 (Series)

© Elsevier Scientific Publishing Company, 1980

All rights reserved. No part of this publication may be reproduced, stored in a retrieval system or transmitted in any form or by any means, electronic, mechanical, photocopying, recording or otherwise, without the prior written permission of the publisher, Elsevier Scientific Publishing Company, P.O. Box 330, 1000 AH Amsterdam, The Netherlands

Printed in The Netherlands

PREFACE

This textbook is a summary of a new graduate course in hydraulic servosystems at the Delft University of Technology. It is in English, because there is vital interest in this subject on the part of scientists, students, engineers and manufacturers in many countries.

The main sources of its contents are:

- o Research and development of the hydraulic research team of Mechanical Control Engineering Laboratory in the Department of Mechanical Engineering of the Delft University of Technology.
- o The thesis *Investigations into the accuracy of hydraulic servomotors*, by T.J. Viersma, Delft University of Technology, 1961.
- o The thesis *Conical hydrostatic bearings* (in German), by P. Blok, Delft University of Technology, 1976.
- o A thesis on *The dynamics of hydraulic supply lines*, by A.A. Ham (in preparation).

An important foundation for the matters dealt with here was laid down by Blackburn, Lee and Shearer in their magnificent book *Fluid Power Control* (Technology Press of M.I.T., 1960). This book of more than 700 pages can no longer give complete and up-to-date information. Nevertheless, it is an almost inexhaustible source of fundamental information.

The next best book is that of Merritt, *Hydraulic Control Systems*, Wiley, New York, 1967. This book of 350 pages is directed more specifically to hydraulic control problems. Except for the last chapters, it is excellent. However, it does not cover the design phase.

I set myself the challenge in this textbook of summing up, in a very brief and compact text, what I know of the analysis, synthesis and design of hydraulic servosystems. Supported by the know-how and experience of the hydraulic research team of the Delft University of Technology, I drew straight lines - as direct as possible - to the design goal!

To the basic material on servosystems, two special chapters have been added. In Chapter 6, the dynamics of hydraulic supply lines for servosystems are treated. In the literature, a constant supply pressure is consistently assumed, without any indication, however, of how to achieve this goal. In Chapter 7, the outstanding work of Blok on hydrostatic bearings is summarized. Hydrostatic bearings eliminate Coulomb friction almost completely, thus avoiding phenomena such as a hydraulic lock, dead zone or slip-stick motions. The resulting and considerable improvements

in the performance of flight simulators in Delft and Amsterdam stimulated many foreign manufacturers to fulfil the high level specifications of some major European airlines.

Readers studying Chapters 1 - 5 on their own can leave out all the sections marked with an asterisk (*). These sections go into details that are not absolutely essential for understanding the remainder. The illustrations in this textbook have a special function. After a thorough orientation, the more than 90 figures on the left-hand pages will serve as an almost complete pictorial summary.

Finally, I have pleasure in thanking the staff members Teerhuis and Reuvekamp for their direct support, Gutteling and Jansen for the drawings and Mrs. Bosch for typing the text.

Delft,
December, 1979

Taco J. Viersma

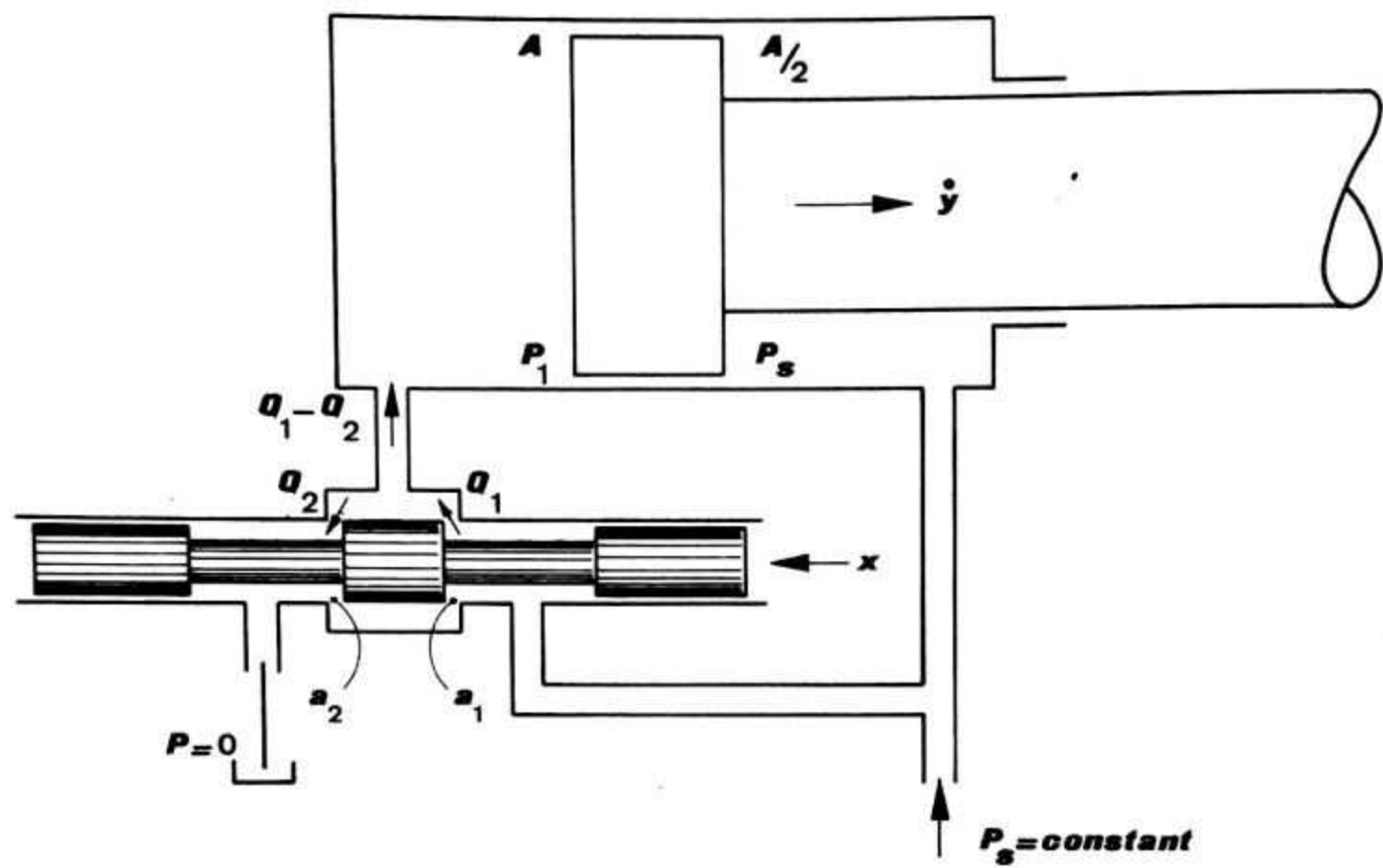
CONTENTS

PREFACE	v
1 INTRODUCTION	1
1.1 Integrator with proportional feedback	1
1.2 Dominant properties of a simple servosystem	7
1.3 Velocity gain K_v and valve configuration	11
* 1.4 Multiple-hole valve	11
2 TOOLS FOR THE ANALYSIS OF HYDRAULIC SERVO'S	15
2.1 Turbulent port flow	15
2.2 Compressibility of oil	21
2.3 Forces and continuity	25
References	31
3 ANALYSIS OF HYDRAULIC SERVOSYSTEMS	33
3.1 Introduction	33
3.2 Asymmetric motors controlled by 3-way valves	39
* 3.3 Load-compensation	45
3.4 Symmetric motor controlled by a symmetric 4-way valve	49
3.5 Asymmetric motor controlled by a symmetric 4-way critical-center valve	53
* 3.6 Asymmetric motor controlled by an asymmetric 4-way valve	55
3.7 From mathematical model to transfer function	59
References	63
4 DYNAMIC ANALYSIS AND SYNTHESIS	67
4.1 Introduction	67
4.2 Proportional control	71
4.3 Dynamic load	73
* 4.4 Dynamic dead zone due to Coulomb friction	75
4.5 Damping methods	77
4.6 Electro-hydraulic servovalve	83
4.7 Block diagram for hardware components	89
* 4.8 PI-control	91
* 4.9 First order system in series	93
* 4.10 Compliant structures	95
* 4.11 LF-control of complex hydraulic servo's	97
* 4.12 Load actuators	99

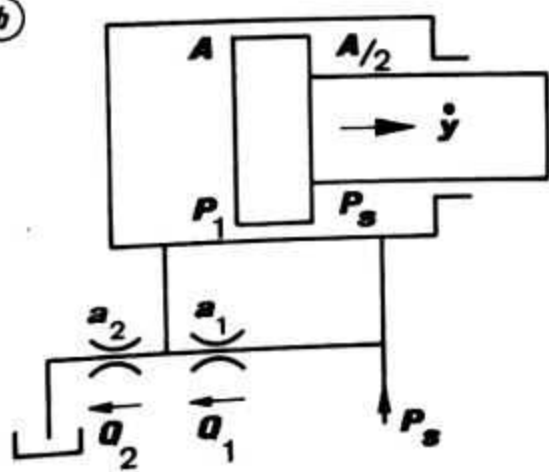
4.13	Coulomb friction and stability	101
*4.14	Analog simulation of Coulomb friction	103
*4.15	Describing function of Coulomb friction	105
	References	113
5	PERFORMANCE DIAGRAM AND DESIGN OF SINUSOIDALLY ACTUATED LINEAR SERVOSYSTEMS LOADED BY INERTIA FORCES AND A CONSTANT PRE-LOAD	115
5.1	Introduction	115
5.2	Relation between max. stroke $S = 2\hat{y}_{\max}$ and bandwidth ω_b	117
5.3	Relation between valve capacity $Q_{\max}$ and maximum speed $\dot{y}_{\max}$	117
5.4	Relation between cavitation boundary and max. acceleration $\ddot{y}_{\max}$	117
5.5	Preliminary performance diagram	119
5.6	Design of symmetric motors	121
5.7	Final performance diagram	121
*5.8	Design of asymmetric servomotors	123
5.9	Design example	125
5.10	Inverse design example	127
6	HYDRAULIC LINE DYNAMICS	129
6.1	Introduction	129
6.2	Basic equations for dynamic behavior of hydraulic lines	135
6.2.1	Assumptions	135
6.2.2	Continuity	135
6.2.3	Momentum	137
6.2.4	Four-pole equations relating Q_i , P_i , Q_o and P_o	145
6.3	Dynamic behavior of a single pipeline	149
6.4	General rules for compound pipelines and components	155
6.4.1	Chain rule	155
6.4.2	Accumulator	157
6.4.3	Long line (T-filter)	151
6.4.4	Two long lines in series	157
6.4.5	Two short lines in series	161
6.4.6	Accumulator with short connection line	163
6.4.7	Pressure control valve	165
6.4.8	Pressure controlled pump	169
6.4.9	Summary	171
6.5	Suppression of pump flow pulsations	173
6.5.1	Introduction	173
6.5.2	Strategy and basic equations	175
6.5.3	Selective filters	181
6.5.4	Pressure control valve and gas-accumulator near the pump	191

6.6	Reducing supply pressure variations due to consumption by the servovalve	201
6.7	Simulation	225
	References	229
7	THE DESIGN OF CONICAL HYDROSTATIC BEARINGS FOR HYDRAULIC CYLINDERS WITH CONSIDERABLE SPEED	231
7.1	Introduction	231
7.2	Single bearings	233
7.3	New designs as a breakthrough to applications	237
7.4	Double bearings	239
7.5	Manufacturing	241
7.6	Applications	243
7.7	Conclusions and summary	245
*7.8	Appendix	245
7.9	Design example	249
*7.10	Derivation of Reynolds' equation	253
*7.11	Derivation of pressure distribution	253
*7.12	Derivation of bearing force	255
*7.13	Derivation of leakage flow	261
*7.14	Derivation of cavitation boundary for double bearings	263
*7.15	Iterative relaxation for conical bearing with two-dimensional flow	267
*7.16	Results of calculations on two-dimensional flow	271
	References	275
	AUTHOR INDEX	277
	SUBJECT INDEX	277

a



b



c

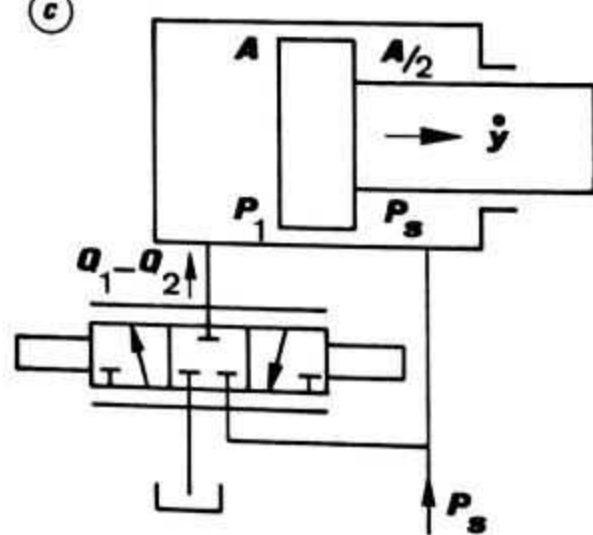


Fig. 1.1 Three-way-valve-controlled asymmetric motor

CHAPTER 1

INTRODUCTION

1.1 Integrator with proportional feedback

Complex electro-hydraulic servomechanisms involve so many facets, elements and components and so many disciplines that a systematic approach from the beginning may take many specialized chapters before some understanding of the essential characteristics of a hydraulic servo is reached.

In an attempt to discover directly some important control features, we shall derive in this Introduction a very simple model of a hydraulic copying system, showing that a hydraulic servosystem behaves like an integrator with proportional position feedback. To show the relationship between mechanical design and control behavior the influence of motor and valve configurations on the loop gain is discussed from the beginning.

In Fig. 1.1 a very simple asymmetric "motor" with a so-called three-way valve is shown. Figure 1.1(a) can be simplified to give (b), while (c) gives the normalized presentation in which the mechanical valve configuration is not included.

Assuming an area ratio of 2:1, at one side of the piston the constant supply pressure P_s prevails, while at the other side of the piston the pressure is $P_1 = 0.5 P_s$ when there is no external or frictional load. Thanks to Newton's law, $P_1 = 0.5 P_s$ not only at a piston speed $\dot{y} = 0$ but also at $\dot{y} = \text{constant}$! Evidently, the magnitude of the speed $\dot{y}$ of the piston (without load) is determined by the inlet opening a_1 and the outlet opening a_2 , which in turn are controlled by the displacement x of the spool valve. Assuming an inlet flow Q_1 and an outlet Q_2 , we find

$$\dot{y} A = Q_1 - Q_2 = a_1 v_1 - a_2 v_2$$

where v_1 and v_2 are the speed of the oil through a_1 and a_2 , respectively. The flows Q_1 and Q_2 depend on the magnitude of the corresponding openings as well as on the pressure drop across these openings.

Assuming *turbulent flow*, the (average) speed v of the oil through an opening is determined exclusively by the pressure drop across that opening. Since the pressure drops across inlet and outlet openings are equal ($P_s - P_1 = 0.5 P_s$ and $P_1 = 0.5 P_s$ respectively), the speeds in the no-load situation are also equal: $v_1 = v_2 = v_o$. Thus $\dot{y} A = v_o (a_1 - a_2)$, and for the valve configuration of Fig. 1.1

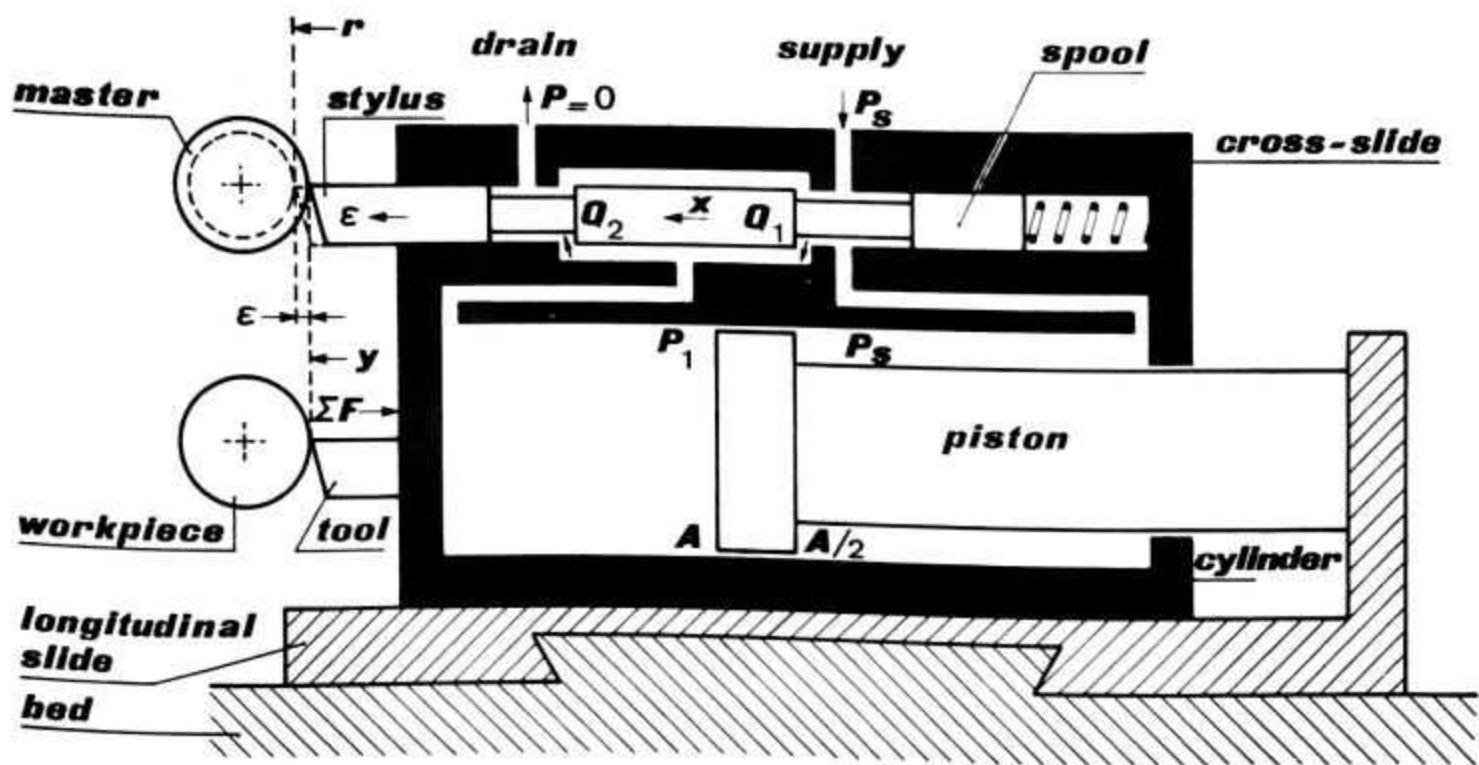


Fig. 1.2 Copying lathe

$$\frac{\partial \dot{y}}{\partial x} = \frac{v_o}{A} \cdot \frac{\partial (a_1 - a_2)}{\partial x} = K_m = \text{constant}$$

Within a limited valve travel the *velocity gain* K_m of our motor really is a constant, so $K_m = \partial \dot{y} / \partial x = \dot{y} / x$, and the input to output transfer function of the open loop system becomes

$$\frac{y}{x} = \frac{K_m}{s}$$

where s is the differential or Laplace operator.

From the control point of view the system is an integrator, meaning that the output signal y can be derived by integrating the input signal x .

The physical implication of this is that *the motor is pushed by flow and not by force!* This is not a contradiction of Newton's law, since external, frictional and inertial forces should be seen as disturbance signals influencing the input of the integrator. This will be clarified later.

Giving the integrator a proportional position feedback, the system described above becomes a real servosystem able to control displacements. By way of illustration, Fig. 1.2 shows very schematically the hydraulic control system of a copying lathe.

Here the piston is stationary, the cylinder movable, while the spool valve can move relative to the cylinder. The input signal r of the control system is given by the master, the (rotary) workpiece to be copied. In perfect copying the tool displacement y (shaping the workpiece being manufactured) equals the master dimension r . Any difference $\epsilon = r - y$ will result in a corresponding spool valve displacement $x = \epsilon$ causing cylinder movements which try to make $x = \epsilon = r - y = 0$ or $y = r$! Here the feedback has been obtained in a purely mechanical way.

The complete cross-slide together with the hydraulic servosystem, is traveling with a constant speed along the bed of the lathe (perpendicular to the plane of drawing). The stylus will follow the variable contours of the master and - controlled by the valve - the cylinder will follow according to

$$\dot{y} = K_m x$$

where

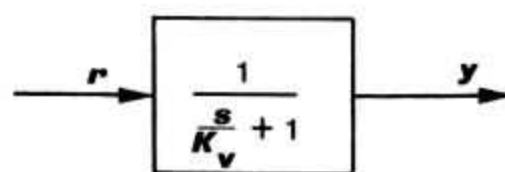
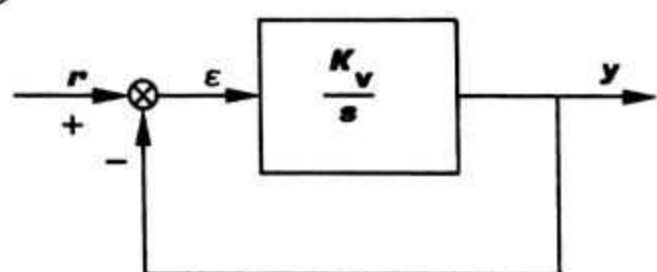
$$x = \epsilon = r - y$$

Of course it is not necessary that $x = \epsilon$, as there could be a lever between stylus and spool or even an electro-hydraulic servovalve, making $x = K_s \epsilon$ and

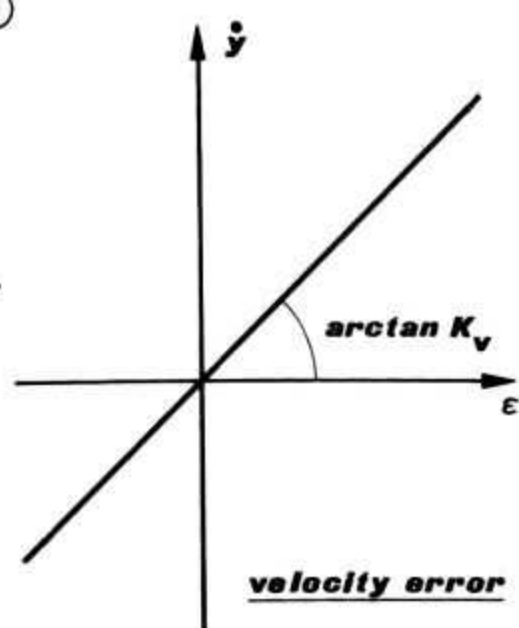
$$\frac{y}{\epsilon} = K_s \frac{y}{x} = \frac{K_s K_m}{s} = \frac{K_v}{s}.$$

But this does not change the essential character of our servo, which is that of an integrator with proportional position feedback.

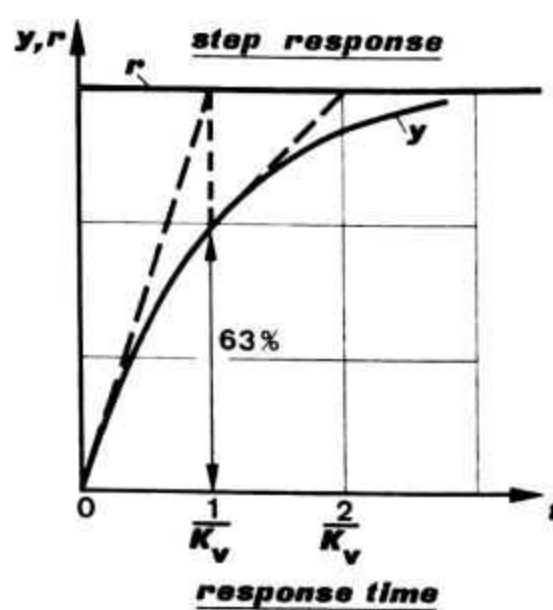
(a)



(b)



(c)



(d)

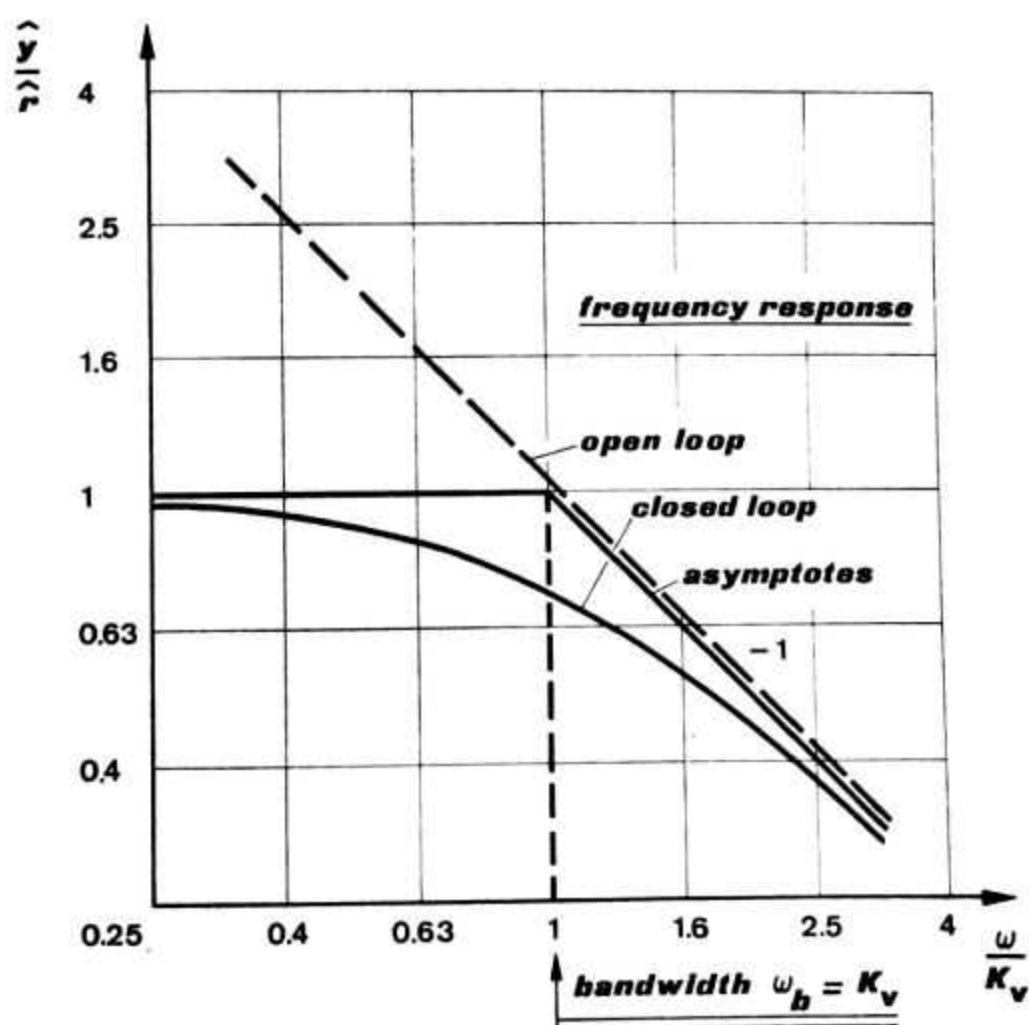
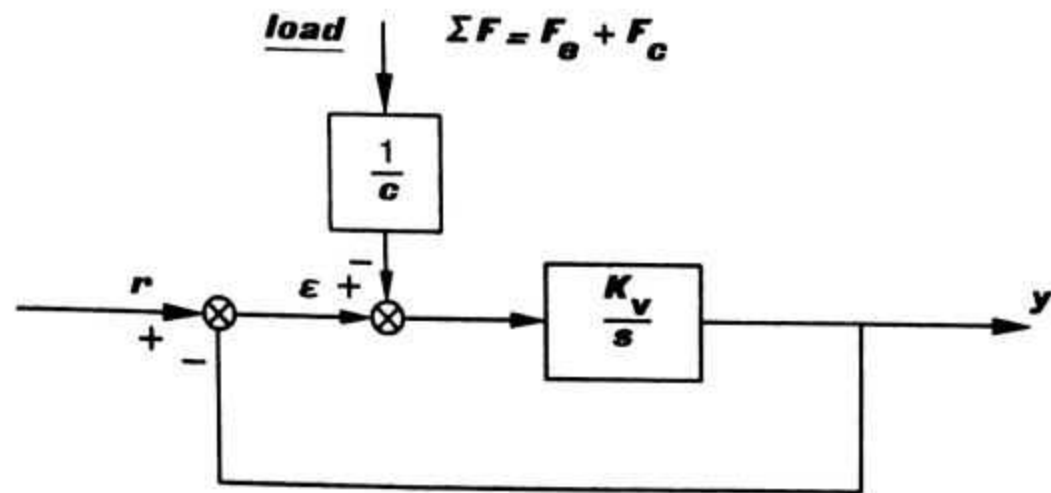
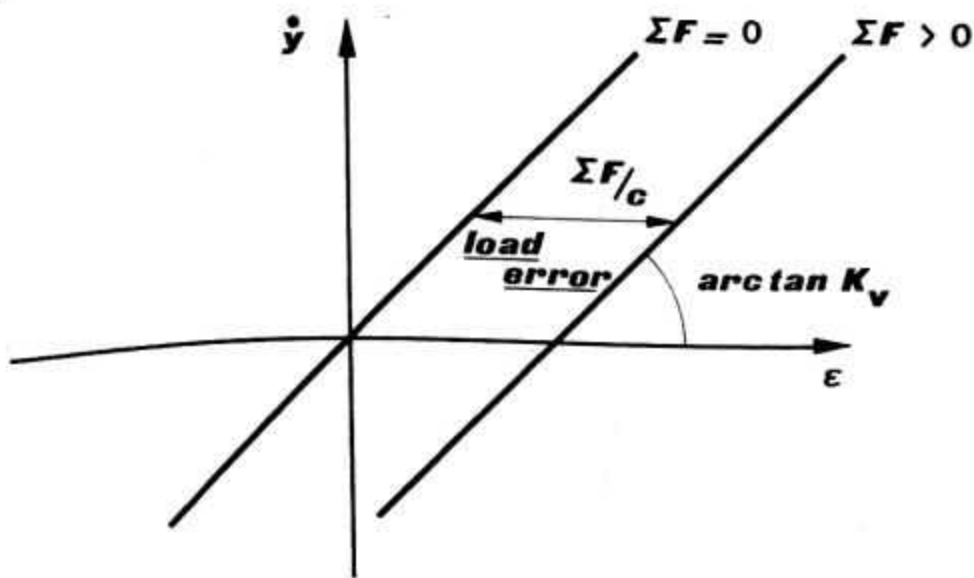


Fig. 1.3 Integrator with proportional feedback

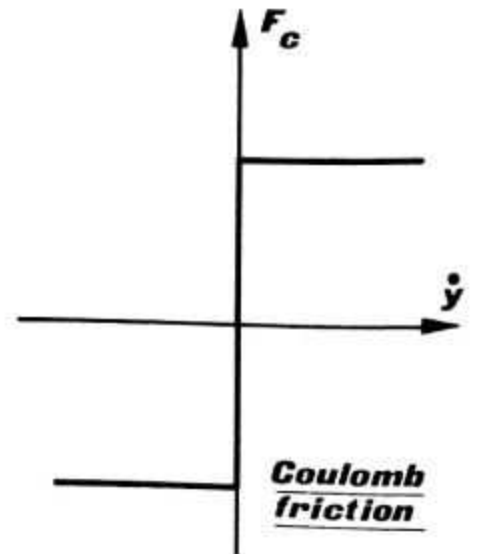
(a)



(b)



(c)



(d)

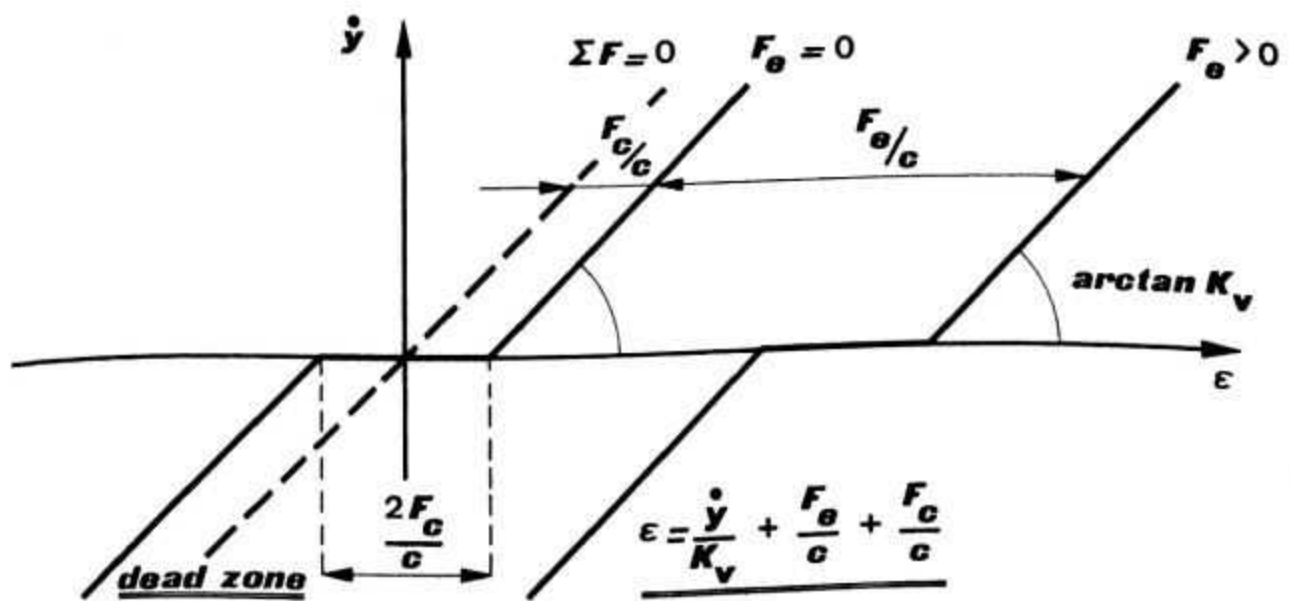


Fig. 1.4 Influence of external and frictional load

we derive

$$\epsilon = \frac{\dot{y}}{K_v} + \frac{\Sigma F}{c} = \frac{\dot{y}}{K_v} + \frac{F_e}{c} + \frac{F_c}{c}.$$

Here c is the load stiffness and ΣF the "total" load including external forces F_e (pay-load) and the (always present) Coulomb friction, see Fig. 1.4(c). $\Sigma F/c$ is called the *load-error*, which must be added to the velocity error $\dot{y}/K_v$ to obtain the complete control error ϵ .

The effect of Coulomb friction on a servosystem with a load stiffness c is shown in Fig. 1.4(d). Coulomb friction, reversing its sign with the speed $\dot{y}$, introduces a *dead zone*, which poses a serious threat to the accuracy of a servosystem.

Unpredictable variations in Coulomb friction will cause corresponding variations in the control error and could have fatal consequences for the "smoothness" of operation, particularly with regard to the speed $\dot{y}$ and the acceleration $\ddot{y}$ of the servosystem.

To measure the performance of a servosystem one is inclined in many cases to measure the result, the output signal y . But, assuming the input signal r to be perfect, it sometimes makes more sense to measure the error signal ϵ during operation or testing. Thus machine tool copying systems should not be tested primarily by measuring the workpieces that have been manufactured, but by measuring the error signal ϵ directly, as is shown in Fig. 1.5, where a micrometer (attached by means of a magnetic clamp) indicates the spool displacement with respect to its housing. In the closed loop situation several points of the characteristic of Fig. 1.4(d) could be measured easily, determining the velocity gain K_v and the load stiffness c , as well as the dead zone. The value of such simple tests, in which relevant system parameters are measured (which is the aim!), can be illustrated by the fact that some firms are using mainly the measurement of the dead zone as a criterion in selecting copying machines to be installed in their workshops. Without any preparation, the measurement of the dead zone of a copying system could - according to Fig. 1.5 - be carried out in a few minutes. It is therefore surprising to observe at machine tool exhibitions that neither sales managers nor machine tool operators understand the essentials of the experiment...

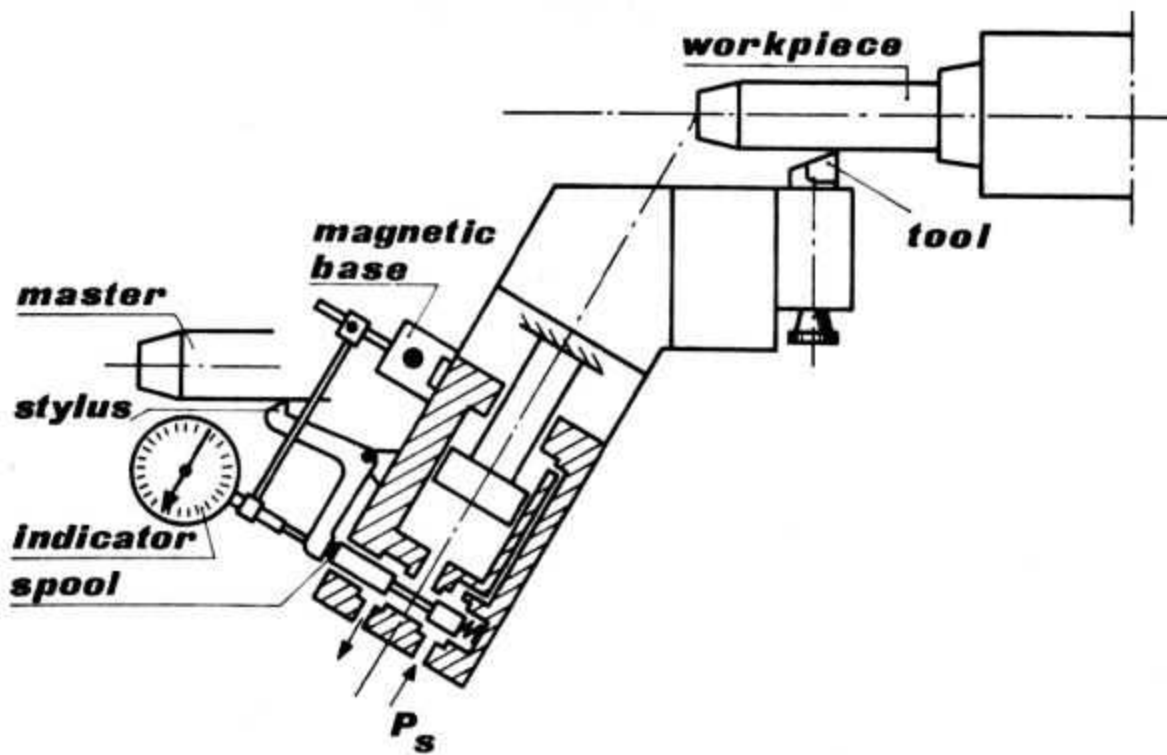
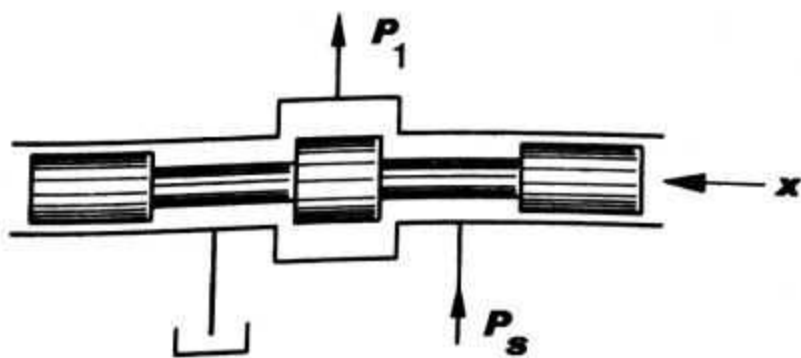
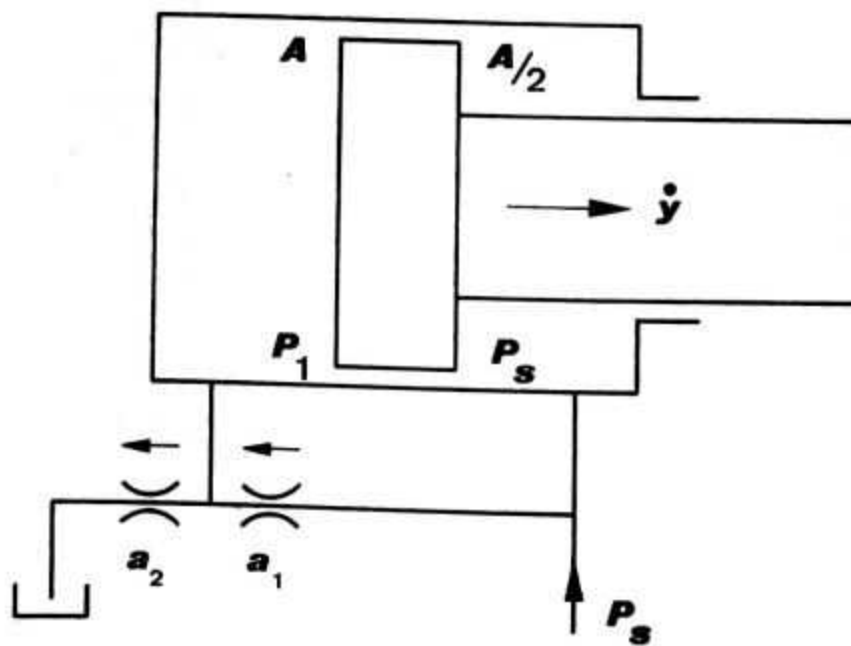
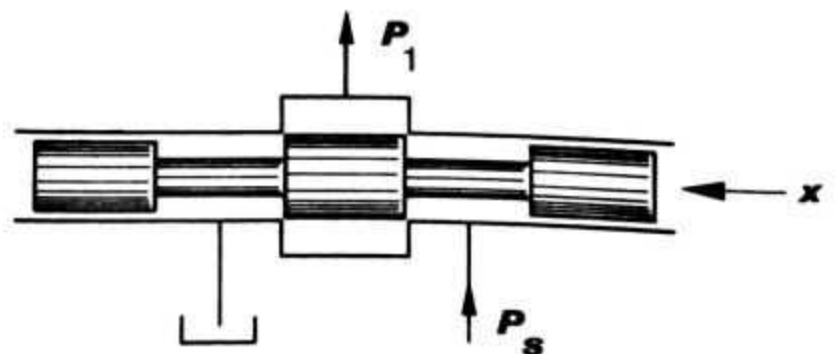


Fig. 1.5 Measuring set-up



(a)



(b)

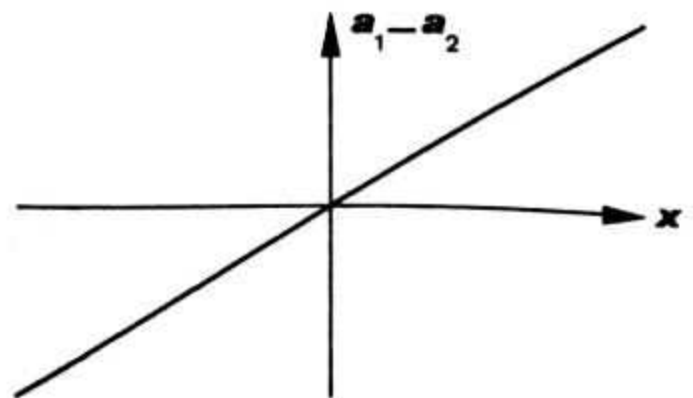
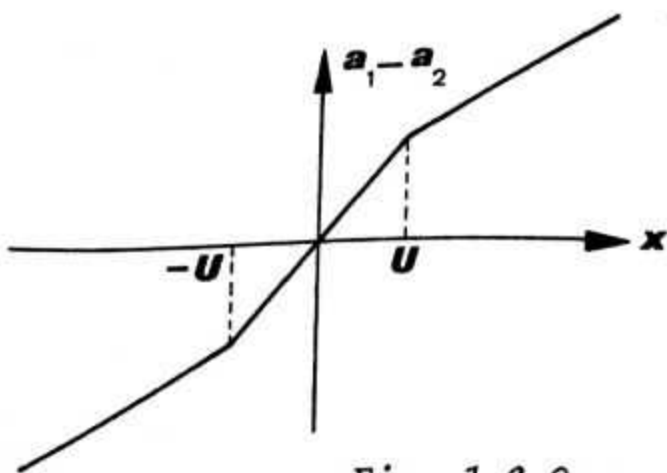


Fig. 1.6 Open-center and critical-center 3-way valve

1.3 Velocity gain K_v and valve configuration

In the previous sections we have found that

$$K_v = \frac{\partial \dot{y}}{\partial \epsilon} = \frac{v_o}{A} \cdot \frac{\partial(a_1 - a_2)}{\partial \epsilon} = K_s \frac{v_o}{A} \cdot \frac{\partial(a_1 - a_2)}{\partial x}.$$

Thus the velocity gain K_v is proportional to the slope of the characteristic, in which the difference between inlet and outlet opening ($a_1 - a_2$) is plotted as a function of the valve displacement x . That characteristic depends completely on the valve configuration and its dimensions. In Fig. 1.6 some examples are given. Figure 1.6(a) shows an open-center 3-way valve and (b) a critical-center valve, both with cylindrical port openings. The corresponding characteristics are given also. It appears that the open-center valve has within the underlap-region a gain twice as large as outside, while the critical-center valve has a constant gain!

Furthermore, it should be noted that the open-center valve implies an oil supply (from the pump) that is practically independent of the spool position in the underlap-region. With a critical-center valve there is no oil consumption at $x = 0$. This absence of "leakage flow" (at least theoretically) could be an advantage, especially in systems (such as aircraft) with a large number of motors connected to one pumpset.

* 1.4 Multiple-hole valve

In Fig. 1.7 a very special 3-way valve is shown, a "multiple-hole" valve, where the inlet a_1 and the outlet a_2 consist of a large number of circular holes arranged in one, two or more rows. Displacing the spool, either the inlet holes (a_1) or the outlet holes (a_2) will be opened. When the "underlap" U is properly chosen, see Fig. 1.7(b), the characteristic of $a_1 - a_2$ versus x (determining K_v !) is rather straight, see (c). Disclosing one single hole with diameter d over a distance h , one finds an opening

$$a = \frac{\pi}{4} d^2 \frac{\arccos(1 - \frac{2h}{d}) - 2(1 - \frac{2h}{d})\sqrt{\frac{h}{d} - \frac{h^2}{d^2}}}{\pi}$$

and

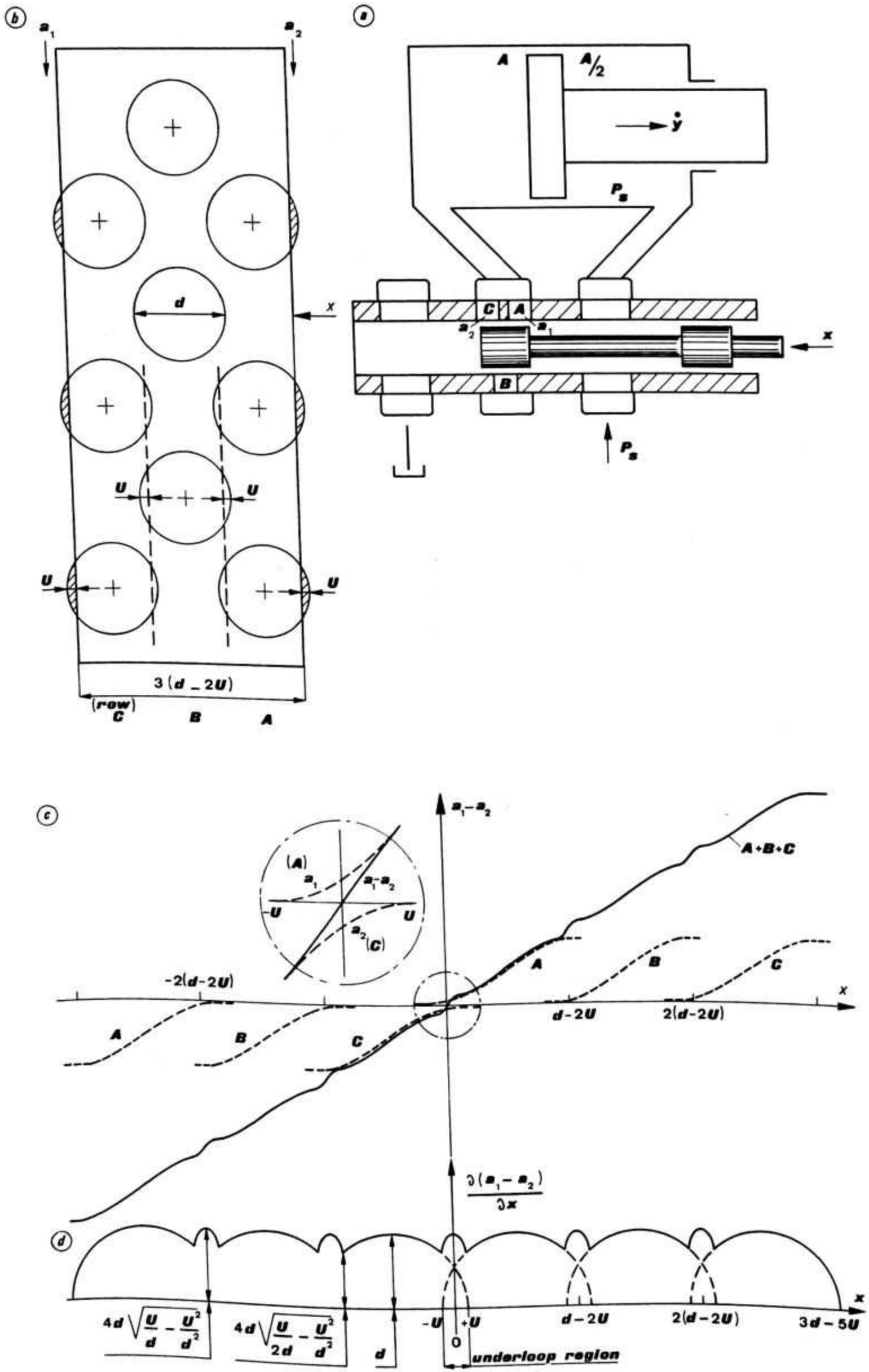


Fig. 1.7 Multiple-hole valve

$$\frac{\partial a}{\partial h} = 2 \sqrt{dh - h^2} = 2d \sqrt{\frac{h}{d} - \frac{h^2}{d^2}}$$

With an underlap U according to Fig. 1.7(b) we find inside the underlap-region at

$$|x| < U \quad \frac{\partial(a_1 - a_2)}{\partial x} = 2d \left[\sqrt{\frac{U+x}{d} - \left(\frac{U+x}{d}\right)^2} + \sqrt{\frac{U-x}{d} - \left(\frac{U-x}{d}\right)^2} \right]$$

$$x = 0 \quad \frac{\partial(a_1 - a_2)}{\partial x} = 4d \sqrt{\frac{U}{d} - \left(\frac{U}{d}\right)^2} = \text{maximum}$$

$$|x| = U \quad \frac{\partial(a_1 - a_2)}{\partial x} = 2 \sqrt{2\frac{U}{d} - 4\left(\frac{U}{d}\right)^2} = \text{minimum}$$

Outside the underlap region we find the maximum gain at

$$x = \frac{d}{2} - U \quad \frac{\partial(a_1 - a_2)}{\partial x} = \frac{\partial a_1}{\partial x} = d.$$

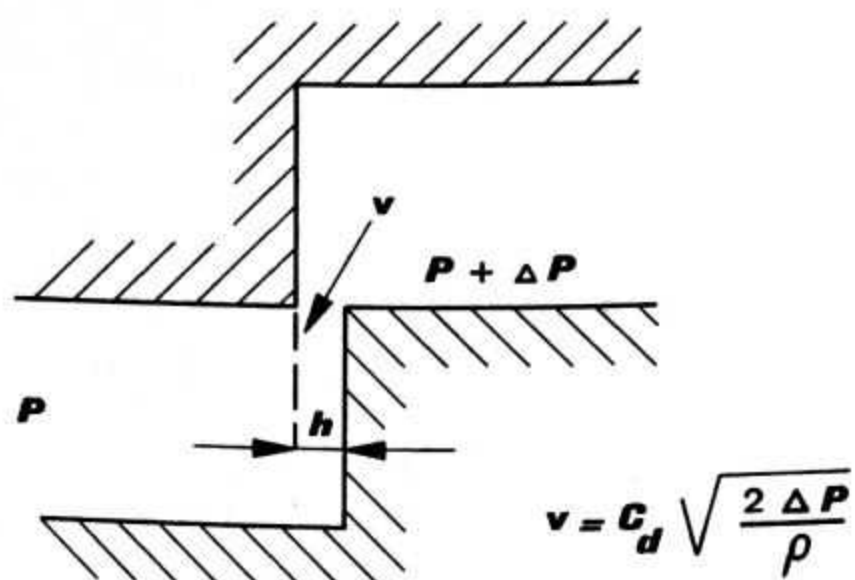
The pattern of $\partial(a_1 - a_2)/\partial x$ is repetitive, as is shown in Figures 1.7(c) and (d). Trying to make variations in $\partial(a_1 - a_2)/\partial x$ as small as possible, we equalize the maxima inside and outside the underlap region, resulting in

$$\frac{U}{d} = \frac{1}{2} - \frac{\sqrt{3}}{4} = 0.067.$$

The maximum and minimum values of $\partial(a_1 - a_2)/\partial x$ then become d and $0.681 d$, respectively.

Leakage flow in a multiple-hole valve depends on the number of rows with holes. In the neutral valve position ($x = 0$), with $U/d = 0.067$, the inlet opening is, in percentage of the maximum opening, 3%, 1½% and 1% when there are, respectively, 1, 2 and 3 rows with holes. Thus the choice of the number of rows depends on the amount of leakage flow being allowed!

(a)



(b)

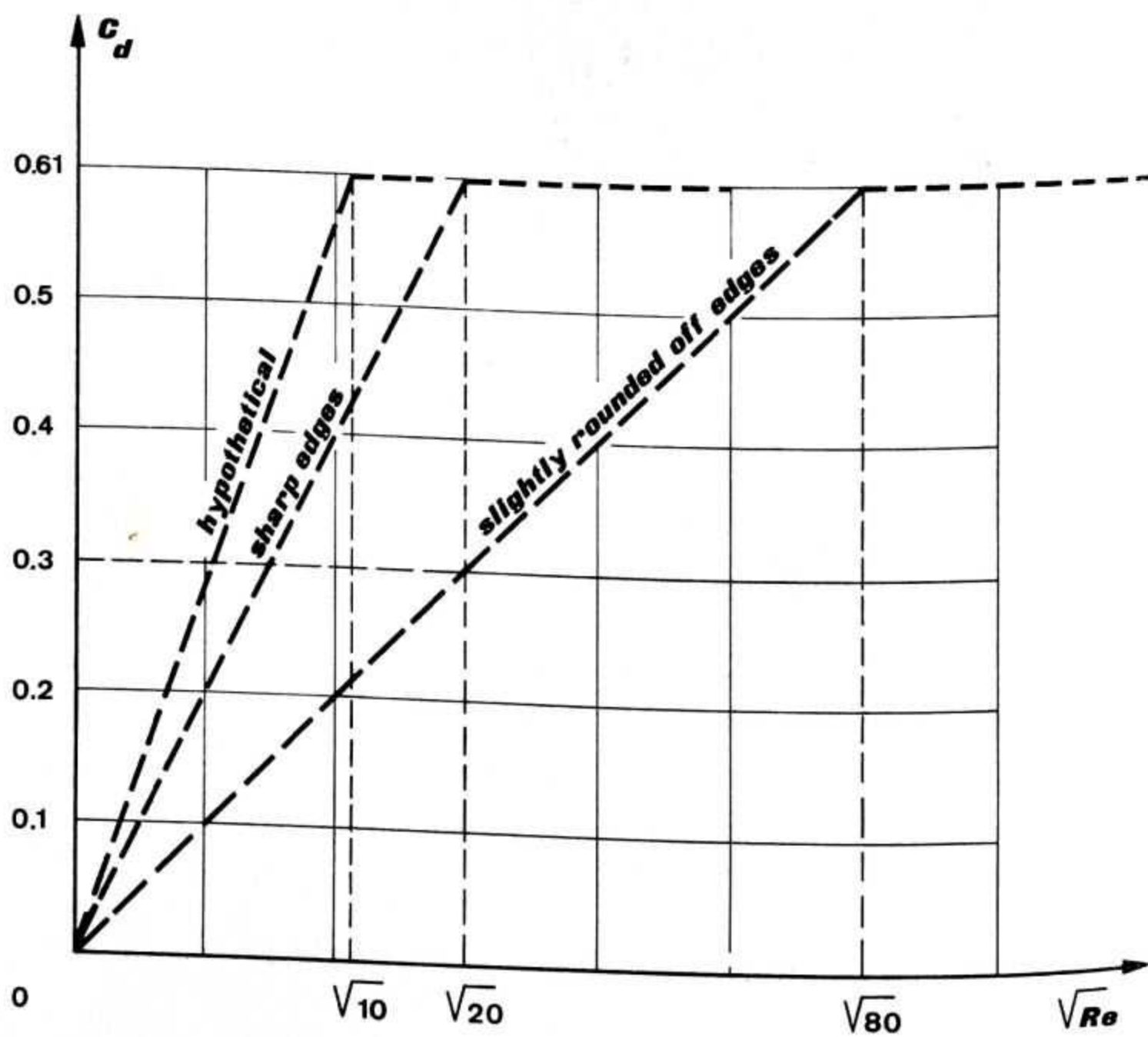


Fig. 2.1 Discharge coefficient C_d

CHAPTER 2

TOOLS FOR THE ANALYSIS OF HYDRAULIC SERVO'S

To analyze hydraulic servosystems

- oil flows through variable port openings (valves),
 - compressibility of oil, and
 - forces and continuity
- must be taken into account.

2.1 Turbulent port flow

The speed v of oil through a narrow orifice is determined by

$$v = C_d \sqrt{\frac{2\Delta P}{\rho}}. \quad (1)$$

In 1917 Von Mises [1] found a constant "discharge" coefficient C_d . Theoretically, $C_d = \pi/(\pi+2) = 0.611$. This type of flow is called turbulent because the pressure drop ΔP across the opening is proportional to ρv^2 .

The flow Q can now be found as the product of port opening a and the speed v .

$$Q = av = bhC_d \sqrt{\frac{2\Delta P}{\rho}}. \quad (2)$$

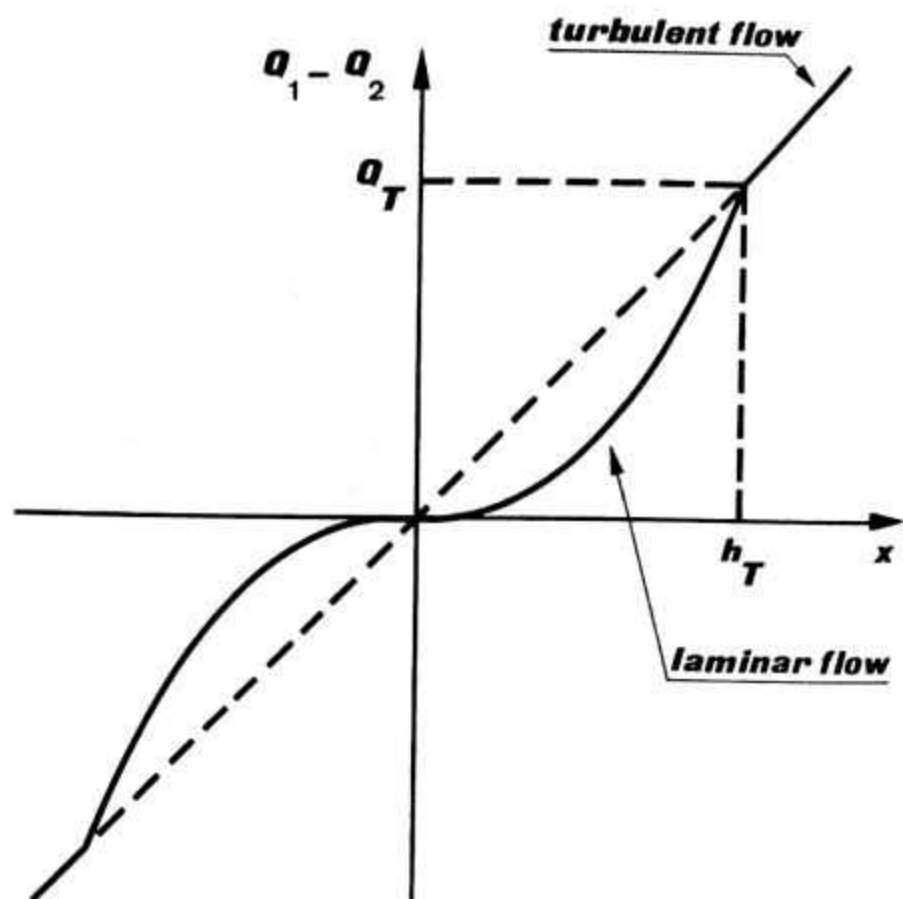
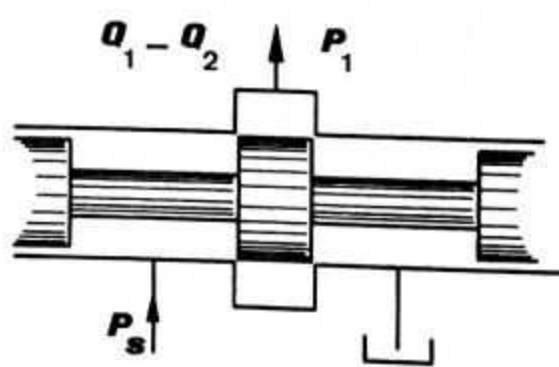
Here h (the variable) is the smallest dimension of the (rectangular) orifice, see Fig. 2.1(a), and b the constant width. Turbulence is ensured only at "large enough" Reynolds numbers.

$$Re = \frac{2vh}{\nu} = \frac{2\rho vh}{\mu}. \quad (3)$$

Here ν is the kinematic viscosity, μ the dynamic viscosity and ρ the specific mass.

Experiments carried out by the author [2] proved that at very sharp edges in narrow orifices the critical value Re_{cr} is as low as 20, while slightly rounded off edges increased Re_{cr} to 80 or higher! Thus at very sharp edges we may assume C_d to be constant at $Re > Re_{cr} \simeq 20$.

(a)



(b)

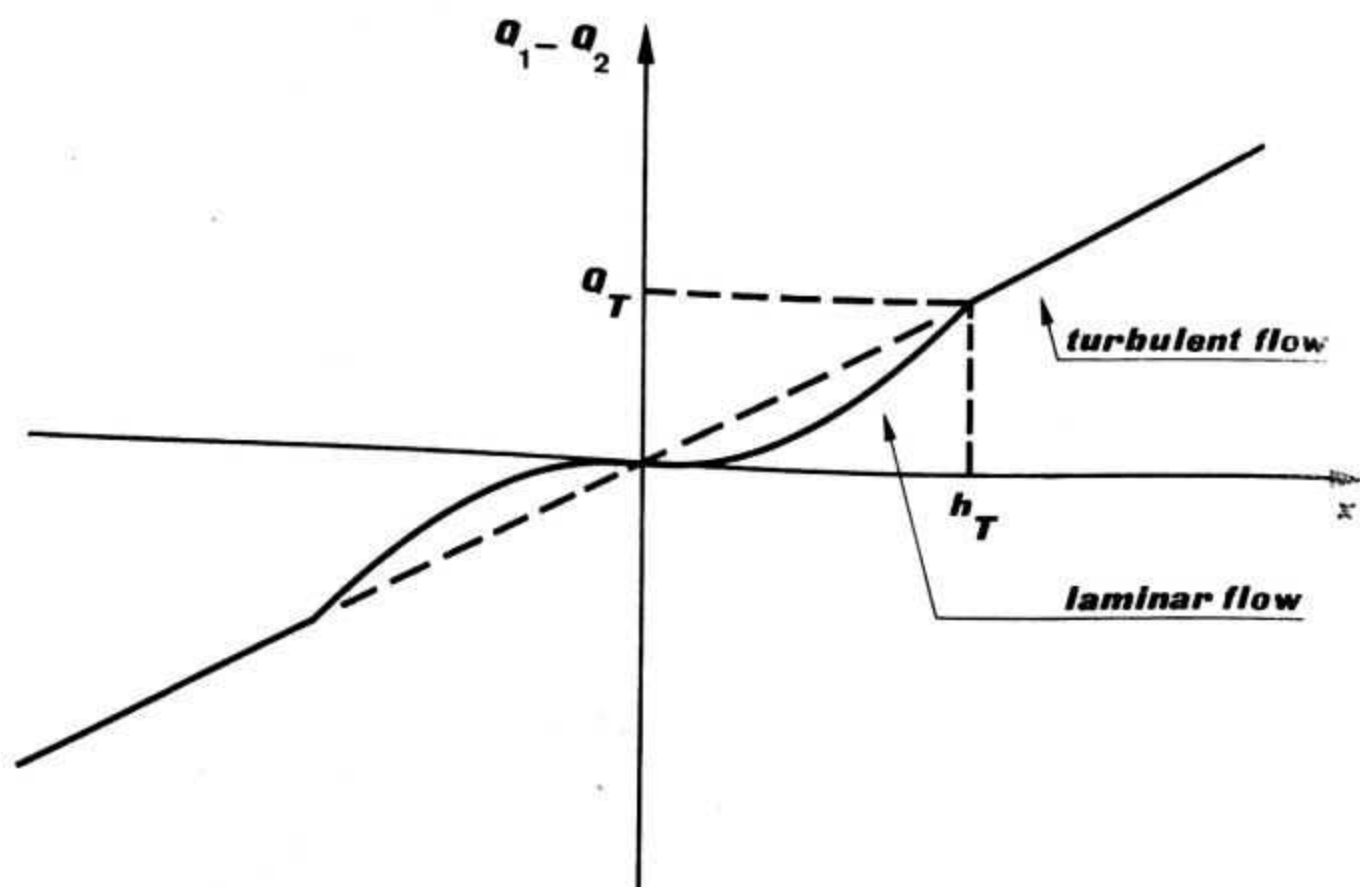
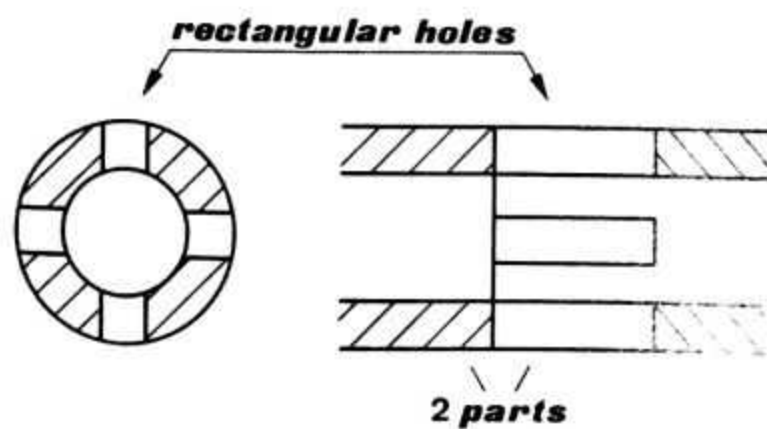
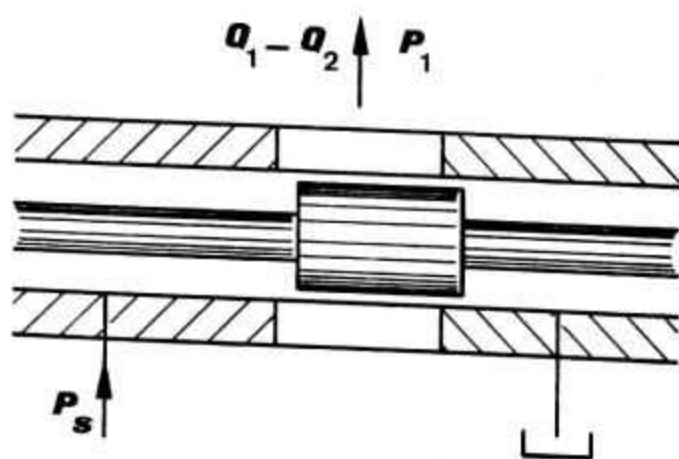


Fig. 2.2 Transition from laminar to turbulent flow

At smaller values of Re a boundary layer will build up. In 1954 Wuest [3] derived, for extreme low values of Re ,

$$C_d = \delta \sqrt{Re}. \quad (4)$$

Here δ is a constant, depending on geometry. Combining eqns. (1), (3) and (4) yields

$$v = C_d \sqrt{\frac{2\Delta P}{\rho}} = \delta \sqrt{Re} \sqrt{\frac{2\Delta P}{\rho}} = \delta \sqrt{\frac{2\rho v h}{\mu}} \cdot \frac{2\Delta P}{\rho}$$

resulting in

$$v = \frac{4\delta^2}{\mu} h \Delta P$$

and

$$Q = av = bhv = \frac{4\delta^2}{\mu} bh^2 \Delta P. \quad (5)$$

Extrapolating laminar flow to larger values of Re and assuming a sudden transition from laminar flow to turbulent flow, we find

$$\delta = \frac{C_{dturb}}{\sqrt{Re_{cr}}} \approx \frac{0.611}{\sqrt{20}} \approx 0.136 \quad (6)$$

This flow model of laminar and turbulent flow as presented by the author in 1961 [2], (see Fig. 2.1(b)) can be used as the basis for calculation of flows.

Applied to a critical-center valve of Fig. 2.2(a) a laminar flow at small port openings will occur with a parabolic relation between Q and x according to eqn. (5). At $x = h_T$ the critical Reynolds number Re_{cr} will be passed, resulting in a constant flow gain according to eqn. (2). Fortunately, the poor flow gain around $x = 0$ is restricted to a small area. Equations (2) - (6) yield

$$h_T = \frac{v Re_{cr}}{2C_{dturb} \sqrt{2\Delta P/\rho}} \quad (7)$$

Letting, for example, $v = 30 \times 10^{-6} \text{ m}^2/\text{s}$, $Re_{cr} = 20$, $C_d = 0.611$, $\Delta P = 50 \times 10^5 \text{ N/m}^2$ and $\rho = 900 \text{ kg/m}^3$, we find

$$h_T = 4.66 \times 10^{-6} \text{ m} = 4.66 \mu\text{m}.$$

The low flow gain area could be restricted in proportion to the whole area by orifices with a smaller width b , as shown in Fig. 2.2(b). Here h_T is the same as before, but owing to the smaller width $b < \pi d$ it takes place at a lower value of the flow. Thus as a percentage of the maximum flow or valve stroke the low gain area is restricted considerably!

In addition it is possible to introduce a very small underlap U in our "critical-center" valve. Assuming inlet and outlet openings $a_1 = bh_1 = b(U+x)$, $a_2 = bh_2 = b(U-x)$, and pressure drops $\Delta P_1 = \Delta P_2 = \Delta P$, we find in the underlap-region

$$Q_1 - Q_2 = \frac{4\delta^2}{\mu} b(h_1^2 \Delta P_1 - h_2^2 \Delta P_2) = \frac{16\delta^2}{\mu} b\Delta P U x$$

So

$$\frac{\partial(Q_1 - Q_2)}{\partial x} = 16 \frac{\delta^2}{\mu} b\Delta P U = \text{constant!!} \quad (8)$$

Requiring this flow gain to be equal to the turbulent flow gain at $x > h_T$ according to eqn. (2), we find

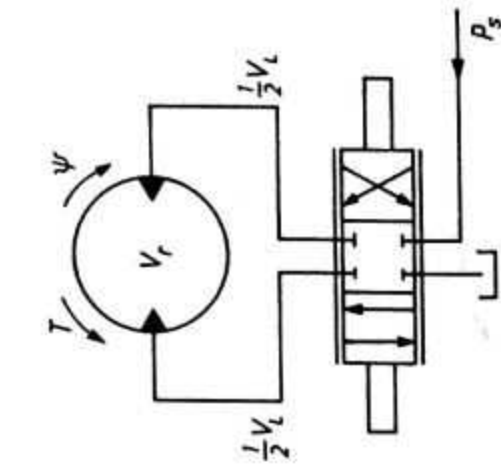
$$U = \frac{vRe_{cr}}{8C_{dturb} \sqrt{2\Delta P/\rho}} = \frac{h_T}{4} \quad (9)$$

Thus - in the example already given - even at an extremely small underlap $U = 1.16\mu\text{m}$, a very good constant flow gain is obtained throughout the whole working range of the valve. This is confirmed by manufacturer's practice, where an underlap of a few microns is usual to provide a constant flow gain. This in spite of the fact that the assumptions leading to eqn. (8) are not fulfilled automatically.

Concluding this section, we state that it is permitted in normal practice to use the turbulent flow equation (2) regardless(!!) of the Reynolds number if we bear in mind that at very small port openings considerable deviations in the calculated flow gain may occur.

(d)

Linear/Rotary
Symmetric
4-way

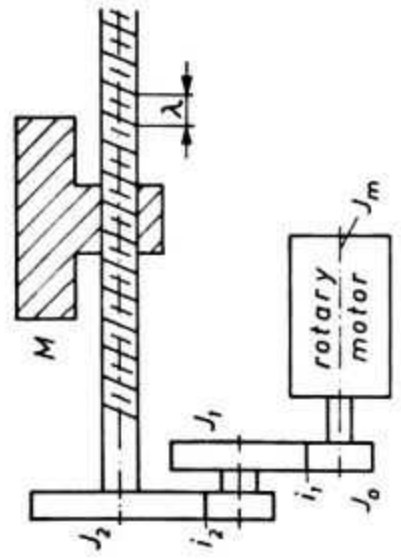


$$c_o = \frac{4V_r^2 E}{\pi V_r + V_L} \quad \left[\frac{Nm}{rad} \right]$$

$$V_L > V_r \quad ?!$$

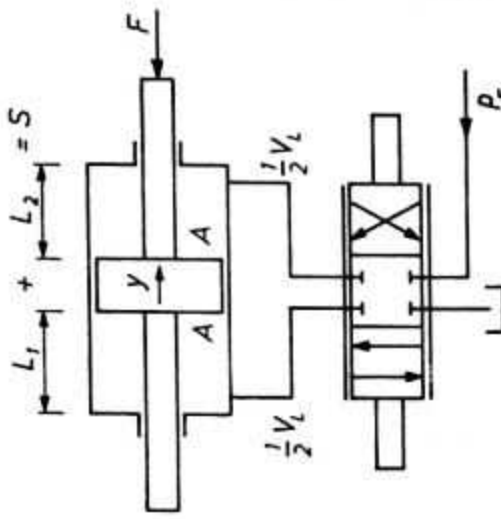
$$\omega_o = \sqrt{\frac{c_o}{J_{tot}}}$$

$$J_{tot} = J_m + J_o + \frac{J_1}{i_1^2} + \frac{J_2}{i_2^2} + M \left[\frac{\lambda}{2\pi i_1 i_2} \right]^2$$



(c)

Linear
Asymmetric
3-way / 4-way



$$c_o = \frac{A^2 E}{AL_1 + \frac{1}{2} V_L} + \frac{A^2 E}{AL_2 + \frac{1}{2} V_L}$$

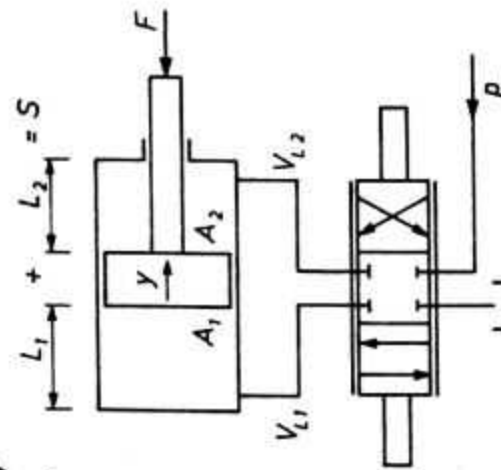
$$c_{o min} = \frac{4A^2 E}{AS + V_L}$$

If $V_L \ll AS$:

$$c_{o min} = \frac{4AE}{S} \quad \left[\frac{N}{m} \right]$$

$$\omega_o = \sqrt{\frac{c_o}{M}} \geq 2 \sqrt{\frac{AE}{MS}}$$

(b)



$$c_o = \frac{A_1^2 E}{A_1 L_1 + V_{L1}} + \frac{A_2^2 E}{A_2 L_2 + V_{L2}}$$

$$c_{o min} = \frac{E(\sqrt{A_1} + \sqrt{A_2})^2}{S + \frac{V_{L1}}{A_1} + \frac{V_{L2}}{A_2}}$$

If $V_{L1} \ll A_1 S, V_{L2} \ll A_2 S$:

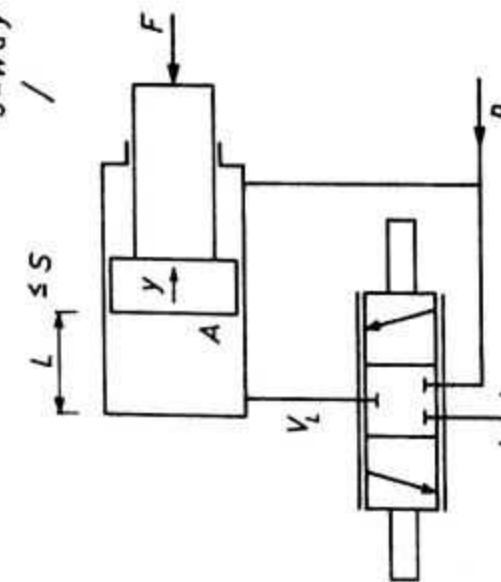
$$c_{o min} = \frac{E(\sqrt{A_1} + \sqrt{A_2})^2}{S} \quad \left[\frac{N}{m} \right]$$

If $A_1 \leq 5A_2$

$$c_{o min} \approx \frac{2E(A_1 + A_2)}{S}$$

$$\omega_o \geq \sqrt{\frac{2E(A_1 + A_2)}{MS}}$$

(a)



$$c_o = \frac{A^2 E}{AL + V_L}$$

$$c_{o min} = \frac{A^2 E}{AS + V_L}$$

If $V_L \ll AS$:

$$c_{o min} = \frac{AE}{S} \quad \left[\frac{N}{m} \right]$$

$$\omega_o = \sqrt{\frac{c_o}{M}} \geq \sqrt{\frac{AE}{MS}}$$

Spring stiffness and natural frequency

2.2 Compressibility of oil

It is quite surprising to outsiders that oil compressibility influences the dynamics of hydraulic servo's significantly. Hydraulic fluids such as oil, even though far less compressible than gas, when compressed in a cylinder compartment act like a spring and introduce therefore a second order mass-spring system whose natural frequency limits the bandwidth of any hydraulic servo abruptly. The compressibility of a hydraulic fluid, trapped in a volume V under pressure P is defined by

$$\frac{\Delta V}{V} = - \frac{\Delta P}{E} \quad (10)$$

E is called the bulk modulus, often indicated by B or β . The bulk modulus has the same dimension (N/m^2) as Young's modulus of elasticity for metals. For reasons of analogy it has been given the same symbol E . For mineral oils, E may be as high as 15,000 bar ($15 \times 10^8 \text{ N/m}^2$). Its effective value in practice, however, is much lower, mainly due to a small percentage of entrained air, which in fact cannot be avoided. However, bad practice in reservoirs (return line splashing, small surface area, clogged suction filters, etc. etc.) as well as blind holes and pockets in the hydraulic system itself could reduce E to less than 10% of its theoretical value! In good engineering practice, values between 6,000 and 12,000 bar can be expected.

Elasticity of steel cylinders and tubing has some effect on the effective E , but could normally be neglected. However, the elasticity of (high pressure) hoses could have a disastrous effect on the effective bulk modulus E .

The oil spring stiffness c_o of various servosystems depicted in Fig. 2.3 has been derived on the basis of a determined bulk modulus E . In Fig. 2.3(a), in terms of eqn. (10), $\Delta V = yA$ and $V = AL + V_L$, where AL is the effective volume of the cylinder compartment and V_L the ineffective one. The pressure increase ΔP is caused by the load $F = A\Delta P$. Thus we find

$$\frac{yA}{AL + V_L} = - \frac{F}{AE} \text{ and } c_o = - \frac{F}{y} = \frac{A^2 E}{AL + V_L} \quad (11)$$

The minus sign is due to conventional (opposing!) positive directions of y and F . In Figures 2.3(b) and (c) the values of c_o have been found by adding up the oil spring stiffnesses on both sides of the piston.

The linear motors (a), (b) and (c) all show a stiffness c_o heavily depending on the piston position. As the lowest possible natural frequency $\omega_o = \sqrt{c_o/M}$ is

most important, c_o min has been calculated for every motor in Fig. 2.3

A special case is the rotary motor of Fig. 2.3(d). Its rotational oil spring stiffness (in Nm/rad) could be derived by analogy with the symmetric linear motor, replacing the piston area A by the motordisplacement V_r per radian and by replacing the stroke S by π radians (half a revolution). Of course c_o can be derived directly also by letting:

$$\text{Compartment volumes: } V_{1,2} = \frac{\pi}{2} V_r + \frac{V_L}{2}, \text{ instead of } V_{1,2} = \frac{S}{2} A + \frac{V_L}{2}.$$

$$\text{Flows} \quad : Q_1 - Q_2 = \dot{\psi} V_r, \text{ instead of } Q_1 - Q_2 = \dot{y} A.$$

$$\text{Load} \quad : T = V_r (P_1 - P_2), \text{ instead of } F = A (P_1 - P_2).$$

Relevant in many applications is the fact that the "ineffective" line volume $0.5V_L$ cannot be neglected with respect to the effective volume $0.5\pi V_r$. Thus one has to be extremely careful in calculating or estimating c_o for rotary motors.

A second complication in rotary motors with their usual transmission is the inertia "felt by the motor". The inertia of gear trains etc. reduced to the motor axis is summarized also in Fig. 2.3(d). Usually the "additional inertia" is 5 - 10 times larger than the inertia J_M of the rotary motor itself. It is therefore almost an understatement to say that careful design of gear trains, etc. is worthwhile in order to obtain acceptable values of the natural frequency

$$\omega_o = \sqrt{c_o / J_{tot}}.$$

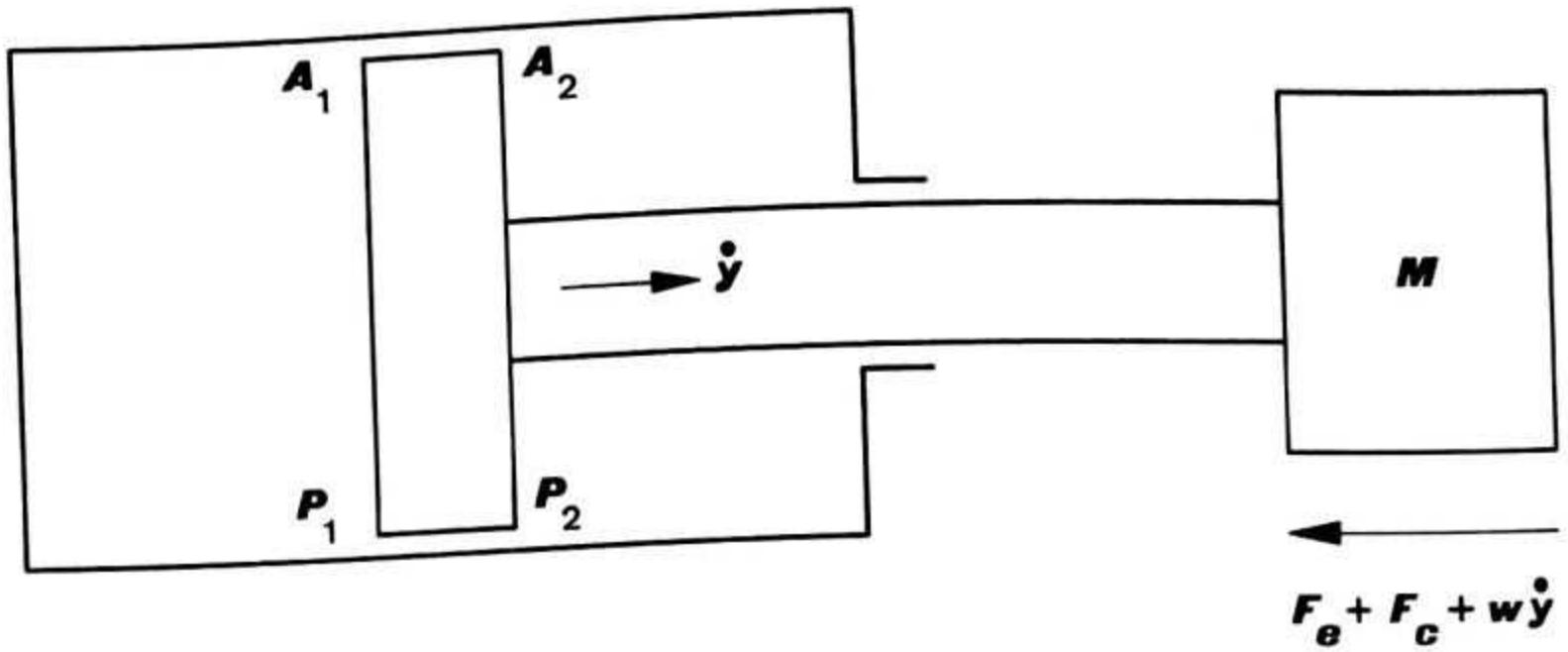


Fig. 2.4 Forces

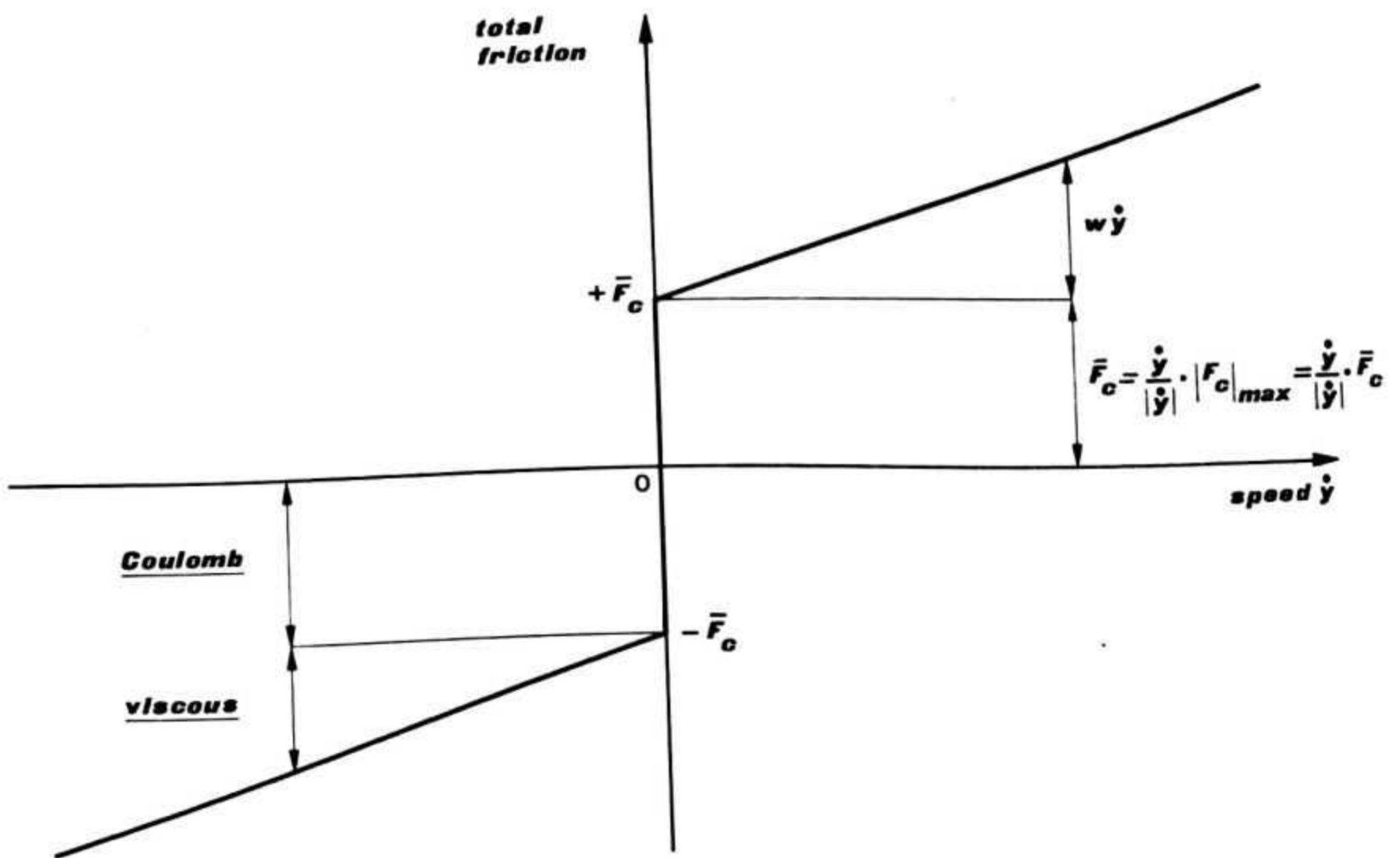


Fig. 2.5 Coulomb friction and viscous friction

2.3 Forces and continuity

The complex situation regarding forces and continuity will be treated here for the most general case of an asymmetric linear motor ($A_1 \neq A_2$) and an asymmetric 4-way valve ($a_1 \neq a_3, a_2 \neq a_4$), see Fig. 2.4. Other cases are far more simple and their results could be found from the general case quite easily.

The most important forces are listed below according to origin and briefly discussed.

- (1) *Oil pressures* P_1 and P_2 causing opposing forces, $P_1 A_1$ and $P_2 A_2$ (see Fig. 2.4).
- (2) *External forces* F_e coming from outside the hydraulic motor, for instance cutting forces of a machine tool.
- (3) *Frictional forces*, opposing the speed $\dot{y}$. One must distinguish between (linear) viscous friction and highly non-linear Coulomb friction or dry friction.
- (4) *Coulomb friction* is the most magic non-linearizing factor in hydraulic servo's and is always suspect when something irregular is happening! In normal practice, Coulomb friction can be 5-15% of the maximum load-capacity. Thus it cannot be neglected without serious consequences, as is done in many textbooks and manufacturers' catalogs. In this book the simplifying assumption is made that in a moving motor Coulomb friction is constant, reversing its sign with the speed $\dot{y}$ (see Fig. 2.5). In spite of some misleading publications it should be stressed here (and will be proved later on) that Coulomb friction itself cannot provide adequate stability in practice.
- (5) *Viscous friction* is proportional to the speed $\dot{y}$ and contributes to stability and damping in many hydraulic systems. Normally, it cannot be predicted sufficiently, its amount cannot be accurately designed and direct measurement is hardly possible. Nevertheless its presence can be proved indirectly. Numerous experiments show a degree of system damping for which the assumption of viscous friction is the only explanation. Thus it should never be neglected in dynamic analysis.

Our experiments showed too, in an indirect way, a strong correlation between viscous and Coulomb friction. Hydrostatic bearings in hydraulic cylinders for instance eliminate not only the Coulomb friction (which is the purpose!) but also the viscous friction, making additional damping measures necessary!

Summarizing so far, we assume a simple (but complicated enough) friction model depicted in Fig. 2.5

$$F_{\text{friction}} = F_c + w\dot{y} = \frac{\dot{y}}{|\dot{y}|} |F_c|_{\text{max}} + w\dot{y}.$$

Combining previous forces of Fig. 2.4 with Newton's law, we get

$$P_1 A_1 - P_2 A_2 - F_e - F_c - w\dot{y} = M\ddot{y}$$

or

$$P_1 A_1 - P_2 A_2 = F_e + F_c + w\dot{y} + M\ddot{y}.$$

Now, defining the total load ΣF to be

$$\Sigma F = F_e + F_c + w\dot{y} + M\ddot{y},$$

we get finally

$$P_1 A_1 - P_2 A_2 = \Sigma F = F_e + F_c + w\dot{y} + M\ddot{y}.$$

So the relation between the oil pressures P_1 and P_2 depends on the total load ΣF . The pressures P_1 and P_2 in turn determine the pressure drops across the port openings in the valve. Using

$$Q = av = aC_d \sqrt{\frac{2\Delta P}{\rho}}$$

the flows through the openings a_1 , a_2 , a_3 and a_4 in Fig. 2.6 become

$$Q_1 = a_1 C_d \sqrt{2 \frac{P_s - P_1}{\rho}}, \quad Q_2 = a_2 C_d \sqrt{\frac{2P_1}{\rho}},$$

$$Q_3 = a_3 C_d \sqrt{\frac{2P_2}{\rho}}, \quad Q_4 = a_4 C_d \sqrt{2 \frac{P_s - P_2}{\rho}}.$$

Considering the compressibility of oil (according to Fig. 2.3) as well as continuity we get two (!) equations for the speed $\dot{y}$ of the motor

$$\dot{y} = \frac{Q_1 - Q_2}{A_1} - \frac{A_1 L_1 + V_{L1}}{A_1 E} \dot{P}_1$$

$$\dot{y} = \frac{Q_3 - Q_4}{A_2} + \frac{A_2 L_2 + V_{L2}}{A_2 E} \dot{P}_2$$

Combining previous equations and neglecting V_{L1} and V_{L2} we get finally

$$\dot{y} = C_d \sqrt{\frac{P_s}{\rho}} \left[\frac{a_1}{A_1} \sqrt{2(1 - \frac{P_1}{P_s})} - \frac{a_2}{A_1} \sqrt{\frac{2P_1}{P_s}} \right] - \frac{L_1}{E} \ddot{P}_1 \quad (12)$$

$$\dot{y} = C_d \sqrt{\frac{P_s}{\rho}} \left[\frac{a_3}{A_2} \sqrt{\frac{2P_2}{P_s}} - \frac{a_4}{A_2} \sqrt{2(1 - \frac{P_2}{P_s})} \right] + \frac{L_2}{E} \ddot{P}_2 \quad (13)$$

where

$$P_1 A_1 - P_2 A_2 = \Sigma F = F_e + F_c + w\dot{y} + M\ddot{y}. \quad (14)$$

From these equations, P_1 and P_2 as well as $\dot{y}$ have to be solved. In general, analytical solutions are impossible. A special solution could be found only for very special conditions.

Any solution must fulfil the requirement that the speed $\dot{y}$ according to eqns. (12) and (13) are the same. A special solution can easily be found by equalizing all corresponding factors in eqns. (12) and (13). The "conditions" are

$$\frac{a_1}{A_1} = \frac{a_3}{A_2} \quad \text{and} \quad \frac{a_2}{A_1} = \frac{a_4}{A_2} \quad (15)$$

$$L_1 = L_2 = \frac{S}{2} \quad (16)$$

$$P_1 + P_2 = P_s. \quad (17)$$

Condition (15) concerns valve and motor dimensions and could be fulfilled theoretically. Condition (16) restricts the solution to only one position of the piston. Condition (17) together with eqn. (14) yields

$$\frac{P_1}{P_s} = \frac{A_2 + \Sigma F/P_s}{A_1 + A_2} \quad \text{and} \quad \frac{P_2}{P_s} = \frac{A_1 - \Sigma F/P_s}{A_1 + A_2}. \quad (18)$$

Ⓐ Asymmetric linear motor Three-way valve Forces $P_c A - P_s \frac{A}{2} - F_e - F_c - w\ddot{y} = M\ddot{y}$ $P_c A - P_s \frac{A}{2} = F_e + F_c + w\ddot{y} + M\ddot{y} = \Sigma F$ $P_c = \frac{1}{2} P_s + \frac{\Sigma F}{A} = \frac{1}{2} P_s + P_L$	Ⓑ Asymmetric linear motor Four-way valve Forces $P_1 A_1 - P_2 A_2 = \Sigma F = F_e + F_c + w\ddot{y} + M\ddot{y}$	Ⓒ Symmetric linear motor Symmetric four-way valve Forces $(P_1 - P_2)A = \Sigma F = F_e + F_c + w\ddot{y} + M\ddot{y}$ $P_1 - P_2 = \frac{\Sigma F}{A} = P_L$	Ⓓ Symmetric rotary motor Symmetric four-way valve Torque $(P_1 - P_2)V_r = \Sigma T = T_e + T_c + w\dot{\psi} + J\ddot{\psi}$ $P_1 - P_2 = \frac{\Sigma T}{V_r} = P_L$
Flows $Q_1 = a_1 C_d \sqrt{2 \frac{P_s - P_c}{\rho}}$ $Q_2 = a_2 C_d \sqrt{2 \frac{P_c}{\rho}}$ $Q_1 - Q_2 = a_1 C_d \sqrt{2 \frac{P_s - P_c}{\rho}} - a_2 C_d \sqrt{2 \frac{P_c}{\rho}}$	Flows $Q_1 - Q_2 = a_1 C_d \sqrt{2 \frac{P_s - P_1}{\rho}} - a_2 C_d \sqrt{2 \frac{P_1}{\rho}}$ $Q_3 - Q_4 = a_3 C_d \sqrt{2 \frac{P_2}{\rho}} - a_4 C_d \sqrt{2 \frac{P_s - P_2}{\rho}}$	Flows $Q_L = \frac{P_1 - P_2}{R} = \frac{P_L}{R}$ $Q_1 - Q_2 = a_1 C_d \sqrt{2 \frac{P_s - P_1}{\rho}} - a_2 C_d \sqrt{2 \frac{P_1}{\rho}}$ $Q_3 - Q_4 = a_3 C_d \sqrt{2 \frac{P_2}{\rho}} - a_4 C_d \sqrt{2 \frac{P_s - P_2}{\rho}}$	Flows $Q_m = \dot{y} A = Q_1 - Q_2 - Q_L - \frac{V_1}{E} \cdot \dot{P}_1 = \dot{\psi} V_r$ $Q_m = \dot{y} A = Q_3 - Q_4 - Q_L + \frac{V_2}{E} \cdot \dot{P}_2 = \dot{\psi} V_r$
Motordisplacement $Q_m = \dot{y} A = Q_1 - Q_2 + \frac{d}{dt} \Delta V_1$ $= Q_1 - Q_2 - V_1 \cdot \frac{d}{dt} \frac{\Delta P_s}{E}$ $Q_m = \dot{y} A = Q_1 - Q_2 - \frac{V_1}{E} \cdot \dot{P}_L$	Motordisplacement $Q_1 - Q_2 - \frac{V_1}{E} \cdot \dot{P}_1 = \dot{y} A$ $Q_3 - Q_4 + \frac{V_2}{E} \cdot \dot{P}_2 = \dot{y} A$	Motordisplacement $Q_m = \dot{y} A = Q_1 - Q_2 - Q_L - \frac{V_1}{E} \cdot \dot{P}_1 = \dot{\psi} V_r$ $Q_m = \dot{y} A = Q_3 - Q_4 - Q_L + \frac{V_2}{E} \cdot \dot{P}_2 = \dot{\psi} V_r$	Motordisplacement $Q_m = \dot{\psi} V_r = Q_1 - Q_2 - Q_L - \frac{V_1}{E} \cdot \dot{P}_1 = \dot{\psi} V_r$ $Q_m = \dot{\psi} V_r = Q_3 - Q_4 - Q_L + \frac{V_2}{E} \cdot \dot{P}_2 = \dot{\psi} V_r$
Solution $Q_m = C_d \sqrt{\frac{P_s}{\rho}} \left[a_1 \sqrt{1 - 2 \frac{P_L}{P_s}} - a_2 \sqrt{1 + 2 \frac{P_L}{P_s}} \right] - \frac{V_1}{E} \cdot \dot{P}_L$ $\dot{y} A \approx C_d \sqrt{\frac{P_s}{\rho}} \left[a_1 - a_2 - (a_1 + a_2) \frac{P_L}{P_s} \right] - \frac{V_1}{E} \cdot \dot{P}_L$	Solution $Q_m = C_d \sqrt{\frac{P_s}{\rho}} \left[a_1 \sqrt{1 - 2 \frac{P_L}{P_s}} - a_2 \sqrt{1 + 2 \frac{P_L}{P_s}} \right] - \frac{V_1}{E} \cdot \dot{P}_L$ $\dot{y} A \approx C_d \sqrt{\frac{P_s}{\rho}} \left[a_1 - a_2 - (a_1 + a_2) \frac{P_L}{P_s} \right] - \frac{V_1}{E} \cdot \dot{P}_L$	Solution $Q_m = \dot{y} A = C_d \sqrt{\frac{P_s}{\rho}} \left[a_1 \sqrt{1 - \frac{P_L}{P_s}} - a_2 \sqrt{1 + \frac{P_L}{P_s}} \right] - \frac{P_L}{R} - \frac{V_1}{2E} \cdot \dot{P}_L = \dot{\psi} V_r$ $Q_m = \dot{y} A \approx C_d \sqrt{\frac{P_s}{\rho}} \left[a_1 - a_2 - (a_1 + a_2) \cdot \frac{P_L}{2P_s} \right] - \frac{P_L}{R} - \frac{V_1}{2E} \cdot \dot{P}_L = \dot{\psi} V_r$	Solution $Q_m = \dot{\psi} V_r = C_d \sqrt{\frac{P_s}{\rho}} \left[a_1 \sqrt{1 - \frac{P_L}{P_s}} - a_2 \sqrt{1 + \frac{P_L}{P_s}} \right] - \frac{P_L}{R} - \frac{V_1}{2E} \cdot \dot{P}_L = \dot{\psi} V_r$ $Q_m = \dot{\psi} V_r \approx C_d \sqrt{\frac{P_s}{\rho}} \left[a_1 - a_2 - (a_1 + a_2) \cdot \frac{P_L}{2P_s} \right] - \frac{P_L}{R} - \frac{V_1}{2E} \cdot \dot{P}_L = \dot{\psi} V_r$

Fig. 2.7 Survey of hydraulic servomotors

Substituting eqns. (15), (16) and (18) into eqn. (12) gives

$$\dot{y} = c_d \sqrt{\frac{2P_s}{\rho}} \left[\frac{a_1}{A_1} \sqrt{\frac{A_1 - \Sigma F/P_s}{A_1 + A_2}} - \frac{a_2}{A_1} \sqrt{\frac{A_2 + \Sigma F/P_s}{A_1 + A_2}} \right] - \frac{\dot{\Sigma F}}{c_o} \quad (19)$$

where $c_o = 2E(A_1 + A_2)/S$.

As is made clear, eqn. (19) represents only one special solution, being valid if conditions (15) and (16) are met. The question is whether or not this solution is a good approximation if the conditions are not fulfilled. In the literature this question is avoided systematically, tacitly assuming that it is not important whether the conditions are met or not.

It will be proved in Chapter 3 that eqns. (18) and (19) do not even show the slightest resemblance to physical reality if condition (15) is not fulfilled, suggesting that an asymmetric motor needs an asymmetric valve. It will be proved that this suggestion is correct but this is not too helpful since asymmetric valves normally are not available commercially.... So special attention is needed to solve this problem.

Symmetric motors with symmetric valves do not cause complications.

Relating the openings a_1 and a_2 to the spool displacement x linearization and approximation will lead to a handsome differential equation that will permit dynamic analysis, synthesis and design. This is done in the following chapters for several types of hydraulic servomotors. Figure 2.7 gives a fundamental survey of hydraulic servomotors as a "summary in advance" in which the above tools for analysis have been applied and the linearization carried out.

REFERENCES

- [1] R. von Mises, Berechnung von Ausfluss- und Ueberfallzahlen, Z. Ver. Deutsch. Ing., 61 (1917) 447.
- [2] T.J. Viersma, Investigations into the accuracy of hydraulic servomotors, Thesis, Delft University of Technology, (1961). Philips Res. Rep. 16 (1961) 507; 17, (1962) 20.
- [3] W. Wuest, Strömung durch Schlitz- und Lochblenden bei kleiner Reynolds-Zahlen, Ing. Arch., 22 (1954) 357

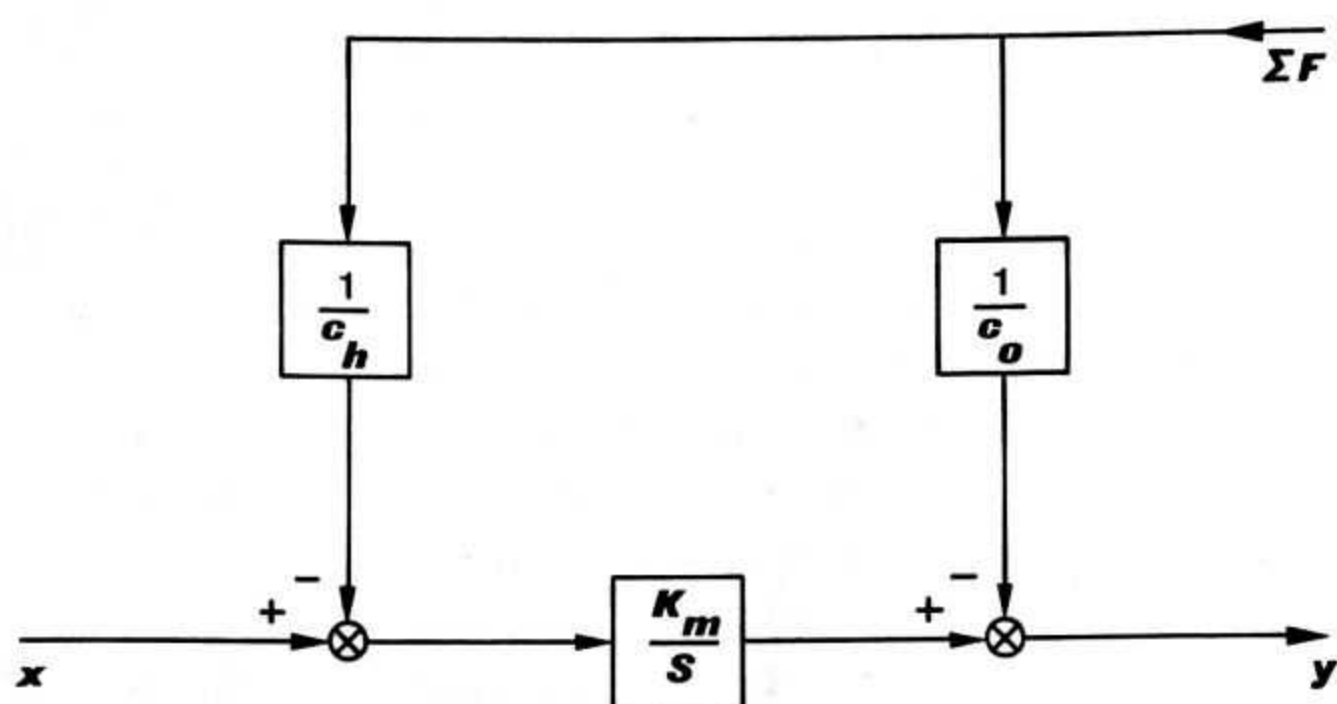


Fig. 3.1 Simple block diagram for the open loop of an hydraulic servo

CHAPTER 3

ANALYSIS OF HYDRAULIC SERVOSYSTEMS

3.1 Introduction

There is a large variety of hydraulic motors and valves to be combined. We distinguish (see also Fig. 2.7):

- (1) Asymmetric linear motor ($A_1 \approx 2A_2$) controlled by a 3-way valve.
- (2) Asymmetric linear motor ($A_1 \neq A_2$) controlled by a symmetric 4-way valve.
- (3) Asymmetric linear motor ($A_1 \neq A_2$) controlled by an asymmetric 4-way valve ($a_1 \neq a_3, a_2 \neq a_4$).
- (4) Symmetric linear motor ($A_1 = A_2$) controlled by a symmetric 4-way valve ($a_1 = a_3, a_2 = a_4$).
- (5) Symmetric rotary motor controlled by a symmetric 4-way valve.

In each of these cases one can distinguish open-center as well as critical-center valves having a positive underlap U and zero lap $U = 0$ respectively. So this analysis could be extended ten times, but this would of course be rather laborious. Therefore the approach in this chapter will be the following.

First the most simple case of an asymmetric linear motor controlled by a 3-way valve will be treated, both for an open-center valve and for a critical-center valve. After linearization and approximation a universal model according to the block diagram shown in Fig. 3.1 will be derived, determining the hydraulic load-stiffness c_h and the velocity gain $K_m = \partial \dot{y} / \partial x$ of the motor-valve combination itself.

To avoid repeating this process endlessly, some short-cuts are made. As 3-way-valve-controlled asymmetric linear motors and 4-way-valve-controlled symmetric motors (linear *and* rotary) differ from each other only by a factor 2 in the load influence (c_h is doubled in symmetric motors, as there are 2 compartments where the pressure can react on the load), open-center valves are treated only in connection with 3-way-valve-controlled asymmetric motors. In all other cases only critical-center valves (having $a_1 = a_2 = a_3 = a_4 = 0$ at $x = 0$) are treated. Pragmatically speaking, (electro-) hydraulic servovalves commercially available normally have zero lap, i.e. critical-center valve. A (sometimes exaggerated) reason for this is the absence of oil consumption in the neutral state

($x = 0, \dot{y} = 0$), which is important in many applications.

However, underlap-valves also have advantages (mainly because systems behavior is far more linear) resulting in good constant values of system parameters. Advanced combinations of feedforward and feedback could result in superior system behavior, especially around $x = 0$ and $\dot{y} = 0$.

After this introduction, we begin with the analysis of various servomotors, aiming at the hydraulic stiffness c_h and the velocity gain $K_m = \partial \dot{y} / \partial x$. Introducing the oil spring stiffness c_o according to Fig. 2.3, we get

$$\dot{y} = K_m \left(x - \frac{\Sigma F}{c_h} \right) - \frac{\dot{\Sigma F}}{c_o}. \quad (1)$$

The mathematical model will be completed in the last section of this chapter, deriving the system transfer function needed for dynamical analysis and synthesis.

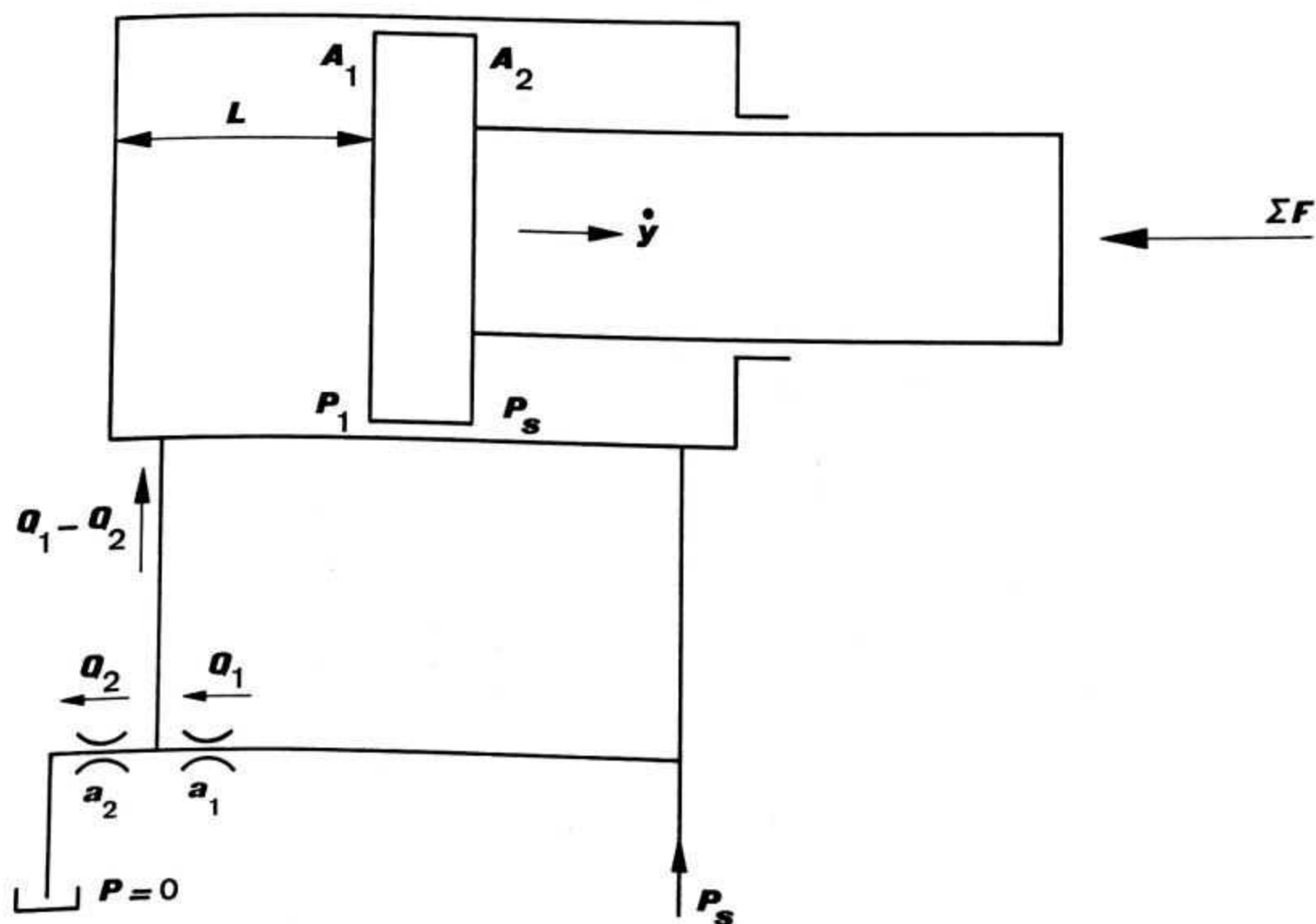


Fig. 3.2 Asymmetric motor controlled by a 3-way valve

3.2 Asymmetric motors controlled by 3-way valves

The most simple servomotors are asymmetric motors controlled by 3-way valves (see Fig. 3.2). They are frequently used in machine tool copying systems with mechanical feedback. Assuming turbulent flow and using the basic formulas of Chapter 2 for the motor behavior, we find

$$P_1 A_1 - P_s A_2 = \Sigma F, \text{ so } \frac{P_1}{P_s} = \frac{A_2}{A_1} + \frac{\Sigma F}{P_s A_1},$$

$$Q_1 = a_1 C_d \sqrt{2 \frac{P_s - P_1}{\rho}} = a_1 C_d \sqrt{\frac{P_s}{\rho}} \sqrt{2 \frac{A_1 - A_2}{A_1} - 2 \frac{\Sigma F}{P_s A_1}},$$

$$Q_2 = a_2 C_d \sqrt{2 \frac{P_1}{\rho}} = a_2 C_d \sqrt{\frac{P_s}{\rho}} \sqrt{2 \frac{A_2}{A_1} + 2 \frac{\Sigma F}{P_s A_1}},$$

$$\dot{y} = \frac{Q_1 - Q_2}{A_1} - \frac{\dot{\Sigma F}}{c_o} = \frac{Q_1 - Q_2}{A_1} - \frac{L}{EA_1} \dot{\Sigma F}.$$

It can be seen that for $A_1/A_2 = 2$ the load-capacity is symmetric, meaning $\Sigma F_{\max} = -\Sigma F_{\min}$. Should ΣF contain a "dc-component" F_{st} (to be considered as a constant pre-load), an area ratio $A_2/A_1 = 0.5 - F_{st}/P_s A_1$ could compensate for it. Since the effect of a constant pre-load F_{st} will be treated again later on, we now assume $F_{st} = 0$ and choose therefore $A_1 = 2A_2 = A$.

At the same time we introduce the so-called *load-pressure*

$$P_L = \frac{\Sigma F}{A}, \text{ so } \frac{P_1}{P_s} = \frac{1}{2} + \frac{\Sigma F}{AP_s} = \frac{1}{2} + \frac{P_L}{P_s}.$$

Now the flows become

$$Q_1 = a_1 C_d \sqrt{\frac{P_s}{\rho}} \sqrt{1 - \frac{2P_L}{P_s}}, \quad Q_2 = a_2 C_d \sqrt{\frac{P_s}{\rho}} \sqrt{1 + \frac{2P_L}{P_s}}$$

and

$$\dot{y} = \frac{C_d}{A} \sqrt{\frac{P_s}{\rho}} \left[a_1 \sqrt{1 - \frac{2P_L}{P_s}} - a_2 \sqrt{1 + \frac{2P_L}{P_s}} \right] - \frac{L}{E} \dot{P}_L. \quad (2)$$

Here the speed $\dot{y}$ of the motor has been solved as a function of the variables a_1 , a_2 and $P_L = \Sigma F/A$. Equation (2) holds for all sorts of valves, open-center as well as critical-center valves. The only assumption made so far is turbulent port flow.

For the open-center valve of Fig. 3.3(a) we can write

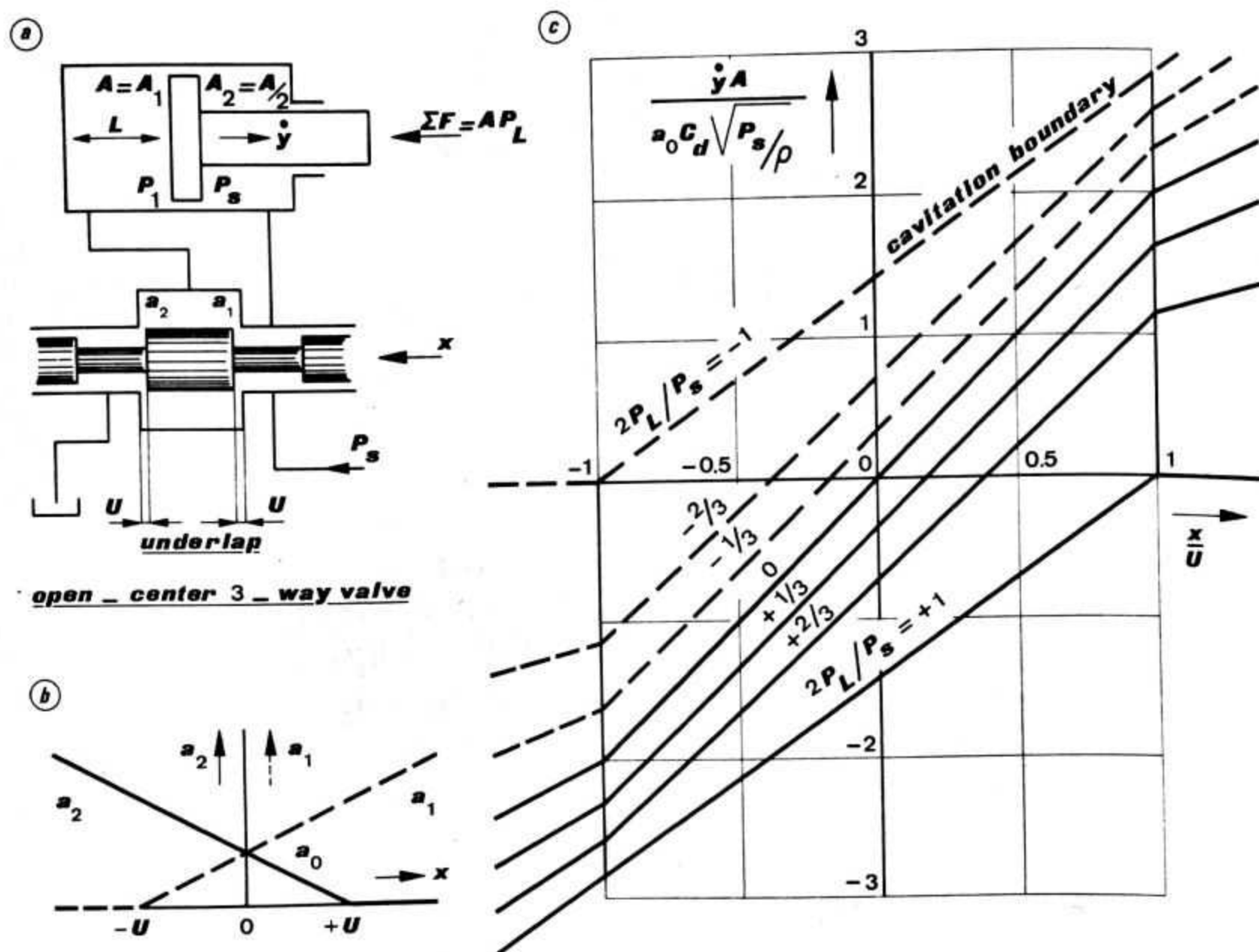


Fig. 3.3 Asymmetric motor controlled by an open-center 3-way valve.

The same velocity-gain characteristic holds for symmetric motors with a 4-way valve if $2P_L/P_s$ is replaced by P_L/P_s

$$\begin{aligned}
 x < -U & \quad a_1 = 0, \quad a_2 = \pi d(U-x) \\
 -U < x < +U & \quad a_1 = \pi d(U+x), \quad a_2 = \pi d(U-x) \\
 x > +U & \quad a_1 = \pi d(U+x), \quad a_2 = 0.
 \end{aligned}$$

Substitution into eqn. (2) yields with $a_0 = \pi dU$

$$\left. \begin{aligned}
 x < -U & \quad \dot{y} = \frac{a_0}{A} C_d \sqrt{\frac{P_s}{\rho}} \left[-\left(1 - \frac{x}{U}\right) \sqrt{1 + \frac{2P_L}{P_s}} \right] - \frac{L}{E} \dot{P}_L \\
 -U < x < +U & \quad \dot{y} = \frac{a_0}{A} C_d \sqrt{\frac{P_s}{\rho}} \left[\left(1 + \frac{x}{U}\right) \sqrt{1 - \frac{2P_L}{P_s}} - \left(1 - \frac{x}{U}\right) \sqrt{1 + \frac{2P_L}{P_s}} \right] - \frac{L}{E} \dot{P}_L \\
 x > +U & \quad \dot{y} = \frac{a_0}{A} C_d \sqrt{\frac{P_s}{\rho}} \left[\left(1 + \frac{x}{U}\right) \sqrt{1 - \frac{2P_L}{P_s}} \right] - \frac{L}{E} \dot{P}_L
 \end{aligned} \right\} \quad (3)$$

The velocity gain characteristic is drawn up in Fig. 3.3(c), showing a remarkably linear character within the underlap region $-U < x < +U$. The load is limited by pressure boundaries $0 < P_1 < P_s$ for obvious reasons. To have some safety margin, we require

$$\frac{1}{6} \leq \frac{P_1}{P_s} = \frac{1}{2} + \frac{P_L}{P_s} \leq \frac{5}{6}, \text{ so } \frac{|P_L|}{P_s} \leq \frac{1}{3}. \quad (4)$$

Now, approximating $\sqrt{1 \pm 2P_L/P_s}$ by $1 \pm P_L/P_s$ we get, *inside the underlap region*,

$$\begin{aligned}
 \dot{y} &= \frac{a_0}{A} C_d \sqrt{\frac{P_s}{\rho}} \left[\left(1 + \frac{x}{U}\right) \left(1 - \frac{P_L}{P_s}\right) - \left(1 - \frac{x}{U}\right) \left(1 + \frac{P_L}{P_s}\right) \right] - \frac{L}{E} \dot{P}_L \\
 &= \frac{2a_0}{AU} C_d \sqrt{\frac{P_s}{\rho}} \left[x - \frac{U}{P_s} P_L \right] - \frac{L}{E} \dot{P}_L.
 \end{aligned}$$

Introducing the portwidth $b = a_0/U = \pi dU/U = \pi d$ and substituting $\Sigma F = AP_L$ and $c_0 = EA/L$, we finally come to our "universal" equation

$$\dot{y} = K_m \left(x - \frac{\Sigma F}{c_h} \right) - \frac{\Sigma F}{c_0} \quad (1)$$

where

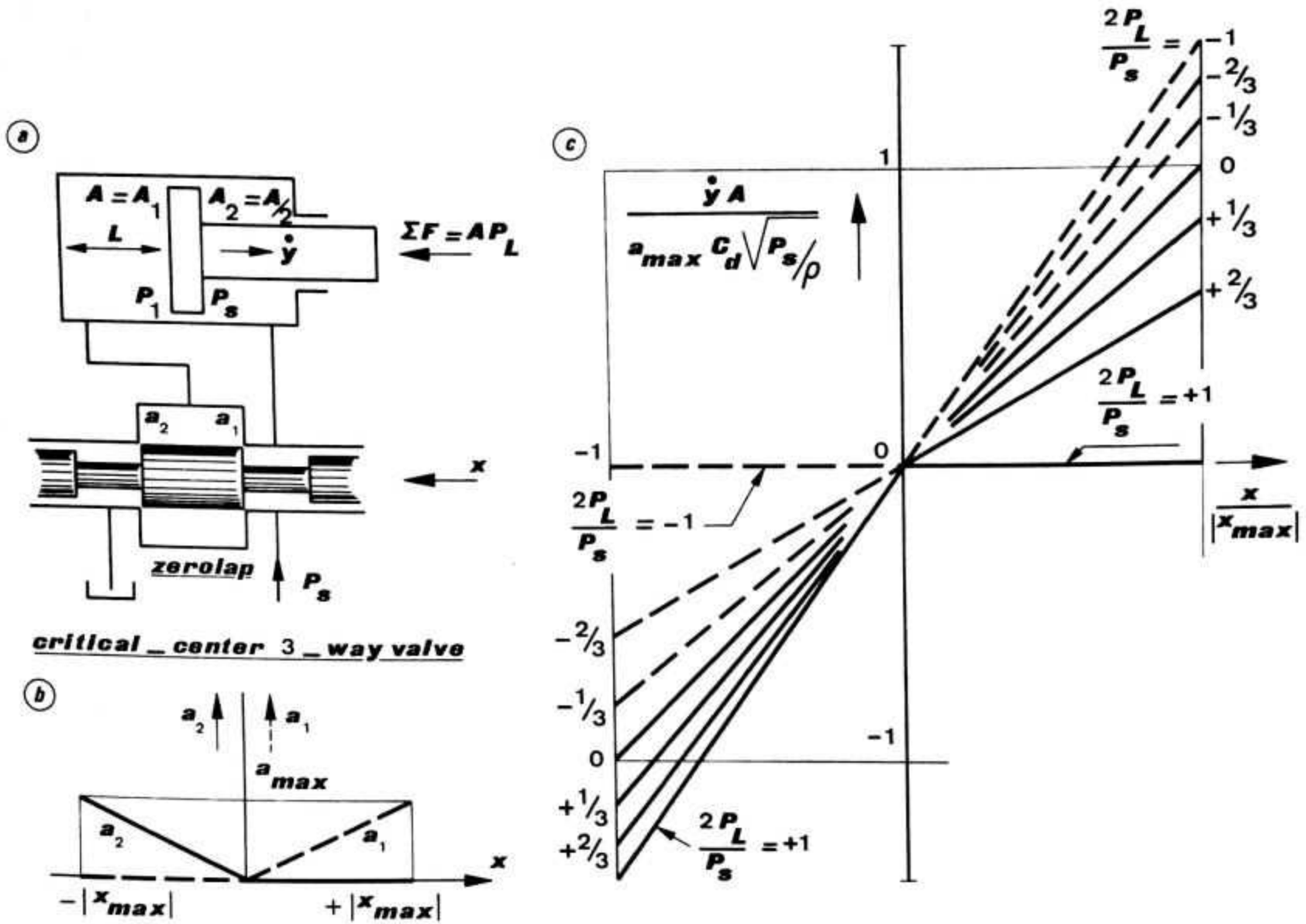


Fig. 3.4 Asymmetric motor controlled by a critical-center 3-way valve.

The same velocity-gain characteristic holds for symmetric motors with a 4-way valve if $2P_L/P_s$ is replaced by P_L/P_s

$$K_m = \frac{2b}{A} C_d \sqrt{\frac{P_s}{\rho}}, \quad c_h = \frac{AP_s}{U}, \quad c_o = \frac{EA}{L}. \quad (5)$$

For the *critical-center valve* of Fig. 3.4(a) we can write

$$x < 0 \quad a_1 = 0 \quad a_2 = - \left| \frac{a_{\max}}{x_{\max}} \right| x = -bx$$

$$x > 0 \quad a_1 = \left| \frac{a_{\max}}{x_{\max}} \right| x = bx \quad a_2 = 0$$

Substitution into eqn. (2) yields

$$\dot{y} = \frac{b}{A} C_d \sqrt{\frac{P_s}{\rho}} \left[x \sqrt{1 - \frac{|x|}{x_{\max}} \frac{2P_L}{P_s}} \right] - \frac{L}{E} \dot{P}_L. \quad (6)$$

The velocity gain characteristic is drawn up in Fig. 3.4(c), showing a far less linear character than Fig. 3.3(c). There is a *jump* at $x = 0$ in the velocity gain $\partial \dot{y} / \partial x$, depending on the load!

Thus approximating more roughly than before, we get

$$\dot{y} = \frac{b}{A} C_d \sqrt{\frac{P_s}{\rho}} \left[x - \frac{|x|}{P_s} P_L \right] - \frac{L}{E} \dot{P}_L.$$

Transformation again into eqn. (1) yields

$$K_m = \frac{b}{A} C_d \sqrt{\frac{P_s}{\rho}}, \quad c_h = \frac{AP_s}{|x|}, \quad c_o = \frac{EA}{L}. \quad (7)$$

It should be noted that in spite of the approximation the critical-center-valve-controlled motor is still a non-linear system, as the parameter c_h depends on the open loop input signal x !

Compared with hydraulic textbooks and manufacturers' catalogs, our presentation by means of velocity gain characteristics is rather unusual. Normally "valve characteristics" are presented, showing the flow as a function of the load, the valve displacement being a parameter. Using eqns. (3) and (7), the valve characteristics of open- and critical-center-valve-controlled motors have been worked out and are shown in Fig. 3.5. These characteristics make sense, as they show the capacity of the valve with respect to motor flow $\dot{y}A$ and the load pressure P_L . In principle, there is no difference between the two types of characteristics,

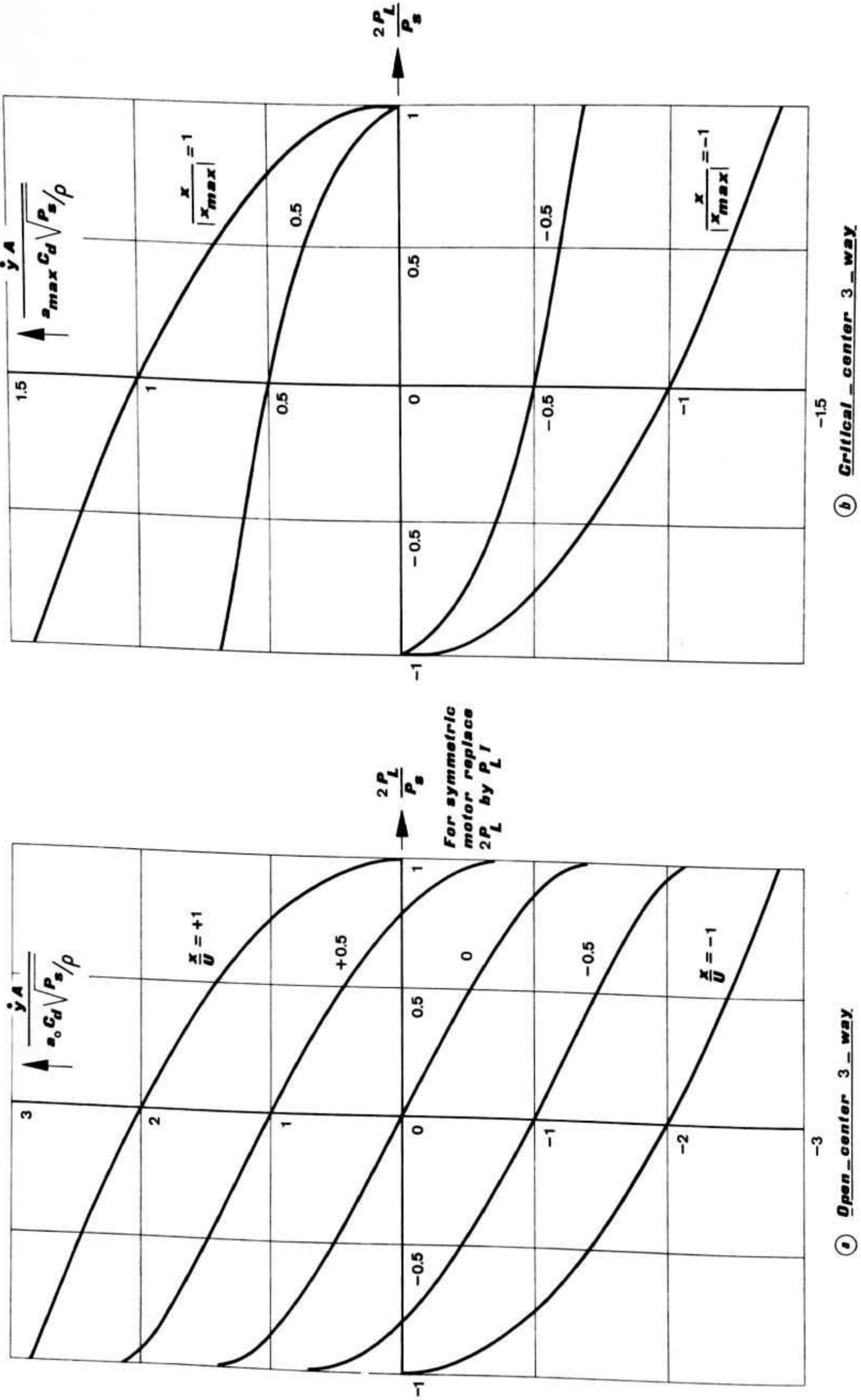


Fig. 3.5 Valve characteristics

because both cover exactly the same behavior. But, from the control point of view the velocity gain characteristic is far more important, as it directly indicates parameters that govern the control error and the dynamic properties of the system.

To investigate a hydraulic servo in practice, one should measure first of all the *velocity gain* K_m , which is proportional to the *slope of the straight lines* in the velocity gain characteristic. The curves of the valve characteristics of Fig. 3.5 do not indicate directly parameters significant for the control error and the dynamic system behavior. Irregularities in the velocity gain characteristic, such as

- decrease of $K_m = \partial \dot{y} / \partial x$ around $x = 0$, due to imperfect turbulent port flow (caused by wear of land edges), more or less compensated for by an increase of K_m due to a small underlap,
- dead zone, due to Coulomb friction,
- local jump in K_m due to backlash in the servovalve feedback path (see Section 4.6)

should be measured carefully, because imperfections could deteriorate the complete system, including its dynamic behavior.

Finally, we summarize some important differences between open- and critical-center valve controlled asymmetric motors:

- (1) The oil consumption of a critical-center valve is proportional to the modulus of the motor speed. $Q_{\text{supply}} = 0.5A|\dot{y}|$. So at stand-still no oil supply is necessary. At an open-center valve the oil consumption is almost independent of the motor speed $\dot{y}$ as long as $|x| < U$. In applications with one single servomotor fed by a constant delivery pump, this is hardly a drawback because the energy will be dissipated anyway, either in the servovalve or in the pressure control valve keeping $P_s = \text{constant}$.
- (2) The velocity gain $K_m = \partial \dot{y} / \partial x$ of an open-center valve operating in the underlap region ($|x| < U$) is twice as large as K_m of a critical-center valve.
- (3) With regard to the load sensitivity the critical-center valve is superior as the hydraulic stiffness $c_h = P_s A / |x|$, especially around $x = 0$, is far better than its counterpart, the open-center valve having $c_h = P_s A / U$. However, the open-center system behaves perfectly linearly around $x = 0$. For this reason the load error could be compensated by feedforward, as is shown in the next section.

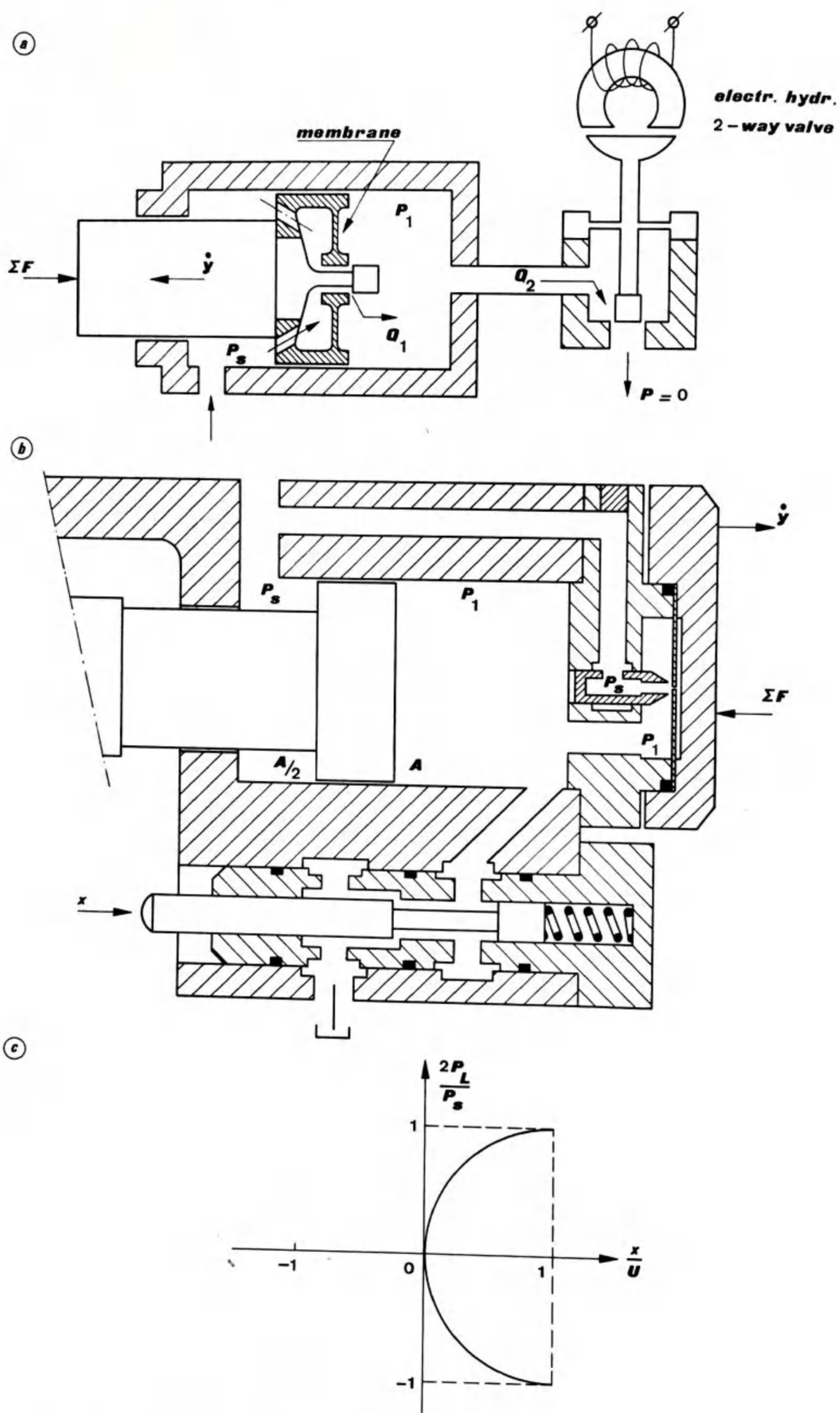


Fig. 3.6 Asymmetric load-compensation. The inlet openings of the systems a and b are influenced by the cylinder pressure $P_1 = 0.5 P_s + P_L$, causing $c_h = \infty$.

*3.3 Load-compensation

The load error $\epsilon = \Sigma F / c_h$ in an open-center-valve-controlled motor can be explained physically. Introducing a load ΣF in Fig. 3.6, the pressure P_1 will rise in order to maintain equilibrium of forces. Consequently the pressure drop $P_s - P_1$ across the inlet port will decrease and so does the inlet flow Q_1 . On the other hand, Q_2 will rise owing to increased pressure drop P_1 across the outlet port. So with $Q_1 < Q_2$, the system will get a speed, unless a change in a_1 and a_2 compensates for the flow due to the load. As $a_1 - a_2$ until now has been proportional to x and ϵ , the implication is a control error. From this diagnosis a therapy could be found. A change in the pressure P_1 (representing the load) must cause a change a_1 and/or a_2 independent of the control error signal ϵ . From eqn. (2) being derived without restrictions

$$\dot{y} = \frac{C_d}{A} \sqrt{\frac{P_s}{\rho}} \left[a_1 \sqrt{1 - \frac{2P_L}{P_s}} - a_2 \sqrt{1 + \frac{2P_L}{P_s}} \right]$$

concerning a_1 or a_2 , it can be seen that by making a_1 solely dependent on P_L and a_2 on x , as shown in Fig. 3.6, the first order effect of the load could be cancelled!

$$\dot{y} = \frac{C_d}{A} \sqrt{\frac{P_s}{\rho}} \left[a_o \left(1 + \frac{2P_L}{P_s}\right) \sqrt{1 - \frac{2P_L}{P_s}} - a_o \left(1 - \frac{x}{U}\right) \sqrt{1 + \frac{2P_L}{P_s}} \right]. \quad (8)$$

Linearizing around $P_L = 0$ and $x = 0$ results in $c_h = \infty$!! Examples of load-compensated systems that have been operating successfully are shown in Fig. 3.6. Figure 3.6(a) is part of a numerically controlled machine [1], [2], (b) of a copying lathe [3]. The compensation method applied in Fig. 3.6(a) and (b) is asymmetric (one port only) and therefore not fully effective at large loads, as can be seen from the "load characteristic" of Fig. 3.6(c) where $2P_L/P_s$ is plotted as a function of x/U at $\dot{y} = 0$. Substitution of $\dot{y} = 0$ in eqn. (8) yields

$$\left(\frac{2P_L}{P_s}\right)^2 + \left(1 - \frac{x}{U}\right)^2 = 1.$$

Only at small loads is the slope of the characteristic infinite, causing $c_h = \infty$. Symmetric load-compensation - though mechanically more complicated - gives better load characteristics. Here the two openings a_1 and a_2 both depend on P_L as well as on x . The optimal design results in

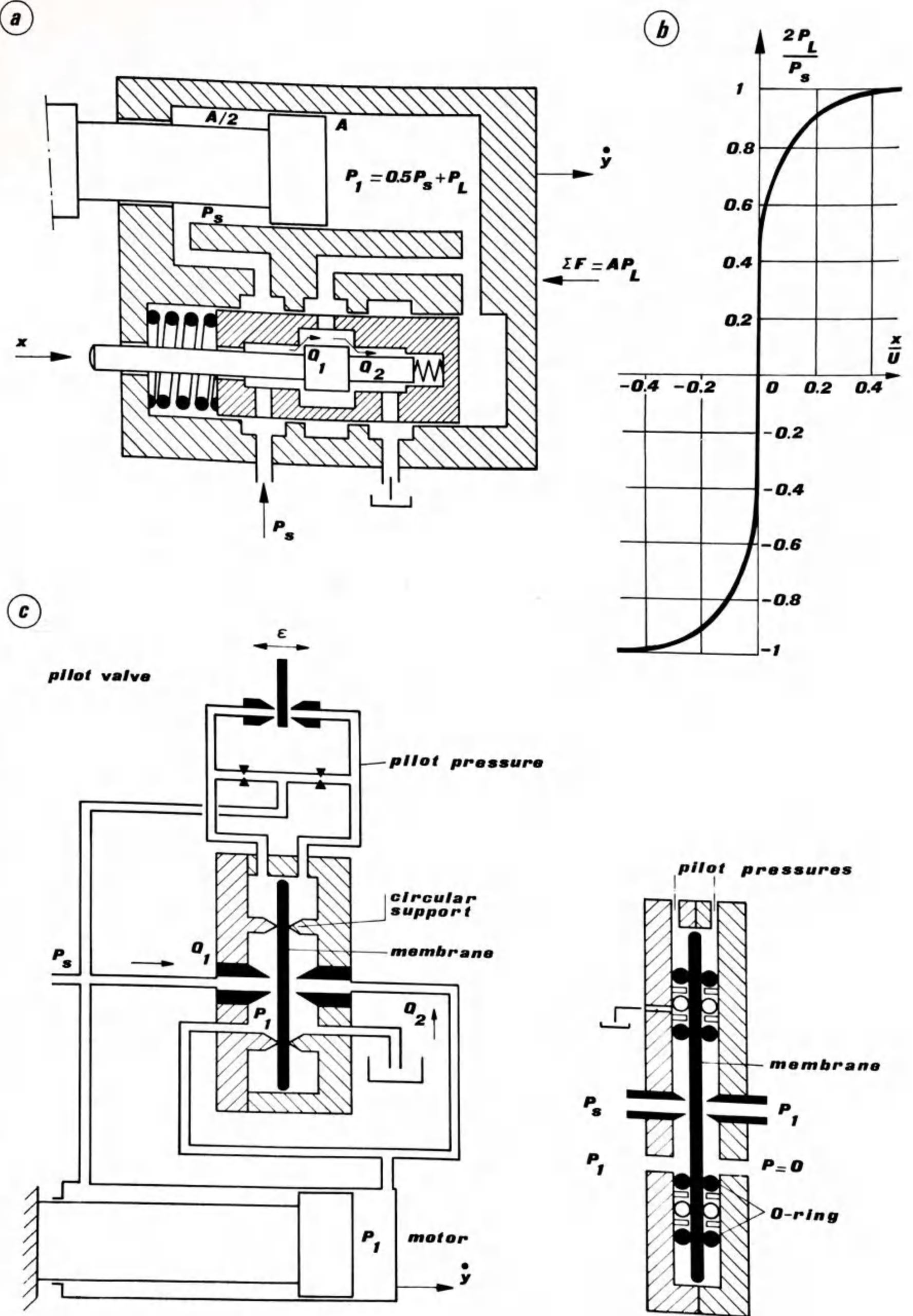


Fig. 3.7 Symmetric load-compensation

$$\dot{y} = \frac{C_d}{A} \sqrt{\frac{P_s}{\rho}} \left[a_o \left(1 + \frac{x}{U} + \frac{P_L}{P_s} \right) \sqrt{1 - \frac{2P_L}{P_s}} - a_o \left(1 - \frac{x}{U} - \frac{P_L}{P_s} \right) \sqrt{1 + \frac{2P_L}{P_s}} \right]. \quad (9)$$

Figure 3.7(a) illustrates the design concept. By superposition the influence of both P_L and x on a_1 and a_2 is effected. Now compensation is effective for larger loads also, as can be seen from the Fig. 3.7(b), derived from eqn. (9) with $\dot{y} = 0$

$$\frac{x}{U} = \frac{1 - 0.5(2P_L/P_s)^2 - \sqrt{1 - (2P_L/P_s)^2}}{2P_L/P_s} = \frac{1}{8} \left(\frac{2P_L}{P_s} \right)^3 + \frac{1}{16} \left(\frac{2P_L}{P_s} \right)^5 + \dots$$

A very compact example of a successful application in practice is shown in (c) where a circularly supported flexible membrane between two nozzles acts like a control flapper. The membrane and the 2 nozzles incorporate the second stage of an open-center 3-way valve. The membrane and the port openings are influenced firstly by the pilot stage pressures of the servovalve, and secondly by the cylinder pressure $P_1 = 0.5 P_s + P_L$ acting on one side of the membrane around the nozzle. The load pressure may change the inlet and outlet pressure drops, but at the same time the inlet and outlet openings are influenced in the opposite direction. Hence, with a properly designed membrane, the inlet Q_1 as well as the outlet Q_2 are almost independent of P_L , P_1 or ΣF ! In the final design, the circular support of the membrane has been established by steel balls, leakage being prevented by O-rings.

In applications involving a linear electro-hydraulic stepping motor [4] and for a copying shaper, careful design of the load-compensated 3-way valve resulted in a membrane with a diameter $\varnothing$ 26 mm and a thickness 0.9 mm.

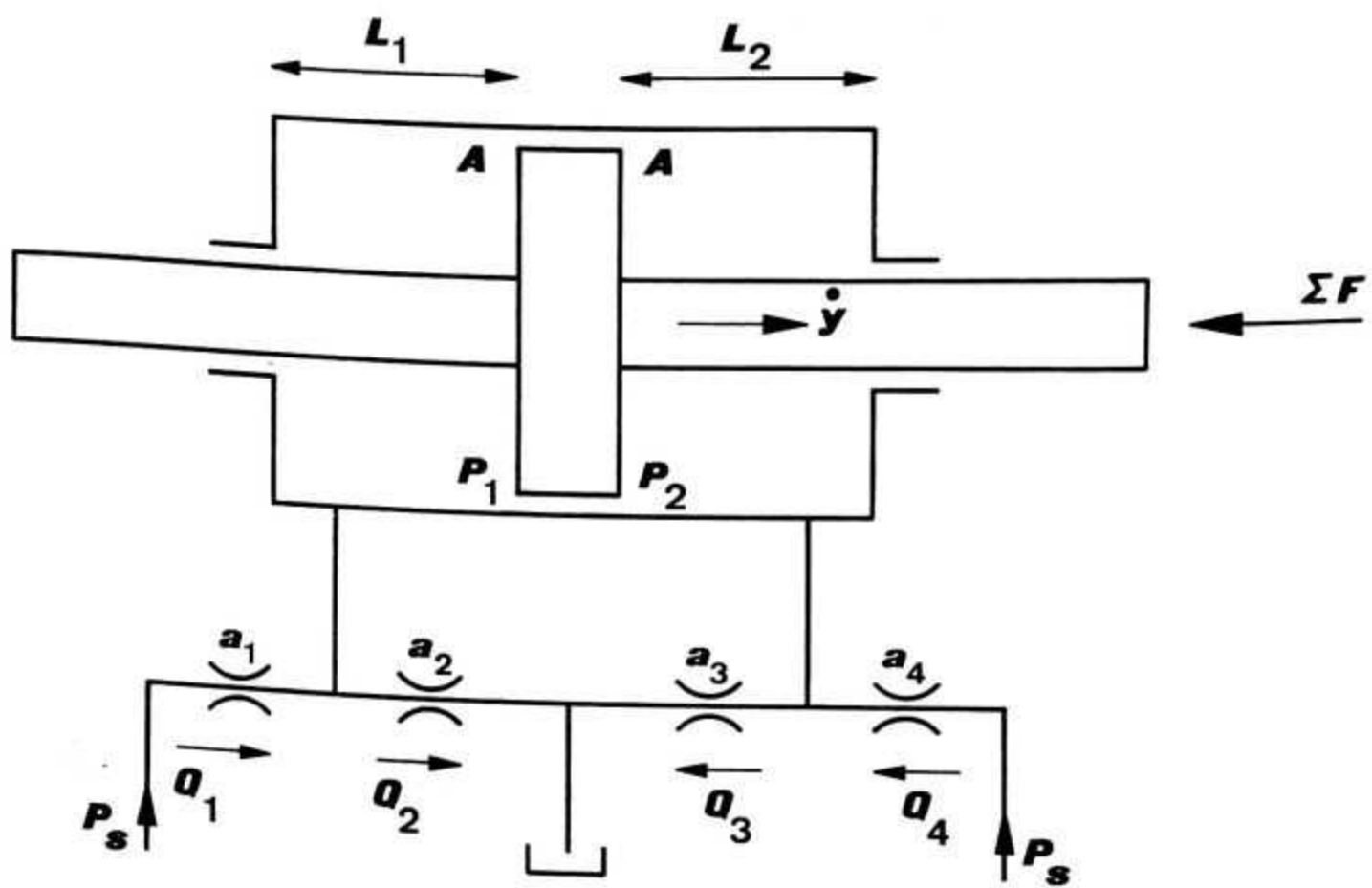


Fig. 3.8 Symmetric motor controlled by a 4-way valve

3.4 Symmetric motor controlled by a symmetric 4-way valve

As there is an almost perfect analogy with an asymmetric motor controlled by a 3-way valve, derivations develop similarly. Using the tools of Section 2.3, the behavior of the symmetric motor of Fig. 3.8 is defined by

$$(P_1 - P_2)A = \Sigma F = F_e + F_c + w\dot{y} + M\ddot{y}$$

$$\dot{y}A = a_1 C_d \sqrt{2 \frac{P_s - P_1}{\rho}} - a_2 C_d \sqrt{\frac{2P_1}{\rho}} - \frac{AL_1}{E} \dot{P}_1$$

$$\dot{y}A = a_3 C_d \sqrt{\frac{2P_2}{\rho}} - a_4 C_d \sqrt{2 \frac{P_s - P_2}{\rho}} + \frac{AL_2}{E} \dot{P}_2$$

Assuming a symmetric valve ($a_1 = a_3$, $a_2 = a_4$) and a piston position with $L_1 = L_2 = L$, we find $P_1 + P_2 = P_s$. Combining this with $P_1 - P_2 = \Sigma F/A$, we find

$$\frac{P_1}{P_s} = \frac{1}{2} + \frac{\Sigma F}{2AP_s} = \frac{1}{2} + \frac{P_L}{2P_s} \quad \frac{P_2}{P_s} = \frac{1}{2} - \frac{\Sigma F}{2AP_s} = \frac{1}{2} - \frac{P_L}{2P_s}$$

Here again, $\Sigma F/A$ is the load pressure P_L .
Substitution yields

$$\dot{y} = \frac{C_d}{A} \sqrt{\frac{P_s}{\rho}} \left[a_1 \sqrt{1 - \frac{P_L}{P_s}} - a_2 \sqrt{1 + \frac{P_L}{P_s}} \right] - \frac{L}{2E} \dot{P}_L \quad (10)$$

Except for a factor 2 in the load pressure $P_L = \Sigma F/A$, eqn. (10) is *identical* to eqn. (2). Consequently, 3-way-valve-controlled asymmetric motors and 4-way-valve-controlled symmetric motors behave (in a stationary condition as well as dynamically) exactly the same, provided the asymmetric motor has a piston area and a spool valve diameter twice as large as the symmetric motor and provided the pistons are half-way along their cylinders. As 3-way-valve-controlled asymmetric motors

are more simple and easier to manufacture than 4-way-valve-controlled symmetric motors it is surprising that asymmetric motors are so seldom used. They deserve more popularity!

Consequently, linearization and derivation of system parameters follow exactly the same lines as in Section 3.2. For a symmetric critical-center valve we can write

$$x > 0 \quad a_1 = a_3 = \left| \frac{a_{\max}}{x_{\max}} \right| x = bx \quad ; \quad a_2 = a_4 = 0$$

$$x < 0 \quad a_1 = a_3 = 0 \quad a_2 = a_4 = - \left| \frac{a_{\max}}{x_{\max}} \right| x = -bx.$$

Substitution into eqn. (10) yields

$$\dot{y} = \frac{b}{A} C_d \sqrt{\frac{P_s}{\rho}} \left[x \sqrt{1 - \frac{|x|}{x_{\max}} \frac{P_L}{P_s}} \right] - \frac{L}{2E} \dot{P}_L.$$

Linearization gives

$$\dot{y} \approx \frac{b}{A} C_d \sqrt{\frac{P_s}{\rho}} \left[x - \frac{|x|}{2P_s} P_L \right] - \frac{L}{2E} \dot{P}_L$$

which can be brought into the standard form

$$\dot{y} = K_m \left[x - \frac{\Sigma F}{c_h} \right] - \frac{\Sigma F}{c_o}$$

where

$$K_m = \frac{b}{A} C_d \sqrt{\frac{P_s}{\rho}}, \quad c_h = \frac{2AP_s}{|x|}, \quad c_o = \frac{2EA}{L}. \quad (11)$$

As could be predicted, the velocity gain K_m is equal to its counter-part for an asymmetric motor, while c_h and c_o have been doubled (see eqn. (7)).

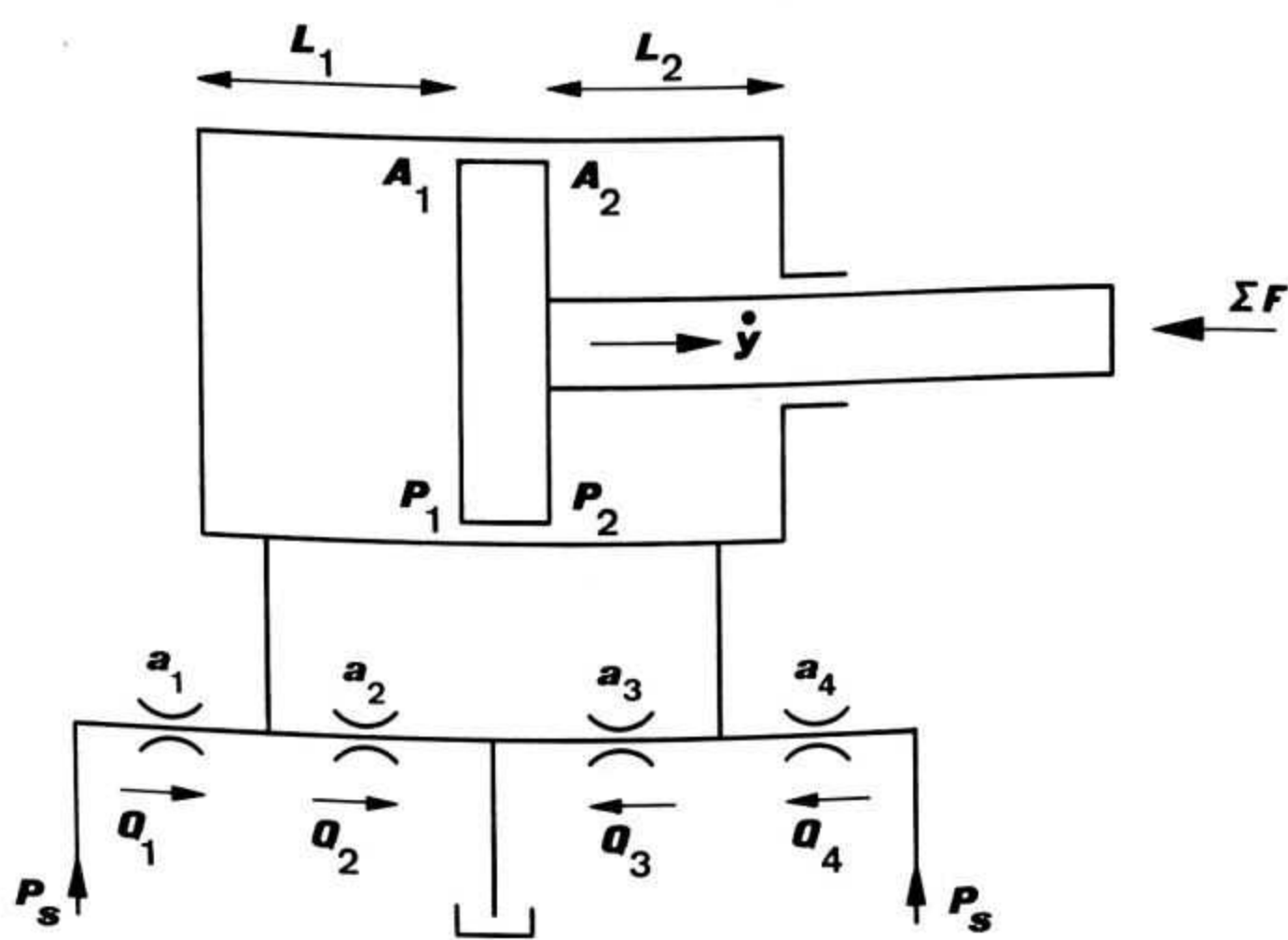


Fig. 3.9 Asymmetric motor controlled by a 4-way valve

3.5 Asymmetric motor controlled by a symmetric 4-way critical-center valve

It will be proved here that an asymmetric linear motor is incompatible with a symmetric 4-way critical-center valve, even under no-load conditions! According to Section 2.3 and Fig. 3.9 we find, for the asymmetric motor, that

$$P_1 A_1 - P_2 A_2 = \Sigma F = F_e + F_c + M\ddot{y} + w\dot{y}$$

$$\dot{y} = \frac{a_1}{A_1} C_d \sqrt{2 \frac{P_s - P_1}{\rho}} - \frac{a_2}{A_1} C_d \sqrt{\frac{2P_1}{\rho}} - \frac{L_1}{E} \dot{P}_1$$

$$\dot{y} = \frac{a_3}{A_2} C_d \sqrt{\frac{2P_2}{\rho}} - \frac{a_4}{A_2} C_d \sqrt{2 \frac{P_s - P_2}{\rho}} + \frac{L_2}{E} \dot{P}_2$$

The implications of a symmetric 4-way critical-center valve are:

Either $a_1 = a_3 > 0$ with $a_2 = a_4 = 0$ or $a_1 = a_3 = 0$ with $a_2 = a_4 > 0$.

The pressures P_1 and P_2 will be calculated for the simple situation $\Sigma F = 0$ and

a_1, a_2, a_3, a_4, P_1 and P_2 are all constants.

At $a_1 = a_3 > 0$ and $\dot{y} > 0$ the pressures P_1 and P_2 can be found from

$$P_1 A_1 = P_2 A_2 \quad \text{and} \quad \frac{a_1}{A_1} \sqrt{P_s - P_1} = \frac{a_3}{A_2} \sqrt{P_2}$$

At $a_2 = a_4 > 0$ and $\dot{y} < 0$ the pressures can be found from

$$P_1 A_1 = P_2 A_2 \quad \text{and} \quad \frac{a_2}{A_1} \sqrt{P_1} = \frac{a_4}{A_2} \sqrt{P_s - P_2}$$

The results have been summarized in the table below.

$\Sigma F = 0$	$\dot{P}_1 = \dot{P}_2 = 0$	$\frac{A_1}{A_2} = 1$	1.71	2	$\dot{P}_1 = \dot{P}_2 = 0$	$\frac{A_1}{A_2} = 1$	1.71	2
$\dot{y} > 0$ $a_1 = a_3 > 0$ $a_2 = a_4 = 0$	$\frac{P_1}{P_s} = \frac{1}{1 + (\frac{A_1}{A_2})^3}$	0.50	0.167	0.111	$\frac{P_2}{P_s} = \frac{\frac{A_1}{A_2}}{1 + (\frac{A_1}{A_2})^3}$	0.50	0.285	0.222
$\dot{y} < 0$ $a_1 = a_3 = 0$ $a_2 = a_4 > 0$	$\frac{P_1}{P_s} = \frac{(\frac{A_1}{A_2})^2}{1 + (\frac{A_1}{A_2})^3}$	0.50	0.487	0.444	$\frac{P_2}{P_s} = \frac{(\frac{A_1}{A_2})^3}{1 + (\frac{A_1}{A_2})^3}$	0.50	0.833	0.889

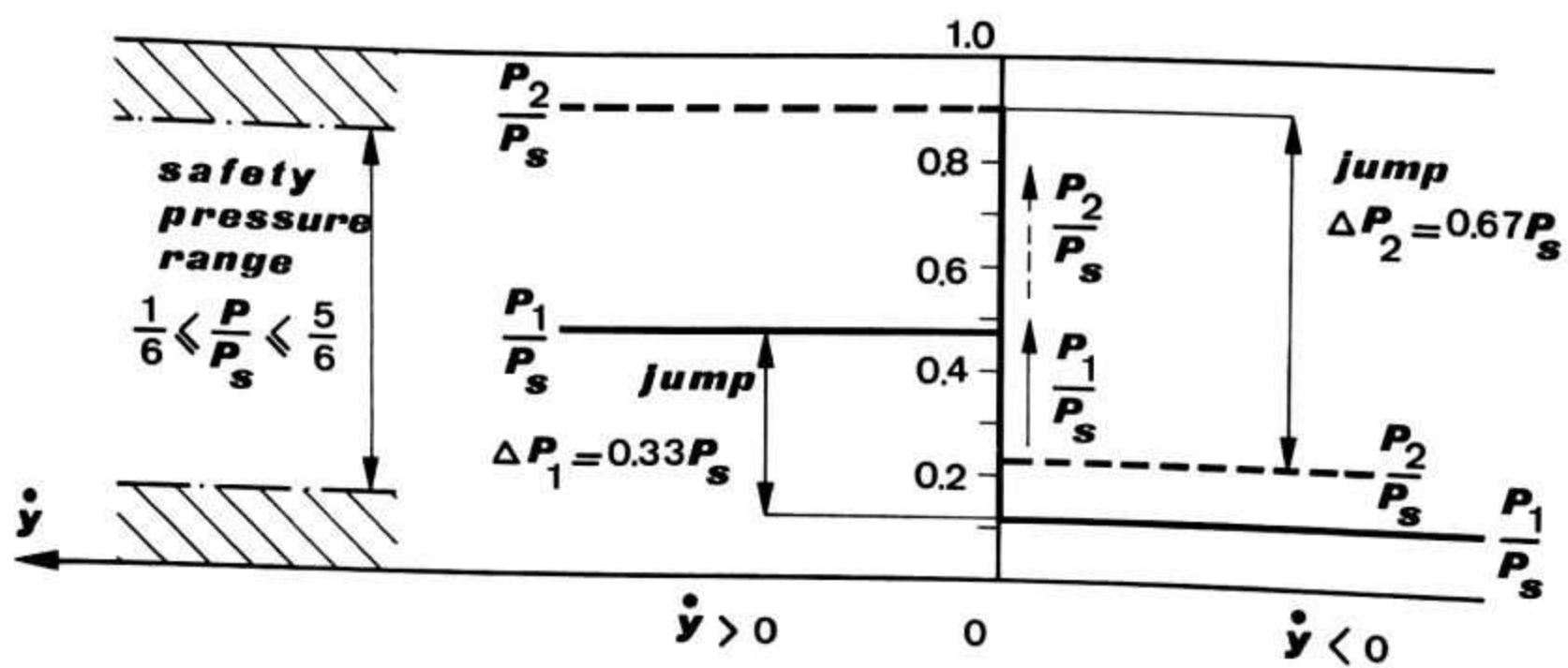


Fig. 3.10 Jumps in the cylinder pressures around $\dot{y} = 0$, due to the asymmetry of the motor ($A_1 \neq A_2$)

To show the astonishing effect of the speed $\dot{y}$ of an unloaded system on its pressures, in Fig. 3.10 P_1 and P_2 have been plotted for $A_1/A_2 = 2$ as a function of the piston speed $\dot{y}$.

The following conclusions can therefore be drawn:

- (1) Except for $A_1/A_2 = 1$, the combinations show enormous pressure jumps around $\dot{y} = 0$, causing "implosions" or "explosions" of oil owing to its compressibility. Consequently smooth operation around $\dot{y} = 0$ is impossible.
- (2) At $A_1/A_2 > 1.71$ the pressures cannot stay within the safety range $\frac{1}{6} P_s \leq P \leq \frac{5}{6} P_s$ even under no-load conditions. Considering the opposing effects of the load ΣF on P_1 and P_2 , motors with $A_1/A_2 < 1.71$ cannot operate within a safe pressure range either.
- (3) For accurate and smooth control the incompatibility of a symmetric valve and an asymmetric motor is clear. To avoid pressure jumps and ensure smooth operation an asymmetric valve (normally not available) must be used, with port openings proportional to the corresponding piston area's

$$\frac{a_1}{A_1} = \frac{a_3}{A_2} \quad \text{and} \quad \frac{a_2}{A_1} = \frac{a_4}{A_2} \quad (12)$$

This combination will be analyzed in Section 3.6.

In the foregoing analysis it has been assumed that $\Sigma F = 0$. It can be shown easily that at $\Sigma F = \text{constant}$, only the locations of the pressure jumps will change, but not their magnitudes!

*3.6 Asymmetric motor controlled by an asymmetric 4-way valve

Since a symmetric valve is incompatible with an asymmetric motor, an asymmetric 4-way critical-center valve in which $a_1/A_1 = a_3/A_2$ and $a_2/A_1 = a_4/A_2$ will be analyzed here. According to Fig. 3.9, the motor behavior is defined by

$$\left. \begin{aligned} P_1 A_1 - P_2 A_2 &= \Sigma F = F_e + F_c + M\ddot{y} + w\dot{y} \\ \dot{y} &= \frac{a_1}{A_1} C_d \sqrt{2 \frac{P_s - P_1}{\rho}} - \frac{a_2}{A_1} C_d \sqrt{\frac{2P_1}{\rho}} - \frac{L_1}{E} \dot{P}_1 \\ \dot{y} &= \frac{a_3}{A_2} C_d \sqrt{\frac{2P_2}{\rho}} - \frac{a_4}{A_2} C_d \sqrt{2 \frac{P_s - P_2}{\rho}} + \frac{L_2}{E} \dot{P}_2 \end{aligned} \right\} \quad (13)$$

The implication of an asymmetric 4-way critical-center valve is:

Either $\frac{a_1}{A_1} = \frac{a_3}{A_2} > 0$ with $a_2 = a_4 = 0$ or $\frac{a_2}{A_1} = \frac{a_4}{A_2} > 0$ with $a_1 = a_3 = 0$.

At $L_1 = L_2 = L$, the pressures P_1 and P_2 can be solved from eqn. (13)

$$\frac{P_1}{P_s} = \frac{A_2 + \Sigma F/P_s}{A_1 + A_2} \quad \frac{P_2}{P_s} = \frac{A_1 - \Sigma F/P_s}{A_1 + A_2} \quad (14)$$

We now introduce the "corrected" total load $\Sigma F'$ and the load pressure P_L of an asymmetric motor, controlled by a 4-way valve

$$\Sigma F' = \Sigma F - \frac{A_1 - A_2}{2} P_s = F_e + F_c + M\ddot{y} + w\dot{y} - \frac{A_1 - A_2}{2} P_s \quad (15)$$

$$P_L = \frac{\Sigma F'}{0.5(A_1 + A_2)} \quad (16)$$

The difference in area $A_1 - A_2$ "causes", in a way, a constant pre-load $F_{st} = 0.5(A_1 - A_2)P_s$, so the introduction of $\Sigma F'$ and P_L makes sense. Combining eqns. (14), (15) and (16) yields

$$\frac{P_1}{P_s} = \frac{1}{2} + \frac{\Sigma F'}{P_s(A_1 + A_2)} = \frac{1}{2} + \frac{P_L}{2P_s} \quad \frac{P_2}{P_s} = \frac{1}{2} - \frac{P_L}{2P_s} \quad (17)$$

Substitution into eqn. (13) gives

$$\dot{y} = \frac{C_d}{A_1} \sqrt{\frac{P_s}{\rho}} \left[a_1 \sqrt{1 - \frac{P_L}{P_s}} - a_2 \sqrt{1 + \frac{P_L}{P_s}} \right] - \frac{L}{2E} \dot{P}_L \quad (18)$$

This equation *exactly* equals the solution (10) derived earlier for a symmetric motor with a symmetric valve, so further derivations will be similar!

Substituting

$$\frac{a_1}{A_1} = \frac{a_3}{A_2} = \left| \frac{a_{1 \max}}{x_{\max}} \right| \frac{x}{A_1} = \frac{b_1}{A_1} x \quad \text{and} \quad a_2 = a_4 = 0 \quad \text{at} \quad x > 0$$

$$a_1 = a_3 = 0 \quad \text{and} \quad \frac{a_2}{A_1} = \frac{a_4}{A_2} = -\frac{b_1}{A_1} x \quad \text{at} \quad x < 0$$

yields

$$\dot{y} = \frac{b_1}{A_1} C_d \sqrt{\frac{P_s}{\rho}} \left[x \sqrt{1 - \frac{|x|}{x} \frac{P_L}{P_s}} \right] - \frac{L}{2E} \dot{P}_L$$

$$\dot{y} \approx \frac{b_1}{A_1} C_d \sqrt{\frac{P_s}{\rho}} \left[x - \frac{|x|}{2P_s} P_L \right] - \frac{L}{2E} \dot{P}_L$$

$$\dot{y} = K_m \left[x - \frac{\Sigma F'}{c_h} \right] - \frac{\Sigma F'}{c_o}$$

where

$$\Sigma F' = \Sigma F - 0.5(A_1 - A_2)P_s$$

$$K_m = \frac{b_1}{A_1} C_d \sqrt{\frac{P_s}{\rho}}$$

$$\frac{b_1}{A_1} = \frac{b_2}{A_2}$$

$$c_h = \frac{(A_1 + A_2)P_s}{|x|}$$

$$c_o = \frac{E(A_1 + A_2)}{L}$$

(19)

Comparison with the results of Section 3.4 shows that the parameters of a symmetric motor with a symmetric 4-way valve are valid for an asymmetric motor with an asymmetric 4-way valve by replacing

$$\Sigma F \quad \text{by} \quad \Sigma F' = \Sigma F - 0.5(A_1 - A_2)P_s$$

$$A \quad \text{by} \quad 0.5(A_1 + A_2)$$

(20)

$$b/A \quad \text{by} \quad b_1/A_1 = b_2/A_2$$

Here $b = \partial a / \partial x$ is the width of the port opening area a .

3.7 From mathematical model to transfer function

The mathematical model derived for various hydraulic servosystems relates the speed $\dot{y}$ to the valve displacement x and to the total load ΣF

$$\dot{y} = K_m \left(x - \frac{\Sigma F}{c_h} \right) - \frac{\Sigma F}{c_o}$$

(21)

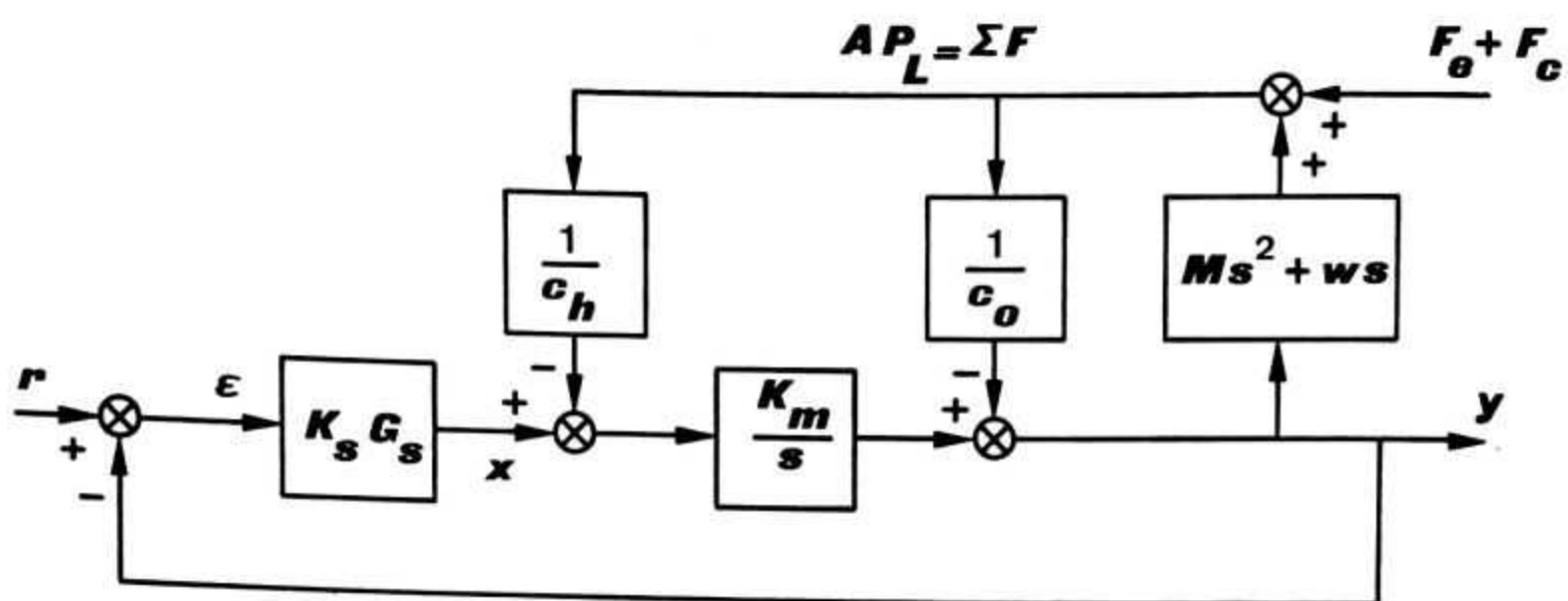


Fig. 3.11 Block diagram for "physical" model of hydraulic servo

As indicated in fig. 3.11 the inertial forces $M\ddot{y}$ and the viscous friction $w\dot{y}$ can be derived from the motor displacement y . The external forces F_e and the Coulomb friction F_c can be seen as disturbance signals

$$\Sigma F = F_e + F_c + M\ddot{y} + w\dot{y} . \quad (22)$$

The velocity gain K_m , the hydraulic stiffness c_h and the oil spring stiffness c_o derived for various valve-motor combinations are summarized in Table 3.1.

Table 3.1 System parameters K_m , c_h and c_o for hydraulic servo's.

Motor	Asymmetric ($A_1 = 2A_2 = A$)		Symmetric ($A_1 = A_2 = A$)	
Valve	3-way		4-way	
Center	Open	Critical	Open	Critical
K_m	$\frac{2b}{A} C_d \sqrt{\frac{P_s}{\rho}}$	$\frac{b}{A} C_d \sqrt{\frac{P_s}{\rho}}$	$\frac{2b}{A} C_d \sqrt{\frac{P_s}{\rho}}$	$\frac{b}{A} C_d \sqrt{\frac{P_s}{\rho}}$
c_h	$\frac{P_s A}{U}$	$\frac{P_s A}{ x }$	$\frac{2P_s A}{U}$	$\frac{2P_s A}{ x }$
c_o	$\frac{EA}{L} \geq \frac{EA}{S}$		$\frac{2EA}{L} \geq \frac{4EA}{S}$	

As indicated in Fig. 3.11, the control loop is closed by

$$\epsilon = r - y \quad (23)$$

and

$$x = K_s G_s(s) \epsilon \quad (24)$$

with

$$\lim_{s \rightarrow 0} G_s(s) = 1$$

$K_s G_s(s) = x/\epsilon$ is the transfer function of the mechanism (for instance a lever or an electro-hydraulic servovalve) transforming the error signal ϵ into the valve displacement x .

Combination of the above equations yields

$$\frac{\dot{\Sigma F}}{c_o} + \frac{K_m}{c_h} \Sigma F + \dot{y} = K_m K_s G_s \epsilon$$

$$\frac{M}{c_o} \ddot{y} + \left(\frac{MK_m}{c_h} + \frac{w}{c_o} \right) \ddot{y} + \left(1 + \frac{wK_m}{c_h} \right) \dot{y} = K_m K_s (G_s \epsilon - \frac{F_e + F_c}{K_s c_h}) - \frac{\dot{F}_e + \dot{F}_c}{c_o} \quad (25)$$

Obviously, new coefficients in the differential equation are desirable in order to express the dynamic character of the system most directly. As new coefficients are chosen ω_o , β , K_v , C_h and C_o , eqn. (25) changes into

$$\frac{\ddot{y}}{\omega_o^2} + 2\beta \frac{\dot{y}}{\omega_o} + y = K_v (G_s \epsilon - \frac{F_e + F_c}{C_h}) - \frac{\dot{F}_e + \dot{F}_c}{C_o} \quad (26)$$

Here

$$C_h = K_s c_h \quad (27)$$

$$C_o = c_o \left(1 + \frac{wK_m}{c_h} \right) \simeq c_o \quad (28)$$

$$K_v = \frac{K_s K_m}{1 + \frac{wK_m}{c_h}} \simeq K_s K_m \quad (29)$$

$$\omega_o = \sqrt{\frac{C_o}{M}} \quad (30)$$

$$\beta = \frac{K_v}{2C_h} \sqrt{MC_o} + \frac{w}{2\sqrt{MC_o}} = \frac{K_v}{2\omega_o} \frac{C_o}{C_h} + \frac{w}{2M\omega_o} \quad (31)$$

C_h , C_o and K_v are the hydraulic stiffness, the oil spring stiffness and the velocity gain of the *complete system*, respectively. It should be noted that in practice $wK_m/c_h \ll 1$, so $C_o \simeq c_o$ and $K_v \simeq K_s K_m$.

Most important for the dynamic behavior are the undamped natural frequency ω_o and the relative damping β , representing the hydraulic mass-spring system of the actuated mass and the oil spring. Equation (26) is depicted visually in the block

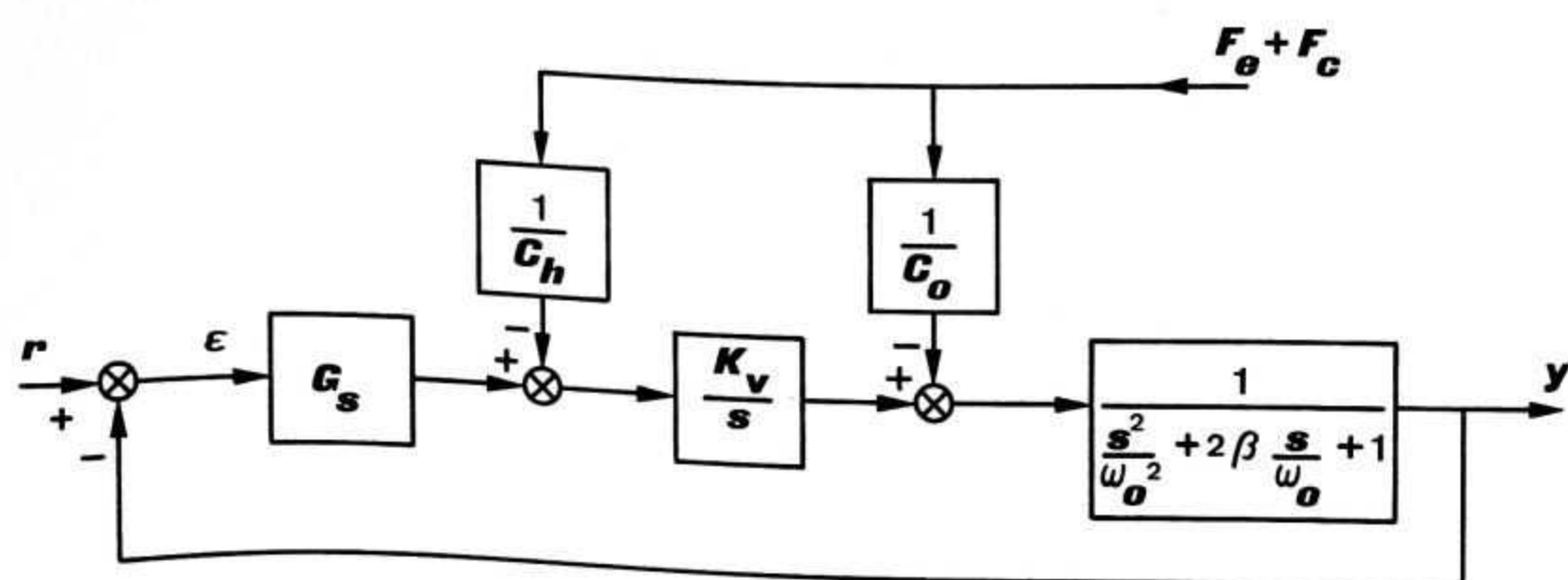


Fig. 3.12 Block diagram for "control" model of hydraulic servo

diagram of Fig. 3.12 which can also be derived directly from Fig. 3.11 by elementary block diagram manipulation. Figure 3.12 shows a *control loop* with an *integrator in series with a second order system*. External load F_e and F_c are disturbing the system.

The transformation of the "physical" model shown in Fig. 3.11, with its parameters K_m , c_h , c_o , M and w into the "control" model shown in Fig. 3.12 with its system parameters K_v , C_h , C_o , ω_o and β will enable us to start dynamic analysis and synthesis in Chapter 4.

Some final remarks regarding the linearity of hydraulic systems must be made. Figures 3.11 and 3.12 suggest firmly a linear system, i.e. a system according to a linear differential equation. However, every system treated so far is far from linear! Not one of the parameters K_m , c_h and c_o in Fig. 3.11 is a real constant. The linearization process is charming, of course, but it suggests too much. Nevertheless, Figs. 3.11 and 3.12 are extremely useful if handled with care.

For example, one has to remember that the oil spring stiffness c_o varies considerably with the piston position (see Fig. 2.3). Thus analysis based on a linear model holds only for small oscillations around *one* position. In another position, another c_o is valid. In practice one takes the position where c_o is a minimum. The velocity gain K_m of a hydraulic motor depends on the load AP_L (see Fig. 3.3). If necessary, adequate corrections must be made.

Regarding the hydraulic stiffness c_h , Table 3.1 shows a highly non-linear character. Only in open-center valves is c_h approximately a constant in the underlap region ($|x| < U$), but in critical-center valves c_h is proportional to $|x|^{-1}$, so $c_h = \infty$ at $x = 0$. In calculations regarding stability and damping, a large c_h up to $c_h = \infty$ will always prove to be a safe assumption.

In these complex matters one could easily forget the main non-linearizing factor. This is Coulomb friction F_c , reversing its sign with the speed $\dot{y}$. Thus if non-linear control analysis has to be carried out, the Coulomb friction should be considered first.

REFERENCES

- [1] T.J. Viersma, Investigations into the accuracy of hydraulic servomotors, Thesis Delft University of Technology, 1961; Philips Res. Rep., 16(1961)507; 17(1962)20.
- [2] T.J. Viersma, Designing load-compensated fast response hydraulic servos, Control Eng., May (1962)111.
- [3] P. Blok and T.J. Viersma, A new construction for a load-compensated fast response hydraulic follow-up system, Proc. Third IFAC Congr., 1966, Vol.1, No.2, Report 25D.
- [4] P. Blok, Linear electrohydraulic stepping motor for NC machines, Hydraul. Pneum., Vol.21, No.10, (1968)117.

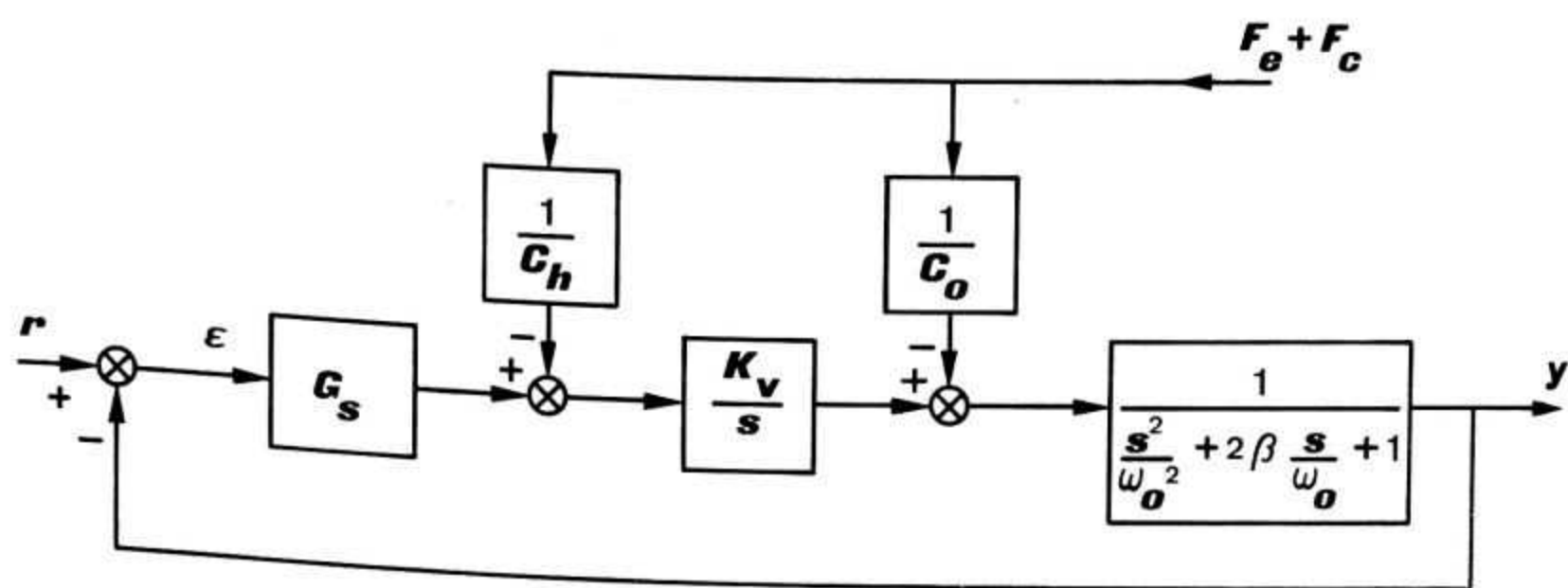


Fig. 4.1 Block diagram for control analysis.

CHAPTER 4

DYNAMIC ANALYSIS AND SYNTHESIS

4.1 Introduction

For the dynamic analysis of hydraulic servo mechanisms the differential equation

$$\frac{\ddot{y}}{\omega_o^2} + 2\beta \frac{\dot{y}}{\omega_o} + y = K_v \left[G_s \varepsilon - \frac{F_e + F_c}{C_h} \right] - \frac{\dot{F}_e + \dot{F}_c}{C_o}, \quad (1)$$

derived in Section 3.7 will be used. Figure 4.1 shows again the block diagram where r = closed loop input signal,

$\varepsilon = r - y$ = open loop input signal,

y = output motion,

F_e = external force, and

F_c = Coulomb friction.

The system parameters are the velocity gain K_v , the hydraulic stiffness C_h and the oil spring stiffness C_o of the complete system as well as the natural frequency ω_o and the damping β of the oil-spring-mass system. They are derived from the "physical" parameters K_s , K_m , c_h , c_o , M and w (see eqns. (27) - (31) and Fig. 3.11 in Section 3.7). The transfer function G_s represents the dynamic part of the mechanism relating the error signal $\varepsilon = r - y$ to the valve displacement x .

Dynamic analysis and adjusting control action will be based mainly on frequency response. A difficulty here is the non-linear behavior of Coulomb friction, making conventional linear frequency analysis impossible. In spite of its large influence on system behavior, Coulomb friction, in the author's experience, hardly affects the adjustment of the control gain in practice. This statement will become more plausible when, at the end of this chapter, it will be proved mathematically in two different situations that Coulomb friction does not influence the system stability in practice.

This at least is some justification for our approach, which ignores Coulomb friction (except for one trivial case) at the beginning of this chapter, enabling us to proceed to classical linear analysis and synthesis. This does not mean that Coulomb friction has no influence on system behavior in general, on the contrary. In many applications, Coulomb friction may deteriorate the system completely with respect to special specifications such as accuracy (owing to dead zone), bandwidth (owing to dynamic dead zone) or slip-stick motion (demonstrating itself by acceleration disturbances caused by irregular friction behavior).

Except for load-compensation (see Section 3.3) the only way to overcome these effects is the elimination of Coulomb friction completely by introducing hydrostatic bearings. This item will be treated separately in Chapter 7.

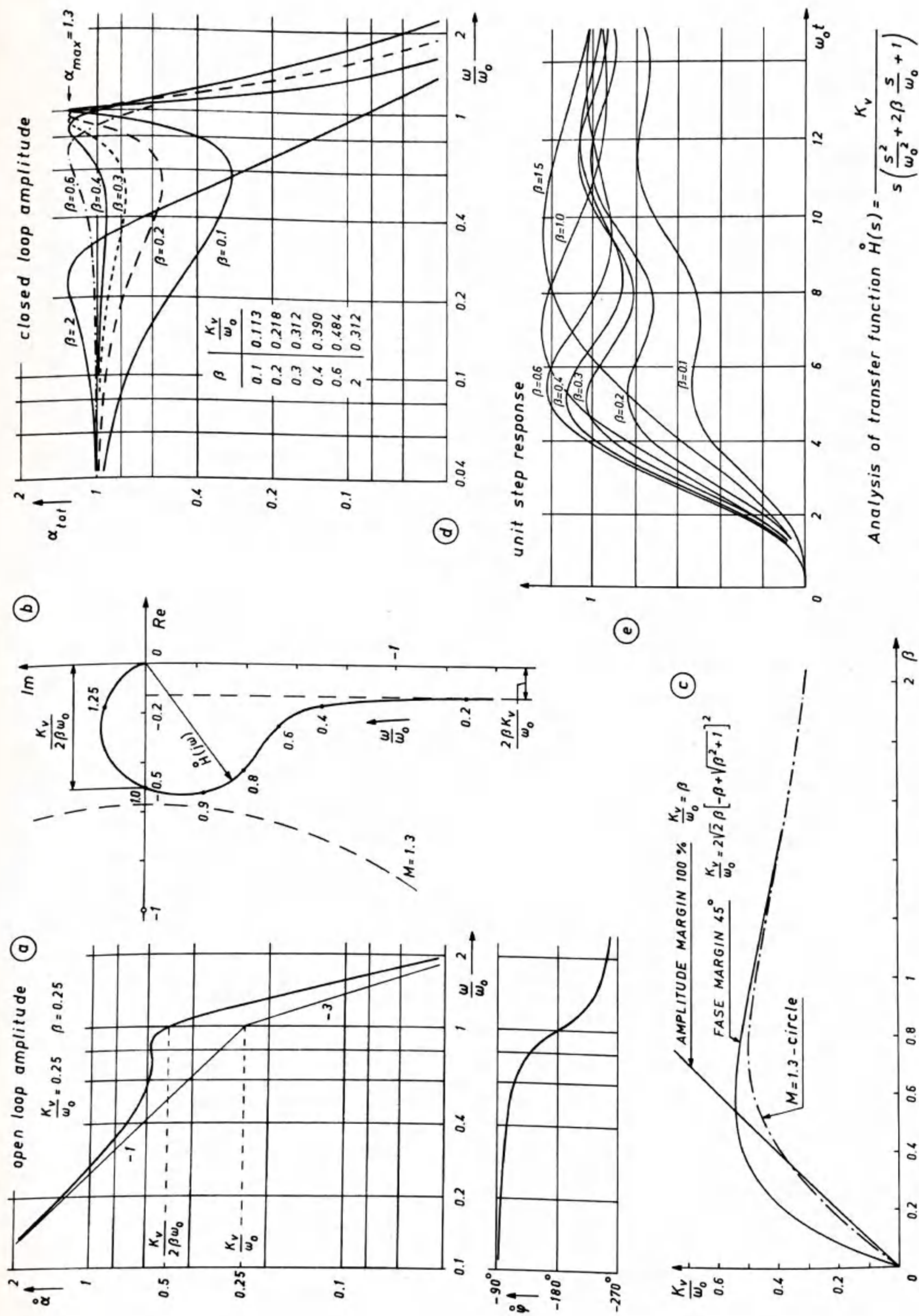


Fig. 4.2 Proportional control

4.2 Proportional control [1]

The open loop transfer function $\hat{H}(s) = y/\varepsilon$ can be found from eqn. (1) or Fig. 4.1 by letting $F_e = 0$ en $F_c = 0$. Assuming further, $G_s(s) = 1$, which is allowed when the bandwidth of the servovalve is much higher than the natural frequency ω_o , we find

$$\hat{H}(s) = \frac{y}{\varepsilon} = \frac{K_v}{s \left[\frac{s^2}{\omega_o^2} + 2\beta \frac{s}{\omega_o} + 1 \right]}. \quad (2)$$

Eqn. (2) shows an integrator in series with a second order system. In the open loop frequency response the amplitude ratio $\hat{\alpha}$ and phase lag $\hat{\phi}$ can be found by letting $s = j\omega$ and

$$\hat{\alpha} = \frac{\hat{y}}{\hat{\varepsilon}} = \left| \hat{H}(j\omega) \right| = \frac{K_v}{\omega \sqrt{\left(1 - \frac{\omega^2}{\omega_o^2}\right)^2 + \left(2\beta \frac{\omega}{\omega_o}\right)^2}} \quad (3)$$

$$\hat{\phi} = \text{Arg } \hat{H}(j\omega) = -\frac{\pi}{2} - \arctan \frac{2\beta \frac{\omega}{\omega_o}}{1 - \frac{\omega^2}{\omega_o^2}}. \quad (4)$$

The Bode diagram is shown in Fig. 4.2(a), the Nyquist plot in (b). Both diagrams should be studied carefully.

A phase $\hat{\phi} = -\pi$ occurs at $\omega = \omega_o$ where $\hat{\alpha} = K_v/2\beta\omega_o$. To be stable the Nyquist plot should not encircle the point $-1,0$. Thus the *stability condition* is

$$\frac{K_v}{\omega_o} < 2\beta \quad (5)$$

and consequently the velocity gain K_v is limited by the product of natural frequency ω_o and damping β . To translate this into physical parameters, ω_o and β of eqns. (30) and (31) of Section 3.7 must be substituted. The result is

$$K_v \left[1 - \frac{C_o}{C_h} \right] < \frac{w}{M}. \quad (6)$$

So the ratio of oil spring stiffness and hydraulic stiffness as well as the ratio of viscous friction coefficient w and mass M are extremely important. Their consequences will be discussed later.

To provide not only stability, but adequate damping as well, the so-called $M = 1.3$ -criterion could be used [2], implying that the closed loop amplitude ratio never exceeds 1.3. Obeying this criterion the closed loop *step response* normally shows an *overshoot* of approximately 23% (for a second order system this is exact!) Thus in the polar diagram(b) the Nyquist plot is not allowed to intersect the $M = 1.3$ -circle. To avoid complicated calculations it is good engineer's practice

to focus attention on two important points of the $M = 1.3$ -circle. They lead to the well known

- *amplitude margin* of about 100%, meaning $\delta < 0.5$ at $\phi = -\pi$,
- *phase margin* of about 45° , meaning $\phi \leq -0.75\pi$ at $\delta = 1$.

Respecting *both* margins implies normally that the closed loop frequency response will show an amplitude ratio of maximally 1.3 and that the step response will have an overshoot smaller than 23%. Following the rules outlined above, the system will be sufficiently damped.

For low values of β , which are usual in hydraulic servo's, the amplitude margin is dominant. At $\omega = \omega_o$ is $\phi = -\pi$, $\delta = K_v / 2\beta\omega_o$. Thus $\delta < 0.5$ gives

$$\frac{K_v}{\omega_o} \leq \beta \quad \text{or} \quad K_v \left[2 - \frac{C_o}{C_h} \right] \leq \frac{W}{M}. \quad (7)$$

For the (unusual) larger values of β the phase margin is dominant. At

$$\frac{\omega}{\omega_o} = -\beta + \sqrt{\beta^2 + 1} \quad \text{is} \quad \phi = -\frac{3}{4}\pi \quad \text{and} \quad \delta = \frac{K_v}{2\beta\omega_o \sqrt{2} [-\beta + \sqrt{\beta^2 + 1}]^2}$$

so $\delta \leq 1$ gives

$$\frac{K_v}{\omega_o} \leq 2\beta\sqrt{2} [-\beta + \sqrt{\beta^2 + 1}]^2. \quad (8)$$

The results of eqns. (7) and (8) have been plotted in Fig. 4.2(c) together with exact results of the $M = 1.3$ -criterion itself, obtained by computer. It shows an almost perfect agreement between the criteria.

Taking values of K_v/ω_o from Fig. 4.2(c) in (d) and (e), the closed loop amplitude ratio ($\leq 1.3!$) and unit step response (maximum overshoot $< 23\%!$) are shown

Satisfactory responses are obtained with $0.3 < \beta < 0.7$. At $\beta < 0.3$ the step response is very nervous and the frequency response shows an amplitude dip lower than 0.7. At $\beta > 0.7$ the step response is slowing down and the bandwidth (i.e. the frequency up to which amplitude ratios do not fall below 0.7) decreases rapidly.

Conclusion: hydraulic control systems with an adequate damping coefficient $0.3 < \beta < 0.7$ and a well adjusted velocity gain K_v show a bandwidth ω_b equal to its natural frequency ω_o .

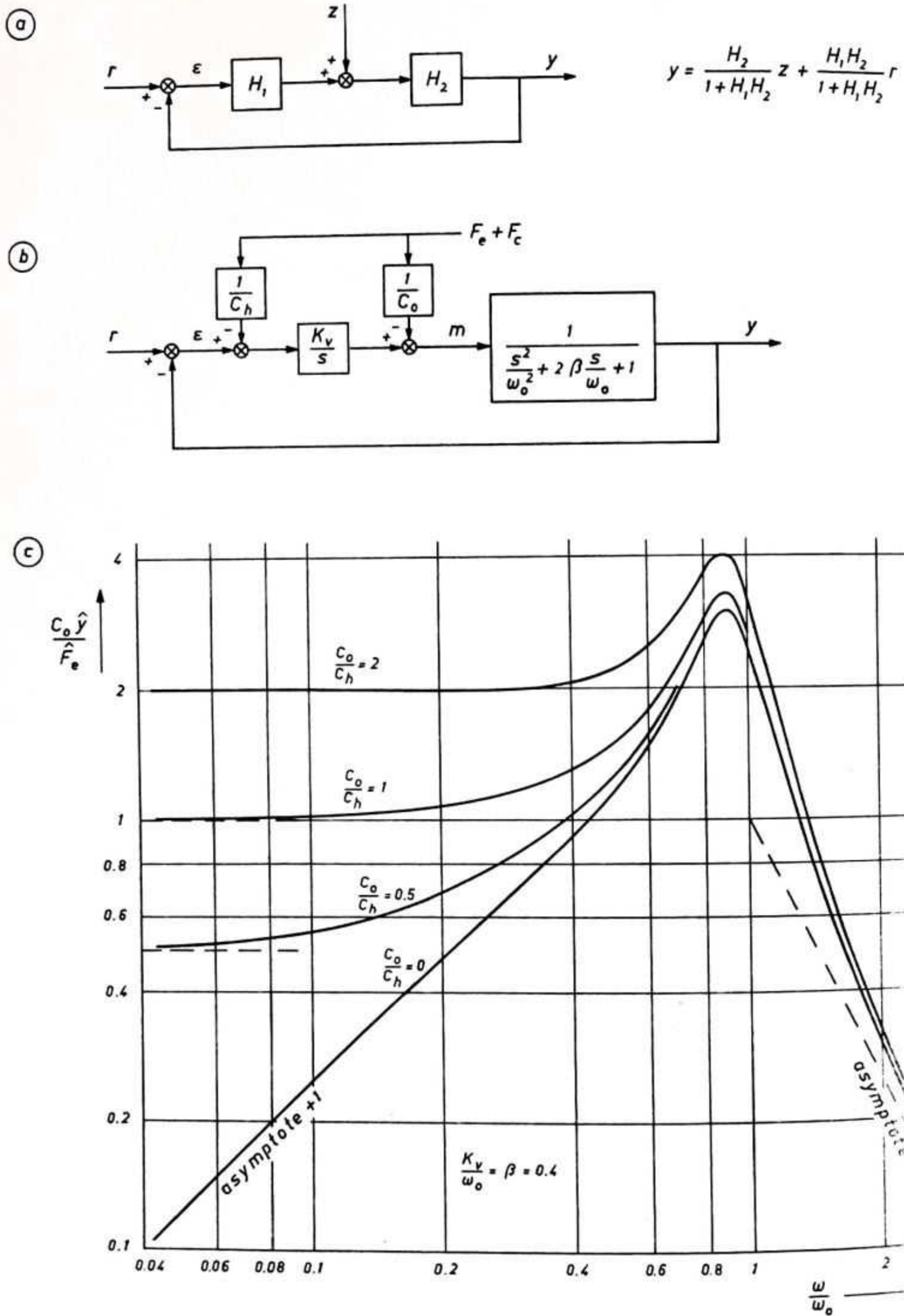


Fig. 4.3 Dynamic load.

4.3 Dynamic load [1]

Following the calculation rule of Fig. 4.3(a), the effect of a sinusoidally varying load F_e on the output y at $r = 0$ and $F_c = 0$ can be found directly.

$$\frac{y}{F_e} = - \frac{\left[\frac{K_v}{C_h s} + \frac{1}{C_o} \right] \left[\frac{s^2}{\omega_o^2} + 2\beta \frac{s}{\omega_o} + 1 \right]^{-1}}{1 + \frac{K_v}{s \frac{s^2}{\omega_o^2} + 2\beta \frac{s}{\omega_o} + 1}} = - \frac{1}{C_o} \cdot \frac{\frac{K_v}{\omega_o} \cdot \frac{C_o}{C_h} + \frac{s}{\omega_o}}{\frac{K_v}{\omega_o} + \frac{s}{\omega_o} \left[\frac{s^2}{\omega_o^2} + 2\beta \frac{s}{\omega_o} + 1 \right]}.$$

$$\frac{C_o \hat{y}}{\hat{F}_e} = \sqrt{\frac{\left(\frac{K_v}{\omega_o} \cdot \frac{C_o}{C_h} \right)^2 + \left(\frac{\omega}{\omega_o} \right)^2}{\left(\frac{K_v}{\omega_o} - 2\beta \frac{\omega^2}{\omega_o^2} \right)^2 + \frac{\omega^2}{\omega_o^2} \left(1 - \frac{\omega^2}{\omega_o^2} \right)^2}}. \quad (9)$$

The stability condition $K_v/\omega_o < 2\beta$, derived earlier, can be recognized easily since $C_o y/F_e = \infty$ at $K_v/\omega_o = 2\beta$ and $\omega = \omega_o$.

For the analysis of the load behavior we assume a velocity gain $K_v = 0.4\omega_o$ adequately adjusted to an acceptable damping $\beta = 0.4$, providing good responses of y on r at $F_e = 0$. With these data however, the load behavior may still differ considerably, as is shown in Fig. 4.3(c) where $C_o \hat{y}/\hat{F}_e$ is plotted as a function of frequency at various values of C_o/C_h . According to eqn. (31) in Section 3.7,

$$\beta = \frac{K_v}{2C_h} \sqrt{MC_o} + \frac{w}{2\sqrt{MC_o}} = \frac{K_v}{2\omega_o} \cdot \frac{C_o}{C_h} + \frac{w}{2M\omega_o} = 0.2 \frac{C_o}{C_h} + \frac{w}{2M\omega_o} = 0.4 \quad (10)$$

both viscous friction w and the ratio C_o/C_h contributing to β . Since it is preferable for the influence of the load F_e on the output to be as small as possible, a hydraulic stiffness $C_h = \infty$ is desirable (the alternative is PI-control, see Section 4.8). In that case there is no load error at low frequencies.

$$\text{With } C_h = \infty, \quad \lim_{\omega \rightarrow 0} \frac{C_o \hat{y}}{\hat{F}_e} = \frac{\omega}{K_v} = 0.$$

When there is, unfortunately, no viscous friction, C_h should be decreased in order to provide adequate damping. But, as Fig. 4.3(c) shows very clearly, this is done at the expense of the load error at low frequencies.

$$\text{With } C_h \neq \infty, \quad \lim_{\omega \rightarrow 0} \frac{C_o \hat{y}}{\hat{F}_e} = \frac{C_o}{C_h} \quad \text{or} \quad \frac{\hat{y}}{\hat{F}_e} = \frac{1}{C_h}$$

corresponding to the load error derived earlier.

*4.4 Dynamic dead zone due to Coulomb friction

In just one trivial case the influences of Coulomb friction on the dynamic behavior can be derived exactly [1]. This is the case of hydraulic lock where Coulomb friction prevents any output motion. The output y being zero, Fig. 4.3(b) directly provides the solution. If $y = 0$, then $m = 0$ as well, thus at $F_e = 0$ we find

$$m = \left[r - \frac{F_c}{C_h} \right] \frac{K_v}{s} - \frac{F_c}{C_o} = 0 \text{ and}$$

$$r = \varepsilon = \frac{F_c}{C_h} + \frac{\dot{F}_c}{K_v C_o} \text{ at } y = 0.$$

This is a *linear* differential equation. Thus at a sinusoidal input signal r or ε the Coulomb friction will be sinusoidal too, staying within its maximum borders

$$- |F_c|_{\max} < F_c < + |F_c|_{\max}.$$

One might say that Coulomb friction just takes the value that is necessary to stop any motion. Using the notation $\bar{F}_c = |F_c|_{\max}$ we find in the frequency domain

$$y = 0 \quad \text{at} \quad \hat{r} = \hat{\varepsilon} < \frac{\bar{F}_c}{C_o} \sqrt{\left(\frac{C_o}{C_h}\right)^2 + \left(\frac{\omega}{K_v}\right)^2}. \quad (11)$$

At low frequencies

$$y = 0 \quad \text{when} \quad \lim_{\omega \rightarrow 0} \hat{r} = \lim_{\omega \rightarrow 0} \hat{\varepsilon} < \frac{\bar{F}_c}{C_h}$$

representing the well-known static dead zone.

At higher frequencies ($\omega > K_v C_o / C_h$)

$$y = 0 \quad \text{when} \quad \hat{r} = \hat{\varepsilon} < \frac{\bar{F}_c}{C_o} \cdot \frac{\omega}{K_v}$$

Thus the dead zone apparently increases with frequency and could therefore be called the *dynamic dead zone*. Its existence has been confirmed experimentally several times.

The other way around, one could say there is a frequency

$$\omega_{\text{stop}} = K_v \sqrt{\left(\frac{C_o \hat{r}}{\bar{F}_c}\right)^2 - \left(\frac{C_o}{C_h}\right)^2} \quad (12)$$

where at $\omega > \omega_{\text{stop}}$ no motion will occur. Hence, Coulomb friction could be a serious threat to the bandwidth of a hydraulic servosystem.

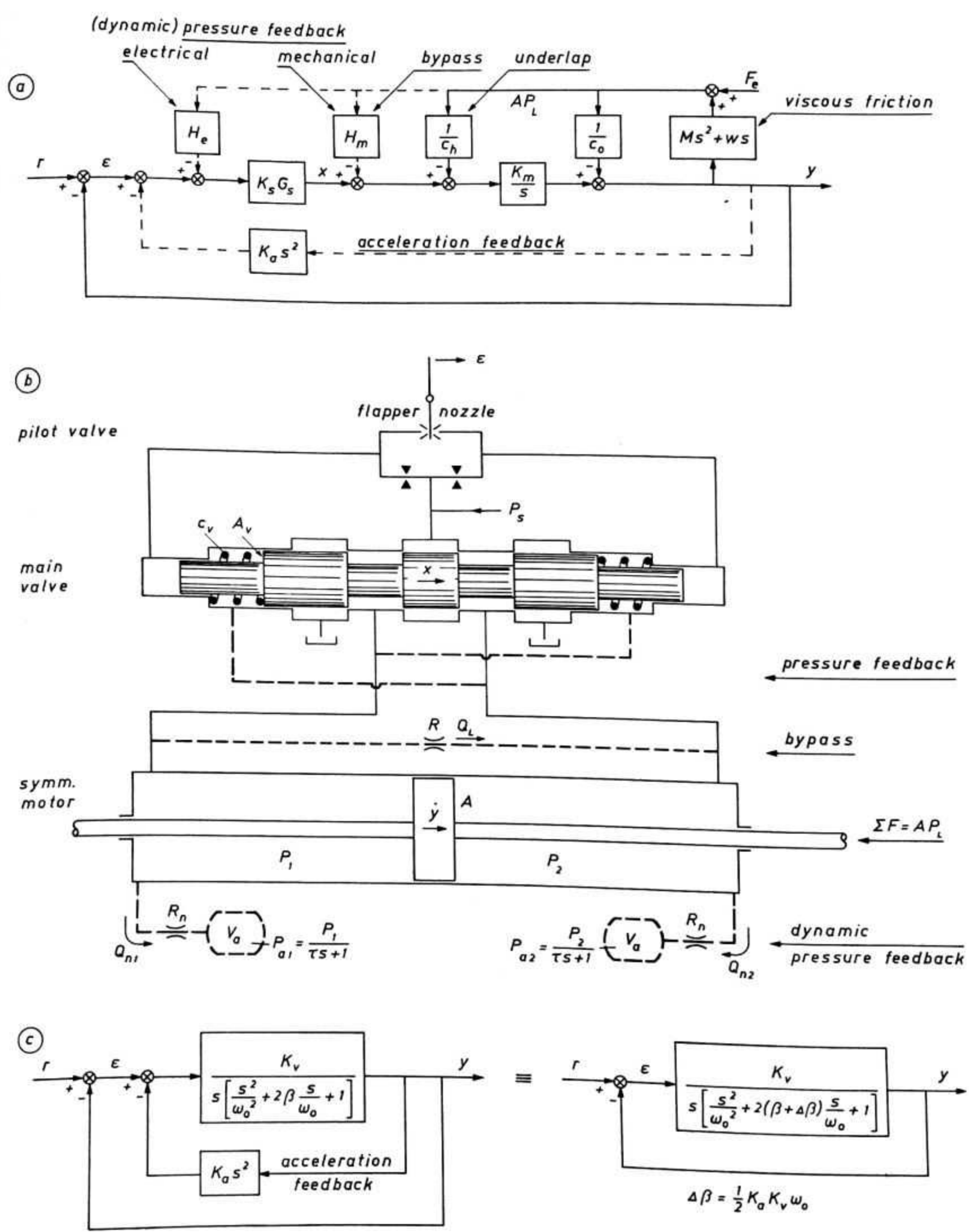


Fig. 4.4 Damping methods.

4.5 Damping methods [1], [3], [4], [5], [6]

In Section 4.2 it was proved that for good proportional control a damping coefficient

$$\beta = \frac{K_v}{2C_h} \sqrt{MC_o} + \frac{w}{2\sqrt{MC_o}} = \frac{K_v}{2\omega_o} \cdot \frac{C_o}{C_h} + \frac{w}{2M\omega_o} \quad (10)$$

ranging from 0.3 to 0.7 is desirable. As the two components of β are proportional to the ratio of oil spring stiffness to hydraulic stiffness C_o/C_h and the viscous friction coefficient w , respectively, one or both of these factors must be present. In Section 4.3 the desirability of a low C_o/C_h to produce a small load sensitivity at low frequencies became clear (in critical-center valves $C_o/C_h = 0$ around $x = 0$, see Table 3.1). Hence an excellent damping method depends completely on the viscous friction coefficient w , indicated in the block diagram of Fig. 4.4. Unfortunately, in practice it is hardly possible to design or to predict beforehand the amount of viscous friction. Careful analysis of shear in oil films of bearings, for instance, shows viscous friction to be neglected completely. Yet viscous friction is present normally, just as a component of the total friction, since Coulomb friction and viscous friction frequently go hand in hand.

To illustrate this, we refer to many experiments in the hydraulic research group of the Delft University of Technology, where in older servomotor designs Coulomb friction and viscous friction were present as well as an acceptable damping coefficient β . This value of β was apparently caused by viscous friction, as load-compensation (according to Section 3.3) with $C_h = \infty$ was provided. In later designs in which Coulomb friction was eliminated almost completely by installing hydrostatic bearings, very poor damping coefficients (as low as $\beta = 0.05$) were measured.

If insufficient viscous friction is present, application of eqn. (10) will suggest increasing the ratio C_o/C_h in order to obtain sufficient damping. Since K_v/ω_o is connected to β to ensure a good control behavior, and C_o (which is proportional to the piston area A) has to govern both load capacity and bandwidth, C_h must be reduced to get adequate damping. However, the consequence is bad load stiffness. Figure 4.4(b) shows several methods to reduce C_h .

A very direct method is the introduction of open-center valves with a large *underlap* U , since C_h is proportional to $P_s A/U$ (see Table 3.1). The unacceptable disadvantage of such a large underlap U is, however, the enormous oil consumption. Even in the neutral position of the underlap valve, the oil consumption can be much higher than that corresponding to the maximum speed $\dot{y}$ of the motor, requiring $A\dot{y}_{\max}$.

A far better method to reduce C_h is therefore the so-called *bypass* method applied in symmetric motors with 4-way valves, see Fig. 4.4(b). A laminar flow resistance R proportional to the load-pressure $P_L = P_1 - P_2$, will produce a "leakage-flow"

$Q_L = P_L/R$, giving the motor (in the open loop situation) an additional speed of $\Delta\dot{y} = Q_L/A = \Delta P_L/RA$. Examining the block diagram of Fig. 4.4(a), we can see that the bypass produces a "shunt" block with H_m parallel to the block $1/c_h$.

$$H_m = \frac{\Delta\dot{y}/K_m}{A\Delta P_L} = \frac{1}{AK_m} \cdot \frac{\Delta\dot{y}}{\Delta P_L} = \frac{1}{RA^2K_m}.$$

The damping coefficient β , due to bypass, can be calculated by replacing $1/c_h$ in eqn. (10) by $1/c_h + H_m$. The bypass method does not have the disadvantage of an enormous extra oil consumption, and in hydraulic servo's with mechanical feedback it is the only damping method that can be installed in a relatively short period of time. A serious disadvantage is the sensitivity of the laminar flow resistance R to temperature changes and its difficult adjustment in practice.

Quite a different but very important damping method is called *pressure feedback*, providing an extra servovalve displacement Δx proportional to the load pressure P_L . A mechanical version of pressure feedback is indicated schematically in Fig. 4.4(b). As with the bypass, its effect can be characterized in a transfer function H_m parallel to $1/c_h$:

$$H_m = \frac{\Delta x}{A\Delta P_L} = \frac{A_v \Delta P_L / c_v}{2A\Delta P_L} = \frac{A_v}{2Ac_v},$$

where A_v is the axial spool area pressed by P_L and springs of stiffness c_v . Mechanical pressure feedback calls for a very special valve, which has to be designed, manufactured, tested, etc. This method is therefore not very attractive, since there is no chance for mass-production. A very useful equivalent is *electrical pressure feedback*, which provides feedback of the electrically measured load pressure $P_L = P_1 - P_2$ to the input of an electro-hydraulic servovalve. The method is indicated in Fig. 4.4(a) by H_e .

Pressure feedback still has the disadvantage of increased load sensitivity. This disadvantage, however, can be avoided by introducing *dynamic pressure feedback* [7] with a transfer function H_m in Fig. 4.4(a) parallel to $1/c_h$.

$$H_m = K_p \frac{\tau s}{\tau s + 1}$$

At low frequencies $H_m = 0$ and the load-stiffness is therefore not spoiled. At higher frequencies, where the effect of C_o/C_h is less (see Section 4.3), $H_m = K_p$ will provide adequate damping if $1/\tau \ll \omega_o$ or $\omega_o \tau \gg 1$.

Dynamic (sometimes called derivative) pressure feedback can be established either mechanically or electrically. Figure 4.4(b) shows schematically two hydro-

mechanical filters connected to the cylinder compartments. The filters incorporate a (laminar) flow resistance R and a very small (gas or fluid) accumulator with a capacity C_a . The time constants indicated in this figure are $\tau = R C_a$. Careful analysis [8] of its effect will result in a transfer function H_m in Fig. 4.4(a)

$$H_m = \frac{1}{2R_n A^2 K_m} \cdot \frac{\tau s}{\tau s + 1}.$$

To design in practice a mechanical dynamic pressure feedback using "restricted" accumulators, is not too easy. To determine R_n and c_h (introduced by pressure feedback) we assume no initial damping at all, so with $w = 0$

$$\beta = \frac{K_v}{2\omega_o} \cdot \frac{C_o}{C_h}.$$

Desiring $K_v/\omega_o \approx \beta$ (≈ 0.4) we require $C_h = 0.5C_o = 0.5c_o$. Assuming a symmetric motor, Fig. 2.3(c) gives $c_{omin} = 4AE/S$, so

$$C_h = 0.5c_o = \frac{2AE}{S}$$

To determine R_n we first look at high frequencies ($\omega\tau \gg 1$)

$$H_m = \frac{1}{2R_n A^2 K_m} = \frac{1}{c_h}$$

Combination gives

$$R_n = \frac{c_h}{2A^2 K_m} = \frac{C_h}{2A^2 K_v} = \frac{2AE/S}{2A^2 K_v} = \frac{E}{K_v AS}$$

To determine the accumulator volume V_a we have to find

$$\omega_o \tau = \omega_o R_n C_a = \omega_o \cdot \frac{E}{K_v AS} \cdot \frac{V_a}{E} = \frac{\omega_o}{K_v} \cdot \frac{V_a}{AS}.$$

The ratio accumulator volume V_a to cylinder volume AS now becomes

$$\frac{V_a}{AS} = \frac{K_v}{\omega_o} \cdot \omega_o \tau.$$

Having $K_v/\omega_o \approx 0.4$ and requiring $\omega_o \tau > 1$, the accumulator volume will roughly be the same as that of the cylinder. Therefore "diaphragm"-accumulator or gas-accumulators ($C_a = V_a/nP_a$) are more appropriate, resulting in considerably smaller volumes. Though elegant practical solutions are known [8], we generally prefer the *electrical* equivalent for mechanical dynamic pressure feedback, which gives scarcely any hardware problems.

Probably the most attractive damping method for electro-hydraulic servosystems is obtained by *acceleration feedback* [1], [3] as shown in the block diagram of Fig. 4.4(c). It is based on electrically measured accelerations of the piston (in rotary motors derived from tachogenerator signals). Rearranging the block diagram to

$$\hat{H}(s) = \frac{\frac{K_v}{s \left[\frac{s^2}{\omega_o^2} + 2\beta \frac{s}{\omega_o} + 1 \right]}}{1 + \frac{K_v K_a s^2}{s \left[\frac{s^2}{\omega_o^2} + 2\beta \frac{s}{\omega_o} + 1 \right]}} = \frac{K_v}{s \left[\frac{s^2}{\omega_o^2} + \left(\frac{2\beta}{\omega_o} + K_a K_v \right) s + 1 \right]} \quad (13)$$

results in an increase $\Delta\beta$ of the damping coefficient β .

$$\underline{\underline{\Delta\beta = 0.5 K_a K_v \omega_o}} \quad (14)$$

This very elegant damping method has been applied numerous times with complete success. Yet some manufacturers of expensive and complex systems seem reluctant to use acceleration feedback and sometimes prefer obsolete damping methods. The validity of eqns. (13) and (14) is limited of course by the validity of the mathematical model as shown in Fig. 4.4(a) and the assumption $G(s) = 1$. In more complicated systems, incorporating compliant structures, etc. (see Section 4.9), the method cannot be used without serious complications. However, this is true for the other damping methods as well.

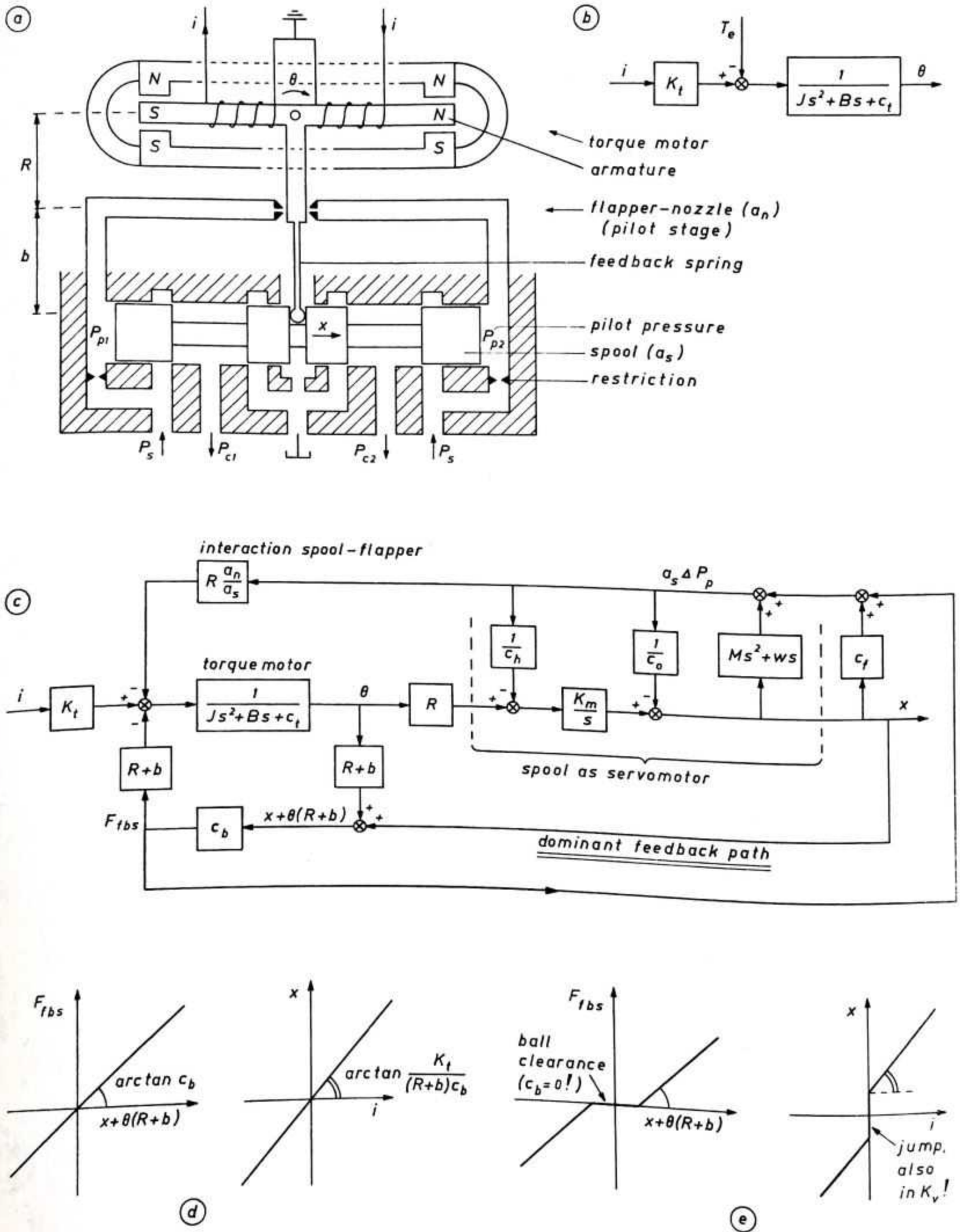


Fig. 4.5 Analysis of two-stage electro-hydraulic servovalve

4.6 Electro-hydraulic servovalve [3], [4], [5]

In Fig. 4.5 the widely used Moog two-stage electro-hydraulic servovalve is depicted schematically. First of all an electrical signal is converted into a mechanical signal, as the control current i results in a torque T which causes a rotation θ of the armature together with the flapper-arm. The (simplified) block diagram of the torque motor is given in Fig. 4.5(b).

The first stage (often called *pilot* stage) of the valve consists of a symmetric flapper-nozzle system, fed by the constant supply pressure via two restrictions. In the stationary situation an excursion θ_R of the flapper results in two different pilot pressures. Assuming a symmetric construction, all orifice area's being equal at $\theta = 0$, an "underlap" u with $-u < \theta_R < +u$, the pilot pressures can be derived directly.

$$\frac{\Delta P_p}{P_s} = \frac{P_{p1} - P_{p2}}{P_s} = \frac{1}{1 + \left(1 - \frac{\theta_R}{u}\right)^2} - \frac{1}{1 + \left(1 + \frac{\theta_R}{u}\right)^2} = \frac{\frac{\theta_R}{u}}{1 + \frac{1}{4}\left(\frac{\theta_R}{u}\right)^4} \approx \frac{\theta_R}{u}. \quad (15)$$

As long as $|\theta_R| < 0.8 u$ its characteristic is almost perfectly linear!

In the dynamic situation the four orifices of the pilot stage control the oil flows to and from the "cylinder" compartments on both sides of the main spool of the servovalve, by which the servomotor itself is controlled. According to the model of hydraulic servomotors derived in Chapter 3, the main spool itself should be considered as a very small servomotor. Figure 4.5(c) gives the block diagram, comprising the well-known parameters K_m , c_h , c_o , M and w .

The dominant feedback path is provided by a spring, usually called the feedback spring. An excursion x of the main valve together with a flapper arm rotation θ results in a force $[x + \theta(R + b)]c_b$ at the end of the feedback spring. The resulting torque acts as a direct feedback indicated in the lower feedback path of Fig. 4.5(b). The stiffness c_b of the feedback spring has a vital function, which will be discussed again later on.

By reaction the feedback force $[x + \theta(R + b)]c_b$ of the spring should be regarded as a load on the spool. Another "spring" force, $c_f x$, is caused by axial flow forces in the port openings of the spool (see Blackburn, Fluid Power Control, Chapter 10, Section 3). The upper left feedback path in Fig. 4.5(c) represents the interaction between main valve and flapper, as the load pressure ΔP_p implies a feedback force on the flapper. Its influence should be small.

The whole block diagram, simplified as it is, still is too complicated to be dealt with here in detail. Briefly we simply state here that the lower feedback path is dominant. Consequently at low frequencies the transfer function becomes

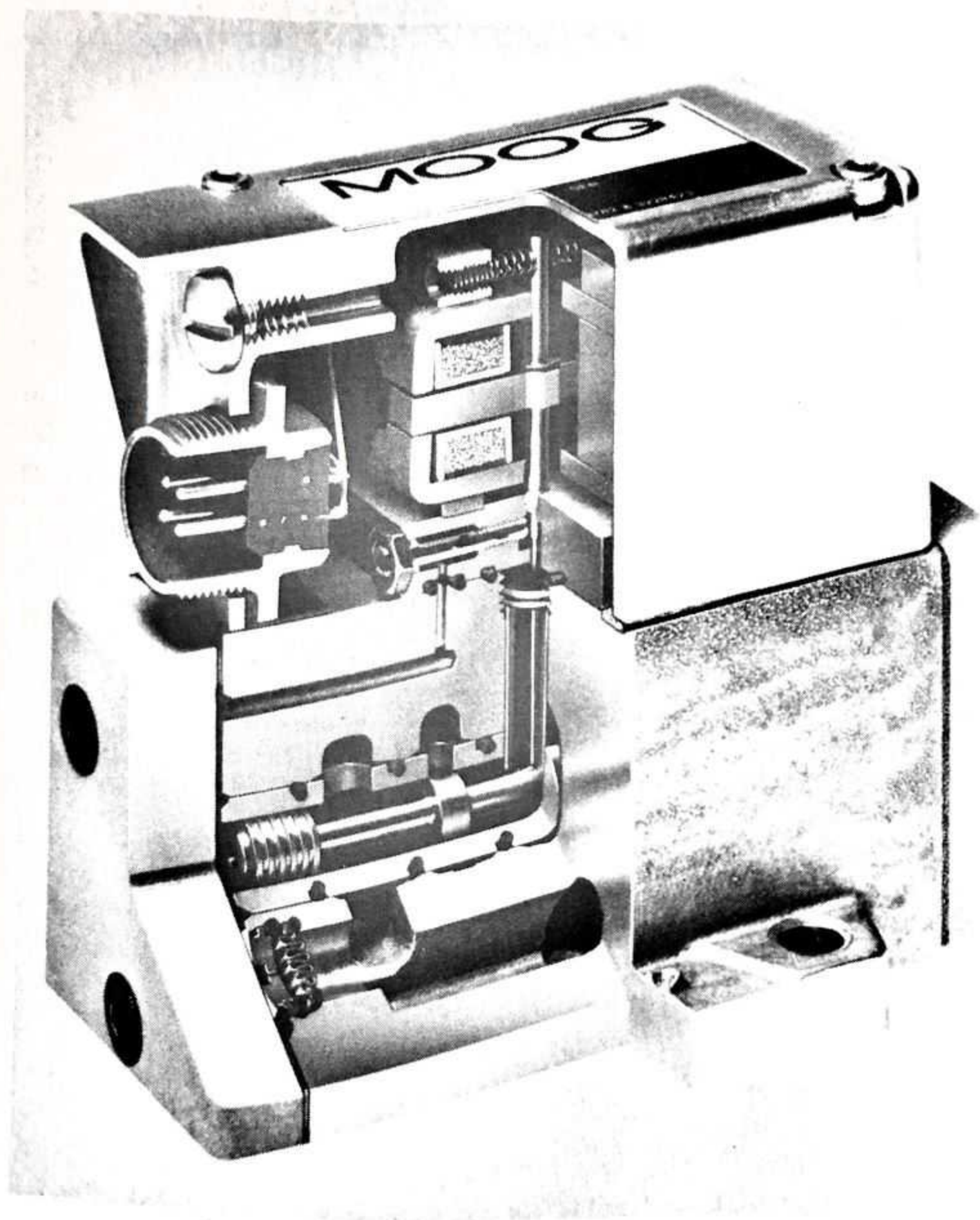


Fig. 4.6 Moog servovalve

$$\lim_{s \rightarrow 0} \frac{x}{i} = \frac{K_t}{(R+b)c_b} \quad (16)$$

It illustrates the well-known fact that the closed loop transfer function of a control system with $\hat{H}(s) \gg 1$ equals (approximately) the reciprocal of the feedback path transfer function.

This knowledge enables us to explain how the servovalve characteristic could deteriorate completely by a clearance or backlash between the ball at the end of the feedback spring and the slot in the spool. Using eqn. (16), Fig. 4.5(d) and (e) show how the presence of such a clearance causes a valve characteristic with a vertical "jump". This is very perilous in closed loop servosystems, as a comparable jump in the velocity gain characteristic of the servo will appear, making the system completely unstable at low speeds!

The construction of the Moog valve is illustrated in Fig. 4.6(a). In the table below some important data have been summarized. On the basis of manufacturer's specifications one could represent the servovalve transfer function in practice by a first or second order system, depending on the frequency range involved [9].

Table 4.1 Properties of Moog servovalves

Series	Q in dm ³ /min	N in kW	ω_{90° (50% rated)
30	1.1 - 7.6	0.38 - 2.6	230 Hz
31	6.5 - 26	2.3 - 9.0	230 Hz
32	26 - 53	9.0 - 18.3	150 Hz
34	48 - 72	16 - 25	120 Hz
35	72 - 113	25 - 39	90 Hz
76	6.5 - 113	2.3 - 39	150 - 90 Hz

4.7 Block diagram for hardware components

Block diagrams for control systems obviously represent hardware components. However, the relations between the parameters in the theoretical block diagram and the characteristics of hardware components are sometimes hard to find. Therefore it makes sense to make two types of block diagram, one for the theoretical control behavior and one for the hardware. As both diagrams should cover the same system, the relations can easily be found.

To illustrate this, Figs. 4.7(a) and (b) present two block diagrams according to control theory, of a simple linear electro-hydraulic servosystem with acceleration feedback. The extra damping $\Delta\beta$ can be found by equalizing both loop gains (see also Section 4.5, Fig. 4.4(c)).

$$\Delta\beta = 0.5 K_a K_v \omega_o$$

In Fig. 4.7(c) the "hardware" block diagram is presented. In a *given* system we have the following components:

- *servovalve amplifier* to convert the control error signal ε (in Volts) to the servovalve input control current x_3 (in Ampères).
- *servovalve* converting x_3 into flow x_4 (in m^3/s). In commercial catalogs available data might indicate milliamps and gallons per minute. These data must be converted into ISO-units!
- *position transducer* (K_3), giving Volts per meter.
- *acceleration transducer* (K_4), giving Volts per m/s^2 .
- *motor*, with piston area A (in m^2) and mass-spring system parameters ω_o and β .

The natural frequency $\omega_o = \sqrt{C_o/M}$ might be determined by calculating (or estimating) C_o from Fig. 2.3. It is also possible to measure ω_o and β experimentally.

The gain V_1 and V_2 in Fig. (c) can be chosen freely to satisfy the desired adjustment of K_v and K_a of Fig. (a). Equalizing the products of transfer functions in the various loops in Fig. (a) and Fig. (c) yields

$$V_1 K_1 K_2 \cdot \frac{1}{A} \cdot K_3 = K_v$$

$$V_1 K_1 K_2 \cdot \frac{1}{A} \cdot K_4 V_2 = K_v K_a$$

The foregoing results can be summarized in the final adjustments

$$K_a = \frac{2\Delta\beta}{K_v \omega_o}, \quad V_1 = \frac{AK_v}{K_1 K_2 K_3}, \quad V_2 = \frac{K_a K_3}{K_4}$$

This example just illustrates how one could try to find the adjustments. However, it is not always possible to translate theoretical diagrams directly into hardware diagrams. By focussing on special aspects, however, in many cases good approximations can be made to enable adequate possibilities for calculation or estimation.

$$\dot{H}(s) = \frac{\tau_i s + 1}{\tau_i s} \cdot \frac{K_v}{s} \cdot \frac{1}{\frac{s^2}{\omega_0^2} + 2\beta \frac{s}{\omega_0} + 1}$$

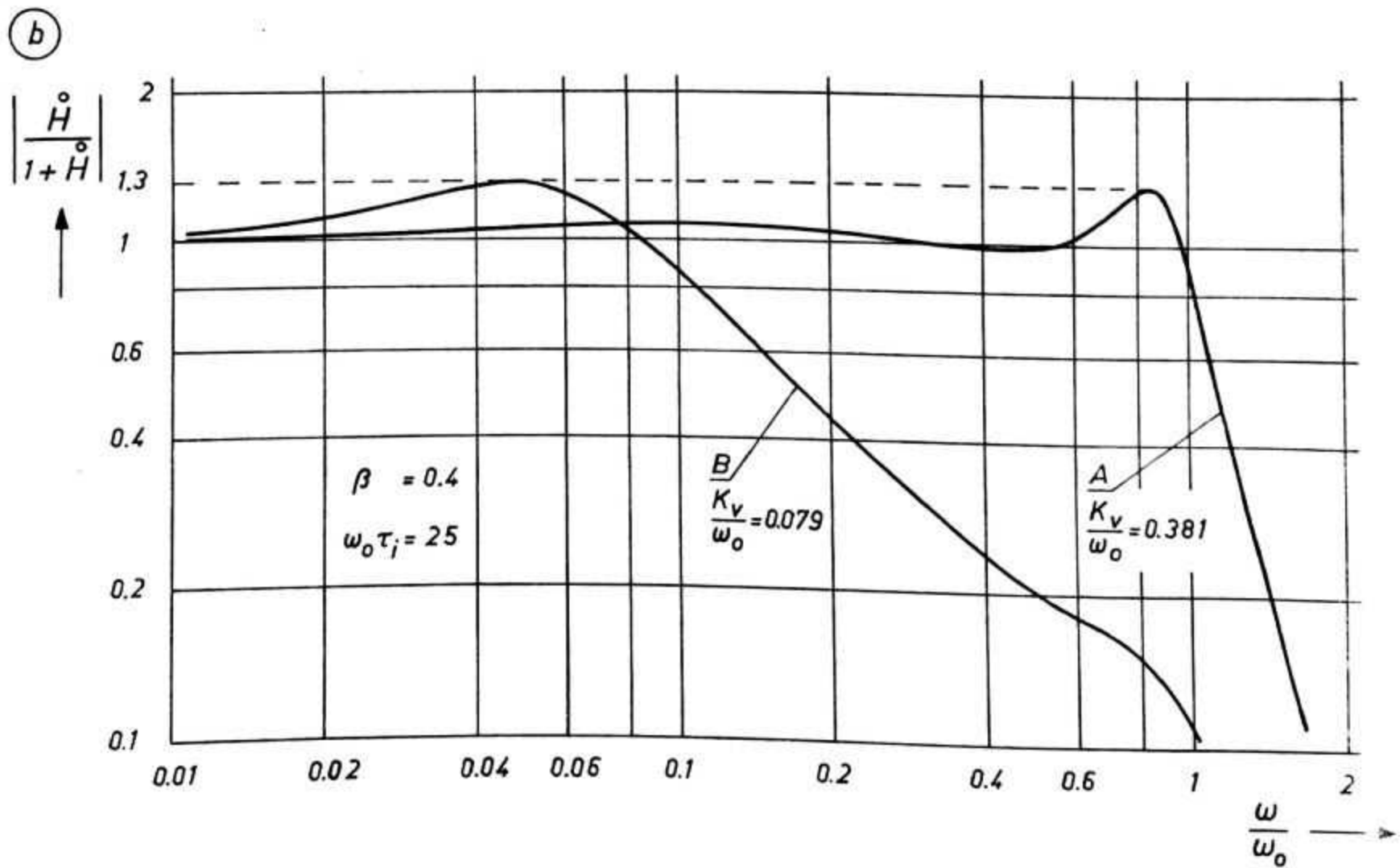
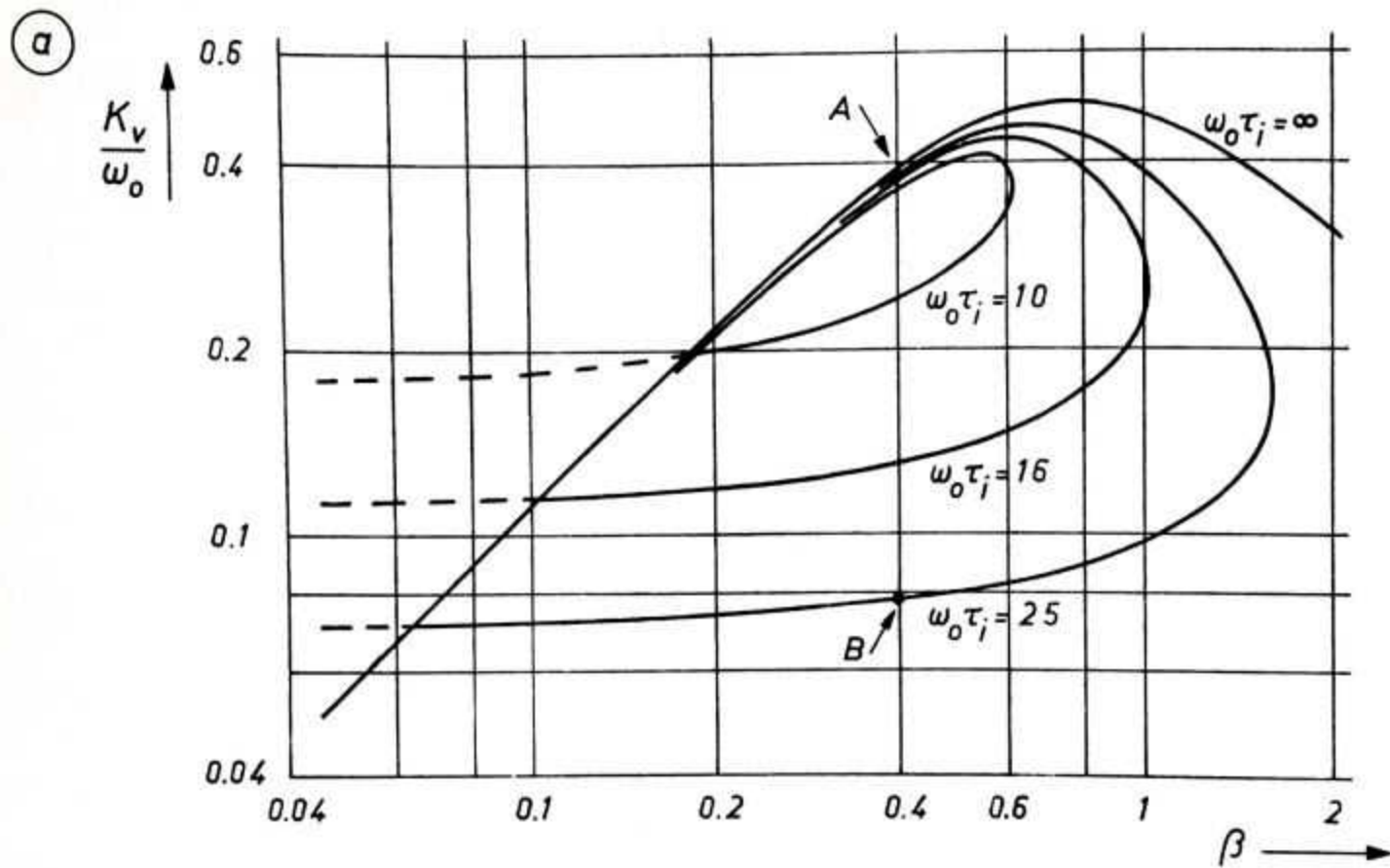


Fig. 4.8 PI-control

*4.8 PI-control

For some applications of hydraulic servosystems a PI-controller might be used in order to increase the open loop gain at low frequencies and to decrease constant-load errors as well as "acceleration" errors. The choice of the "reset time" τ_i of the PI-controller should be considered carefully, as the transfer function

$$\hat{H}(s) = \frac{\tau_i s + 1}{\tau_i s} \cdot \frac{K_v}{s \left[\frac{s^2}{\omega_o^2} + 2\beta \frac{s}{\omega_o} + 1 \right]} \quad (17)$$

contains a double integrator, causing serious stability problems. To see this we ignore for a moment the second order system (so $\omega_o = \infty$) and choose the value of K_v on the basis of a phase margin of 45° . The result is

$$K_v \tau_i \geq 0.5\sqrt{2} \quad \text{or} \quad K_v / \omega_o \geq 0.7 / \omega_o \tau_i \quad (18)$$

so the gain K_v should be chosen not too small (!) in order to provide an adequate damping.

Results of computer calculations, using the $M = 1.3$ -criterion and eqn. (17), have been plotted in Fig. 4.8(a), where K_v / ω_o as a function of β with $\omega_o \tau_i$ as a parameter shows that upper boundaries as well as lower boundaries have been respected. As accurate knowledge of the values of ω_o and β cannot be expected, one should be very careful in adjusting K_v and τ_i . To avoid negative influences or risks due to the I-action $\omega_o \tau_i > 25$ is recommended.

Figure 4.8(b) makes more clear that one should choose the value of K_v / ω_o corresponding to the upper boundary in order to obtain a maximum bandwidth.

Since PI-control introduces real risks with regard to stability, in the large majority of applications P-control is preferable!

$$\dot{H}(s) = \frac{1}{\tau s + 1} \cdot \frac{K_v}{s} \cdot \frac{1}{\frac{s^2}{\omega_0^2} + 2\beta \frac{s}{\omega_0} + 1}$$

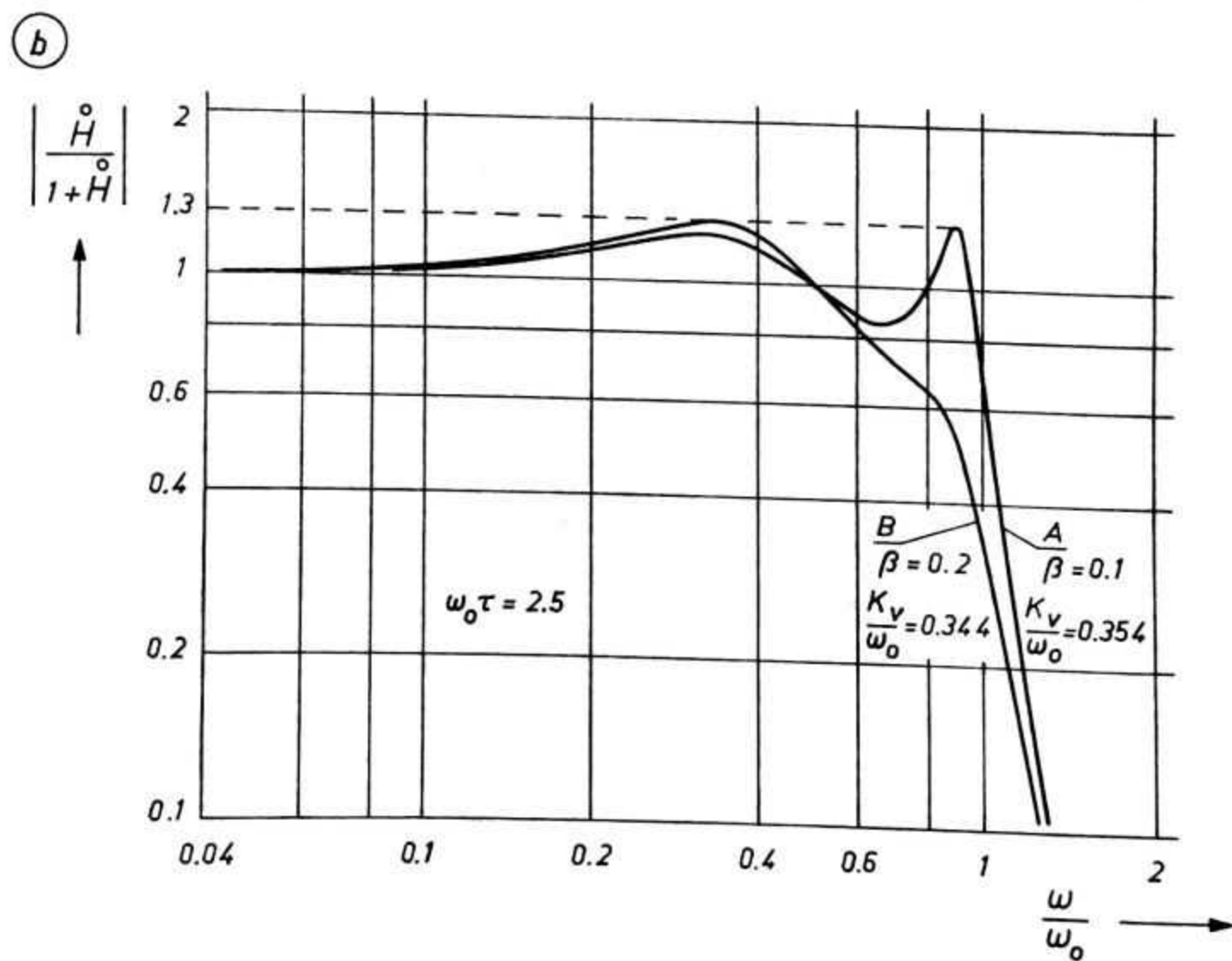
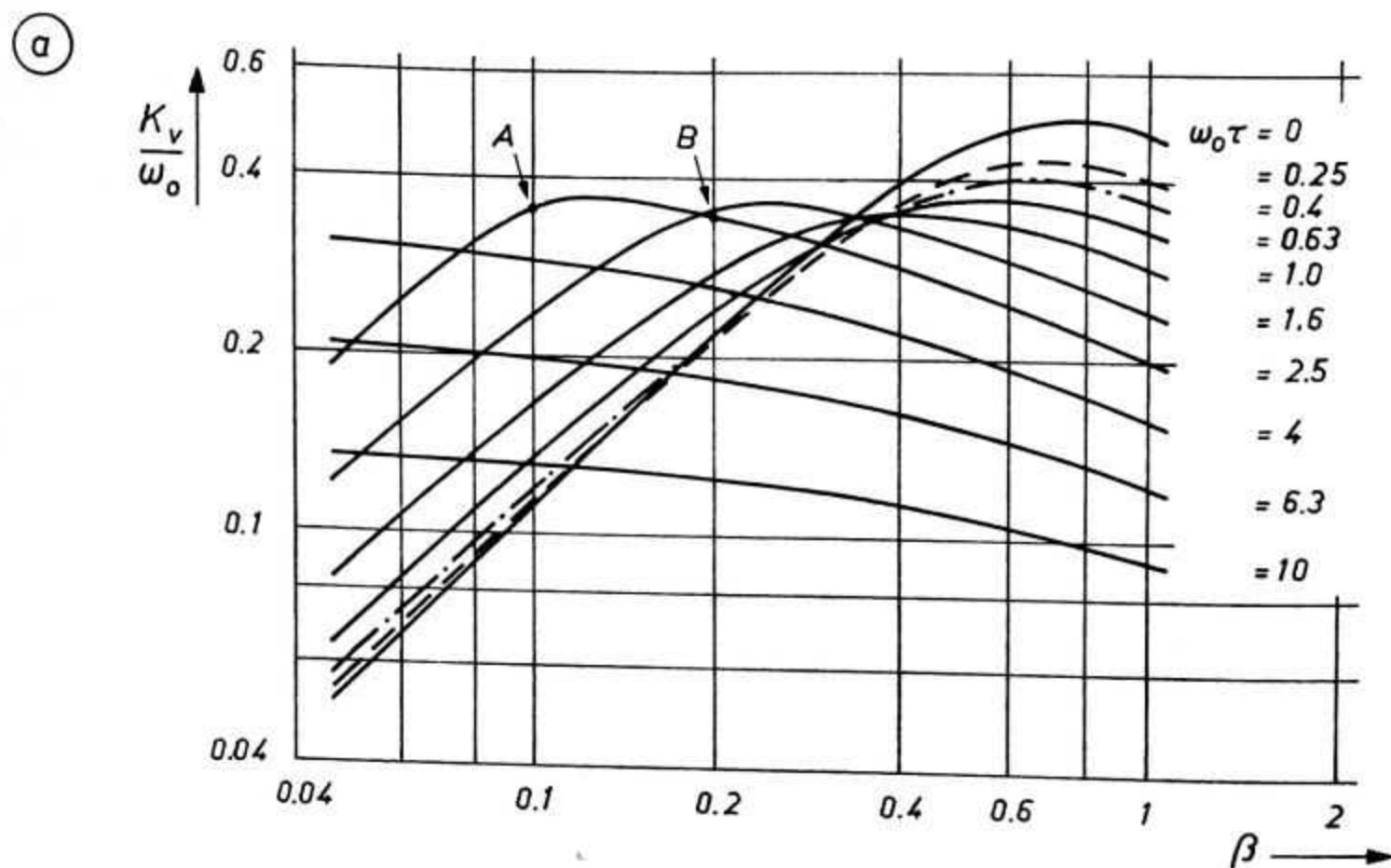


Fig. 4.9 First order system in series

*4.9 First order system in series

To give a more complete picture of the dynamic behavior of hydraulic servosystems, the influence of a first order system in series, implying a transfer function

$$\bar{H}(s) = \frac{1}{\tau s + 1} \cdot \frac{K_v}{s \left[\frac{s^2}{\omega_o^2} + 2\beta \frac{s}{\omega_o} + 1 \right]}, \quad (19)$$

is discussed. Such a first order system could be introduced in a "natural" way (for instance, by bringing into account some influence of the servovalve transfer function) or deliberately by a simple RC-network in series with an electro-hydraulic servovalve.

The influence of the time constant τ on the maximum allowable velocity gain K_v has been investigated systematically. Allowing - just as before - a closed loop amplitude ratio $M = |\bar{H}/(1 + \bar{H})|$ never exceeding 1.3, the velocity gain K_v could be calculated. The results are presented in Fig. 4.9(a), showing a remarkable increase of K_v/ω_o at poor damping coefficients ($\beta < 0.2$). Assigning value to the bandwidth $1/\tau$ of the first order system considerably lower than the natural frequency ω_o (i.e. $\omega_o \tau > 2$), the velocity gain could be increased easily by a factor 2. This improvement is mainly due to the extra phase lag of the first order system. At $\omega_o \tau \gg 1$ the influence of the second order system could even be neglected, since K_v could be as large as $1.4/\tau$ or $K_v/\omega_o \approx 1.4/\omega_o \tau$. Approaching that extreme situation however, the bandwidth ω_b of the controlled system is decreased sharply.

Figure 4.9(b) shows that for values of the damping coefficient β as low as 0.1, a system bandwidth approaching the natural frequency ω_o is possible. Thus compared with proportional control (see Fig. 4.2), the influence of a proper first order system is a positive one!

*Hydraulic servo's
with 'parasitic'
mass-spring systems*

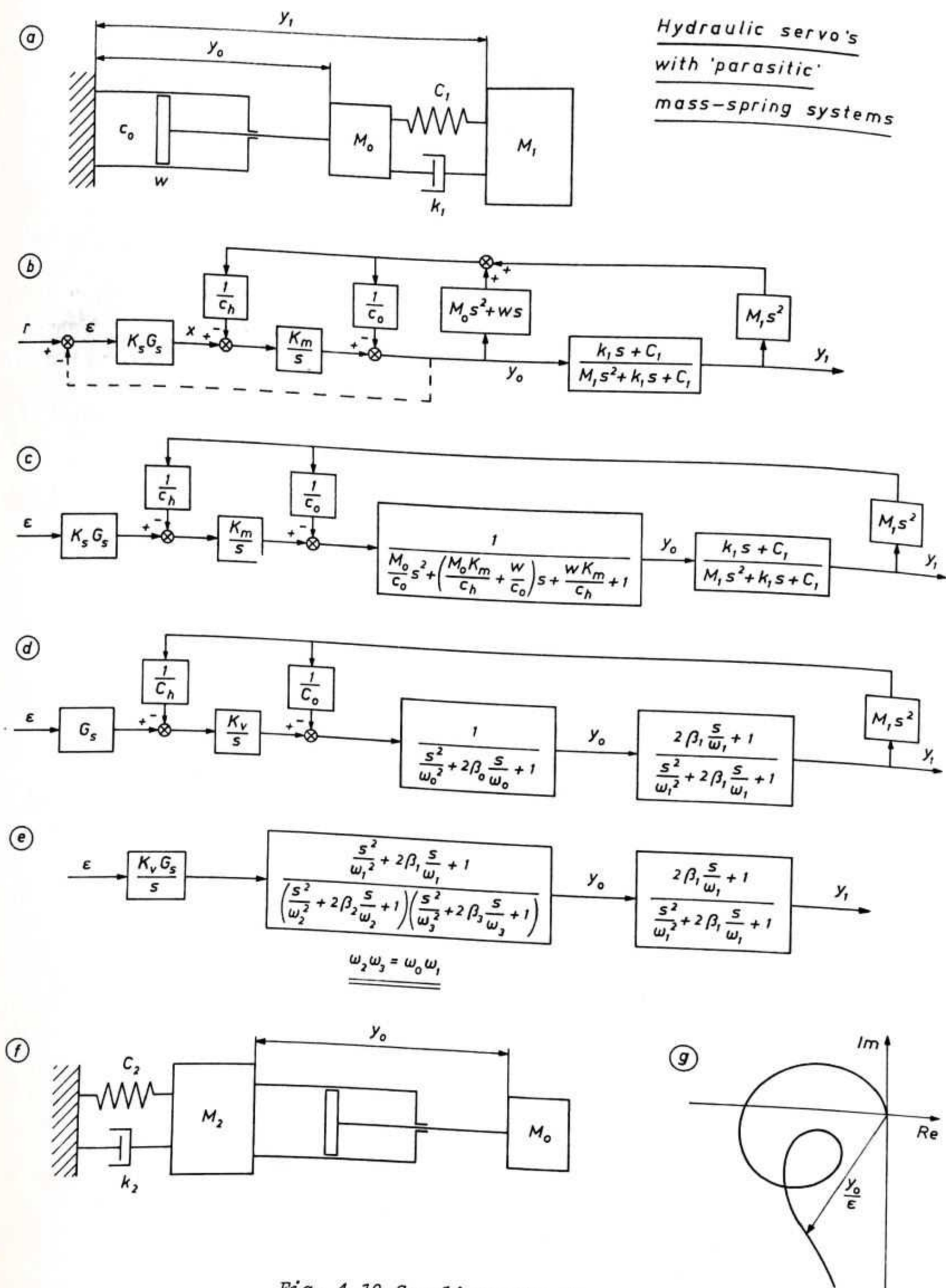


Fig. 4.10 Compliant structures

*4.10 Compliant structures

A way to change dynamic properties of hydraulic servo's dramatically is by the introduction of compliant structures. Figure 4.10(a) gives a (simplified) example, where the piston is actuating a mass-spring system. Such a situation can be expected in larger constructions where part of the actuated mass is connected more or less flexibly to the piston. Analysis of Fig. 4.10(a) shows

$$\left. \begin{aligned} F &= C_1(y_0 - y_1) + K_1(\dot{y}_0 - \dot{y}_1) = M_1\ddot{y}_1 \\ (K_1s + C_1)y_0 &= (M_1s^2 + K_1s + C_1)y_1 \end{aligned} \right\}$$

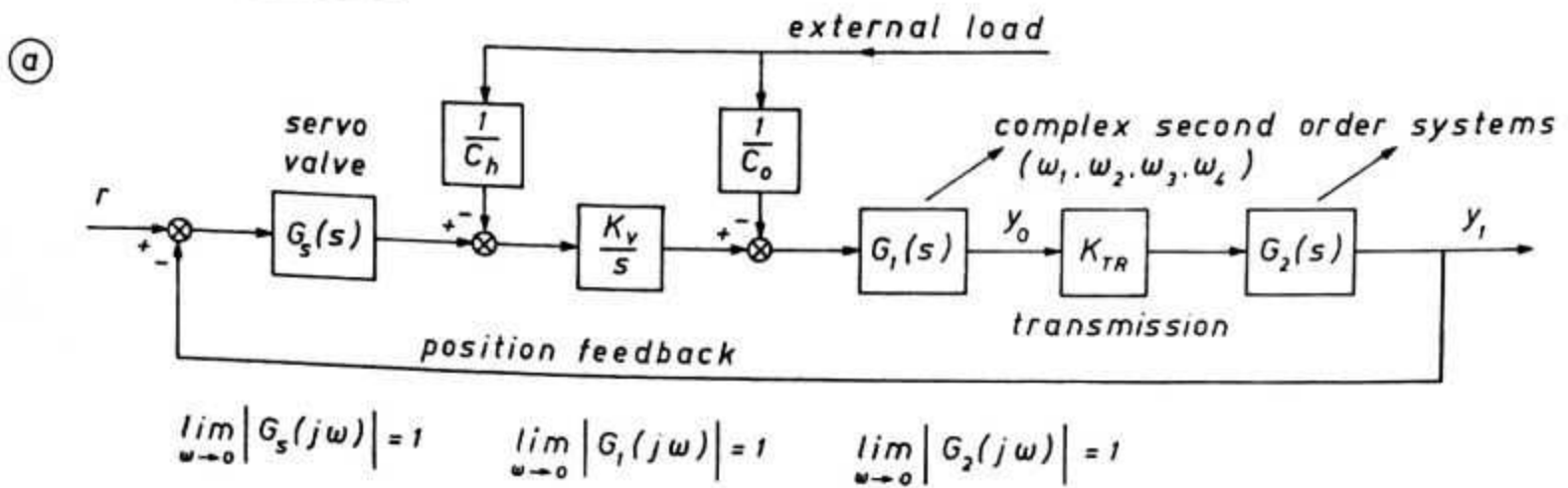
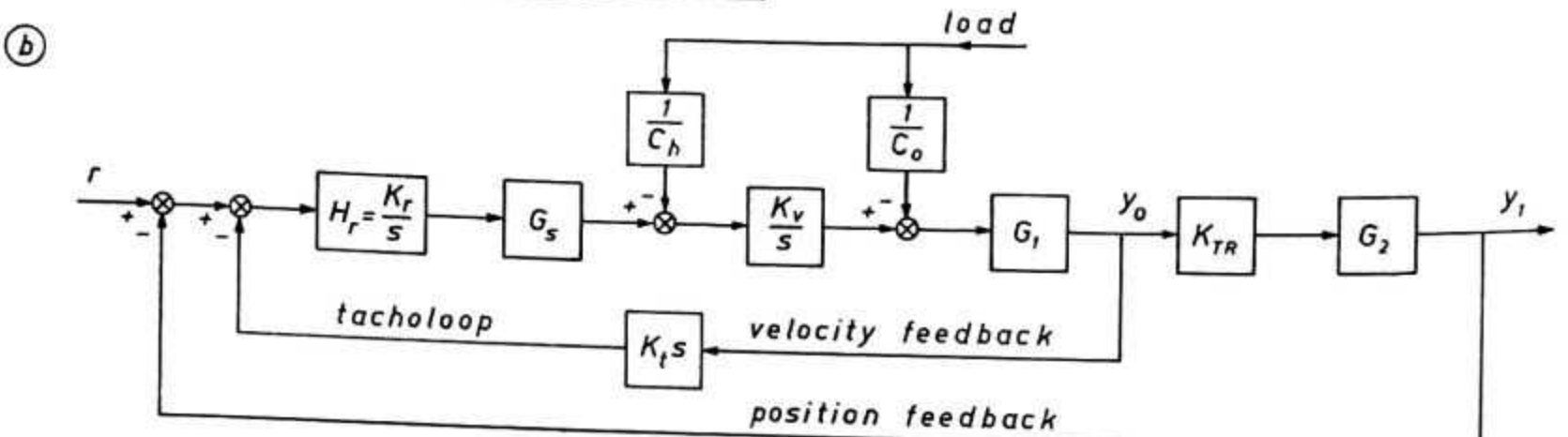
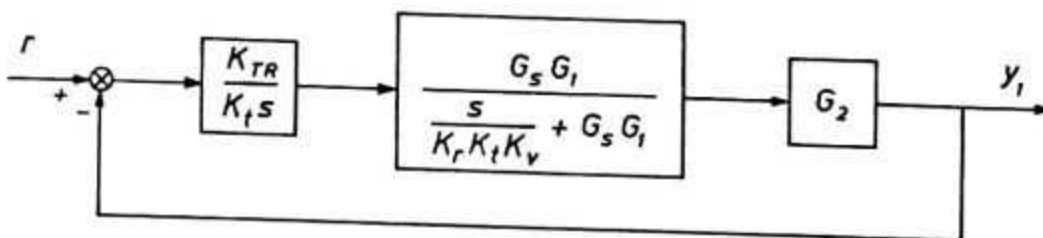
$$\frac{y_0}{y_1} = \frac{K_1s + C_1}{M_1s^2 + K_1s + C_1}.$$

The block diagrams (b), (c), (d) and (e) indicate how the transfer function could be found. The result is a rather complex transfer function with *three* natural frequencies ω_1 , ω_2 and ω_3 !

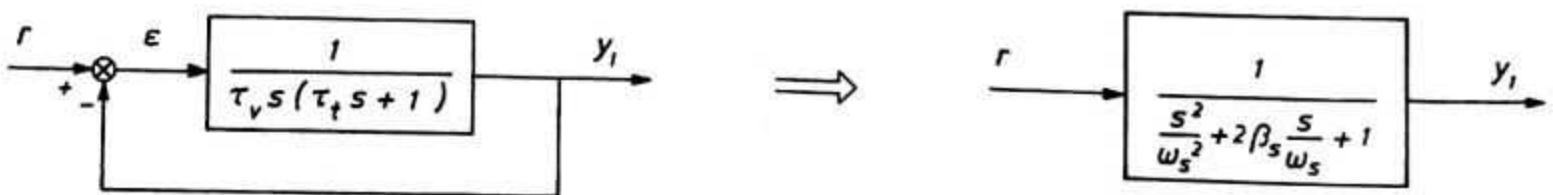
Its frequency response is usually rather wild, especially when two of the natural frequencies (for example ω_1 and ω_2) are very close, as happens frequently. Then in a small frequency range (around ω_1 and ω_2) heavy changes in the open loop amplitude ratio and phase lag could be expected, especially at low values of β_1 and/or β_2 . The Nyquist plot in (g) gives a typical example showing a characteristic curl!

As the systematic analysis of compliant structure will lead us too far now, we restrict ourselves in a more qualitative way to one important aspect. Because compliant structures could deteriorate the dynamic properties of a servosystem completely they are sometimes indicated as "parasitic" mass-spring systems. To limit its influence, the natural frequency of the parasitic system itself should be designed as high as possible, preferably 10 times higher (or even more) than the designed bandwidth of the complete system.

A comparable complex situation is depicted in Fig. 4.10(f), arising for instance when an actuator stands on a floor, having a natural frequency itself. Both types of compliant structures frequently occur in hydraulically actuated machine tools. One simple solution of the control problem is shown in the next section.

Proportional controlWith tacholoop and integrating controllerNeglecting external load

Assuming $\left. \begin{array}{l} |G_s(j\omega)| = 1 \\ |G_1(j\omega)| = 1 \\ |G_2(j\omega)| = 1 \end{array} \right\} \text{ at } \omega \approx \frac{1}{\tau_t} = K_r K_t K_v$



$\tau_v = \frac{K_t}{K_{TR}}$ $\tau_t = \frac{1}{K_r K_t K_v}$ $\tau_v \geq 4\tau_t \Rightarrow \beta_s \geq 1$

Fig. 4.11 LF-control

*4.11 LF-control of complex hydraulic servo's

In Fig. 4.11 a proportionally controlled hydraulic servo with a parasitic mass-spring system is represented by its simplified block diagram corresponding to Fig. 4.10(e). Figure 4.11(b) shows the way in which many hydraulically actuated NC machine tools are controlled. In those machines the screw spindle and machine tool slide represent the mass-spring system and a rotary hydraulic motor the actuator.

In Fig. 4.11(b) a tacho feedback loop has been introduced as well as an integrating controller ($H_r = K_r/s$). Neglecting the external load, the block diagram can be simplified if the natural frequencies of G_s , G_1 and G_2 are very large compared with the desired bandwidth of the complete system. That bandwidth usually is very low in NC machines.

In this low bandwidth control system, transformed to diagram (c), we may assume $G_s(j\omega)$, $G_1(j\omega)$ and $G_2(j\omega) = 1$ at $\omega = 1/\tau_t = K_r K_t K_v$. So this LF-control system is simplified to an integrator in series with a first order system, fed back proportionally. A major purpose of this method of control is that the velocity time constant $\tau_v = K_t/K_{TR}$ (inverse of the velocity gain!) is determined completely by constant and very reliable linear elements. It can be adjusted easily as well as very precisely to a desired fixed value, and that is precisely what should be done in NC machine with a servo for each axis of motion. Its importance can be demonstrated by assuming an NC machine milling with constant speed along a straight line, in which two servo's participate. The velocity errors of the servo's do not cause any workpiece error if the velocity gains are exactly equal. In that particular case the combination of the velocity errors results in a cutter path tangential to the work piece. For this reason the velocity gain of all servo's should be examined and adjusted experimentally to have equal values within a few percentages.

The implication of this method of LF-control is that *the maximum obtainable bandwidth is not reached at all*, but in many NC machines a large bandwidth is considered to be far less important than equal velocity gains. Thus Fig. 4.11 shows a safe and practical solution for a complex control problem.

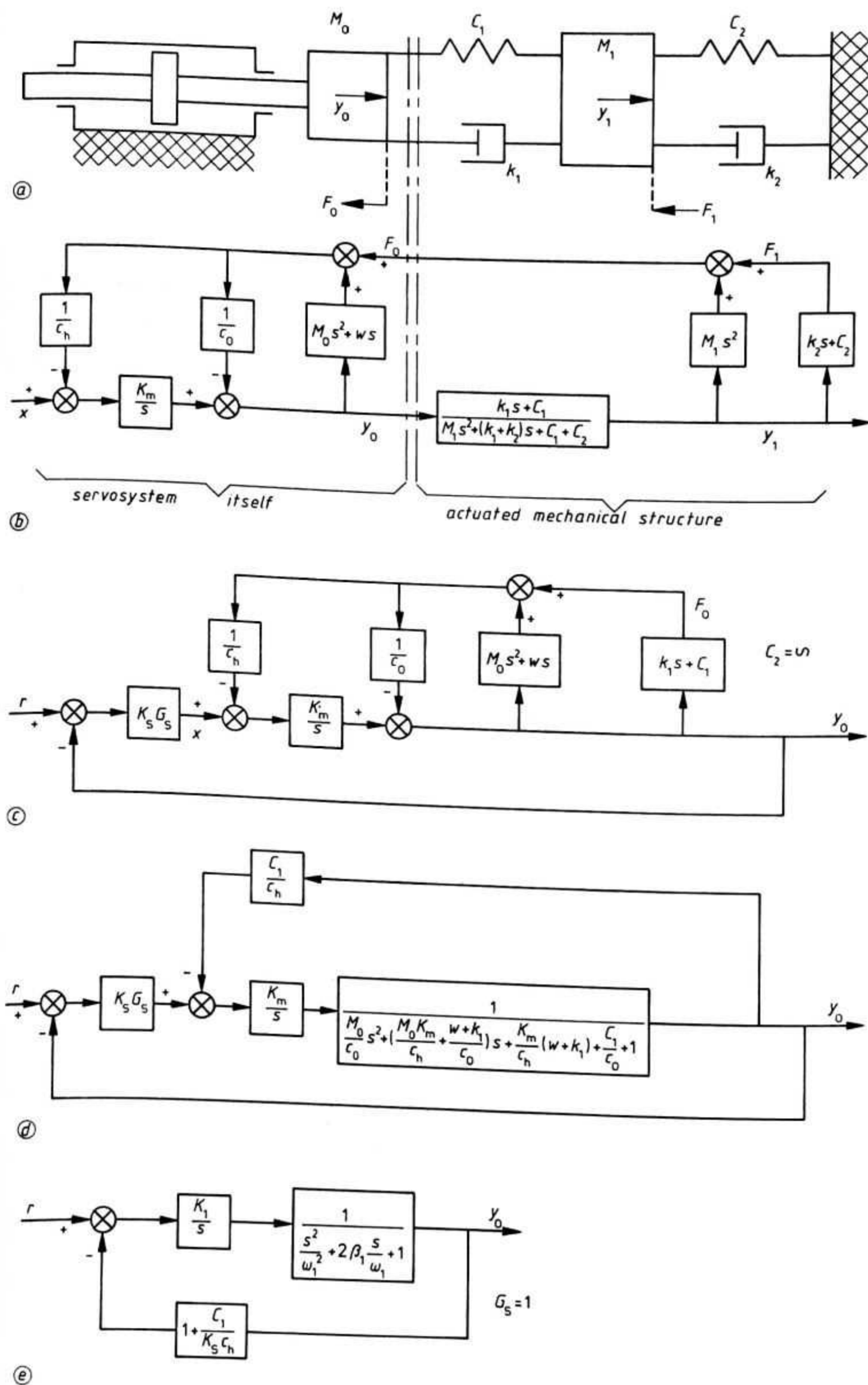


Fig. 4.12 Simplified model of a load actuator

*4.12 Load actuators

Hydraulic servosystems find widespread applications in materials and fatigue testing equipment where the forces (= load being introduced) are essential. There is a remarkable lack of clear information concerning the dynamic behavior of load actuators. One of the reasons for this is the enormous variety of the mechanical structures in which load actuators are incorporated. Obviously, the structure of a hydraulic press, of fatigue testing equipment for aeroplane wings, and of a simple tensile-strength probe tester are completely different. Normally, each special equipment requires thorough analysis, revealing in many cases distributed parameter systems as well as lumped parameter (mass-spring) systems.

Figure 4.12(a) represents a simplified model of a load actuator in a compliant mechanical structure. Figure (b) gives the block diagram in which the servosystem itself as well as the mechanical structure can be recognized easily. Introducing two "concentrated" masses M_0 and M_1 , two springs (C_1 and C_2) and two dampers (k_1 and k_2) we find:

$$\left. \begin{aligned} M_1 \ddot{y}_1 &= C_1(y_0 - y_1) + k_1(\dot{y}_0 - \dot{y}_1) - C_2 y_1 - k_2 \dot{y}_1 \\ [M_1 s^2 + (k_1 + k_2)s + C_1 + C_2] y_1 &= (k_1 s + C_1) y_0 \end{aligned} \right\}$$

$$\frac{y_1}{y_0} = \frac{k_1 s + C_1}{M_1 s^2 + (k_1 + k_2)s + C_1 + C_2}$$

The *leftward* forces F_0 and F_1 on the piston and on the mass M_1 , respectively, are:

$$F_0 = k_1(\dot{y}_0 - \dot{y}_1) + C_1(y_0 - y_1) = (k_1 s + C_1)(y_0 - y_1) = (M_1 s^2 + k_2 s + C_2) y_1$$

$$F_1 = k_2 \dot{y}_1 + C_2 y_1 = (k_2 s + C_2) y_1$$

The transfer function y_1/y_0 as well as the forces F_0 and F_1 are indicated directly in diagram (b). It should be noted that - in reaction - the forces F_0 and F_1 act in *rightward* direction on the structure.

There are at least four possibilities for feedback:

y_0 and y_1 by means of a displacement transducer,

F_0 and F_1 by means of a force transducer (load-cell).

The choice to be made depends largely on the nature of the application and the values of the various parameters. Since generalisations are hardly possible, we take as examples two extremely simplified cases.

o In the case that $C_2 = \infty$ (and consequently $y_1 = 0$), diagram (b) can be simplified to (c) and (d). If $G_s = 1$, the resulting diagram (e) is similar to that of an usual servo with position feedback. From the control point of view it hardly matters whether there is a position or a force feedback!

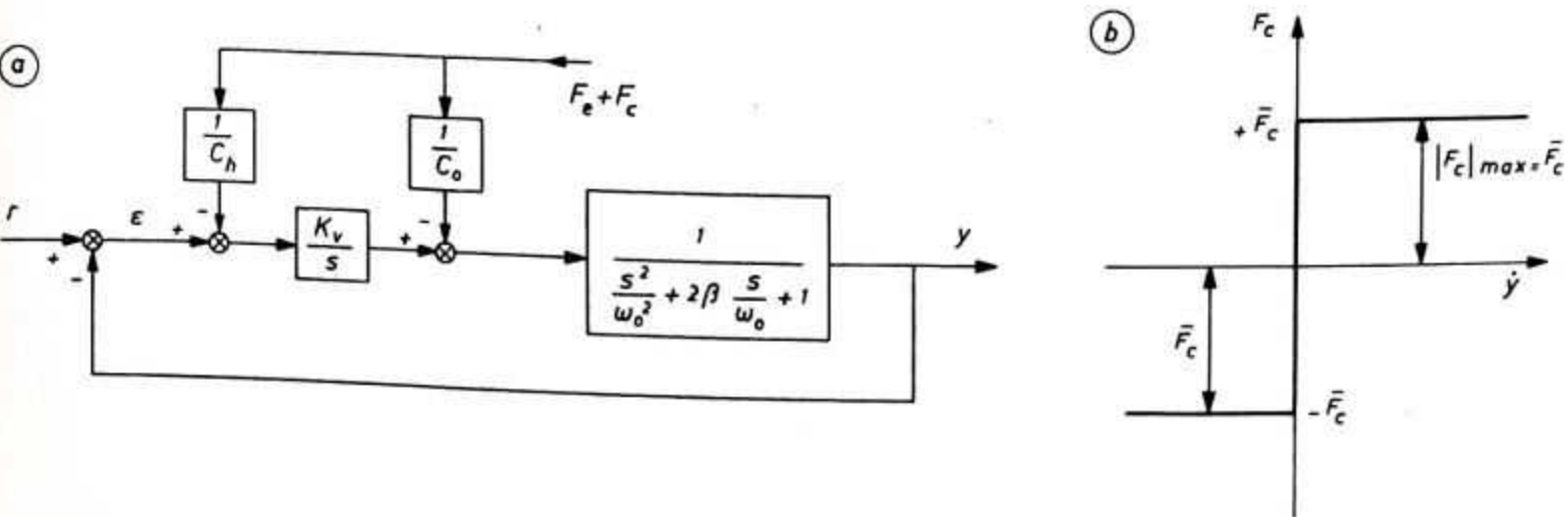


Fig. 4.13 Hydraulic servosystem with Coulomb friction

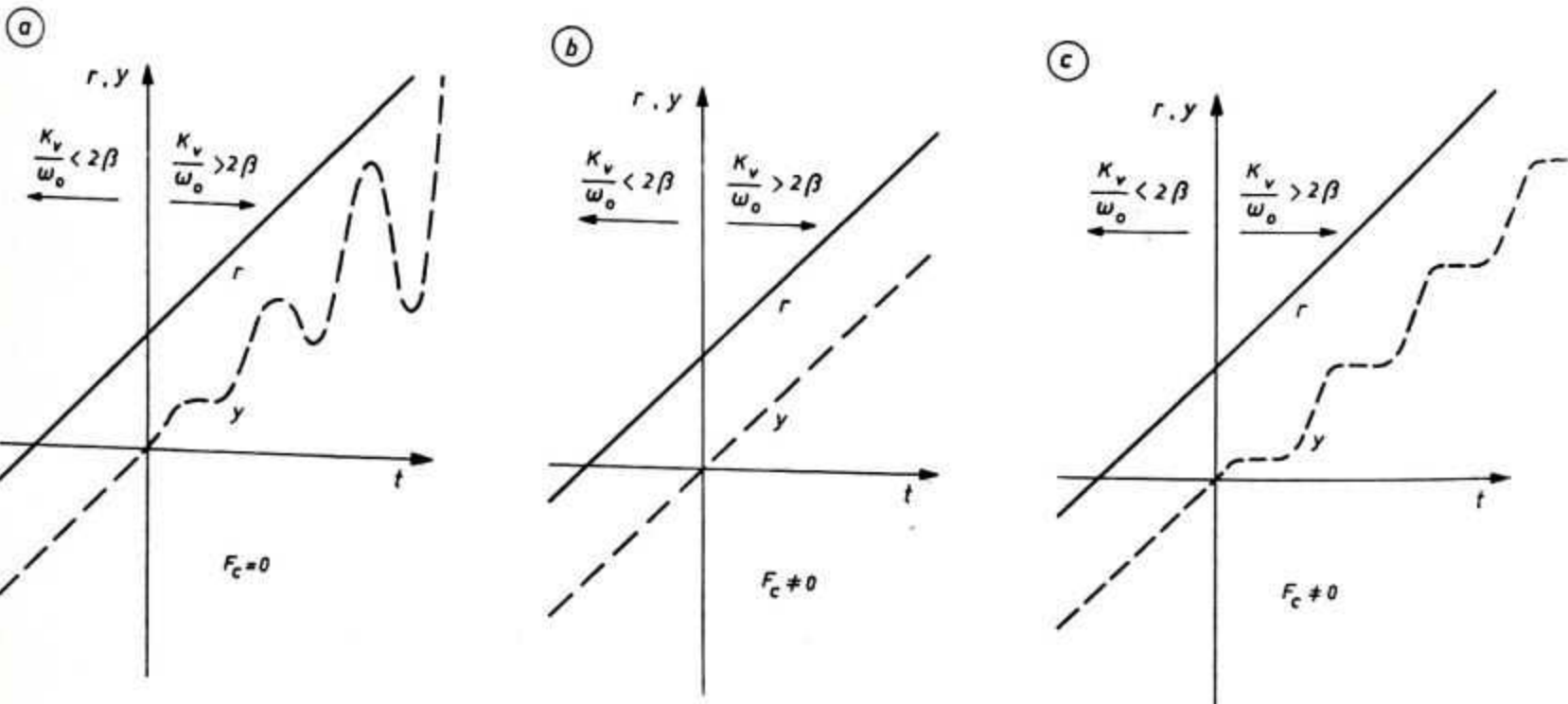


Fig. 4.14 Stability and Coulomb friction

o In the case $C_2 = 0$ and $k_2 = 0$, we find the compliant structure being dealt with in Section 4.10. Now position and force feedback differ essentially!

These two simple cases should make very clear that the nature of the mechanical structure has an overwhelming influence on the overall dynamic behavior.

*4.13 Coulomb friction and stability

As stability and damping are crucial problems in controlling hydraulic servo-systems, it is worth while to investigate whether Coulomb friction contributes to stability and damping or not. Simply by intuition (or by misleading references like [10]) one might expect one beneficial consequence of Coulomb friction, as it behaves in more or less the same way as viscous friction, which does provide stability and damping.

To get a one-sided but nevertheless clarifying insight in these matters, we make the following hypothetical experiment with a hydraulic servosystem according to Fig. 4.13, in which Coulomb friction is the only non-linearity. We assume the system to be adjusted so that it is stable also at $F_c = 0$. Consequently $K_v/\omega_o < 2\beta$.

We give this system a constant speed input signal $\dot{r} = v$ from $t = -\infty$ to $t = +\infty$ and at the same time increase the velocity gain "infinitely slowly" to $K_v/\omega_o > 2\beta$.

A system without Coulomb friction ($F_c = 0$) would show instability, exactly when passing $K_v/\omega_o = 2\beta$, manifested by gradually arising oscillations with ever increasing amplitudes, suggested (in an exaggerated way) in Fig. 4.14(b).

Let us assume for a moment that a system with Coulomb friction remains stable at $K_v/\omega_o > 2\beta$ and then prove that this assumption is wrong and thereby prove indirectly that Coulomb friction has no stabilizing effect.

The initial assumption implies that the same experiment as before, with a system where $F_c \neq 0$, should show a response like Fig. 4.14(b). Stability due to Coulomb friction implies at the very least that the servo at $K_v/\omega_o > 2\beta$ continues its response with $\dot{y} > 0$ permanently! However, such a response is impossible because its implication is that $F_c = +\bar{F}_c$ permanently, see Fig. 4.14(b). But a constant frictional force ($F_c = +\bar{F}_c = \text{constant}$) implies a perfect linear system behavior, which is inherently unstable, see Fig. 4.14(a)! So our initial assumption was wrong!

The most likely response of a system with Coulomb friction to $\dot{r} = v = \text{constant}$, where K_v/ω_o is somewhat larger than 2β , is illustrated in (c), which shows a hunting (stick-slip) motion with periodical stops ($\dot{y} = 0$). Analog simulations confirmed this possibility. Although one could say that a response according to (c) is stable, because oscillations have a limited amplitude, such a response in practice is simply not acceptable. Thus we conclude that Coulomb friction will not improve stability in practice.

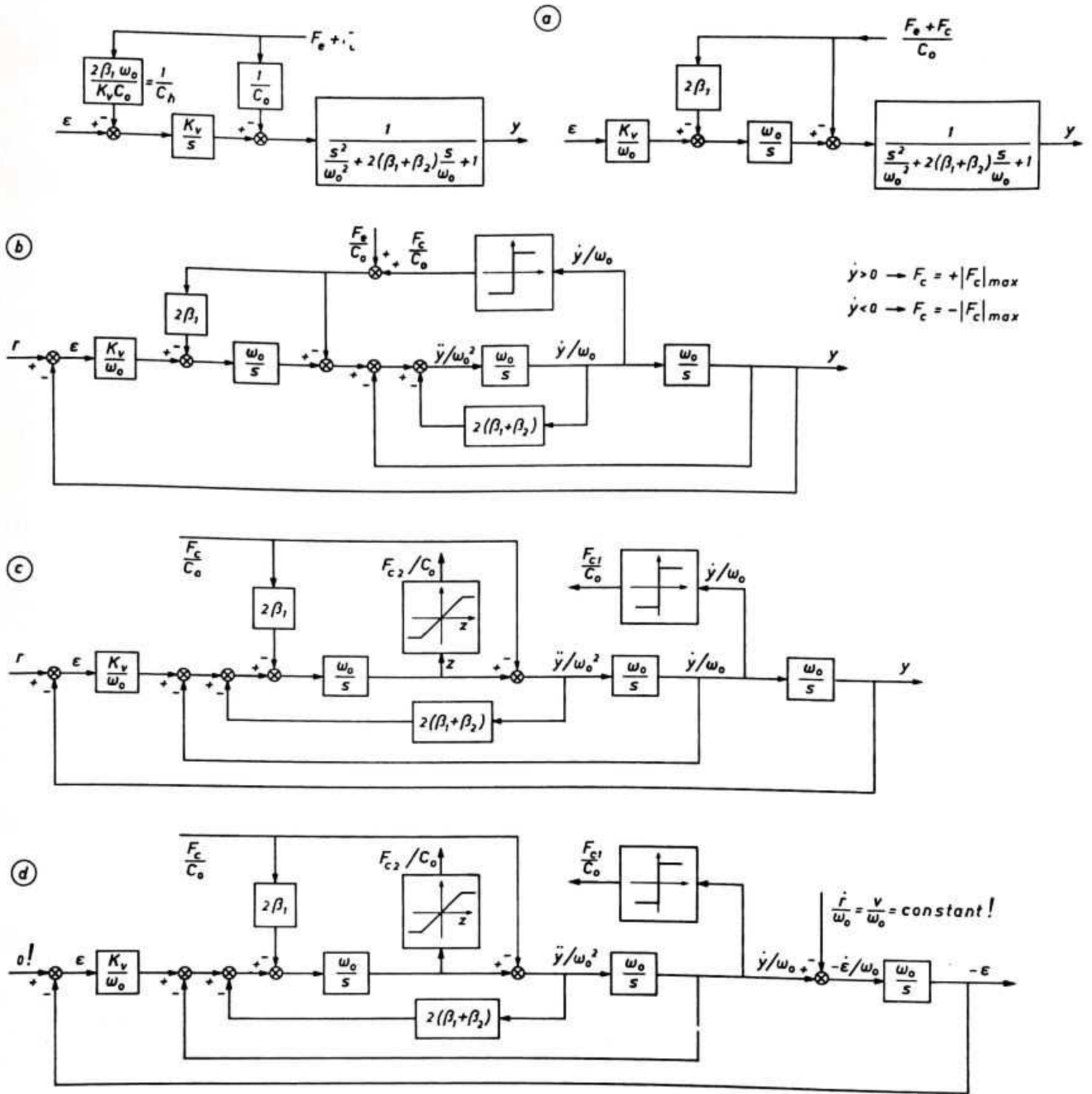


Fig. 4.15 Analog simulation of Coulomb friction

*4.14 Analog simulation of Coulomb friction

The influence of Coulomb friction on the dynamic behavior of hydraulic servo-systems can be analyzed by analog simulation. To permit this, Figs. 4.13(a) and (b) have been combined and worked out completely to the "analog block diagram" of Fig. 4.15(a). To avoid improper parameter adjustments, the two components of the damping coefficient β have been separated. According to eqn. (10)

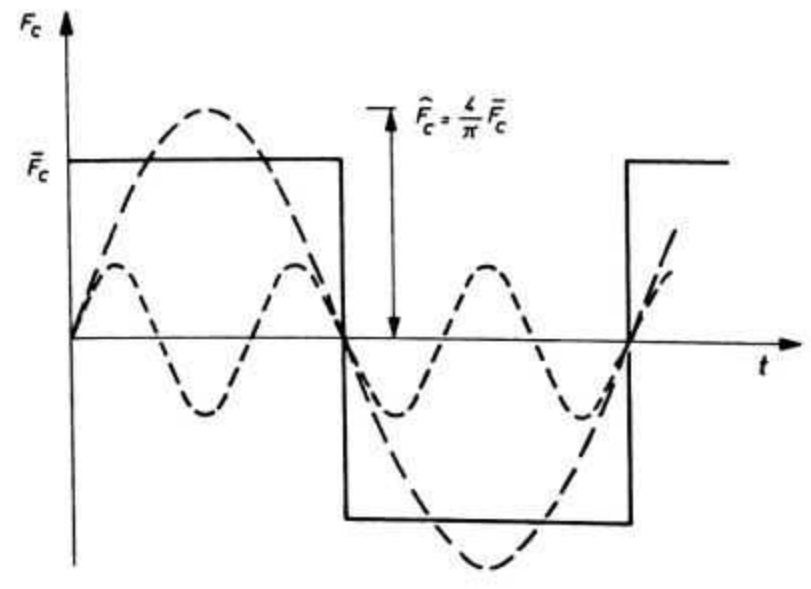
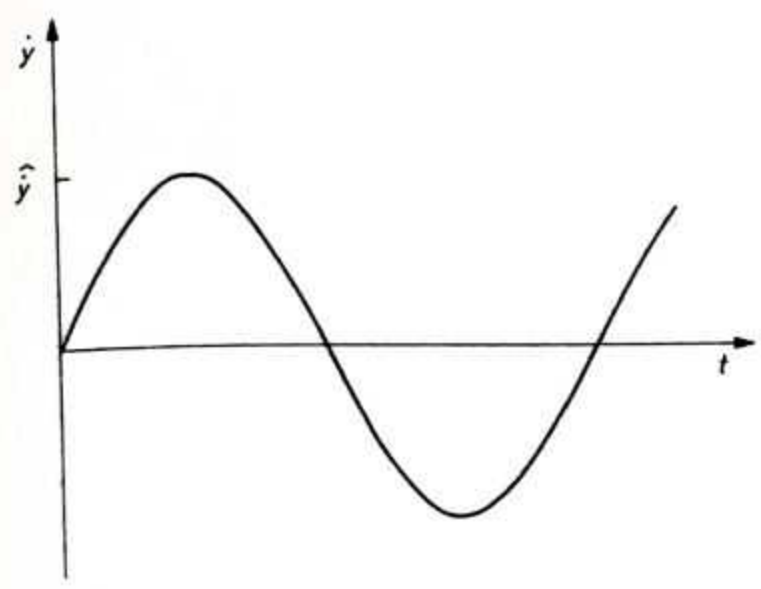
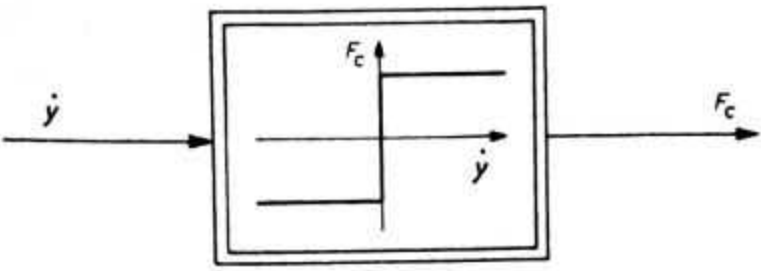
$$\beta = \frac{K_v}{2\omega_o} \cdot \frac{C_o}{C_h} + \frac{w}{2M\omega_o} = \beta_1 + \beta_2 .$$

Figure 4.15(b) seems to be sufficient for analog simulation, as Coulomb friction according to Fig. 4.14(b) is included. But it is not, because at $\dot{y} = 0$ the Coulomb friction is not determined!

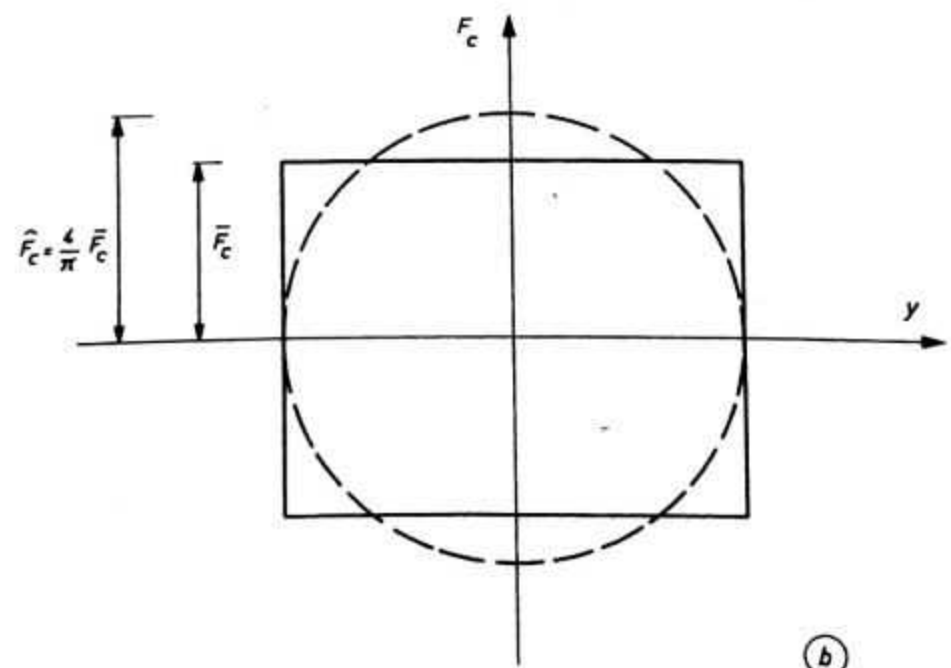
In physical reality, however, Coulomb friction at $\dot{y} = 0$ is determined by the fact that it will simply take whatever value is necessary to *keep* $\dot{y} = 0$. As long as the required friction is $-\bar{F}_c < F_c < +\bar{F}_c$, the speed $\dot{y}$ will *remain* zero. As soon as $F_c = +\bar{F}_c$ (at $dF_c/dt > 0$) or $F_c = -\bar{F}_c$ (at $dF_c/dt < 0$) is reached, the system starts moving again. This procedure has been implemented in Fig. 4.15(c), where $F_c = F_{c1}$ as long as $\dot{y} \neq 0$ and $F_c = F_{c2}$ while $\dot{y} = 0$.

To simulate a constant and permanent speed ($\dot{r} = v = \text{constant}$, see Section 4.13) some rearrangements must be made, see diagram (d). In this set-up the statements made in Section 4.13 are completely confirmed. Analog experiments with $\dot{r} = v = \text{constant}$ showed two stability boundaries. The boundary between stable and semi-unstable is reached at $K_v/\omega_o = 2\beta$. The boundary between semi-unstable and completely unstable is determined by three dimensionless quantities, K_v/ω_o , β and $\omega_o \bar{F}_c / v C_o$.

The influence of Coulomb friction on sinusoidally actuated systems was investigated also. Experiments confirmed our suspicion that Coulomb friction has no effective stabilizing influence. In the Section 4.15 of this chapter an elegant method is introduced to prove this theoretically. At the same time, more insight is obtained into some very peculiar phenomena.



(a)



(b)

Fig. 4.16 Describing function of Coulomb friction

*4.15 Describing function of Coulomb friction [1]

In the method of the Describing Function (DF-method) all signals are supposed to be sinusoidal. This may be realistic in linear systems, but in non-linear systems it is an approximation. The non-linear element converting speed $\dot{y}$ into Coulomb friction requires special attention. Assuming the input signal $\dot{y}$ of the non-linear element to be sinusoidal, the output F_c will be block-type (see Fig. 4.16) which can be expanded in a Fourier series.

$$\text{Input: } \dot{y} = \hat{\dot{y}} \sin \omega t$$

$$\text{Output: } F_c = \frac{4}{\pi} \bar{F}_c \left(\sin \omega t + \frac{1}{3} \sin 3 \omega t + \frac{1}{5} \sin 5 \omega t + \dots \right).$$

In the DF-method one takes the first harmonic only, ignoring the higher harmonics, which should have considerably smaller amplitudes. So the DF-method gives a linearized transfer function

$$\frac{F_c}{\dot{y}} = \frac{4}{\pi} \cdot \frac{\bar{F}_c}{\hat{\dot{y}}} = \frac{4}{\pi} \cdot \frac{\bar{F}_c}{\omega \hat{y}}.$$

With the notation $\hat{F}_c = 4\bar{F}_c/\pi = \text{constant}$, we get

$$\underline{\underline{F_c = \frac{\hat{F}_c}{\omega \hat{y}} \cdot \dot{y}}}. \quad (20)$$

The DF-approximation is also justified by the so-called *harmonic balance* (see Fig. 4.16(b)). At one given sinusoidal signal y , the "exact" Coulomb friction and its first harmonic (having 90° phase lead to y) *withdraw the same amount of energy* from the servosystem.

Introducing the DF-friction into our differential equation of the open loop, we get

$$\frac{\ddot{y}}{\omega_o^2 K_v} + 2\beta \frac{\dot{y}}{\omega_o K_v} + \frac{y}{K_v} = \epsilon - \frac{F_e + F_c}{C_h} - \frac{\dot{F}_e + \dot{F}_c}{K_v C_o}$$

$$\begin{aligned} \left[\frac{s^3}{\omega_o^2 K_v} + 2\beta \frac{s^2}{\omega_o K_v} + \frac{s}{K_v} \right] y &= \epsilon - \frac{F_e}{C_o} \left[\frac{C_o}{C_h} + \frac{s}{K_v} \right] - \frac{F_c}{C_o} \left[\frac{C_o}{C_h} + \frac{s}{K_v} \right] \\ &= \epsilon - \frac{F_e}{C_o} \left[\frac{C_o}{C_h} + \frac{s}{K_v} \right] - \frac{\hat{F}_c}{C_o \hat{y}} \left[\frac{C_o}{C_h} + \frac{s}{K_v} \right] \frac{sy}{\omega}. \end{aligned}$$

All signals being sinusoidal, the operator s can be replaced by $j\omega$. Thus after some calculating with $F_e = 0$, we get

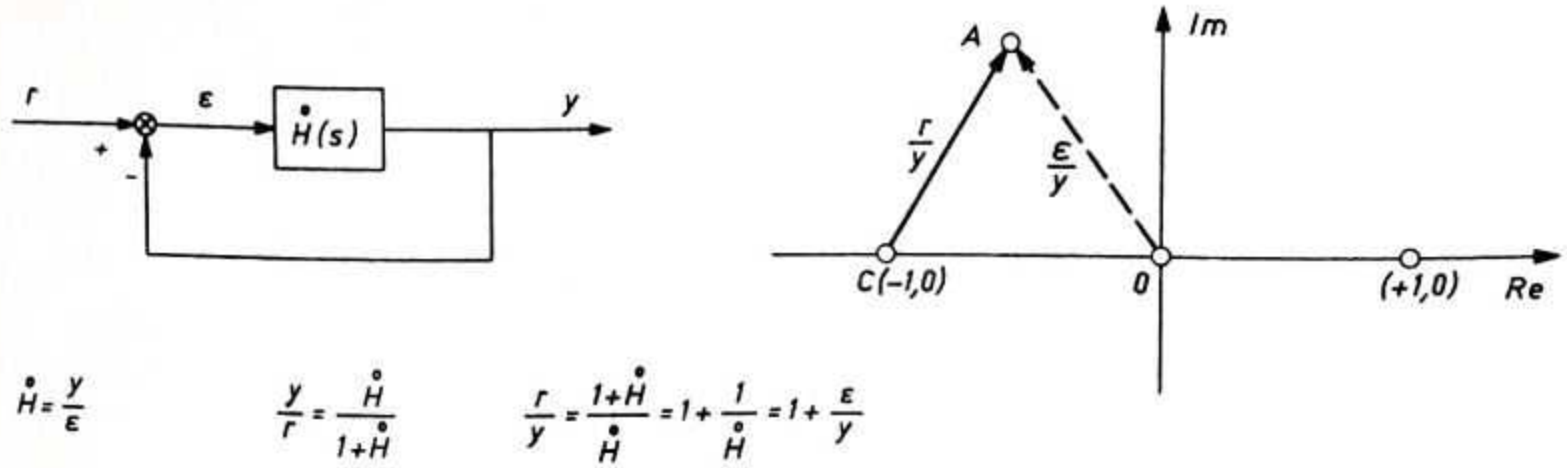


Fig. 4.17 Inverse Nyquist plot

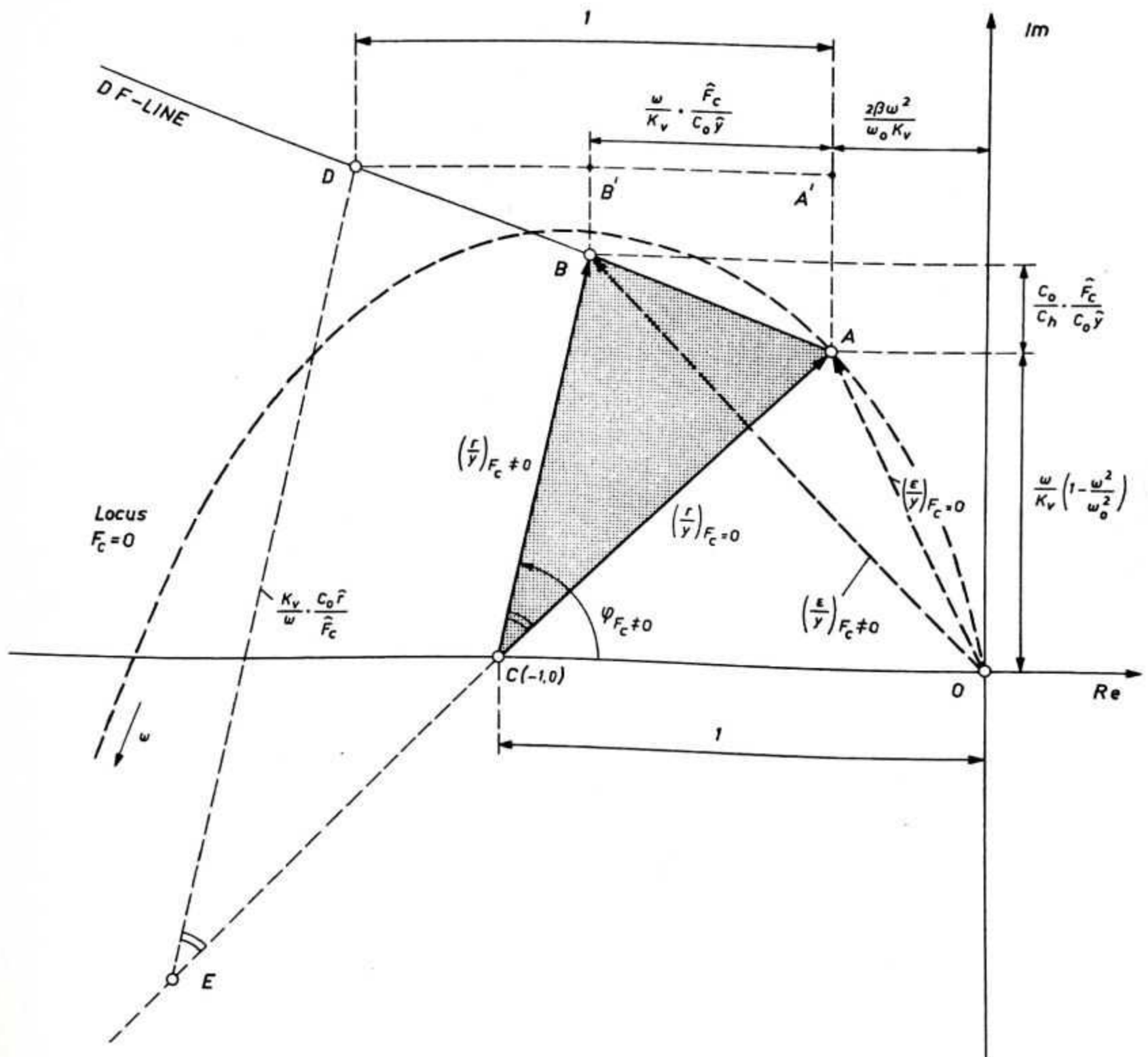


Fig. 4.18 Describing function inverse Nyquist plot of hydraulic servosystem with Coulomb friction

$$\frac{\epsilon}{y} = -2\beta \frac{\omega^2}{\omega_o^2 K_v} - \frac{\omega}{K_v} \cdot \frac{\hat{F}_c}{C_o \hat{y}} + j \left[\frac{\omega}{K_v} \left(1 - \frac{\omega^2}{\omega_o^2} \right) + \frac{C_o}{C_h} \cdot \frac{\hat{F}_c}{C_o \hat{y}} \right]$$

Here a not too complicated equation has been derived for the *inverse open loop transfer function*. It is very profitable to make use here of an inverse Nyquist plot, because it gives the most simple relation between open loop and closed loop presentation (see Fig. 4.17). In the case of eqn. (21) the advantage is even larger as the influence of Coulomb friction on the inverse transfer function is *linear* (see Fig. 4.18). It shows first of all the locus for $\hat{F}_c = 0$ (thick dashed line, starting at 0). At a given frequency the vector $OA = \epsilon/y$ and the vector $CA = r/y$ represent the inverse transfer functions of the open loop and closed loop respectively.

Equation (21) directly indicates the influence of $\hat{F}_c/C_o \hat{y}$ on the (complex) vector $OA = \epsilon/y$. Owing to Coulomb friction the vector OA is changed into the vector OB , where B is traveling along a *straight line* (the DF-line) as $\hat{F}_c$ increases. The inverse Nyquist plot now makes possible a direct switch from the open loop to the closed loop, simply by replacing the origin 0 by the point $C(-1,0)$. So the *same locus and the same DF-line can be used for open loop and closed loop*. Consequently the closed loop vector CA is changed into CB by Coulomb friction. Thus Fig. 4.18 enables us to analyze the influence of Coulomb friction graphically.

As the final analysis of eqn. (21) will be made analytically, we restrict ourselves to three observations on Fig. 4.18.

- (a) To find the amplitude $\hat{y}$ of the signal y from a given $\hat{r}$ and a given $\hat{F}_c$, the construction of $\triangle ADE$ is useful, where $A'D = 1$ and $DE \parallel CB$.

$$\frac{DE}{CB} = \frac{DA}{AB} = \frac{DA'}{B'A'} = \frac{1}{A'B'} \quad \text{so} \quad |DE| = \frac{|CB|}{|A'B'|} = \frac{\frac{\hat{r}}{\hat{y}}}{\frac{\omega}{K_v} \cdot \frac{\hat{F}_c}{C_o \hat{y}}} = \frac{K_v}{\omega} \cdot \frac{C_o \hat{r}}{\hat{F}_c}$$

The length of DE being known, the point E can be found, and drawing the line $CB \parallel DE$ we find the point B also: As $|CB| = \hat{r}/\hat{y}$, the amplitude $\hat{y}$ can be found as well.

- (b) Coulomb friction will stop any motion ($\hat{y} = 0$) if $CB = \infty$, and as $DE \parallel CB$ this may occur if $|DE| = |DA|$. Analysis of Fig. 4.18 yields

$$\left| \frac{DE}{DA} \right| = \left| \frac{CB}{BA} \right| = \frac{C_o \hat{r}}{\hat{F}_c \sqrt{\left(\frac{\omega}{K_v} \right)^2 + \left(\frac{C_o}{C_h} \right)^2}}$$

So $|DE| : |DA| = 1$ and $\hat{y} = 0$ if

$$\frac{C_o \hat{r}}{\hat{F}_c} = \sqrt{\left(\frac{\omega}{K_v}\right)^2 + \left(\frac{C_o}{C_h}\right)^2} . \quad (22)$$

In fact, this is the *dynamic dead zone*, derived earlier in Section 4.4!

The only difference from eqn. (11) is a factor $\pi/4$, due to the DF-approximation.

- (c) A linear system is considered to be unstable ($\hat{y}/\hat{r} = \infty$) if the point C(-1,0) is on the locus. If $F_c = 0$ this will occur here at $K_v/\omega_o = 2\beta$ when $\omega = \omega_o$. If $F_c \neq 0$ and $K_v/\omega_o = 2\beta$, again $\hat{y}/\hat{r} = \infty$ at $\omega = \omega_o$ is a (steady-state) solution, confirming that Coulomb friction is not influencing stability.

To study some very peculiar phenomena, we continue further analysis of eqn. (21) analytically. Substitution of $\epsilon = r - y$ yields

$$\frac{r}{y} = 1 - 2\beta \frac{\omega^2}{\omega_o K_v} - \frac{\omega}{K_v} \cdot \frac{\hat{F}_c}{C_o \hat{y}} + j \left[\frac{\omega}{K_v} \left(1 - \frac{\omega^2}{\omega_o^2} \right) + \frac{C_o}{C_h} \cdot \frac{\hat{F}_c}{C_o \hat{y}} \right] .$$

Taking the modulus gives

$$\left[\frac{\hat{r}}{\hat{y}} \right]^2 = \left[1 - 2\beta \frac{\omega^2}{\omega_o K_v} - \frac{\omega}{K_v} \cdot \frac{\hat{F}_c}{C_o \hat{y}} \right]^2 + \left[\frac{\omega}{K_v} \left(1 - \frac{\omega^2}{\omega_o^2} \right) + \frac{C_o}{C_h} \cdot \frac{\hat{F}_c}{C_o \hat{y}} \right]^2 .$$

Multiplication with $(\hat{y}/\hat{r})^2$ and working out finally gives

$$\begin{aligned} \left[\frac{\hat{y}}{\hat{r}} \right]^2 \cdot \left[\left(1 - 2\beta \frac{\omega^2}{\omega_o K_v} \right)^2 + \frac{\omega^2}{K_v^2} \left(1 - \frac{\omega^2}{\omega_o^2} \right)^2 \right] + \left[\frac{\hat{F}_c}{C_o \hat{r}} \right]^2 \cdot \left[\left(\frac{\omega}{K_v} \right)^2 + \left(\frac{C_o}{C_h} \right)^2 \right] + \\ + 2 \frac{\hat{y}}{\hat{r}} \cdot \frac{\hat{F}_c}{C_o \hat{r}} \cdot \frac{\omega}{K_v} \left[\frac{C_o}{C_h} \left(1 - \frac{\omega^2}{\omega_o^2} \right) - \left(1 - 2\beta \frac{\omega^2}{\omega_o K_v} \right) \right] - 1 = 0 . \end{aligned} \quad (23)$$

Evidently the second degree equation in $\hat{y}/\hat{r}$ and $\hat{F}_c/C_o \hat{r}$ represents conic sections, normally even ellipses having a center coinciding with the origin! In Fig. 4.19(a) several *amplitude describing functions* are drawn up at different frequencies at $K_v/\omega_o = 0.25$, $\beta = 0.25$ and $C_h = \infty$.

Substituting $\hat{y}/\hat{r} = 0$ in eqn. (23) we again meet the dynamic dead zone.

$$\hat{y} = 0 \quad \text{if} \quad \frac{C_o \hat{r}}{\hat{F}_c} = \sqrt{\left(\frac{\omega}{K_v}\right)^2 + \left(\frac{C_o}{C_h}\right)^2} . \quad (22)$$

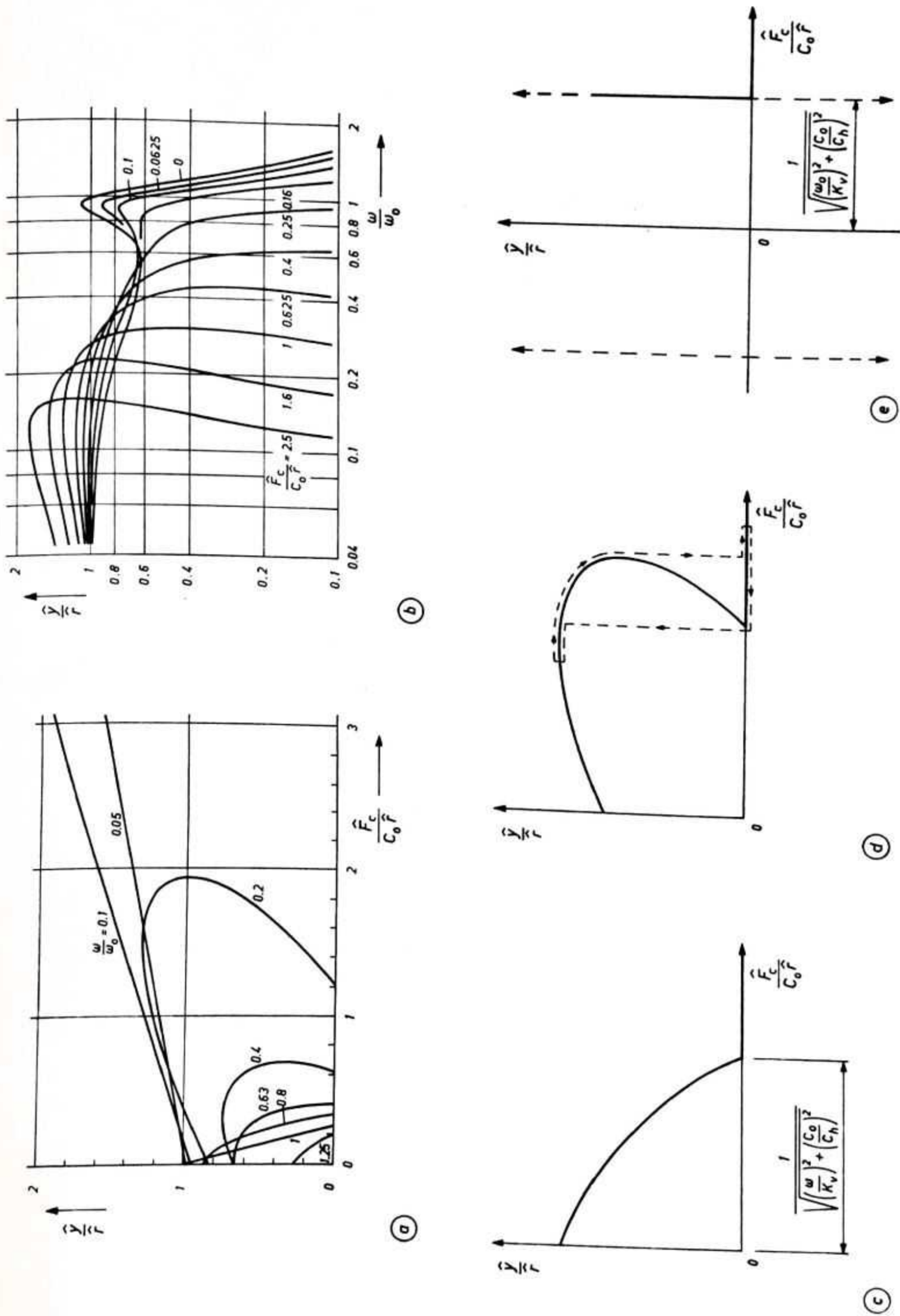


Fig. 4.19 Amplitude describing functions and frequency characteristic

From eqn. (22) the so-called stop frequency can also be derived.

$$\hat{y} = 0 \quad \text{if} \quad \omega = \omega_{\text{stop}} = K_v \sqrt{\left(\frac{C_o \hat{r}}{\hat{F}_c}\right)^2 - \left(\frac{C_o}{C_h}\right)^2} \quad (24)$$

This stop frequency has been derived earlier also (in Section 4.4) and is illustrated in the amplitude characteristic (b).

The interpretation of the amplitude describing functions is explained in more detail, according to Figs. 4.19(c) and (d). The relation between $\hat{y}/\hat{r}$ and $\hat{F}_c/C_o \hat{r}$ will not stop at $\hat{y}/\hat{r} = 0$ and $\hat{F}_c/C_o \hat{r} = \{(\omega/K_v)^2 + (C_o/C_h)^2\}^{-1/2}$. As Coulomb friction $-\bar{F}_c < F_c < +\bar{F}_c$ might take smaller values than $\hat{F}_c = 4\bar{F}_c/\pi$, if it is sufficient to stop motion, the horizontal axis with

$$\frac{\hat{y}}{\hat{r}} = 0 \quad \text{and} \quad \frac{\hat{F}_c}{C_o \hat{r}} > \frac{1}{\sqrt{\left(\frac{\omega}{K_v}\right)^2 + \left(\frac{C_o}{C_h}\right)^2}}$$

should be seen as a substantial part of the picture. A very peculiar situation is now presented in (d), where (according to our interpretation just made) three different values of $\hat{y}/\hat{r}$ might occur at one single value of $\hat{F}_c/C_o \hat{r}$! Analog simulation revealed however that one part of the ellipse does not provide a steady-state solution. Figure 4.19(d) shows how $\hat{y}/\hat{r}$ can change discontinuously with varying $\hat{F}_c/C_o \hat{r}$. These discontinuities have been observed during analog simulations. A kind of paradox is included in Fig. 4.19(d) since under certain conditions the system can move at a friction level that might stop any motion.

Considering stability for the last time, substitution of $K_v/\omega_o = 2\beta$ and $\omega = \omega_o$ in eqn. (23) gives

$$\left[\frac{\hat{F}_c}{C_o \hat{r}}\right]^2 = \frac{1}{\left(\frac{\omega_o}{K_v}\right)^2 + \left(\frac{C_o}{C_h}\right)^2}$$

Its representation in Fig. 4.19(e) should be seen as an ellipse with one infinitely long axis, $\hat{F}_c/C_o \hat{r} = 0$ being an axis of symmetry. Conform the foregoing interpretation is

$$\frac{\hat{y}}{\hat{r}} = \infty \quad \text{at} \quad \frac{\hat{F}_c}{C_o \hat{r}} < \frac{1}{\sqrt{\left(\frac{\omega_o}{K_v}\right)^2 + \left(\frac{C_o}{C_h}\right)^2}} \quad .$$

$$\frac{\hat{y}}{\hat{r}} = 0 \quad \text{at} \quad \frac{\hat{F}_c}{C_o \hat{r}} > \frac{1}{\sqrt{\left(\frac{\omega_o}{K_v}\right)^2 + \left(\frac{C_o}{C_h}\right)^2}}.$$

Unless $\hat{y}/\hat{r} = 0$ is seen as a satisfactory stabilization, Coulomb friction does not stabilize hydraulic servosystems, since the stability condition $K_v/\omega_o < 2\beta$ is valid for systems with $F_c = 0$ as well as $F_c \neq 0$. This confirms previous statements as well as various practical experiments. Controversial publications [10], [11] and [12] could be ignored.

REFERENCES

- [1] T.J. Viersma, Investigation into the accuracy of hydraulic servomotors, Thesis Delft University of Technology, 1961; Philips Research Report, 16(1961)507; 17(1962)20.
- [2] G.S. Brown and A.C. Hall, Dynamic behavior and design of servomechanisms, Trans. ASME, (1946)503.
- [3] H.E. Merrit, Hydraulic Control Systems, Wiley, New York, 1967.
- [4] M. Guillon, Etude et détermination des systèmes hydrauliques, Dunod, Paris, 1961.
- [5] A.C. Morse, Electrohydraulic Servomechanisms, McGraw-Hill, New York, 1963.
- [6] R. Walters, Hydraulic and Electro-hydraulic Servosystems, Iliffe, London, 1967.
- [7] L.H. Geyer, Controlled damping through dynamic pressure feedback, Dowty Equipment, Cheltenham, June 1958.
- [8] J.J. 't Mannetje, Stabilizing networks for hydraulic motors, Control Eng., June(1974)55.
- [9] W.J. Thayer, Transfer functions for Moog servovalves, Moog Technical Bulletin 103, New York, 1965.
- [10] P. Lenssen, The influence of dry friction and mechanical parameters on the stability and accuracy of an hydraulic copying system, Proc. 10th M.T.D.R. Conference, 1969.
- [11] P. Lenssen, Analog simulation of hydraulic servosystems with stabilizing influence of dry friction, (in Dutch), Thesis, University of Leuven, Belgium, April, 1970.
- [12] R. Chiappulini, Friction and its effects on linear positional hydraulic and electro-hydraulic servosystems, Annals CIRP, 19(1971)125.

CHAPTER 5

PERFORMANCE DIAGRAM AND DESIGN OF SINUSOIDALLY ACTUATED LINEAR SERVOMOTORS LOADED BY INERTIA FORCES AND A CONSTANT PRE-LOAD

5.1 Introduction

The design of linear servomotors involves the determination of the piston area A , the stroke S , the valve capacity $Q_{\max}$ and the supply pressure P_s . The mechanism to be actuated normally requires certain specifications such as the stroke S and the actuated mass M . Other specifications, such as the maximum speed $\dot{y}_{\max}$, the maximum acceleration $\ddot{y}_{\max}$ and the required bandwidth or speed of response, are sometimes less determined beforehand. In many cases they are subject to compromises and their consequences. Frequently, it is the price of the installation that makes specifications more realistic.

In this chapter, the relationships between specifications and design data are shown for a sinusoidally-actuated linear motor loaded by inertia forces and a constant pre-load. An enormous help in the design process is the *performance diagram*, which shows some fundamental relationships in a frequency diagram.

As the design process is simpler for a symmetric linear motor controlled by a critical-center 4-way servovalve, this motor will be treated first. It will be proved that, with a few simple corrections, the results are valid also for an asymmetric linear motor controlled by an asymmetric 4-way valve. In practice, this asymmetric motor has the advantage that the space needed for the motor is considerably smaller, because there is only one piston rod. But then an asymmetric 4-way valve is desirable, to obtain smooth system behavior (see Sections 3.5 and 3.6). This is a disadvantage (systematically overlooked in literature), because such an asymmetric valve is normally not available. Commercially, it can be obtained only on special request and at a special price.

An alternative is the 3-way-valve-controlled asymmetric motor. Here the lower natural frequency ω_0 (see Fig. 2.3) could be a decisive drawback. Relevant formulas for the design of a 3-way-valve-controlled asymmetric motor can easily be found as the design process follows exactly the same lines as for the symmetric linear motor.

The same is true for 4-way-valve-controlled rotary motors. Here, however, the moment of inertia J is subject to design as well, as the transmission is part of it. In many cases the optimal design of the transmission to obtain a minimum inertia will introduce major complications (see Fig. 2.3).

5.2 Relation between max. stroke $S = 2\hat{y}_{\max}$ and bandwidth ω_b

For well-designed servosystems (without "parasitic" mass-spring systems or poor servovalves) the obtainable bandwidth ω_b roughly equals the natural frequency ω_o caused by the oil-spring with stiffness C_o and the moving mass M . For instance, Fig. 4.2 shows an open loop transfer function $\hat{H}(s) = K_v/s \left[\frac{s^2}{\omega_o^2} + 2\beta \frac{s}{\omega_o} + 1 \right]$ where a proper choice of K_v enables $\omega_b \simeq \omega_o$, provided the damping β has, or has been given (see Fig. 4.4), values between 0.3 and 0.7. According to Fig. 2.3(c), where $\omega_o = \sqrt{C_o/M}$ and $C_o \geq 4AE/S = 2AE/\hat{y}_{\max}$, we find

$$\hat{y}_{\max} \leq \frac{2AE}{C_o} = \frac{2AE}{M\omega_o^2} \simeq \frac{2AE}{M\omega_b^2}. \quad (1)$$

Here we have a fundamental relation between the stroke $S = 2\hat{y}_{\max}$ of the servomotor and its bandwidth ω_b . The relevant parameters are M , A and E .

5.3 Relation between valve capacity $Q_{\max}$ and maximum speed $\dot{y}_{\max}$

A given servovalve has - determined by its maximum port openings - a given maximum flow $Q_{\max}$. As $Q_{\max} = A \dot{y}_{\max}$ we find $Q_{\max} = A \hat{y}_{\max} \geq A \omega \hat{y}$ at sinusoidal movements. So

$$\hat{y} \leq \frac{\hat{y}_{\max}}{\omega} = \frac{Q_{\max}}{A\omega}. \quad (2)$$

5.4 Relation between cavitation boundary and max. acceleration $\ddot{y}_{\max}$

In Section 3.4, the cylinder pressure P_1 and P_2 on both sides of the piston were derived. Assuming a constant pre-load F_{st} and neglecting friction here we find

$$P_{1,2} = \frac{1}{2} P_s \pm \frac{1}{2} P_L, \quad \text{where} \quad P_L = \frac{\Sigma F}{A} = \frac{M\ddot{y} + F_{st}}{A}.$$

To avoid cavitation and/or saturation, $P_L < \frac{2}{3} P_s$ must be ensured, so

$$P_L = \frac{M\ddot{y}_{\max} + F_{st}}{A} = \frac{M\omega^2 \hat{y} + F_{st}}{A} < \frac{2}{3} P_s$$

and

$$\hat{y} < \frac{\hat{\ddot{y}}_{\max}}{\omega^2} = \frac{\frac{2}{3} P_s A - F_{st}}{M\omega^2} \quad (3)$$

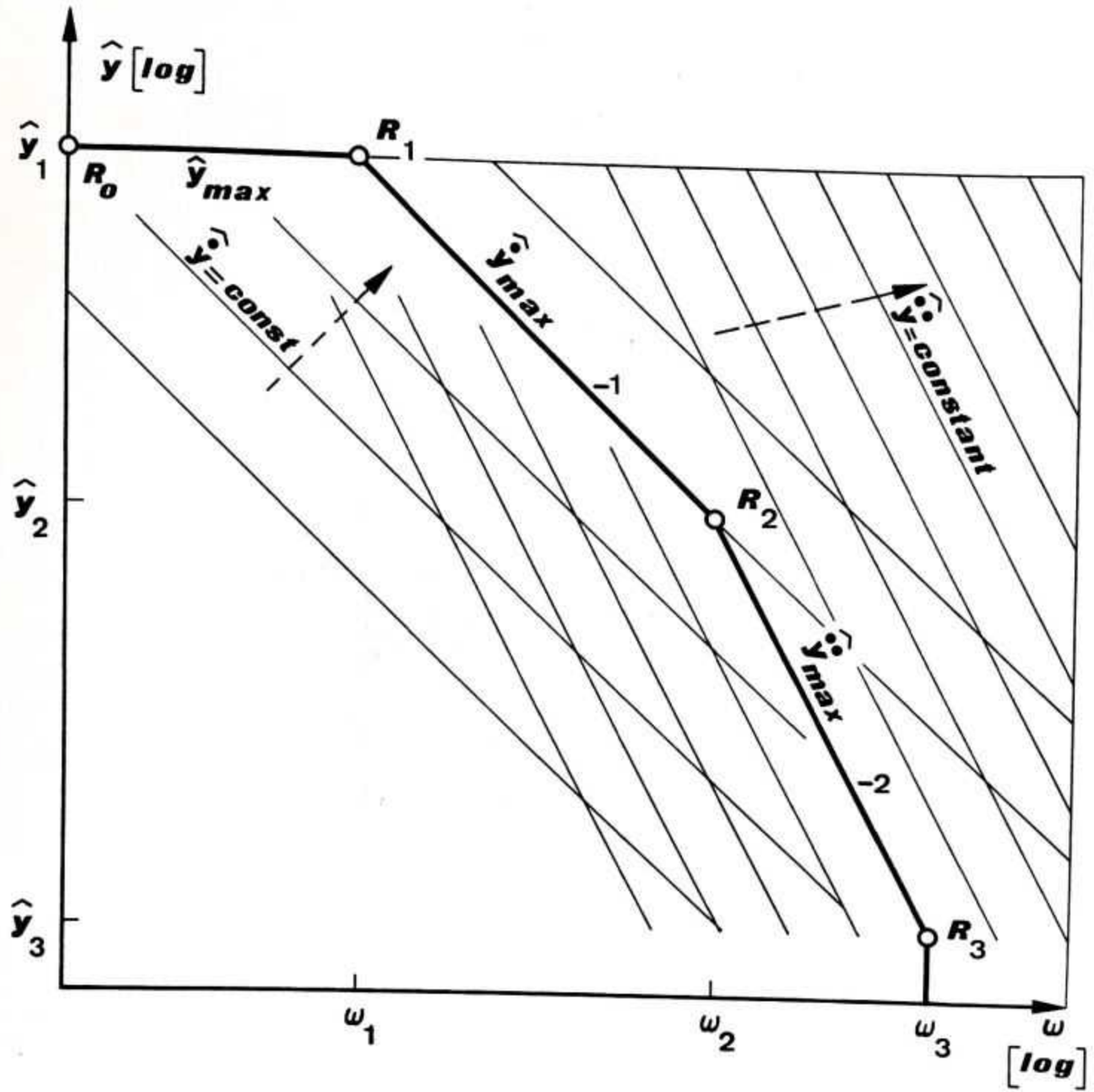


Fig. 5.1 Preliminary performance diagram

5.5 Preliminary performance diagram

The obtainable amplitude $\hat{y}$ of a given motor-valve combination is now determined in the low, middle and high frequency ranges.

$$\text{LF} \quad \hat{y} < \frac{2AE}{M\omega_b^2} \quad \omega < \omega_1 \quad \hat{y}_1 = \hat{y}_{\max} \quad (4)$$

$$\text{MF} \quad \hat{y} \leq \frac{\dot{\hat{y}}_{\max}}{\omega} = \frac{Q_{\max}}{A\omega} \quad \omega_1 < \omega < \omega_2 \quad \hat{y}_1 > y > \hat{y}_2 \quad (5)$$

$$\text{HF} \quad \hat{y} \leq \frac{\ddot{\hat{y}}_{\max}}{\omega^2} = \frac{\frac{2}{3} P_s A - F_{st}}{M\omega^2} \quad \omega_2 < \omega < \omega_3 \quad \hat{y}_2 > y > \hat{y}_3. \quad (6)$$

In the preliminary performance diagram of Fig. 5.1(a) (with logarithmic scales for $\hat{y}$ and ω), the lines for the middle frequencies have a slope -1, the HF-lines a slope -2. These lines at the same time indicate values of $\dot{\hat{y}}_{\max}$ and $\ddot{\hat{y}}_{\max}$.

The positions of the points $R_1(\hat{y}_1, \omega_1)$, $R_2(\hat{y}_2, \omega_2)$ and $R_3(\hat{y}_3, \omega_3)$ are characteristic for the performance of the servosystem.

Concerning the point $R_1(\hat{y}_1, \omega_1)$, eqns. (4) and (5) give

$$\hat{y}_1 = \hat{y}_{\max} = \frac{2AE}{M\omega_b^2} \quad \omega_1 = \frac{\dot{\hat{y}}_{\max}}{\hat{y}_{\max}} = \frac{M\omega_b^2 \cdot Q_{\max}}{2EA^2} = \frac{Q_{\max}}{A\hat{y}_{\max}}$$

where

$$\omega_1 \hat{y}_1 = \dot{\hat{y}}_{\max} = \frac{Q_{\max}}{A} = \omega_2 \hat{y}_2.$$

Concerning the point $R_2(\hat{y}_2, \omega_2)$, eqns. (5) and (6) give

$$\hat{y}_2 = \frac{\dot{\hat{y}}_{\max}^2}{\ddot{\hat{y}}_{\max}} = \frac{MQ_{\max}^2}{A^2 \left[\frac{2}{3} P_s A - F_{st} \right]} \quad \omega_2 = \frac{\ddot{\hat{y}}_{\max}}{\dot{\hat{y}}_{\max}} = \frac{A \left[\frac{2}{3} P_s A - F_{st} \right]}{MQ_{\max}}$$

where

$$\omega_2 \hat{y}_2 = \dot{\hat{y}}_{\max} = \frac{Q_{\max}}{A} = \omega_1 \hat{y}_1$$

and

$$\omega_2^2 \hat{y}_2 = \ddot{\hat{y}}_{\max} = \frac{\frac{2}{3} P_s A - F_{st}}{M} = \omega_3^2 \hat{y}_3.$$

Point $R_3(\hat{y}_3, \omega_3)$ can be found by substitution of $\omega_3 = \omega_b = \sqrt{2AE/M\hat{y}_{\max}}$ into eqn. (6).

$$\hat{y}_3 = \frac{\dot{\hat{y}}_{\max}^2}{\omega_b^2} = \frac{\hat{y}_{\max} \left[\frac{2}{3} P_s A - F_{st} \right]}{2AE}.$$

$R_0 \quad R_1$ $\hat{y} = \frac{2 AE}{M \omega_b^2}$ $\omega < \omega_1 \quad \hat{y}_1 = \hat{y}_{\max}$	$\hat{y}_1 = \hat{y}_{\max} = \frac{2 AE}{M \omega_b^2}$ $\omega_1 = \frac{\hat{y}_{\max}}{\hat{y}_{\max}} = \frac{Q_{\max}}{A \hat{y}_{\max}}$	$\omega_1 \hat{y}_1 = \omega_2 \hat{y}_2$ $= \hat{y}_{\max} = \frac{Q_{\max}}{A}$
$R_1 \quad R_2$ $\hat{y} = \frac{\hat{y}_{\max}}{\omega} = \frac{Q_{\max}}{A \omega}$ $\omega_1 < \omega < \omega_2 \quad \hat{y}_1 > \hat{y} > \hat{y}_2$	$\hat{y}_2 = \frac{\hat{y}_{\max}^2}{\hat{y}_{\max}} = \frac{M Q_{\max}^2}{A^2 [\frac{2}{3} P_s A - F_{st}]}$ $\omega_2 = \frac{\hat{y}_{\max}}{\hat{y}_{\max}} = \frac{A [\frac{2}{3} P_s A - F_{st}]}{M Q_{\max}}$	
$R_2 \quad R_3$ $\hat{y} = \frac{\hat{y}_{\max}}{\omega^2} = \frac{\frac{2}{3} P_s A - F_{st}}{M \omega^2}$ $\omega_2 < \omega < \omega_3 \quad \hat{y}_2 > \hat{y} > \hat{y}_3$	$\hat{y}_3 = \frac{\hat{y}_{\max}}{\omega_b^2} = \frac{\hat{y}_{\max} [\frac{2}{3} P_s A - F_{st}]}{2 AE}$ $\omega_3 = \omega_b = \omega_0 = \sqrt{\frac{2 AE}{M \hat{y}_{\max}}}$	$\omega_2^2 \hat{y}_2 = \omega_3^2 \hat{y}_3$ $= \hat{y}_{\max} = \frac{\frac{2}{3} P_s A - F_{st}}{M}$
Design of symmetric Servomotor with 4-way servovalve	$A = \frac{M \omega_b^2 \cdot \hat{y}_{\max}}{2 E}$ $P_s = \frac{3 E [\hat{y}_{\max} + \frac{F_{st}}{M}]}{\omega_b^2 \cdot \hat{y}_{\max}}$	$Q_{\max} = \frac{M \omega_b^2 \cdot \hat{y}_{\max} \cdot \hat{y}_{\max}}{2 E}$ $N_p = Q_{\max} P_s = \frac{3}{2} M [\hat{y}_{\max} + \frac{F_{st}}{M}] \cdot \hat{y}_{\max}$
Asymmetric Servomotor with Asymmetric 4-way servovalve	$P_s = 3 E \cdot \frac{M \hat{y}_{\max} + F_{st} - \frac{A_1 - A_2}{2} P_s }{M \omega_b^2 \hat{y}_{\max}}$ $A = \frac{A_1 + A_2}{2} = \frac{M \omega_b^2 \hat{y}_{\max}}{2 E}$ $\text{Minimum } P_s = \frac{3 E \hat{y}_{\max}}{\omega_b^2 \hat{y}_{\max}} \text{ at } A_{1,2} = \frac{M \omega_b^2 \hat{y}_{\max}}{2 E} [1 \pm \frac{2 F_{st}}{3 M \hat{y}_{\max}}]$	
	Servovalve portopenings: $\frac{a_{1\max}}{A_1} = \frac{a_{2\max}}{A_1} = \frac{a_{3\max}}{A_2} = \frac{a_{4\max}}{A_2}$	

Fig. 5.2 Design chart

5.6 Design of symmetric motors

All the above results have been summarized in the design chart of Fig. 5.2. It enables us to design the motor and valve, i.e. to derive A , P_s , $Q_{\max}$ and N_{peak} from the specified E , M , ω_b , $\hat{y}_{\max}$, $\hat{\ddot{y}}_{\max}$, $\hat{\ddot{\ddot{y}}}_{\max}$ and F_{st} ,

Eqn. (4) gives the piston area
$$A = \frac{M\omega_b^2 \hat{y}_{\max}}{2E} \quad (7)$$

Eqn. (6) gives the supply pressure
$$P_s = \frac{3E \left[\hat{\ddot{y}}_{\max} + \frac{F_{st}}{M} \right]}{\omega_b^2 \hat{y}_{\max}} \quad (8)$$

Eqn. (5) gives the valve capacity
$$Q_{\max} = A \hat{\dot{y}}_{\max} = \frac{M\omega_b^2 \hat{y}_{\max} \hat{\dot{y}}_{\max}}{2E} \quad (9)$$

Max. peak-power now becomes
$$N_p = Q_{\max} P_s = \frac{3}{2} M \left[\hat{\ddot{y}}_{\max} + \frac{F_{st}}{M} \right] \hat{\dot{y}}_{\max} \quad (10)$$

So the design data A , P_s , $Q_{\max}$ and N_p can be expressed explicitly in the specifications. It is clear that, if the specifications lead to unacceptable values of A , P_s , $Q_{\max}$ or N_p , a compromise must be made.

5.7 Final performance diagram

Easier to handle than the preliminary performance diagram is the final performance diagram (Fig. 5.3), where the speed $\hat{y}$ and the frequency are the ordinates. Now the LF-lines of constant amplitude $\hat{y} = \hat{\dot{y}}/\omega$ have a slope +1, while the HF-lines of constant acceleration $\hat{\ddot{y}} = \hat{\dot{y}}\omega$ have a slope -1. So the lines of constant $\hat{y}$ and those of constant $\hat{\ddot{y}}$ are perpendicular. Just knowing the so-called 25%-series

1 - 1.25 - 1.60 - 2.00 - 2.50 - 3.15 - 4.00 - 5.00 - 6.30 - 8.00 - 10.-

(which divides the range 1 to 10 into increments of equal-percent), the performance diagram can be put together easily by anyone from linear-scaled paper in any size!

*5.8 Design of asymmetric servomotors ($A_1 \neq A_2$)

The results obtained so far are relevant for symmetric linear motors with symmetric 4-way valves. In practice, asymmetric motors must sometimes be used, for instance to avoid one of the two piston rods. Such a motor must be combined with an asymmetric 4-way valve, as a symmetric valve is *incompatible* (see Sections 3.5 and 3.6).

It can be found directly from the conclusions in Section 3.6 that the above results also cover asymmetric motors, provided that some typical corrections are made. In eqn. (7) the piston area A should be replaced by $0.5(A_1 + A_2)$ and in eqn. (8) for the supply pressure, F_{st} should be replaced by $F_{st} - 0.5(A_1 - A_2)P_s$.

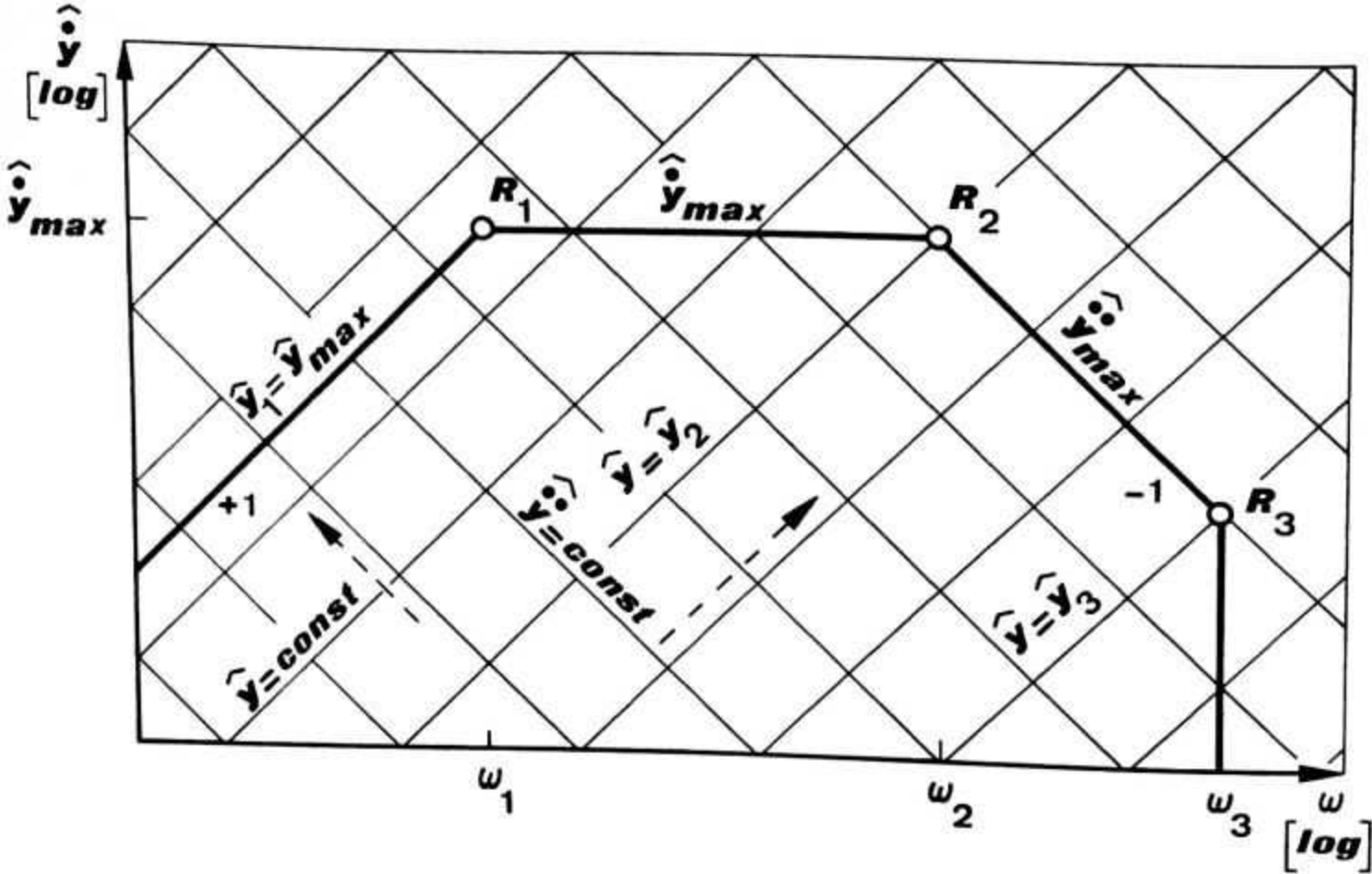


Fig. 5.3 Final performance diagram

So

$$A = 0.5(A_1 + A_2) = \frac{M\omega_b^2 \hat{y}_{\max}}{2E} \quad (11)$$

$$P_s = 3E \cdot \frac{\hat{M}\ddot{y}_{\max} + |F_{st} - 0.5(A_1 - A_2)P_s|}{M\omega_b^2 \hat{y}_{\max}} \quad (12)$$

$Q_{\max}$ and N_p should be corrected consequently.

Equation (12) makes clear that the lowest possible supply pressure P_s will be obtained at $F_{st} = 0.5(A_1 - A_2)P_s$. Combination of eqns. (11) and (12) gives the area's A_1 and A_2 .

$$A_{1,2} = A \pm \frac{F_{st}}{P_s} = \frac{M\omega_b^2 \hat{y}_{\max}}{2E} \left[1 \pm \frac{2F_{st}}{3\hat{M}\ddot{y}_{\max}} \right]. \quad (13)$$

$$P_{s \min} = \frac{3E\hat{y}_{\max}}{\omega_b^2 \hat{y}_{\max}}. \quad (14)$$

Maximum valve capacity and peak power now become

$$Q_{1 \max} = A_1 \hat{y}_{\max}, \quad (15)$$

$$N_{1p} = P_s A_1 \hat{y}_{\max} = \left[\frac{3}{2} \hat{M}\ddot{y}_{\max} + F_{st} \right] \hat{y}_{\max}. \quad (16)$$

In comparison with the symmetric motor, the energy consumption of a pre-loaded asymmetric motor could be considerably lower, especially when the "integrated" power consumption (in kWh) is considered.

In many cases an "optimal" area ratio A_1/A_2 , making $F_{st} = 0.5(A_1 - A_2)P_s$, is not possible for practical reasons. Taking the ratio $A_1/A_2 = \lambda$ as a system parameter, eqns. (11) and (12) yield, after considerable working out:

$$P_s = 3E \cdot \frac{\hat{M}\ddot{y}_{\max} + F_{st}}{M\omega_b^2 \hat{y}_{\max}} \cdot \frac{2(\lambda + 1)}{5\lambda - 1} \quad \text{at} \quad \lambda \leq \frac{3\hat{M}\ddot{y}_{\max} + 2F_{st}}{3\hat{M}\ddot{y}_{\max} - 2F_{st}}, \quad (17)$$

$$P_s = 3E \cdot \frac{\hat{M}\ddot{y}_{\max} - F_{st}}{M\omega_b^2 \hat{y}_{\max}} \cdot \frac{2(\lambda + 1)}{5 - \lambda} \quad \text{at} \quad \lambda \geq \frac{3\hat{M}\ddot{y}_{\max} + 2F_{st}}{3\hat{M}\ddot{y}_{\max} - 2F_{st}}, \quad (18)$$

$$P_s = \frac{3E\hat{y}_{\max}}{\omega_b^2 \hat{y}_{\max}} \quad \text{at} \quad \lambda = \frac{3\hat{M}\ddot{y}_{\max} + 2F_{st}}{3\hat{M}\ddot{y}_{\max} - 2F_{st}}. \quad (19)$$

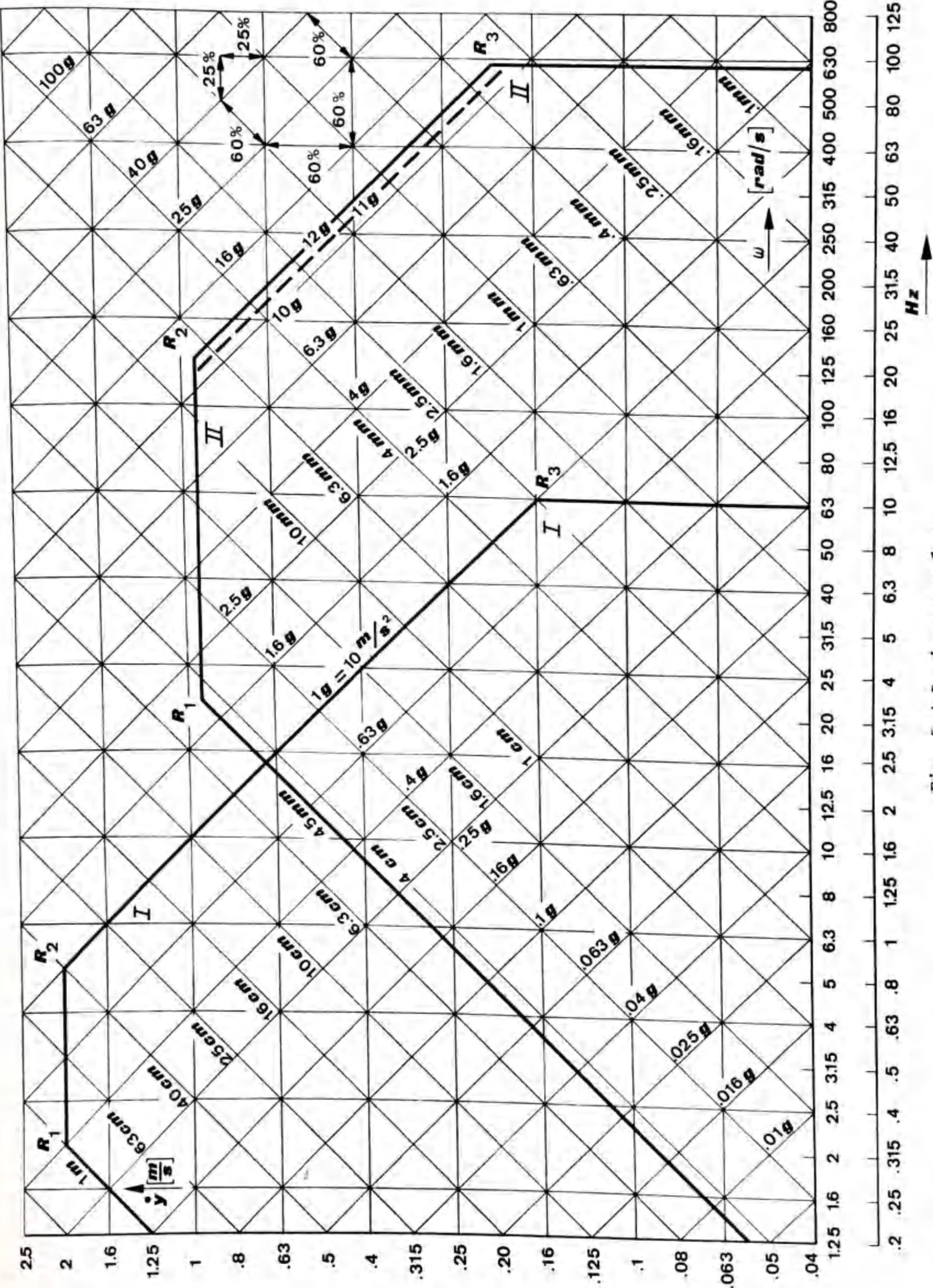


Fig. 5.4 Design examples

5.9 Design example (I in Fig. 5.4)

Specifications of a symmetric motor to be designed:

$$M = 1500 \text{ kg}, \quad \omega_b = 10 \text{ Hz} = 63 \text{ rad/s}, \quad E = 10^9 \text{ N/m}^2, \quad F_{st} = 0$$

$$\hat{y}_{\max} = 1 \text{ m}, \quad \hat{\dot{y}}_{\max} = 2 \text{ m/s}, \quad \hat{\ddot{y}}_{\max} = 10 \text{ m/s}^2 = 1 \text{ g}.$$

The design data now become

$$A = \frac{M \omega_b^2 \hat{y}_{\max}}{2E} = \frac{1500 \cdot 63^2 \cdot 1}{2 \cdot 10^9} = 3 \cdot 10^{-3} \text{ m}^2 = 30 \text{ cm}^2,$$

$$P_s = \frac{3E \hat{\ddot{y}}_{\max}}{\omega_b^2 \hat{y}_{\max}} = \frac{3 \cdot 10^9 \cdot 10}{63^2 \cdot 1} = 75 \cdot 10^5 \frac{\text{N}}{\text{m}^2} = 75 \text{ bar},$$

$$Q_{\max} = \frac{M \omega_b^2 \hat{y}_{\max} \hat{\dot{y}}_{\max}}{2E} = \frac{1500 \cdot 63^2 \cdot 1 \cdot 2}{2 \cdot 10^9} = 6 \cdot 10^{-3} \frac{\text{m}^3}{\text{s}} = 360 \frac{\text{dm}^3}{\text{min}},$$

$$N_p = Q_{\max} \cdot P_s = 6 \cdot 10^{-3} \cdot 75 \cdot 10^5 = 45000 \text{ Watt} = 45 \text{ kW}.$$

Further we find

$$\omega_1 = \frac{\hat{\dot{y}}_{\max}}{\hat{y}_{\max}} = \frac{2}{1} = 2 \frac{\text{rad}}{\text{s}} = 0.32 \text{ Hz}, \quad \hat{y}_2 = \frac{\hat{\ddot{y}}_{\max}}{\hat{\dot{y}}_{\max}} = \frac{2^2}{10} = 0.4 \text{ m} = 40 \text{ cm},$$

$$\omega_2 = \frac{\hat{\ddot{y}}_{\max}}{\hat{\dot{y}}_{\max}} = \frac{10}{2} = 5 \frac{\text{rad}}{\text{s}} = 0.8 \text{ Hz}, \quad \hat{y}_3 = \frac{\hat{\ddot{y}}_{\max}}{\omega_b^2} = \frac{10}{63^2} = \frac{1}{400} \text{ m} = 2.5 \text{ mm}.$$

The results (I) are plotted in the performance diagram of Fig. 5.4.

Introducing a constant pre-load $F_{st} = Mg = 15000 \text{ N}$, the design of motor and valve (A , $Q_{\max}$) remain the same, but P_s and N_p must be doubled, see eqns. (8) and (10).

The constant pre-load $F_{st} = 15000 \text{ N}$ may lead to the design of an asymmetric motor with an asymmetric 4-way valve. Equations (17), (18) and (19) lead to:

$A_1/A_2 = 5$	gives	$A_1 = 50 \text{ cm}^2$	$P_s = 75 \text{ bar}$	$Q_{1 \max} = 600 \frac{\text{dm}^3}{\text{min}}$	$N_{1p} = 75 \text{ kW}$
2		40 cm^2	100 bar	$480 \frac{\text{dm}^3}{\text{min}}$	80 kW
1		30 cm^2	150 bar	$360 \frac{\text{dm}^3}{\text{min}}$	90 kW

This example shows very clearly the enormous influence of the choice of A_1/A_2 on the supply pressure P_s , the valve capacity $Q_{\max}$ and the peak-power N_p to fulfil the specifications.

5.10 "Inverse" design example (II in Fig. 5.4)

It is also possible of course to calculate the performance of an existing motor, following the "inverse" design procedure. As an example, we envisage a symmetric motor, having

$$\begin{aligned} M &= 105 \text{ kg} , & \hat{y}_{\max} &= 0.045 \text{ m} , & A &= 9 \cdot 10^{-4} \text{ m}^2 , \\ E &= 10^9 \text{ N/m}^2 , & P_s &= 210 \cdot 10^5 \text{ N/m}^2 , & Q_{\max} &= \frac{5}{6} \cdot 10^{-3} \text{ m}^3/\text{s} . \end{aligned}$$

We now find

$$\omega_1 = \frac{Q_{\max}}{A \hat{y}_{\max}} = 20.58 \frac{\text{rad}}{\text{s}} = 3.28 \text{ Hz} ,$$

$$\omega_2 = \frac{2P_s A^2}{3MQ_{\max}} = 129.6 \frac{\text{rad}}{\text{s}} = 20.6 \text{ Hz} , \quad \hat{y}_2 = \frac{3MQ_{\max}^2}{2P_s A^3} = 0.00714 \text{ m} = 7.14 \text{ mm} ,$$

$$\omega_3 = \sqrt{\frac{2AE}{M\hat{y}_{\max}}} = 617 \frac{\text{rad}}{\text{s}} = 98 \text{ Hz} , \quad \hat{y}_3 = \frac{P_s \hat{y}_{\max}}{3E} = 0.000315 \text{ m} = 0.315 \text{ mm} ,$$

$$\hat{\ddot{y}}_{\max} = \frac{Q_{\max}}{A} = 0.926 \frac{\text{m}}{\text{s}} , \quad \hat{\ddot{y}}_{\max} = \frac{2P_s A}{3M} = 120 \frac{\text{m}}{\text{s}^2} = 12 \text{ g} .$$

The results (II) have been plotted also in the performance diagram, Fig. 5.4.

Tacitly it was assumed here that the mass M is moving in a horizontal plane. When it is moving vertically, a constant pre-load $F_{st} = Mg$ is introduced. The results remain the same except that

$$\hat{\ddot{y}}_{\max} + g = \frac{2P_s A}{3M} = 120 \frac{\text{m}}{\text{s}^2} = 12 \text{ g} , \quad \text{so} \quad \hat{\ddot{y}}_{\max} = 11 \text{ g} .$$

Consequently, $\omega_2 = 18.9 \text{ Hz}$, $\hat{y}_2 = 7.79 \text{ mm}$ and $\hat{y}_3 = 0.289 \text{ mm}$. The dashed line in Fig. 5.4 shows the influence of g .

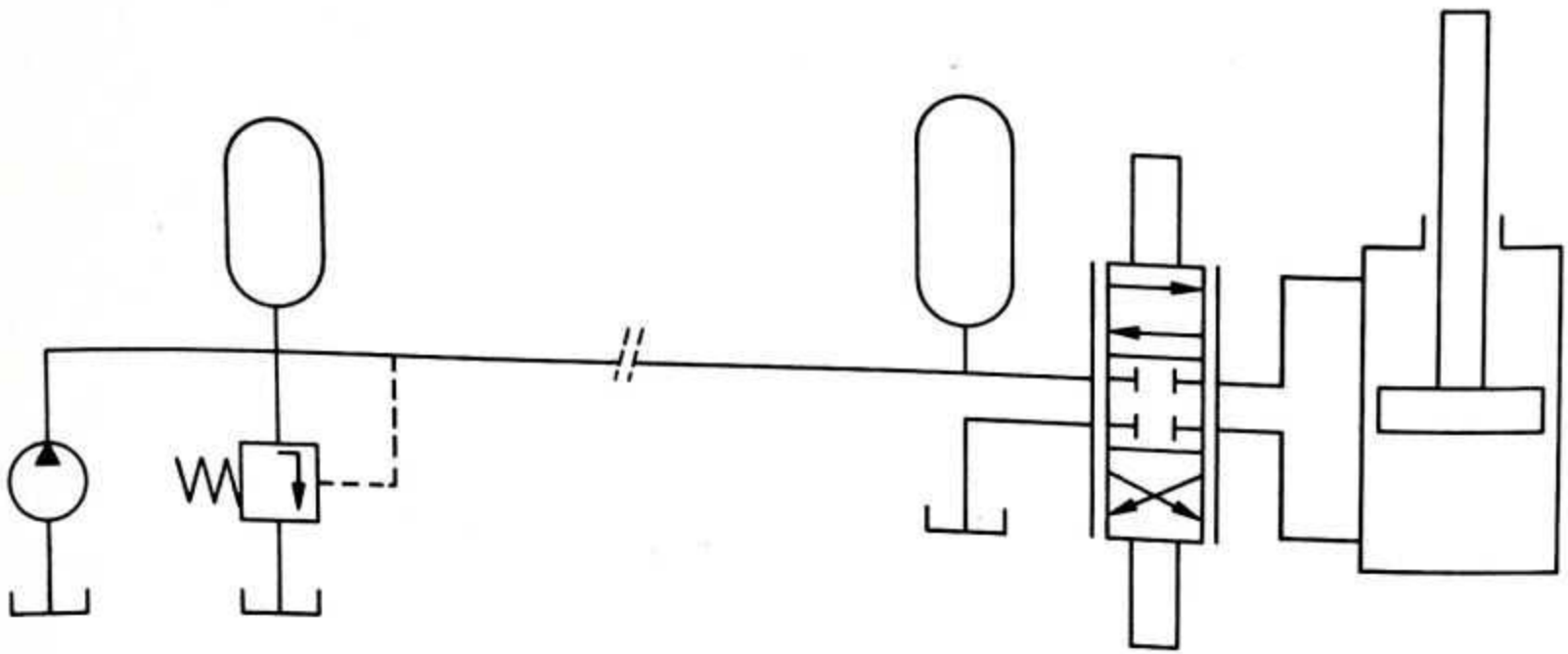


Fig. 6.1 High pressure supply line between pump and servosystem. In the line two gas-accumulators and a pressure control valve have been installed to keep the supply pressure as constant as possible.

CHAPTER 6

HYDRAULIC LINE DYNAMICS

6.1 Introduction

In current literature as well as in previous chapters it has been suggested firmly that the hydraulic supply pressure is and should be constant. In practice the assumption of a constant supply pressure is frequently a fair one, provided "adequate measures" have been taken to ensure this. This gives rise to the question of what is precisely meant by adequate measures.

In addition to this question there is the observation that in some applications small fluctuations in the supply pressure act as a disturbance on the servosystem. In that case the question of how to "stabilize" the supply pressure is vital. As the reader has noticed, both questions are the same. This chapter tries to answer them.

To illustrate the practical situation we refer to Fig. 6.1, where a high pressure line (pipe or hose) between a fixed displacement pump and a hydraulic servosystem is installed with a pressure control valve (relief valve) and two gas-accumulators. The pressure control valve is supposed to keep the pressure constant; the first accumulator should cancel the pump flow pulsations and the second accumulator (right) should compensate for fast flow fluctuations via the servovalve and motor.

In literature and handbooks one looks in vain for directives and design rules for size and location of accumulators and valves. Furthermore, one cannot avoid the impression that some of the practical and/or fundamental knowledge of manufacturers is carefully being kept secret! So, how does one resolve complaints about pump flow pulsations, and what does one do about pressure oscillations due to a combination of line dynamics and the size and location of accumulators, triggered by the servosystem's oil consumption?

Recent research in the Mechanical Control Engineering Laboratory of Delft University of Technology tried to answer some of these questions. The goal of our efforts was to keep the pressure in the supply line as constant as possible. Based on the well known mathematical model of hydraulic lines and the lesser known models of components, a *chain structure* is set up in which the lines and the components are the elements. This more comprehensive model structure will be reshaped, simplified and approximated several times to force out of it the design rules we want. To justify this risky process of simplification and approximation numerous tests have been carried out. Some of the results will be presented in this chapter.

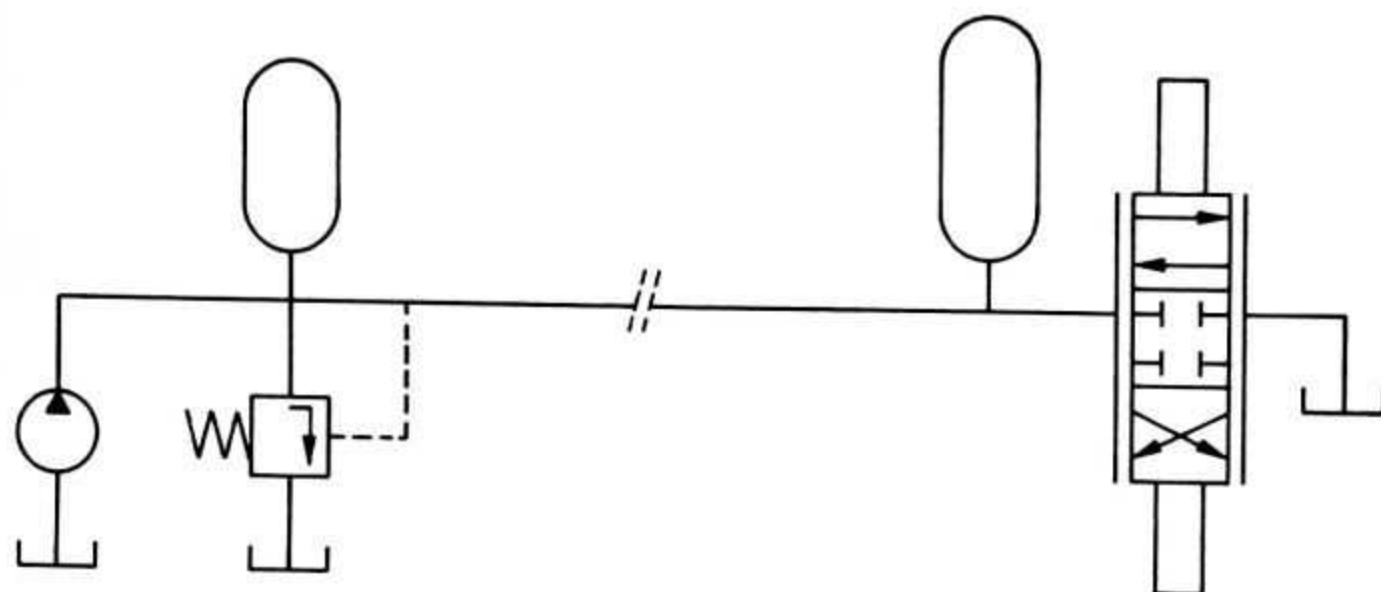


Fig. 6.2 Laboratory test set-up to measure hydraulic line dynamics. To avoid some of the interactions between the hydraulic line and the servosystem the servomotor itself is left out.

As will be shown later, some results and design rules seem trivial, simple as they are. For instance:

- Two-stage pressure control valves can only maintain some DC-level of the pressure. Dynamically, the valve is scarcely doing anything.
- A gas-accumulator can compensate for pump flow pulsations in a wide frequency range if
 - the accumulator is placed within 30 cm of the pump,
 - the connection line between main line and accumulator is as short as possible, preferably less than 5 cm. If its volume exceeds $0.5 - 1 \text{ dm}^3$, its influence is no longer determined by its volume but by its connection line.
- To wipe out a specific frequency of the pump flow pulsation, a Helmholtz-resonator is more effective than any other dynamic filter or accumulator.
- Pressure-controlled variable displacement pumps could easily oscillate at very low frequencies, showing intolerable pressure fluctuations. A large gas-accumulator, very close to the servovalve, could be an excellent remedy, provided the steady-state losses between pump and servovalve are not too low.

The above statements are somewhat provocative. They are intended to be so to stimulate a lively discussion between scientists and designers in order to improve fundamental knowledge applicable to practice and to bridge the gap between theory and practice. As long as theoretical experts refuse to come down now and then and dirty their hands in solving problems that are both fundamental and practical, this gap will continue to exist.

In addition to the structure and dynamics of hydraulic lines and its components, there is the problem of the boundary conditions. At the end of the pipeline, interaction will occur between

- servovalve openings, motor load and motor speed and
- pressure P_o and flow Q_o at the end of the pipeline.

In many cases one could try to simplify the interaction to one of two extremes:

- a - Q_o is an independent variable, given by the servomotor excursions. So here interaction is supposed to work in one direction only, which is approximately correct as long as the variations in P_o are small.
- b - Q_o is determined completely by P_o . Constant servovalve openings as well as leakage flows could enhance a situation where, around a given working point, the flow Q_o is proportional to the pressure P_o . This situation could be simulated by a laminar flow resistance $R_o = P_o/Q_o$ at the end of the pipeline.

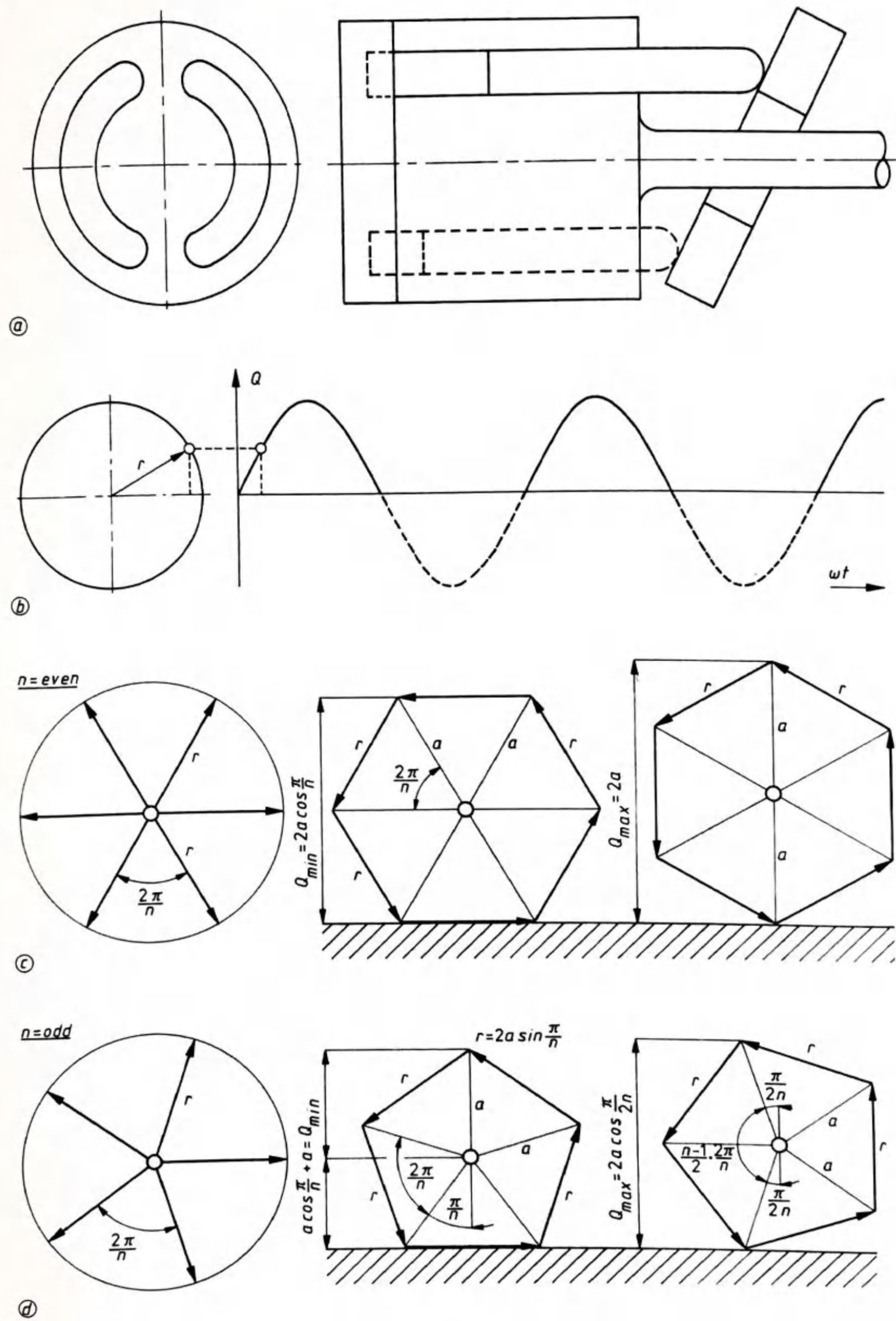


Fig. 6.3 Flow pulsations of axial plunger pumps.

To reduce complications due to interaction, in most of our laboratory experiments only a servovalve (without servomotor) has been used at the end of the pipeline (see Fig. 6.2). In this case we have

$$Q_o = a_v \cdot C_d \sqrt{\frac{2P_o}{\rho}}$$

which can be linearized around the nominal values of the valve opening a_v and the pressure P_o . In the case of constant valve openings, this leads directly to case b.

At the other end of the pipeline the pump is giving the boundary conditions. In our considerations the pump is regarded as a flow source. In general, however, a pump is not a *constant* flow source, since positive displacement pumps inherently show flow fluctuations. To illustrate this, in Fig. 6.3(a) an axial plunger pump is indicated. The flow due to one single plunger in the rotating cylinder block is given in Fig. 6.3(b): the positive halves of a sinusoidal signal.

In an axial plunger pump several plungers contribute phase-shifted flow patterns that should be summed. Clearly, this sum does not yield a perfect constant flow, and thus there are pulsations.

Figures 6.3(c) and (d) show how the sum of the flows can be found at an even and an odd number of plungers. At n plungers a continuously rotating set of n vectors (equally spaced at $2\pi/n$ radians) enables us to find the total flow by summation of the *positive vertical components of n vectors*. This sum can be found easily by placing these vectors *head to tail in a polygon*. The varying height of the polygon, rolling on a horizontal line, represents the varying flow exactly!

The maximum and minimum flow of polygons with n even and odd respectively is shown in Figs. 6.3(c) and (d). These flows can be expressed directly in the individual plunger flow, represented by its vector length r . With some simple trigonometry the result can be summarized in the following table.

Flow pulsations axial plunger pumps		$n = \text{even}$	$n = \text{odd}$
Maximum flow	$Q_{\max}$	$2a$	$2a \cos \frac{\pi}{2n}$
Minimum flow	$Q_{\min}$	$2a \cos \frac{\pi}{n}$	$2a \cos^2 \frac{\pi}{2n}$
Deviation	$\frac{1}{2}(Q_{\max} - Q_{\min})$	$2a \sin^2 \frac{\pi}{2n}$	$2a \cos \frac{\pi}{2n} \cdot \sin^2 \frac{\pi}{4n}$
Nominal flow	$\frac{1}{2}(Q_{\max} + Q_{\min})$	$2a \cos^2 \frac{\pi}{2n}$	$2a \cos \frac{\pi}{2n} \cdot \cos^2 \frac{\pi}{4n}$
Specific deviation	$\frac{\frac{1}{2}(Q_{\max} - Q_{\min})}{\frac{1}{2}(Q_{\max} + Q_{\min})}$	$\tan^2 \frac{\pi}{2n} \approx \left(\frac{\pi}{2n}\right)^2$	$\tan^2 \frac{\pi}{4n} \approx \left(\frac{\pi}{4n}\right)^2$

The table shows that, for instance, at $n = 6$ and $n = 7$ the specific deviations from the nominal flow are 6.85% and 1.26%, respectively. So the specific deviation at an odd number of plungers is approximately four times lower than at an even number. For this reason axial plunger pumps should have an odd number of plungers!

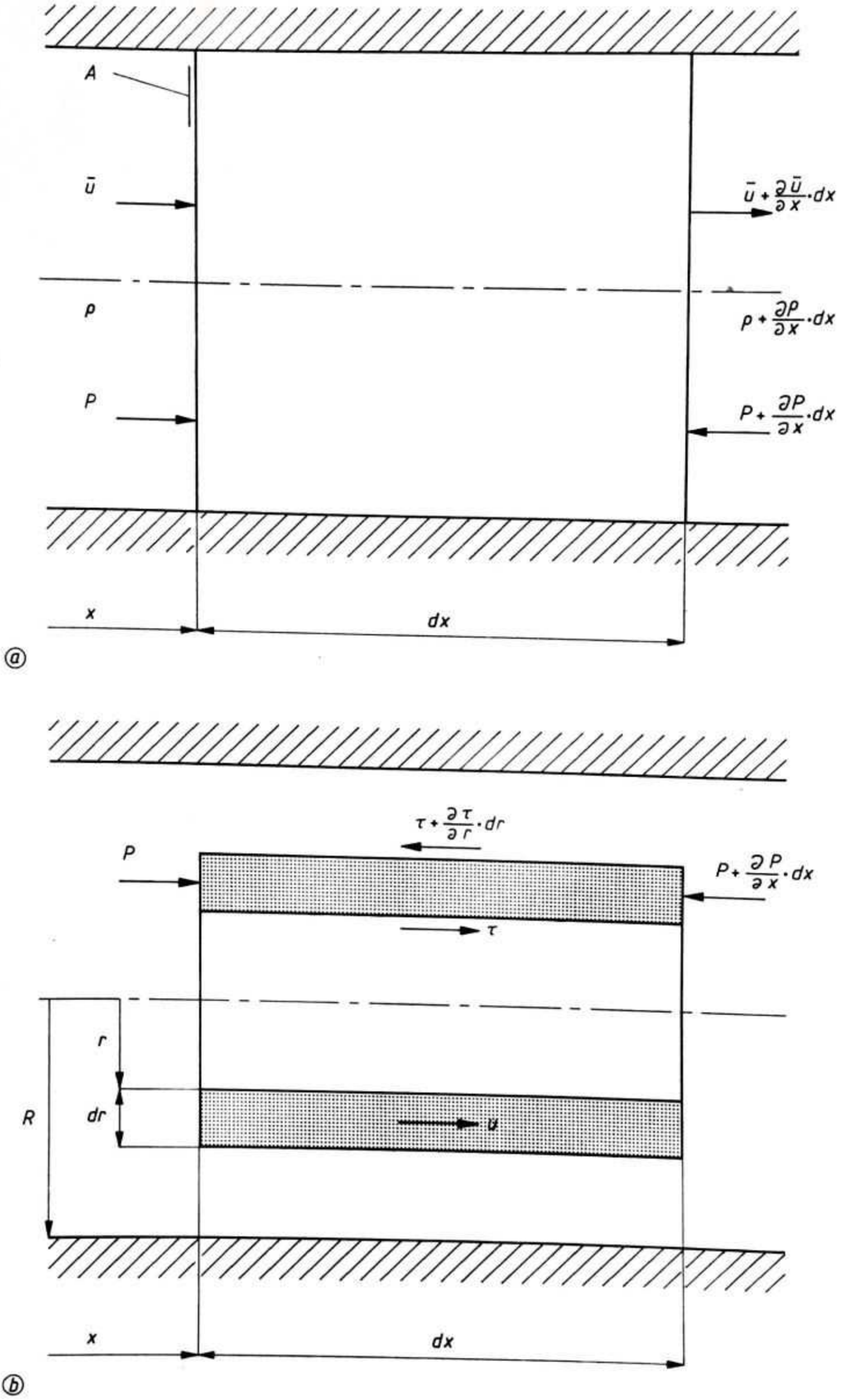


Fig. 6.4 Control volumes concerning the conservation of mass (a) and of momentum (b).

6.2 Basic equations for the dynamic behavior of hydraulic lines

6.2.1 Assumptions

For the analysis of the dynamic behavior of hydraulic pipelines having distributed parameters, we start with the continuity equation and the Navier-Stokes equation that will be derived here in a simplified fashion. The results of d'Souza and Oldenburger [1], who gave a far stricter derivation, guarantee that these simplifications are justified.

We assume a *laminar flow* ($Re < 2000$) in a *cylindrical pipe with non-elastic walls*. Thus tangential flow is neglected. In the pipe, the pressure P , the specific mass ρ , the axial speed U of the oil and the radial speed V fundamentally are functions of the axial direction x , the radial distance r to the center line of the pipe and time t . In addition, the temperature is supposed to be constant.

A major simplification can be made on the basis that the changes of P and ρ in the radial direction are far less than in the axial direction. Thus we neglect differences of P and ρ in the radial direction. Consequently, we also neglect the radial speed V . Summarizing so far, we have the following variables (see Fig. 6.4):

$$P(x, r, t) = P(x, t), \quad \rho(x, r, t) = \rho(x, t), \quad U(x, r, t) \quad \text{and} \quad V(x, r, t) = 0$$

As a final result, we require relationships between pressures and flows. The flow Q can be found by determining the mean value of the axial speed U in a pipe with cross-section A .

$$Q(x, t) = A \cdot \bar{U}(x, t) = \frac{A}{\pi R^2} \int_0^R 2\pi r U(x, t, r) \cdot dr = 2\pi \int_0^R r U(x, t, r) \cdot dr$$

6.2.2 Continuity

Owing to the conservation of mass the difference between the inflow and the outflow of mass equals the rate of mass increase inside the control volume of Fig. 6.4(a).

$$A\bar{U}\rho - A(\bar{U} + \frac{\partial \bar{U}}{\partial x} dx)(\rho + \frac{\partial \rho}{\partial x} dx) = \frac{\partial m}{\partial t} = A dx \frac{\partial \rho}{\partial t}$$

so

$$\frac{1}{\rho} \cdot \frac{\partial \rho}{\partial t} = - \frac{\partial \bar{U}}{\partial x} - \frac{\bar{U}}{\rho} \cdot \frac{\partial \rho}{\partial x}$$

With the equation of state $\partial \rho / \rho = \partial P / E$, we get

$$\frac{1}{E} \cdot \frac{\partial P}{\partial t} + \frac{\partial \bar{U}}{\partial x} + \frac{\bar{U}}{E} \cdot \frac{\partial P}{\partial x} = 0$$

We may assume $\bar{U} \cdot \partial P / \partial x \ll \partial P / \partial t$ [1], so we find

$$\frac{1}{E} \cdot \frac{\partial P}{\partial t} + \frac{\partial \bar{U}}{\partial x} = \frac{1}{E} \cdot \frac{\partial P}{\partial t} + \frac{1}{A} \cdot \frac{\partial Q}{\partial x} = 0$$

To solve this equation, together with the Navier-Stokes equation, particularly for sinusoidal fluctuations, Laplace transformation must be carried out. Using $L[P(x,t)] = P(x,s)$ and $L[Q(x,t)] = Q(x,s)$ and neglecting the initial state, we finally get

$$\boxed{\frac{As}{E} \cdot P(x,s) + \frac{dQ(x,s)}{dx} = 0} \quad (1)$$

6.2.3 Momentum

Owing to the conservation of momentum of the mass inside the control volume of Fig. 6.4(b), we find

$$\begin{aligned} F_x &= m \frac{dU}{dt} = 2\pi r dr dx \cdot \rho \frac{dU}{dt} \\ &= 2\pi r dr \left[P - \left(P + \frac{\partial P}{\partial x} dx \right) \right] + 2\pi r dx \cdot \tau - 2\pi(r + dr) \left(\tau + \frac{\partial \tau}{\partial r} dr \right) dx \end{aligned}$$

So

$$\rho \frac{dU}{dt} = - \frac{\partial P}{\partial x} - \frac{\tau}{r} - \frac{\partial \tau}{\partial r}$$

With

$$\frac{dU}{dt} = \frac{\partial U}{\partial t} + U \frac{\partial U}{\partial x} \approx \frac{\partial U}{\partial t}$$

and with

$$\tau = - \mu \frac{\partial U}{\partial r}$$

we get

$$\rho \frac{\partial U}{\partial t} = - \frac{\partial P}{\partial x} + \frac{\mu}{r} \cdot \frac{\partial U}{\partial r} + \mu \frac{\partial^2 U}{\partial r^2}$$

In this case, before introducing the mean value $\bar{U}$, Laplace transformation should be carried out.

$$\rho s U(x,r,s) = - \frac{dP(x,s)}{dx} + \frac{\mu}{r} \cdot \frac{dU(x,r,s)}{dr} + \mu \frac{d^2 U(x,r,s)}{dr^2}$$

After rearranging and introducing

$$\gamma^2 = -\frac{\rho s}{\mu} = -\frac{s}{\nu} \quad \text{or} \quad \gamma = j\sqrt{\frac{s}{\nu}} \quad (2)$$

we find Bessel's differential equation of order zero.

$$\gamma^2 r^2 \frac{d^2 U}{d(\gamma r)^2} + \gamma r \frac{dU}{d(\gamma r)} + \gamma^2 r^2 U = +\frac{r^2}{\mu} \cdot \frac{dP}{dx} = -\frac{\gamma^2 r^2}{\rho s} \cdot \frac{dP}{dx}$$

Using, for example, the *Mathematical Handbook of Formulas and Tables* by M.R. Spiegel (McGraw-Hill, 1968), especially the equations 24.1, 24.2, 24.14, 24.17, 24.21, 24.22 and Figs. 24.1 and 24.2, we can find the following solution and elaborations

$$U(x, r, s) = C_1 J_0(\gamma r) + C_2 Y_0(\gamma r) - \frac{1}{\rho s} \cdot \frac{dP}{dx}$$

In this, J_0 and Y_0 are Bessel and Weber functions respectively.

C_1 and C_2 are constants to be solved by boundary conditions.

At $r = 0$ is $U \neq \infty$, $J_0(\gamma r) = 1$ and $Y_0(\gamma r) = -\infty$, so $C_2 = 0$.

At $r = R$ is $U = C_1 J_0(\gamma R) - \frac{1}{\rho s} \cdot \frac{dP}{dx} = 0$, so $C_1 = \frac{1}{J_0(\gamma R)} \cdot \frac{1}{\rho s} \cdot \frac{dP}{dx}$

So the complete solution becomes

$$U(x, r, s) = \frac{1}{\rho s} \cdot \frac{dP}{dx} \left[\frac{J_0(\gamma r)}{J_0(\gamma R)} - 1 \right]$$

Now the mean value $\bar{U}$ of U can be determined.

$$\begin{aligned} \bar{U} &= \frac{Q}{A} = \frac{2}{R^2} \int_0^R U r dr = \frac{1}{\rho s} \cdot \frac{dP}{dx} \cdot \frac{2}{R^2} \int_0^R \left[\frac{r J_0(\gamma r)}{J_0(\gamma R)} - r \right] dr \\ &= \frac{1}{\rho s} \cdot \frac{2}{(\gamma R)^2} \cdot \frac{dP}{dx} \int_0^{\gamma R} \left[\frac{\gamma r J_0(\gamma r)}{J_0(\gamma R)} - \gamma r \right] d\gamma r = \frac{1}{\rho s} \cdot \frac{dP}{dx} \cdot \frac{2}{\gamma R^2} \left[\frac{\gamma r J_1(\gamma r)}{J_0(\gamma R)} - \frac{1}{2} \gamma^2 r^2 \right]_0^{\gamma R} \end{aligned}$$

$$Q = \frac{A}{\rho s} \cdot \frac{dP}{dx} \left[\frac{2 J_1(\gamma R)}{\gamma R J_0(\gamma R)} - 1 \right] = \frac{A}{\rho s} \cdot \frac{dP}{dx} \cdot \frac{J_2(\gamma R)}{J_0(\gamma R)}$$

This brings the Navier-Stokes equation in the following form

$$\boxed{\frac{dP(x, s)}{dx} + \frac{\rho s}{A} \cdot Q(x, s) \cdot N(s) = 0} \quad (3)$$

$$\text{In this, } N(s) = -\frac{J_0(\gamma R)}{J_2(\gamma R)} \quad (4)$$

With the definition of Bessel functions of the order n ($n =$ positive integer)

$$J_n(\gamma R) = \sum_{k=0}^{\infty} \frac{(-1)^k (\gamma R/2)^{n+2k}}{k!(n+k)!} = \left(\frac{\gamma R}{2}\right)^n \sum_{k=0}^{\infty} \frac{(-\gamma^2 R^2/4)^k}{k!(n+k)!}$$

and the introduction of the "viscosity factor"

$$\alpha = \frac{8\nu}{R^2} = \frac{32\nu}{D^2} \quad (5)$$

the final result becomes

$$N(s) = -\frac{J_0(\gamma R)}{J_2(\gamma R)} = \frac{\alpha}{s} \cdot \frac{\sum_{k=0}^{\infty} \frac{2^k}{(k!)^2} \cdot \left(\frac{s}{\alpha}\right)^k}{\sum_{k=0}^{\infty} \frac{2^{k+1}}{k!(k+2)!} \cdot \left(\frac{s}{\alpha}\right)^k} \quad (6a)$$

$$= \frac{\alpha}{s} \cdot \frac{1 + \frac{2s}{\alpha} + \frac{s^2}{\alpha^2} + \frac{2s^3}{9\alpha^3} + \frac{s^4}{36\alpha^4} + \dots}{1 + \frac{2s}{3\alpha} + \frac{s^2}{6\alpha^2} + \frac{s^3}{45\alpha^3} + \frac{s^4}{540\alpha^4} + \dots} \quad (6b)$$

$$= \frac{\alpha}{s} + \frac{4}{3} - \frac{s}{18\alpha} + \frac{2s^2}{135\alpha^2} - \frac{7s^3}{1620\alpha^3} + \frac{11s^4}{8505\alpha^4} - \dots \quad (6c)$$

The asymptotes ($\alpha/s \rightarrow \infty$; $\alpha/s \rightarrow 0$) of this function can be found easily. Evidently

$$\lim_{\frac{\alpha}{s} \rightarrow \infty} N(s) = \lim_{\frac{s}{\alpha} \rightarrow 0} N(s) = \frac{\alpha}{s}$$

Substitution in eqn. (3) gives

$$\frac{dP}{dx} = -\frac{\rho\alpha}{A} \cdot Q = -\frac{32\nu\rho}{AD^2} \cdot Q = -\frac{128\mu}{\pi D^4} \cdot Q$$

This corresponds to the well known steady laminar flow with a parabolic velocity profile. The other asymptote becomes

$$\lim_{\frac{\alpha}{s} \rightarrow 0} N(s) = \lim_{\frac{s}{\alpha} \rightarrow \infty} N(s) = 1$$

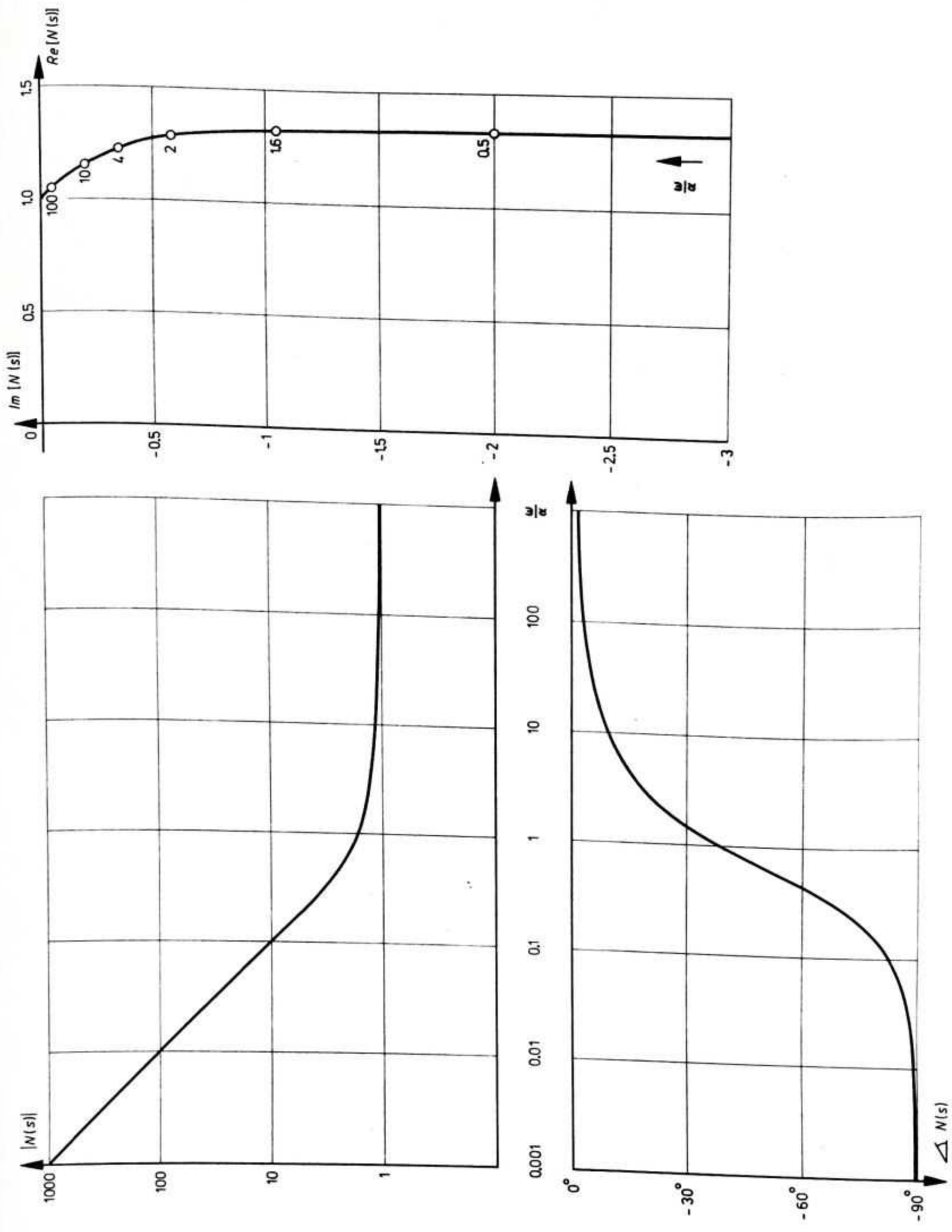


Fig. 6.5 Bode diagram and polar plot of the function $N(s)$.

Substitution in eqn. (3) now gives

$$A \frac{dP}{dx} = - \rho s Q$$

This corresponds to a frictionless flow with a uniform speed $\bar{U} = U$.

Analysis of $N(s)$ in the frequency domain by letting $s = j\omega$ has been carried out also. The results are shown in the Bode diagram and the polar diagram of Fig. 6.5. The above asymptotes can be recognized directly.

A rough approximation over the whole frequency range is

$$N(s) = \frac{\alpha}{s} + 1 \quad (6d)$$

which includes the asymptotes for low and high frequencies. Trikha [2] derived a far better approximation (via transformation and optimization in the time domain followed by re-transformation into the frequency domain).

$$N(s) = \frac{\alpha}{s} + 1 + \frac{0.1515}{1 + 0.3030 \frac{s}{\alpha}} + \frac{0.1620}{1 + 0.04 \frac{s}{\alpha}} + \frac{0.020}{1 + 0.001 \frac{s}{\alpha}} \quad (6e)$$

The approximation is perfect as long as $|s| = \omega < 300\alpha$.

The results obtained here so laboriously enable us to find elegant and practical relationships between pressures and flows at both ends of the pipeline.

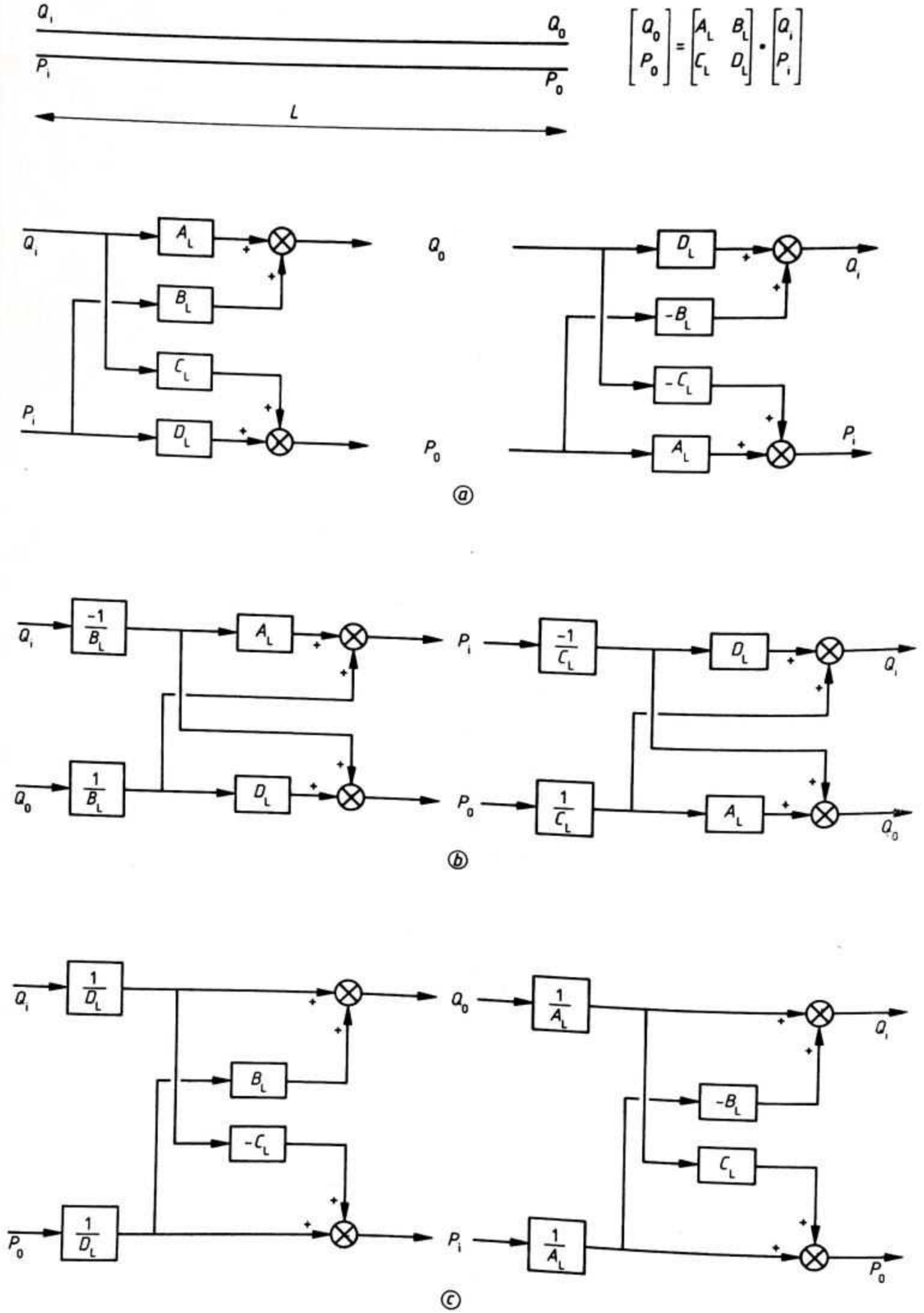


Fig. 6.6 Representations of the dynamic behavior of one single pipeline.
For summary see Appendix to Fig. 6.6 on next page.

6.2.4 Four-pole equations relating Q_i , P_i , Q_o and P_o

The continuity equation (1) and the Navier-Stokes equation (3) can be brought into the following form

$$P = -\frac{E}{As} \cdot \frac{dQ}{dx} = -\frac{aZ_c}{s} \cdot \frac{dQ}{dx}$$

$$\frac{dP}{dx} = -\frac{\rho s}{A} \cdot Q \cdot N = -\frac{Z_c s}{a} \cdot Q \cdot N$$

The introduction of the *characteristic impedance* Z_c and the *sonic velocity* a according to

$$Z_c = \frac{\rho a}{A} \quad (7)$$

$$a = \sqrt{\frac{E}{\rho}} \quad (8)$$

is justified by the fact that the number of relevant system parameters is reduced from three (E , A , ρ) to two (Z_c , a).

Elimination of the pressure P from the above equations gives

$$\frac{d^2 Q}{dx^2} - \frac{s^2}{a^2} \cdot Q \cdot N = 0$$

The solution of this second-order differential equation is

$$Q(x, s) = C_1 e^{\frac{sx}{a} \sqrt{N}} + C_2 e^{-\frac{sx}{a} \sqrt{N}}$$

$$\text{so } P(x, s) = -Z_c \sqrt{N} \left[C_1 e^{\frac{sx}{a} \sqrt{N}} - C_2 e^{-\frac{sx}{a} \sqrt{N}} \right]$$

The boundary conditions lead to the solution of the constants C_1 and C_2 . Letting $P = P_i$ and $Q = Q_i$ at $x = 0$, we find

$$C_1 = \frac{1}{2} \left(Q_i - \frac{P_i}{Z_c \sqrt{N}} \right) \quad C_2 = \frac{1}{2} \left(Q_i + \frac{P_i}{Z_c \sqrt{N}} \right)$$

Letting $P = P_o$ and $Q = Q_o$ at $x = L$ (end of the pipeline) and introducing the wave-propagation time along the line

$$T = \frac{L}{a} \quad (9)$$

we find at last the desired result: the four-terminal network equation or, for short, *four-pole equations*.

Appendix to Figure 6.6 - SUMMARY

$$a = \sqrt{E/\rho}, \quad T = L/a, \quad Z_c = \rho a/A, \quad \alpha = 32/D^2, \quad \gamma^2 = -s/\nu$$

$$\text{Continuity} \quad P(x, s) + \frac{aZ_c}{s} \cdot \frac{dQ(x, s)}{dx} = 0$$

$$\text{Navier-Stokes} \quad \frac{dP(x, s)}{dx} + \frac{Z_c s}{a} \cdot Q(x, s) \cdot N(s) = 0$$

$$\left. \begin{aligned} A_L Q_i + B_L P_i &= Q_o \\ C_L Q_i + D_L P_i &= P_o \end{aligned} \right\} \quad \left. \begin{aligned} D_L Q_o - B_L P_o &= Q_i \\ -C_L Q_o + A_L P_o &= P_i \end{aligned} \right\} \quad \text{diagrams (a)}$$

$$\left. \begin{aligned} A_L &= D_L = \cosh Ts \sqrt{N} \\ B_L &= -[\sinh Ts \sqrt{N}]/Z_c \sqrt{N} \\ C_L &= -Z_c \sqrt{N} \cdot \sinh Ts \sqrt{N} \end{aligned} \right\} \quad A_L D_L - B_L C_L = 1$$

This enables transformation to diagrams (b) and (c):

$$\left. \begin{aligned} -\frac{A_L}{B_L} Q_i + \frac{1}{B_L} Q_o &= P_i \\ -\frac{1}{B_L} Q_i + \frac{D_L}{B_L} Q_o &= P_o \end{aligned} \right\} \quad \left. \begin{aligned} -\frac{D_L}{C_L} P_i + \frac{1}{C_L} P_o &= Q_i \\ -\frac{1}{C_L} P_i + \frac{A_L}{C_L} P_o &= Q_o \end{aligned} \right\}$$

$$\left. \begin{aligned} \frac{1}{D_L} Q_i + \frac{B_L}{D_L} P_o &= Q_o \\ -\frac{C_L}{D_L} Q_i + \frac{1}{D_L} P_o &= P_i \end{aligned} \right\} \quad \left. \begin{aligned} \frac{1}{A_L} Q_o - \frac{B_L}{A_L} P_i &= Q_i \\ \frac{C_L}{A_L} Q_o + \frac{1}{A_L} P_i &= P_o \end{aligned} \right\}$$

$$N(s) = -\frac{J_o(\gamma R)}{J_2(\gamma R)} \approx 1 + \frac{\alpha}{s} + \sum_{i=1}^3 \frac{K_i}{\tau_i s + 1} = 1 - \frac{j\alpha}{\omega} + \sum_{i=1}^3 \frac{K_i(1 - j\omega\tau_i)}{\omega^2\tau_i^2 + 1}$$

Prescription to calculate $\sqrt{N}$, $\sinh Ts \sqrt{N}$ and $\cosh Ts \sqrt{N}$.

$$\sqrt{N} = \sqrt{a + jb} = c + jd$$

$$\sinh Ts \sqrt{N} = -\cos c\omega T \sinh d\omega T + j \sin c\omega T \cosh d\omega T$$

$$\cosh Ts \sqrt{N} = \cos c\omega T \cosh d\omega T - j \sin c\omega T \sinh d\omega T$$

$$c^2 = 0.5(a + \sqrt{a^2 + b^2}) \quad d^2 = 0.5(-a + \sqrt{a^2 + b^2})$$

$$a = \operatorname{Re}[N] = 1 + \sum_{i=1}^3 \frac{K_i}{\omega^2\tau_i^2 + 1}$$

$$b = \operatorname{Im}[N] = -\frac{\alpha}{\omega} - \sum_{i=1}^3 \frac{K_i\omega\tau_i}{\omega^2\tau_i^2 + 1}$$

$$K_1 = 0.5/3.3, \quad K_2 = 4.05/25, \quad K_3 = 2/100, \quad \tau_1 = 1/3.3\alpha, \quad \tau_2 = 1/25\alpha, \quad \tau_3 = 1/1000\alpha$$

$$Q_o = \cosh Ts \sqrt{N} \cdot Q_i - \frac{\sinh Ts \sqrt{N}}{Z_c \sqrt{N}} \cdot P_i, \quad P_i = A_L Q_i + B_L P_i$$

$$P_o = -Z_c \sqrt{N} \cdot \sinh Ts \sqrt{N} \cdot Q_i + \cosh Ts \sqrt{N} \cdot P_i = C_L Q_i + D_L P_i$$

(10)

$N(s)$ is given in eqn. (6)

The four-pole equations, relating the signals at the ends of the pipeline, will prove to be extremely powerful tools for the analysis, synthesis and design of hydraulic pipelines. First, however, some remarks must be made regarding the four-pole equations themselves.

- Deriving the continuity equation and the Navier-Stokes equation strictly will show some extra factors (mainly with regard to the radial speed V and kinetic energy). D'Souza and Oldenburger [1] convincingly proved, however, that it is permissible to neglect these extra factors.
- The four-pole equations derived here correspond exactly to those of D'Souza and Oldenburger [1]. Notations used by them, as well as the lack of approximations for $N(s)$, scarcely permit a practical analysis of complex pipeline systems (including components like accumulators) if the viscosity is taken into account.
- As inverse Laplace transformation is scarcely possible, the four-pole equations give only practical solutions in the frequency domain. For analysis in the *time domain*, the complete pipeline (not just the ends) should be considered. In Section 6.7 a possible approach by way of *simulation* will be shown.
- The four-pole equations imply that only two of the four variables Q_i , P_i , Q_o and P_o should be regarded as independent variables. Blackburn [3] pointed out that the two independent variables are *not* located at the same end of the pipe.
- Notation of the four-pole equations in matrix form

$$\begin{bmatrix} \cosh Ts \sqrt{N} & -\frac{\sinh Ts \sqrt{N}}{Z_c \sqrt{N}} \\ -Z_c \sqrt{N} \cdot \sinh Ts \sqrt{N} & \cosh Ts \sqrt{N} \end{bmatrix} \cdot \begin{bmatrix} Q_i \\ P_i \end{bmatrix} = \begin{bmatrix} A_L & B_L \\ C_L & D_L \end{bmatrix} \cdot \begin{bmatrix} Q_i \\ P_i \end{bmatrix} = \begin{bmatrix} Q_o \\ P_o \end{bmatrix}$$

makes it clear that the determinant of the systems transfer function equals unity.

$$A_L D_L - B_L C_L = \cosh^2 Ts \sqrt{N} - \sinh^2 Ts \sqrt{N} = 1 \quad (11)$$

Thanks to this remarkable property the relationships between Q_i , P_i , Q_o and P_o can be presented quite differently. Figure 6.6 shows some interesting presentations. A summary of relevant formulas is given also, including a "prescription" to calculate N , $\sinh Ts \sqrt{N}$ and $\cosh Ts \sqrt{N}$ as functions of frequency.

6.3 Dynamic behavior of a single pipeline

To see how the above theory works out in a very simple case, we consider a single pipeline with a laminar flow resistance R_o at the end. At the entrance one could try to establish a sinusoidally fluctuating P_i or Q_i , by using an electro-hydraulic servovalve (with a constant supply pressure, $P_s = \text{constant}$). Using the four-pole equations

$$A_L Q_i + B_L P_i = Q_o$$

$$C_L Q_i + D_L P_i = P_o$$

and the boundary condition

$$P_o = R_o Q_o$$

at the end of the pipeline, one can derive no less than five transfer functions, namely P_i/Q_i , P_o/Q_i , Q_o/Q_i , P_o/P_i and Q_o/P_i by eliminating each time two of the variables from the above equations. The result is shown in Table 6.1

Table 6.1 *Transfer functions and amplitude ratios of a single pipeline*

Elimination	Result	Amplitude ratio at $R_o = \infty$, $\alpha = 0$
Q_o, P_o	$\frac{P_i}{Q_i} = \frac{A_L R_o - C_L}{D_L - B_L R_o}$	$\left \frac{P_i}{Z_c Q_i} \right = \frac{1}{ \tan \omega T }$
Q_o, P_i	$\frac{P_o}{Q_i} = \frac{R_o}{D_L - B_L R_o}$	$\left \frac{P_o}{Z_c Q_i} \right = \frac{1}{ \sin \omega T }$
P_o, P_i	$\frac{Q_o}{Q_i} = \frac{1}{D_L - B_L R_o}$	$\left \frac{Q_o}{Q_i} \right = 0$
Q_o, Q_i	$\frac{P_o}{P_i} = \frac{R_o}{A_L R_o - C_L}$	$\left \frac{P_o}{P_i} \right = \frac{1}{ \cos \omega T }$
P_o, Q_i	$\frac{Q_o}{P_i} = \frac{1}{A_L R_o - C_L}$	$\left \frac{Z_c Q_o}{P_i} \right = 0$

Taking A_L , B_L , C_L and D_L from Fig. 6.6, and substituting $s = j\omega$, the amplitude ratios have been calculated in the special case where $R_o = \infty$ (pipe closed at the end; $Q_o = 0$) and there is no viscosity.

Figure 6.7 gives the amplitude ratios $P_i/Z_c Q_i$, $P_o/Z_c Q_i$ and P_o/P_i as functions of the frequency ω . This figure makes it clear that the interpretation in practice is not simple at all. If one tries to establish experimentally a sinusoidally fluctuating flow Q_i at the entrance of the pipe, then at $\omega T = 0; \pi; 2\pi$, etc. the

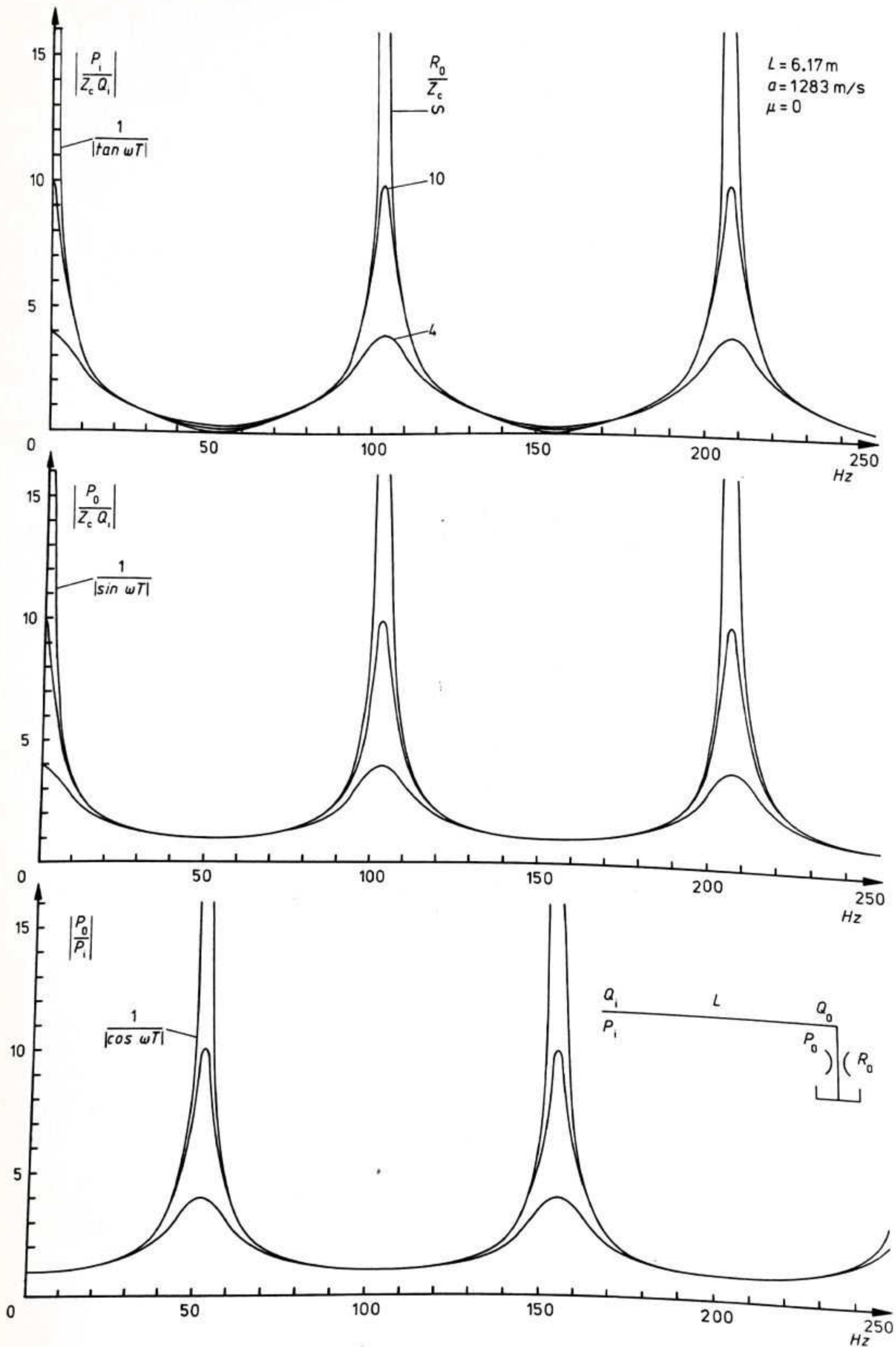


Fig. 6.7 Dynamics of a single pipeline (viscosity excluded).

corresponding amplitude of the pressure P_i should be infinitely large. Naturally, this cannot be effected. If we try, on the other hand, to establish a sinusoidal P_i , then at $\omega t = \pi/2; 3\pi/2; 5\pi/2$, etc. the corresponding Q_i should be infinite. Such a situation cannot be achieved in practice either.

So experiments according to this model cannot be carried out perfectly. These types of limitations frequently occur in practice, and one has to consider them beforehand.

If the flow resistance $R_o \neq \infty$, the above extremes will be moderated. Introducing $R_o \neq \infty$ limits the maximum values of the amplitude ratios to R_o/Z_c , as can be derived easily from the results in Table 6.1. In Fig. 6.7 the result is shown for $R_o/Z_c = 4$ and $R_o/Z_c = 10$.

The viscosity, neglected so far, is another factor limiting the extremes. To investigate this we assume again $R_o = \infty$ and calculate the amplitude ratios with the formulae of the previous section. Figure 6.8 gives an example for the following data:

$L = 6.17 \text{ m}$	$a = 1360 \text{ m/s}$	$T = 0.0045 \text{ s}$
$R_o = \infty$	$D = 0.013 \text{ m}$	$Z_c = 0.922 \cdot 10^{10} \text{ Ns/m}^5$
$\mu = 67.5 \cdot 10^{-3} \text{ Ns/m}^2$	$\nu = 75 \cdot 10^{-6} \text{ m}^2/\text{s}$	$\alpha = 14.2 \text{ s}^{-1}$

Owing to the viscosity, the infinitely large amplitude ratios (of Fig. 6.7) are moderated to more realistic levels (see Fig. 6.8). As it is very difficult to measure Q_i in a case like this, only P_o/P_i has been measured experimentally. The test set-up is indicated in Fig. 6.8(c). The measuring results fit the theoretical curve reasonably well, provided the bulk modulus E has been chosen correctly. In fact, the experiments indicated in Fig. 6.8(c) can and should be used to determine the bulk modulus E , because a direct measurement of E is scarcely feasible!

Experiments show that the bulk modulus E depends on the nominal pressure being used (see Table 6.2). A 100% explanation for this is not available, but it certainly is related to the amount of air in the oil. Thus it can be understood that at higher pressures its effect is less, resulting in higher values of E .

Another factor influencing the effective bulk modulus is the elasticity of the wall of the hydraulic pipeline. Correction factors for E due to the elasticity of the pipewall can be found theoretically [3] but can be neglected in the case of a metal pipe. In the case of high pressure rubber-based hoses, reinforced with metal threads in a textile type of structure, the influence of the elasticity is considerable. It is

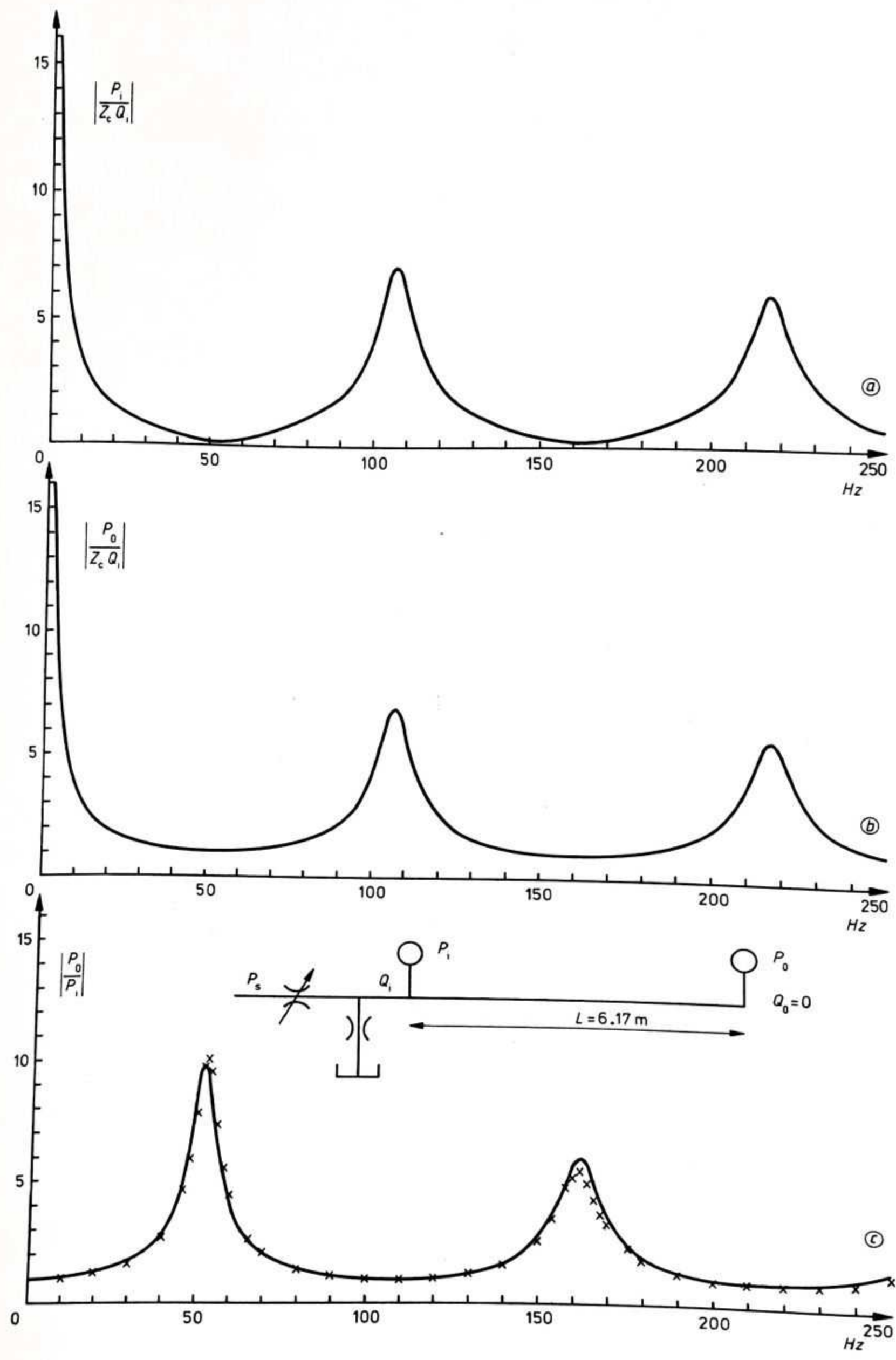


Fig. 6.8 Dynamics of a single pipeline (viscosity included).

Table 6.2 Factors influencing the bulk modulus E and the sonic velocity a

Nominal pressure	Steel pipeline $D_i = 12.5 \text{ mm}$, $D_u = 16 \text{ mm}$	High pressure hose $D_i = 12.5 \text{ mm}$
$P_s = 50 \text{ bar}$	$E = 14600 \text{ bar}$, $a = 1270 \text{ m/s}$	$E = 5000 \text{ bar}$, $a = 746 \text{ m/s}$
90 bar	15100 bar, 1296 m/s	5370 bar, 772 m/s
130 bar	15700 bar, 1320 m/s	5680 bar, 795 m/s
225 bar	18900 bar, 1450 m/s	

Sinusoidally varying pressures and flows show one more phenomenon to be observed. As stated before, pressures and flows are functions of both time and place. In this chapter, most attention is paid to the entrance and the exit of pipelines, the intermediate locations being of far less importance. Nevertheless, it should be clear that along a pipeline the signals also vary sinusoidally (as a function of place).

Pressure and flow being known at one place (P_1 and Q_1), the signals P_2 and Q_2 can be found directly at another place (distance L_d) by using

$$Q_2 = A_L Q_1 + B_L P_1 = Q_1 \cos(\omega L_d / a) - j P_1 / Z_c \cdot \sin(\omega L_d / a)$$

$$P_2 = C_L Q_1 + D_L P_1 = -j Z_c Q_1 \sin(\omega L_d / a) + P_1 \cos(\omega L_d / a)$$

From these equations it is clear that at a given frequency ω the signals are the same at certain intervals

$$\lambda = L_d = 2\pi a / \omega \quad (12)$$

These intervals are called *wavelengths* and correspond fully to classical wave theory. So, using one fixed frequency, one should be able to detect locations along the pipeline with a maximum amplitude of the pressure, as well as places where no (varying) pressure signal at all is present. Touching the pipeline with fingers (!), it is quite possible to "measure" the wavelength.

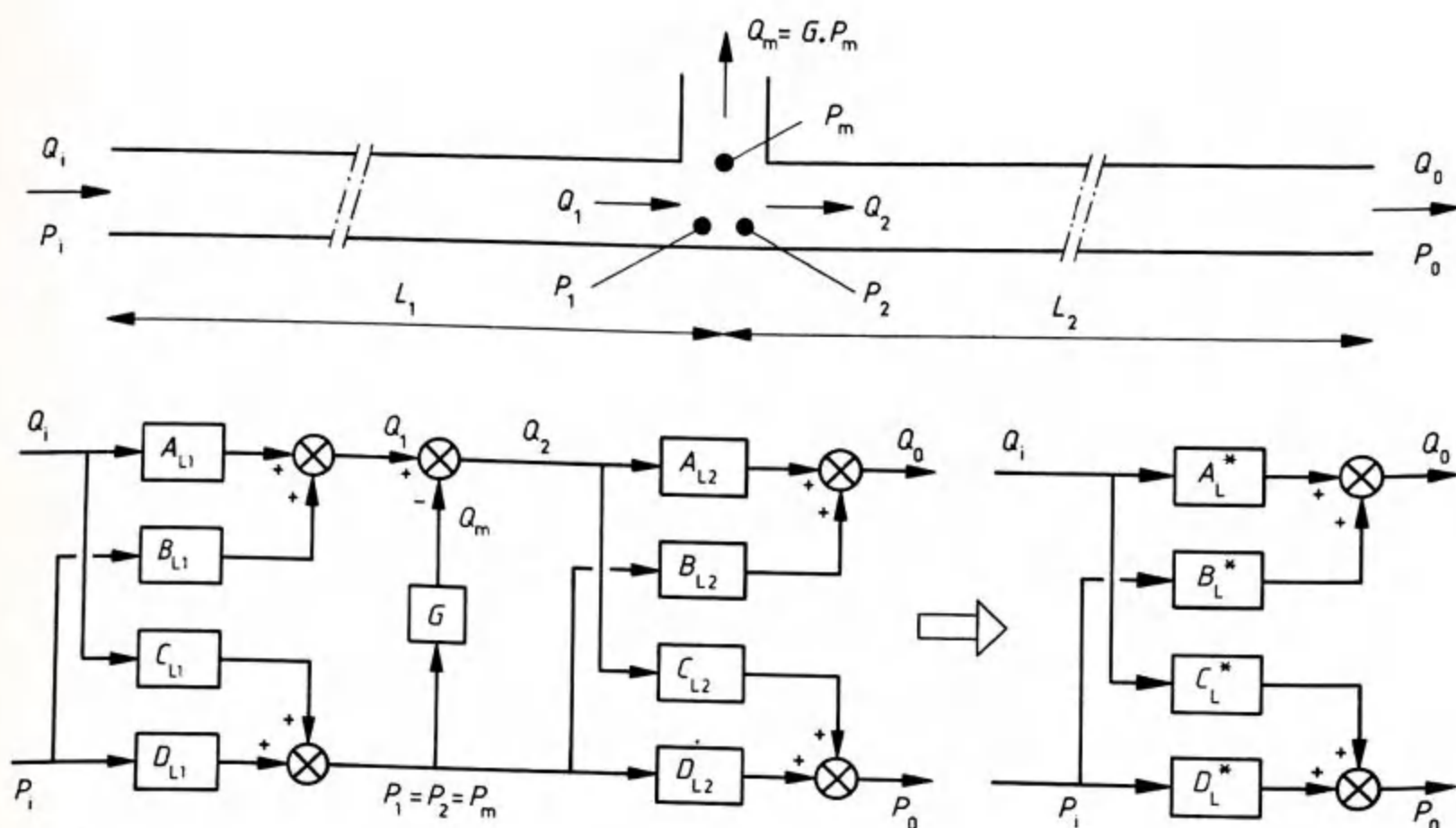


Fig. 6.9 Chain rule

$$\begin{bmatrix} A_{L2} & B_{L2} \\ C_{L2} & D_{L2} \end{bmatrix} \cdot \begin{bmatrix} 1 & -G \\ 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} A_{L1} & B_{L1} \\ C_{L1} & D_{L1} \end{bmatrix} \cdot \begin{bmatrix} Q_i \\ P_i \end{bmatrix} = \begin{bmatrix} A_L^* & B_L^* \\ C_L^* & D_L^* \end{bmatrix} \cdot \begin{bmatrix} Q_i \\ P_i \end{bmatrix} = \begin{bmatrix} Q_o \\ P_o \end{bmatrix}$$

$$A_L^* = A_{L1} A_{L2} + C_{L1} B_{L2} - G C_{L1} A_{L2}$$

$$A_{L1} = \cosh T_1 s$$

$$A_{L2} = \cosh T_2 s$$

$$B_L^* = B_{L1} A_{L2} + D_{L1} B_{L2} - G D_{L1} A_{L2}$$

$$Z_{c1} B_{L1} = -\sinh T_1 s$$

$$Z_{c2} B_{L2} = -\sinh T_2 s$$

$$C_L^* = A_{L1} C_{L2} + C_{L1} D_{L2} - G C_{L1} C_{L2}$$

$$C_{L1}/Z_{c1} = -\sinh T_1 s$$

$$C_{L2}/Z_{c2} = -\sinh T_2 s$$

$$D_L^* = D_{L1} D_{L2} + B_{L1} C_{L2} - G D_{L1} C_{L2}$$

$$D_{L1} = \cosh T_1 s$$

$$D_{L2} = \cosh T_2 s$$

$T_1 + T_2 = T_L$. If $Z_{c1} = Z_{c2} = Z_c$ then

$$A_L^* = \cosh T_L s + Z_c G \sinh T_1 s \cosh T_2 s$$

$$Z_c B_L^* = -\sinh T_L s - Z_c G \cosh T_1 s \cosh T_2 s$$

$$C_L^*/Z_c = -\sinh T_L s - Z_c G \sinh T_1 s \sinh T_2 s$$

$$D_L^* = \cosh T_L s + Z_c G \cosh T_1 s \sinh T_2 s$$

Viscosity can be accounted for by replacing

$$T_1 \text{ and } T_2 \quad \text{by} \quad T_1 \sqrt{N} \text{ and } T_2 \sqrt{N}$$

$$Z_c \quad \text{by} \quad Z_c \sqrt{N}$$

Herein $N(s) = \frac{\alpha}{s} + 1 + \dots = \frac{32\nu}{D^2 s} + 1 + \dots$, see eqn. (6).

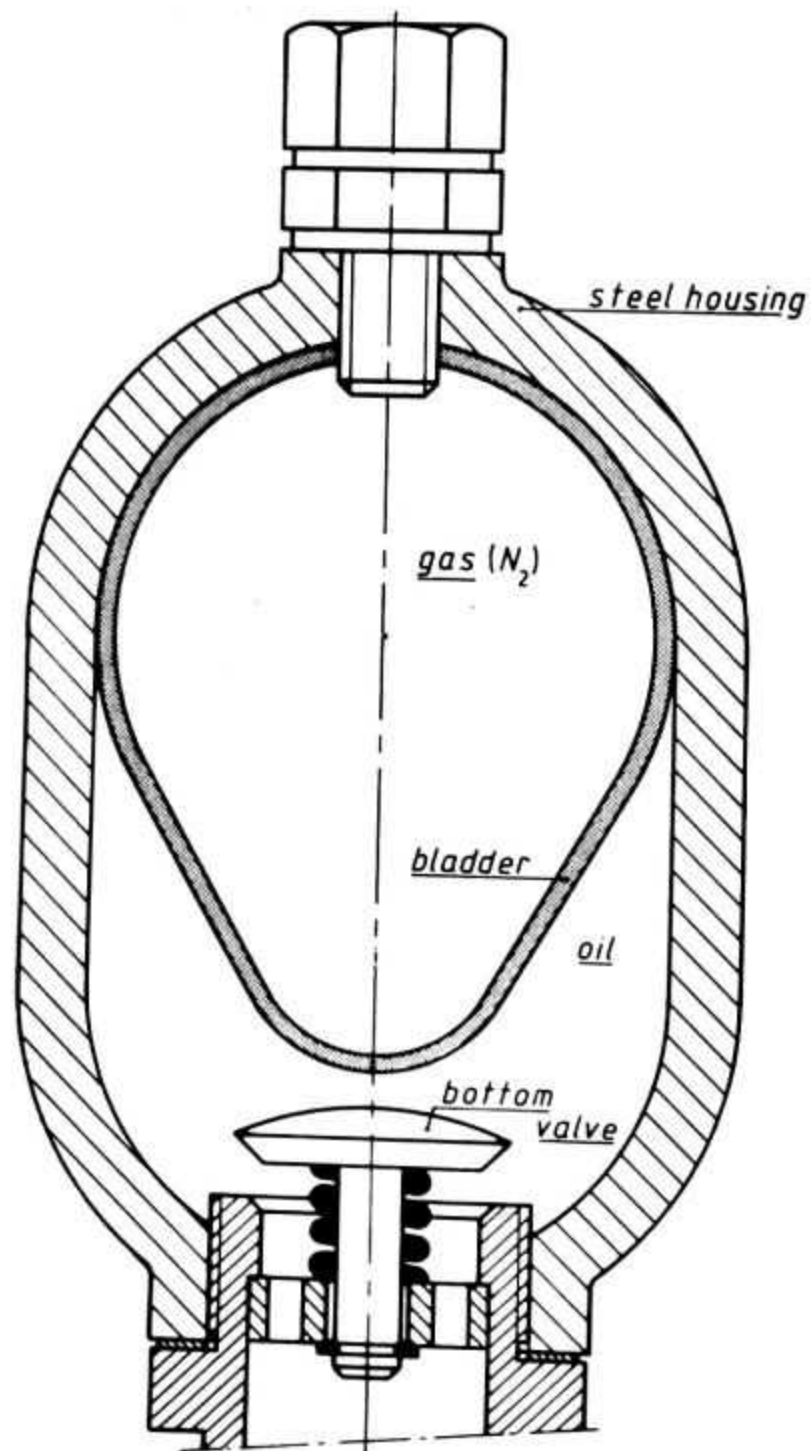


Fig. 6.10 Gas-accumulator

The caption under Fig. 6.9 summarizes the chain rule. The compound pipeline is characterized by the system parameters A_L^* , B_L^* , C_L^* and D_L^* , which can be directly derived from line element parameters and the component's G-function.

From the matrix notation in Fig. 6.9, it can be seen directly that the determinant of the overall matrix

$$\begin{vmatrix} A_L^* & B_L^* \\ C_L^* & D_L^* \end{vmatrix} = A_L^* D_L^* - B_L^* C_L^* = 1 \quad (15)$$

Thanks to this property, the same transformation (with respect to Q_i , P_i , Q_o and P_o) can be carried out, as is done in Fig. 6.6.

Naturally, the chain rule can be extended quite easily for more line elements and components. Therefore, the same procedure (and transformations) can be used, enabling us to analyse compound pipeline systems with several components in a very simple and direct way. To do this the G-functions for the components are indispensable, of course. They are derived in the next section.

6.4.2 Accumulator

As its name indicates, an accumulator is able to accumulate an amount of oil branched away from the main line. In Fig. 6.10 a typical example of a gas-accumulator is depicted. A compressed gas (mostly N_2) is separated from the oil by an elastic bladder inside a steel housing. The bottom valve prevents the bladder from coming out. The accumulator is prepared for operation by filling it up to about 80% of the nominal pressure in the main line.

Assuming an adiabatic process in a perfect gas with pressure P_a and volume V_a , we have $P_a V_a^n = \text{constant}$. The capacity C_a of the gas-accumulator is

$$C_a = - \frac{dV_a}{dP_a} = \frac{V_a}{nP_a} \quad (16a)$$

In an accumulator completely filled with oil, we have

$$C_a = - \frac{dV_a}{dP_a} = \frac{V_a}{E} \quad (16b)$$

where E is the bulk modulus.

If we neglect the influence of a (very short) supply line between the main line and the accumulator, then $P_a = P_m$ and

$$V_a(t) = - \int Q_m(t) dt \quad \text{or} \quad V_a(s) = - \frac{Q_m(s)}{s}$$

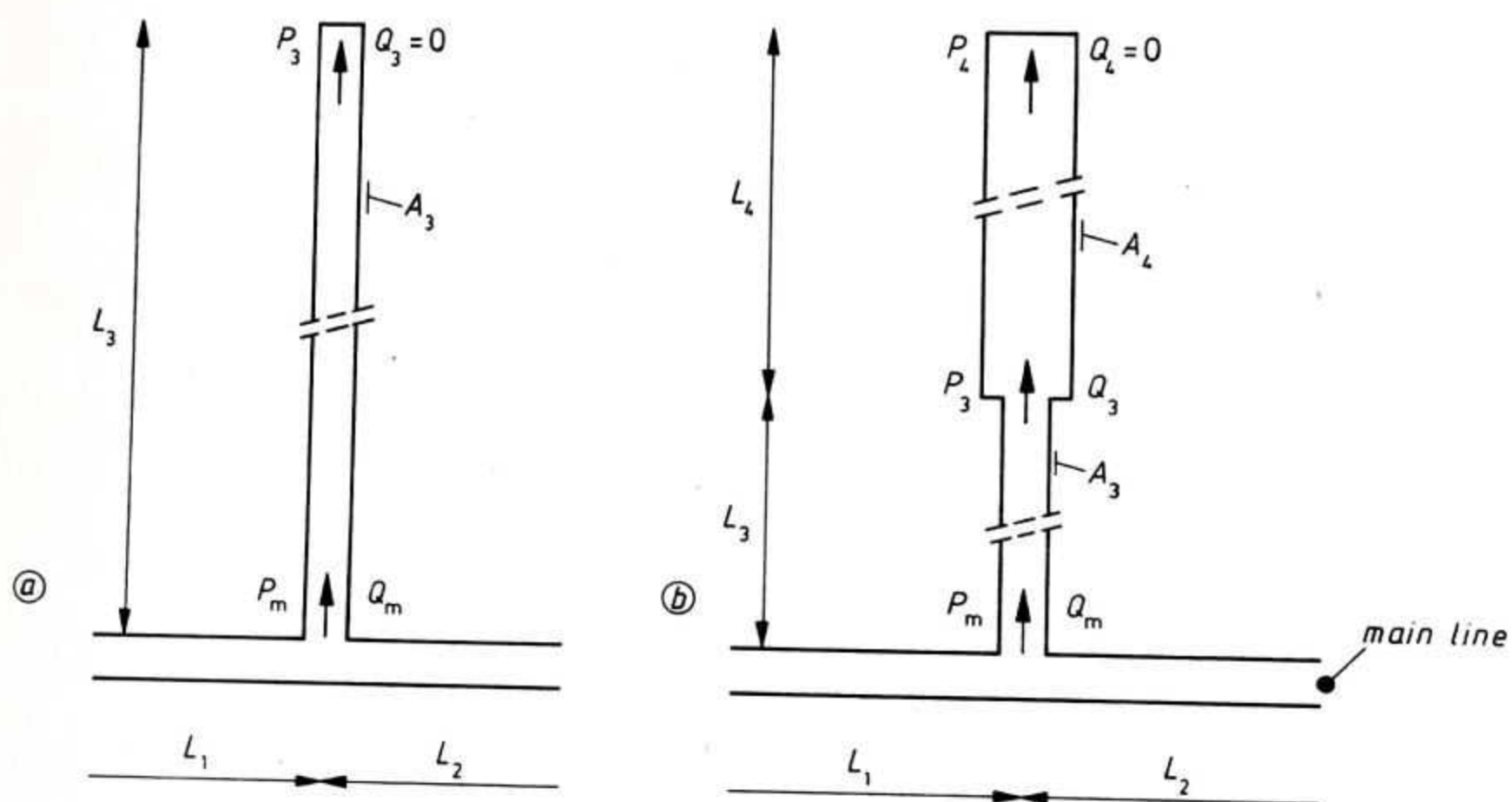


Fig. 6.11 Simple elements branched away from the main line.

(a) - Long line (T-filter). (b) - Two long lines in series.

$$G(s) = \frac{Q_m(s)}{P_m(s)} = C_a s \quad (16c)$$

In Section 6.5.4 it will be proved that in most cases the influence of the supply to a gas-accumulator cannot be neglected. In many cases its influence is dominant!

6.4.3 Long line (T-filter)

Probably the most simple component one can think of is a side line closed off at the end (see Fig. 6.11(a)). With P_m and Q_m at the entrance of the side line with length $L_3 = T_3 a$ and characteristic impedance Z_{c3} , we have $Q_3 = 0$ at the end, thus

$$A_{L3} Q_m + B_{L3} P_m = \cosh T_3 s \cdot Q_m - \frac{1}{Z_{c3}} \sinh T_3 s \cdot P_m = Q_3 = 0$$

thus

$$G(s) = \frac{Q_m}{P_m} = -\frac{B_{L3}}{A_{L3}} = \frac{\tanh T_3 s}{Z_{c3}} \quad (17)$$

The remarkable character of this component appears from sinusoidal signals.

At $s = j\omega$

$$G(j\omega) = j \frac{\tan \omega T_3}{Z_{c3}}$$

At $\omega T_3 = \pi/2, 3\pi/2$, etc. $|G|$ becomes infinitely large. It is on this property that the action of the so-called T-filter is based. As will be shown later, pump flow pulsations with a frequency coinciding with $\omega T_3 = \pi/2$ will be filtered out completely.

6.4.4 Two long lines in series

If the branch away two long lines (L_3 and L_4) in series and close the second line at the end ($Q_4 = 0$, see Fig. 6.11(b)) then

$$\begin{bmatrix} A_{L4} & B_{L4} \\ C_{L4} & D_{L4} \end{bmatrix} \cdot \begin{bmatrix} A_{L3} & B_{L3} \\ C_{L3} & D_{L3} \end{bmatrix} \cdot \begin{bmatrix} Q_m \\ P_m \end{bmatrix} = \begin{bmatrix} A_{L3}A_{L4} + C_{L3}B_{L4} & B_{L3}A_{L4} + D_{L3}B_{L4} \\ A_{L3}C_{L4} + C_{L3}D_{L4} & D_{L3}D_{L4} + B_{L3}C_{L4} \end{bmatrix} \cdot \begin{bmatrix} Q_m \\ P_m \end{bmatrix} = \begin{bmatrix} Q_4 \\ P_4 \end{bmatrix}$$

Working it out with $Q_4 = 0$ and substituting the parameters of the line elements yields

$$G(s) = \frac{Q_m}{P_m} = -\frac{B_{L3}A_{L4} + D_{L3}B_{L4}}{A_{L3}A_{L4} + C_{L3}B_{L4}} = \frac{\frac{\tanh T_3 s}{Z_{c3}} + \frac{\tanh T_4 s}{Z_{c4}}}{\frac{Z_{c3}}{Z_{c4}} \tanh T_3 s \tanh T_4 s + 1} \quad (18)$$

This complicated G-function is unimportant in itself, but it enables a simple transition to the following component.

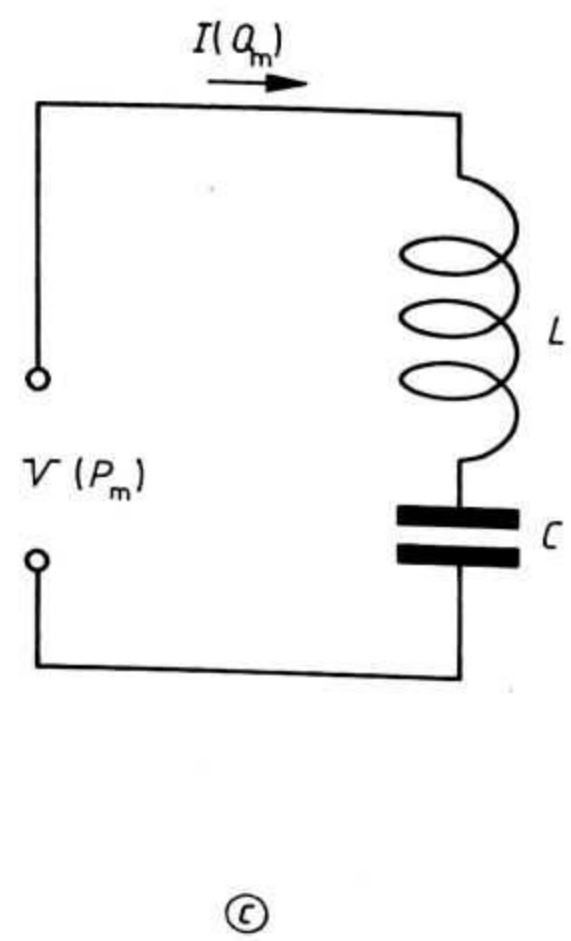
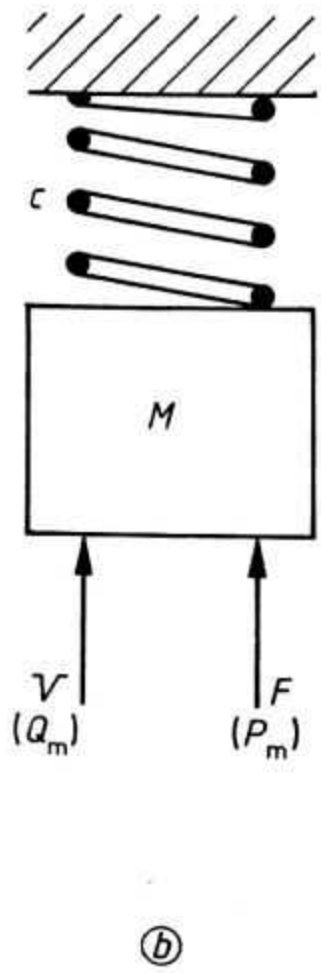
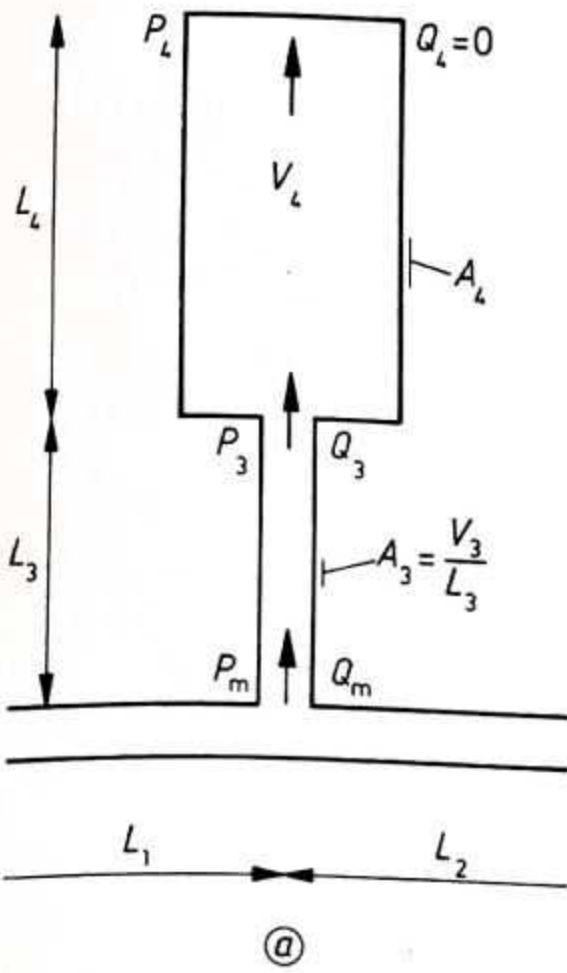


Fig. 6.12 Helmholtz-resonator a (H-filter) and its mechanical and electrical analogies b and c.

6.4.5 Two short lines in series (Helmholtz-resonator)

If the two lines in series of the previous section are very short, drastic simplifications can be made. First we define relations regarding volume, length and cross section of the two line elements (see Fig. 6.12(a))

$$V_3 = L_3 A_3 \quad V_4 = L_4 A_4$$

If L_3 and L_4 are so short that $\omega T_3 \ll \pi/2$ and $\omega T_4 \ll \pi/2$, in the relevant frequency range we may approximate

$$\tanh T_3 s \simeq T_3 s \quad \tanh T_4 \simeq T_4 s$$

Using $T_3 = L_3/a$, $T_4 = L_4/a$, $Z_{c3} = \rho a/A_3$ and $Z_{c4} = \rho a/A_4$

substitution in eqn. (18) yields

$$G = \frac{\left(\frac{T_3}{Z_{c3}} + \frac{T_4}{Z_{c4}}\right)s}{\frac{Z_{c3}T_3T_4}{Z_{c4}}s^2 + 1} = \frac{\frac{V_3 + V_4}{E} \cdot s}{\frac{L_3V_4}{A_3} \cdot \left(\frac{s}{a}\right)^2 + 1} = \frac{C_a s}{\left(\frac{s}{\omega_H}\right)^2 + 1} \quad (19)$$

In this

$$C_a = \frac{V_3 + V_4}{E} = \text{accumulator capacity of the two lines}$$

$$\omega_H = a \sqrt{\frac{A_3}{L_3 V_4}} = \text{resonance-frequency}$$

At $\omega = \omega_H$ we have $|G| = \infty$. The capacity to filter out the pump flow pulsation is based on this property of the Helmholtz-resonator (for short: H-filter), provided the filter is tuned properly.

In Figs. 6.12(b) and (c) mechanical and electrical analogies are shown. A mass M supported by a spring with stiffness c , actuated by a force F with a speed V , shows a transfer function

$$\frac{V}{F} = \frac{s/c}{\frac{M}{c} s^2 + 1}$$

Here the speed V represents the flow Q_m and the force F the pressure P_m . In the electrical circuit, with an inductance L and a capacity C , the actuating voltage V represents P_m and the current I the flow Q_m .

6.4.6 Accumulator with short connection line

To branch away an accumulator from the main line, a connection line, however short, is of course indispensable. To investigate the influence of its length L_3 we assume the signals Q_m and P_m at the entrance of the connection line and Q_3 and P_3 at the end where the accumulator is situated.

Elimination of $Q_3 = Q_a$ and $P_3 = P_a$ from the following equations yields after working it out completely

$$\left. \begin{aligned} A_{L3}Q_m + B_{L3}P_m &= Q_3 \\ C_{L3}Q_m + D_{L3}P_m &= P_3 \\ C_a s &= \frac{Q_a}{P_a} = \frac{Q_3}{P_3} \end{aligned} \right\}$$

$$G = \frac{Q_m}{P_m} = \frac{C_a s - \frac{B_{L3}}{D_{L3}}}{\frac{A_{L3}}{D_{L3}} - \frac{C_{L3}}{D_{L3}} C_a s} = \frac{C_a s + \frac{\tanh T_3 s}{Z_{c3}}}{Z_{c3} C_a s \tanh T_3 s + 1} \quad (20)$$

There are three interesting cases of degeneration of this G-function.

- In case $C_a = 0$ we get a T-filter (see eqn. (17))
- In case $T_3 = 0$ we get a "pure" accumulator (see eqn. (16))
- In the case of a *short line* ($\tanh T_3 s \approx T_3 s$ because $\omega T_3 \ll \pi/2$) we get in fact the G-function of an H-filter (see eqn. (19))

$$G \approx \frac{C_a s}{Z_{c3} C_a T_3 s^2 + 1} = \frac{C_a s}{\left(\frac{s}{\omega_H}\right)^2 + 1} \quad (20a)$$

Using a gas-accumulator, the resonance-frequency ω_H is normally rather low. So at higher frequencies we get

$$G \approx \frac{1}{Z_{c3} T_3 s} = \frac{A_3}{\rho L_3 s} \quad \text{at} \quad \omega \gg \omega_H = \frac{1}{\sqrt{Z_{c3} C_a T_3}} \quad (20b)$$

It should be noted that this G-function equals that of an open side-line with $P_3 = 0$ at the end! This should make it clear that a gas-accumulator becomes very effective at high frequencies if the connection line is made so short that $\omega T_3 \ll \pi/4$.

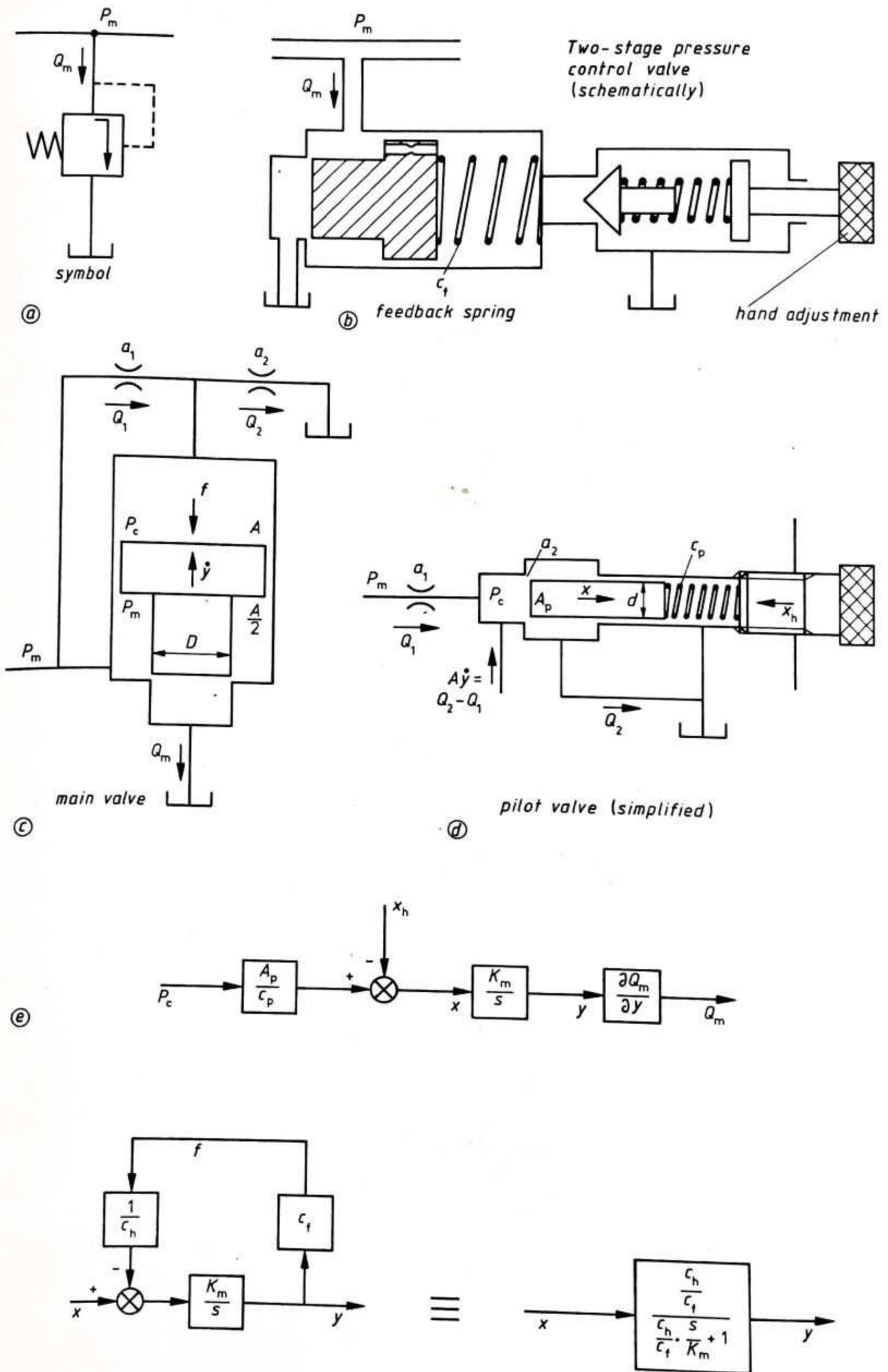


Fig. 6.13 Simplified model of a two-stage pressure control valve.

6.4.7 Pressure control valve

A pressure control valve is supposed to keep the pressure in the main supply line between pump and servovalve as constant as possible. In the steady state the surplus of the pump delivery over the servovalve "consumption" is drained via the pressure control valve. To investigate how fluctuations in delivery and consumption influence the supply line pressure, one has to determine first of all the dynamics of a pressure control valve, i.e. its G-function. Based on the analysis and measurements of Ham [4], in Fig. 6.13 a very simple model of the valve is presented in which a considerable number of effects have been neglected completely, such as forces due to flow, friction and inertia, as well as the compressibility. The resulting model, seemingly oversimplified, is justified by our practice of hydraulic line analysis and experiments, for which there has never been a greater need for a more accurate (but more complicated) model.

It is vital to regard the main spool in Figs. 6.13(b) and (c) as an asymmetric motor (see Chapter 3!) with an area ratio of 2:1. Neglecting (initially) the effect of the feedback spring, we get

$$P_c A = P_m \cdot A/2 \quad \text{or} \quad P_c = 0.5 P_m$$

$$Q_1 = a_1 C_d \sqrt{2 \frac{P_m - P_c}{\rho}} = a_1 C_d \sqrt{\frac{P_m}{\rho}}$$

$$Q_2 = a_2 C_d \sqrt{2 \frac{P_c}{\rho}} = a_2 C_d \sqrt{\frac{P_m}{\rho}}$$

$$\dot{y}A = Q_2 - Q_1$$

$$\frac{\partial \dot{y}}{\partial x} = \frac{\partial(a_2 - a_1)}{\partial x} \cdot \frac{C_d}{A} \sqrt{\frac{P_m}{\rho}}$$

With the strongly simplified model of the pilot valve of Fig. 6.13(d) we find

$$\left. \begin{array}{l} a_1 = \text{constant} \\ a_2 = \pi d(U + x) \end{array} \right\} \frac{\partial(a_2 - a_1)}{\partial x} = \pi d$$

Combination gives us the first important interim result.

$$\frac{y}{x} = \frac{\pi d}{A} \cdot C_d \sqrt{\frac{P_m}{\rho}} \cdot \frac{1}{s} = \frac{K_m}{s}$$

The change x in the pilot valve position is determined by the changes P_c of the pressure and by the change x_h in the hand adjustment

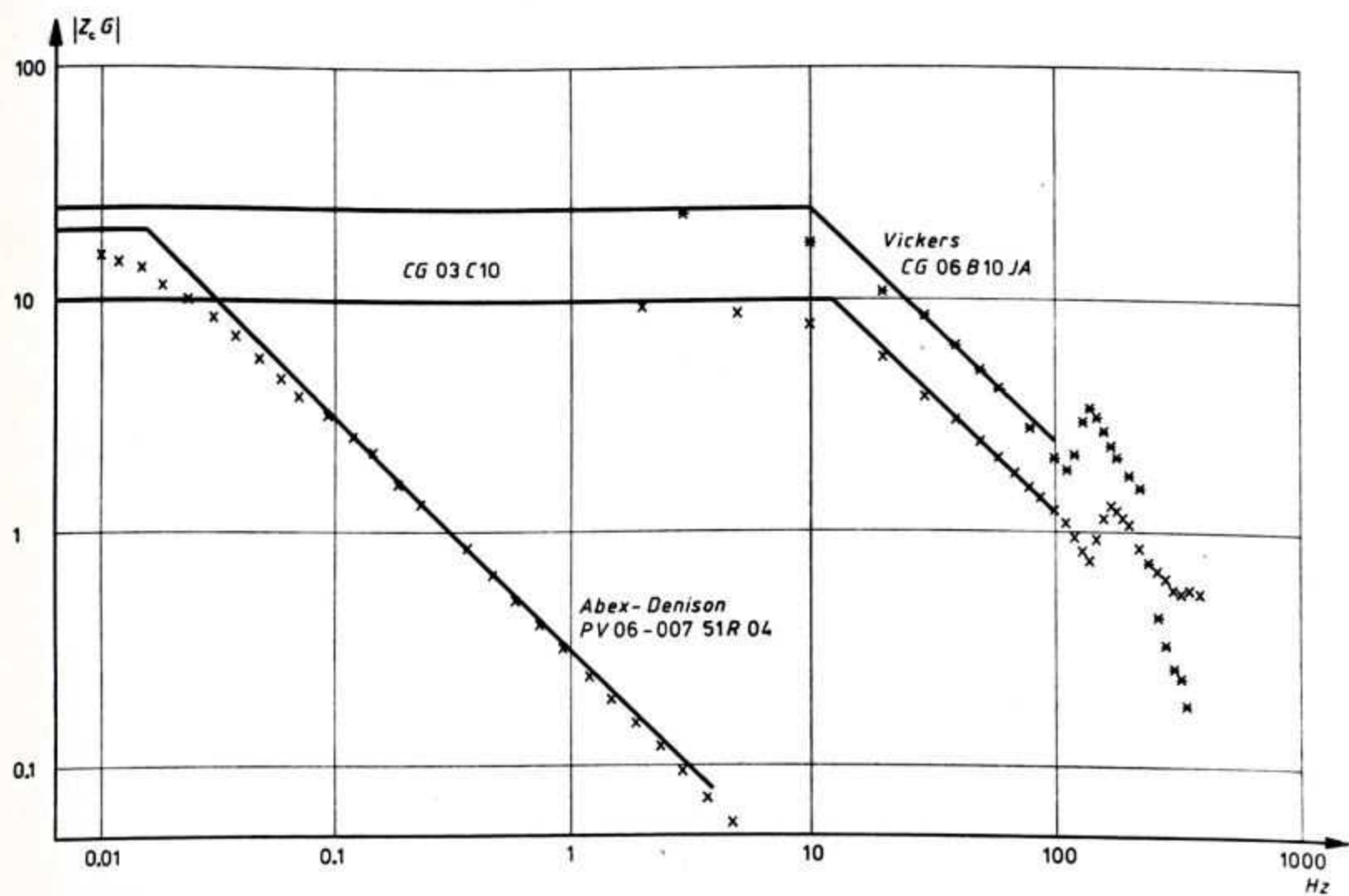


Fig. 6.14 Measured G-functions of two Vickers pressure control valves and of a Denison pressure-controlled pump.

$$x = \frac{A_p}{c_p} \cdot P_c - x_h$$

Finally, changes in the flow Q_m follow from the main spool displacement y

$$\frac{\partial q_m}{\partial y} = \pi D C_d \sqrt{\frac{2P_m}{\rho}}$$

Previous results are summarized in the first block diagram Fig. 6.13(e), showing an *integrator* as the dominant characteristic of the pressure control valve!

The effect of the feedback-spring (see Fig. 6.13(b) and (c) with $f = c_f \cdot y$) is shown in the block diagram of Fig. 6.13(f). The integrator gets a feedback loop, resulting in a first order system!

The derivations given here are rather approximate and compact. In spite of this it should be clear that a two-stage pressure control valve can be characterized roughly by

$$G(s) = \frac{Q_m}{P_m} = \frac{K_p}{\tau_p s + 1} \quad (21)$$

It should be perfectly clear too that the system is strongly nonlinear as the parameters K_m , c_h and $\partial q_m / \partial y$ as well as K_p and τ_p depend heavily on the pressure P_m .

In spite of numerous simplifications and limitations, measurements carried out by Ham [4] confirm reasonably well the nature of the G-function derived. Figure 6.14 shows measurements on two different Vickers pressure control valves. The smaller valve (Vickers CG 03C10), with a maximum flow of $0.5 \text{ dm}^3/\text{sec}$ has values of $Z K_p = 10$ and $\tau_p \approx 0.013 \text{ sec}$. The large valve (Vickers CG03B10JA), with a maximum nominal flow of $1.0 \text{ dm}^3/\text{sec}$, has values of $Z K_p = 25$ and $\tau_p \approx 0.016 \text{ sec}$. The experiments were carried out at a nominal pressure of 90 bar. Further data: $D = 13.0 \text{ mm}$, $a = 1280 \text{ m/sec}$, $\rho = 895 \text{ kg/m}^3$, $Z_c = 0.863 \cdot 10^{10} \text{ Ns/m}^5$.

Figure 6.14 shows deviations from the theoretical curves at higher frequencies. Inertia forces and compressibility of the oil certainly have something to do with those deviations, but we were not successful in deriving an accurate G-function nicely confirming measurements. This being so, we stick to the simple first-order G-function, knowing that at higher frequencies the accuracy is gradually becoming poor.

It should be stressed here that dynamic data on pressure control valves are not normally available, as manufactures have completely different (empirical) criteria with which to evaluate the performance of pressure control valves. So here we touch an important part of the gap between theory and practice in hydraulic line dynamics!

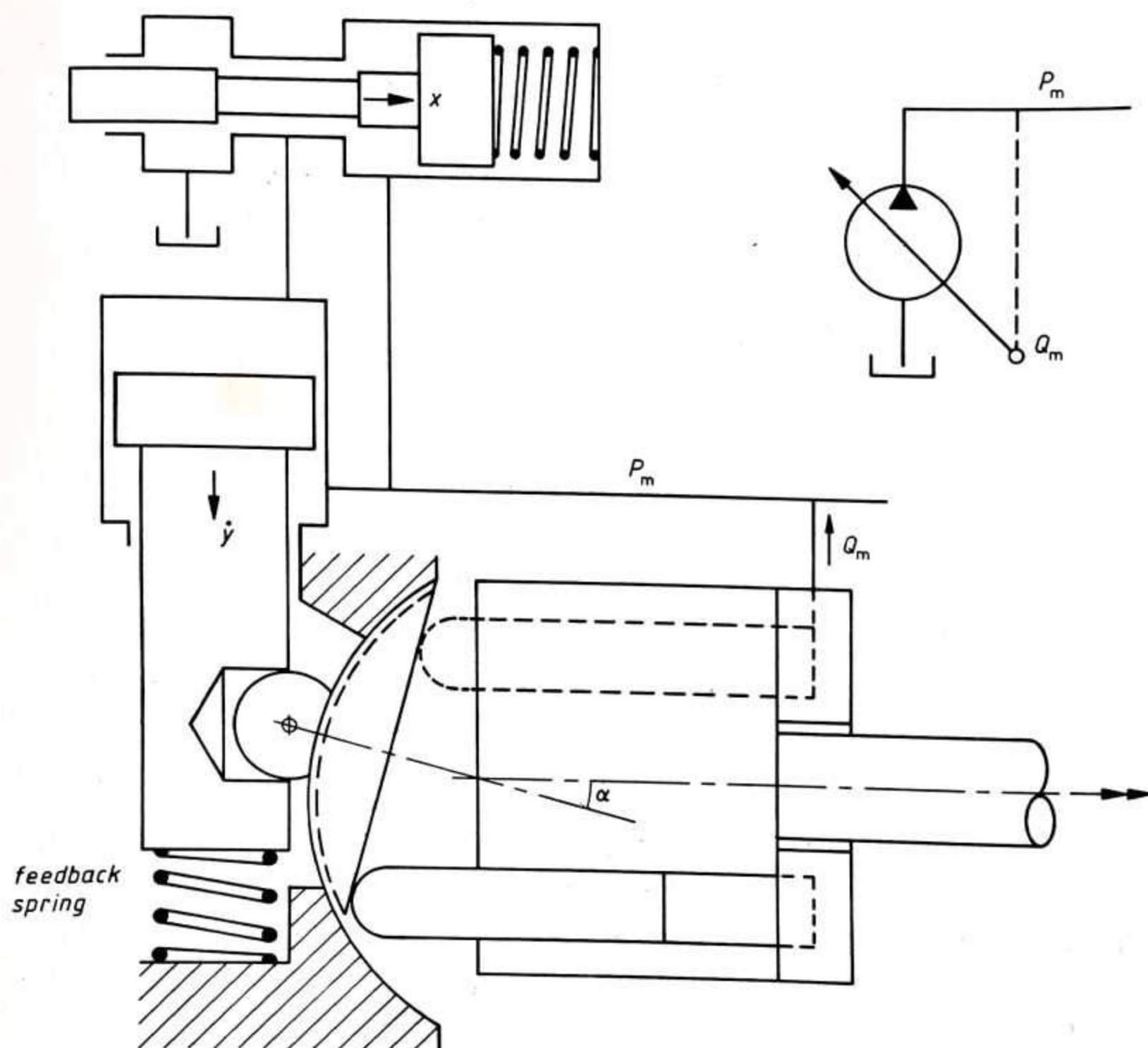


Fig. 6.15 Pressure-controlled pump.

6.4.8 Pressure-controlled pumps

Hydraulic servosystems requiring large oil flows during short time intervals and considerably smaller flows most of the time imply problems regarding the pumpset. A constant delivery pump has to meet maximum demands and is therefore wasting its energy most of the time, the surplus flow being drained via the pressure control valve, introducing a cooling problem as well. For this reason, the pressure controlled pump has been developed, adjusting delivery to the demand.

In Fig. 6.15 a simple example is depicted. Although, strictly speaking, pressure controlled pumps are not "branch-away components", it is quite possible to regard them as a combination of a fixed displacement pump plus a branch-away component, where the component is placed directly on the pump. This becomes clear from Fig. 6.15, where we observe the pump flow to be dependent on the angle α between the spherical swash plate bearing and the center line of the rotating cylinder block. Starting at a given position, the pump flow decreases approximately proportional to the change of α , which in turn is proportional to the piston displacement y . The piston is part of an asymmetric servomotor whose transfer function is $y/x = K_m/s$.

The functional analogy with the pressure control valve is striking. It goes so far that most of the derivation of the transfer function is even identical. For this reason we shall not repeat it and shall simply state the same transfer function.

$$G(s) = \frac{Q_m}{P_m} = \frac{K_p}{\tau_p s + 1} \quad (22)$$

Here Q_m is *not* the flow branched away but the change of the pump flow due to changes in α , y , x and P_m . Thus, functionally, the pump has a built-in G-function. The main difference from the pressure control valve, having the same G-function, is the value of the time constant τ_p . For some pumps we noticed a time constant τ_p about 100 times (!) larger than the τ_p of a pressure control valve. This has something to do with the involvement of the inertia.

Measurements carried out on a Abex-Denison pump (type PV06-007-51R-04) are shown in Fig. 6.14.

A very interesting alternative to the pressure-controlled pump of Fig. 6.15 actuated by a servomotor with hydro-mechanical pressure feedback is a pump actuated by an electro-hydraulic servomotor, taking its input signal from a supply pressure transducer. Here, considerable *flexibility* is introduced to adjust the parameters K_p and τ_p or even the very nature of the G-function. As the time constant τ_p is highly critical for stabilizing the supply pressure (which is proved in Section 6.6) the alternative of a variable displacement pump actuated by an electro-hydraulic servomotor is recommended here without any hesitation.

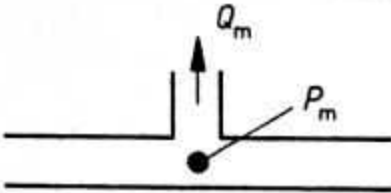
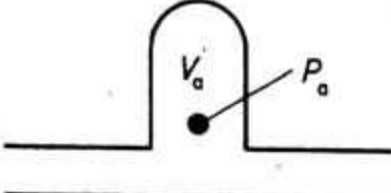
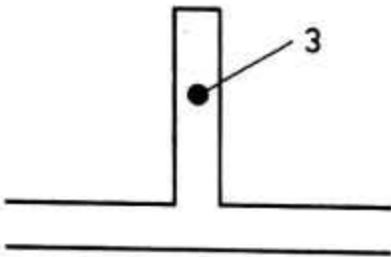
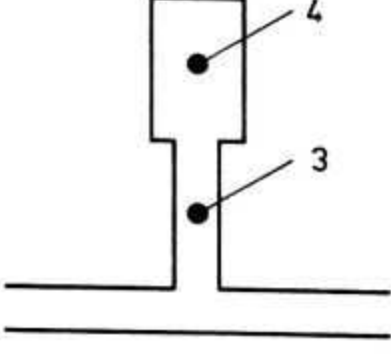
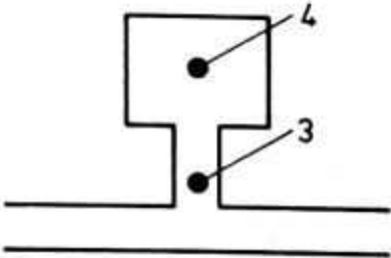
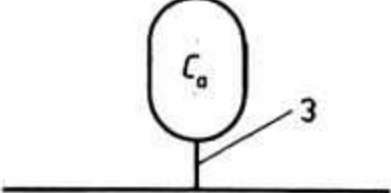
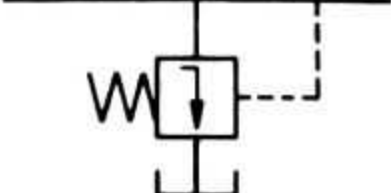
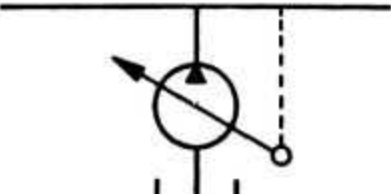
Definition		$G(s) = \frac{Q_m}{P_m}$
Accumulator without supply line	 Gas Oil	$G = C_a s = \frac{V_a}{n P_a}$ $G = C_a s = \frac{V_a}{E}$
Long line (T-filter)		$G = \frac{\tanh T_3 s}{Z_{c3}}$
Two long lines in series		$G = \frac{\frac{\tanh T_3 s}{Z_{c3}} + \frac{\tanh T_4 s}{Z_{c4}}}{\frac{Z_{c3}}{Z_{c4}} \tanh T_3 s \tanh T_4 s + 1}$
Two short lines (H-filter)		$C_a = \frac{V_3 + V_4}{E}$ $\omega_H = a \sqrt{\frac{A_3}{L_3 V_4}}$ $G = \frac{C_a s}{(\frac{s}{\omega_H})^2 + 1}$
Accumulator + supply line		$G = \frac{C_a s + \frac{\tanh T_3 s}{Z_{c3}}}{Z_{c3} C_a s \tanh T_3 s + 1} \approx \frac{C_a s}{Z_{c3} T_3 C_a s^2 + 1}$
Pressure control valve		$G = \frac{K_p}{\tau_p s + 1}$
Pressure controlled pump		$G = \frac{K_p}{\tau_p s + 1}$

Fig. 6.16 Survey of branch-away components and their G -functions.

6.4.9 Summary

In Fig. 6.16 the G-functions of the components that can be placed in a junction of the main line between the pump and the servovalve of a servosystem are summarized.

Figures 6.6 (with its appendix), 6.9 and 6.16 together summarize this chapter up to this point. In the remaining text we shall, in fact, refer back only to these three figures.

The next sections will be devoted to:

- o filtering out the first harmonic of the pump flow pulsation completely by means of T-filters and H-filters.
- o suppressing the effects of pump flow pulsations by a small accumulator, whose location and length of connection line will be shown to be very critical.
- o suppressing pressure fluctuations, due to changing flows via the servovalve, by means of large accumulators placed near the servovalve.
- o stabilizing supply pressure oscillations due to pressure controlled pumps connected to a long hydraulic line.

These cases will be treated one by one. It will be clear that, without the G-functions of Fig. 6.16, it is not possible to analyze the dynamic behavior of normal compound hydraulic lines, let alone the synthesis and the design.

6.5 Suppression of pump flow pulsations

6.5.1 Introduction

Pressure pulsation in hydraulic pipelines can have their origin in pump flow pulsations. Axial plunger pumps, for instance, (as discussed in Section 6.1, Fig. 6.3) can show flow pulsations with an amplitude equal to 1% of the nominal flow. With an odd number of plungers (Fig. 6.3(d)) the frequency of the first harmonic is twice the plunger "repetition frequency". Thus for an axial plunger pump with 1500 rpm and 7 plungers the first harmonic has a frequency $2 \times 7 \times 1500 / 60 = 350$ Hz. Naturally, higher harmonics are present also. In practice we measured first of all 350 Hz but also 25 and 175 Hz. The latter should not be there theoretically. Geometrical imperfections may provide an explanation for it. In addition, the dynamics of the pressure build-up in the seven mini-cylinders in the rotating cylinder block have not been taken into account. Effects like this will certainly affect the higher harmonics.

Vane pumps and gear pumps also have flow pulsations. Their calculation, however, is considerably more complicated. But the first harmonic can be derived similarly from the rpm and the number of vanes or gears.

As far as we know, there is no simple, direct method to measure pump flow pulsations. By measuring pressure pulsations in a simple hydraulic line one could try to determine pump flow pulsations indirectly. But such an indirect method is not known either.

In this section we assume pump flow pulsations between 0.5% and 5% of the nominal flow. We need this assumption (which is pessimistic and therefore on the safe side) to calculate the product $Z_c Q_i$ which appears in the transfer functions $P_o / Z_c Q_i$ and $P_i / Z_c Q_i$. So when we want to judge whether measured or calculated amplitude ratios $P_o / Z_c Q_i$ and $P_i / Z_c Q_i$ are acceptable or not, we have to know the magnitude of $Z_c Q_i$.

$$Z_c Q_i = \frac{\rho a}{A} \cdot Q_i$$

The specific mass ρ and the sonic velocity a are given by the medium and therefore cannot be influenced. The pipeline cross section A , however, is determined in practice by the maximum oil speed $\bar{U}$ being permitted. Usually

$$\bar{U}_{\max} = \frac{Q_{\text{pump}}}{A} \approx 5 \frac{\text{m}}{\text{sec}}$$

Taking Q_i 0.5% $\hat{a}$ 5% of Q_{pump} , $\rho = 900 \text{ kg/m}^3$ and $a = 1200 \text{ m/sec}$, we find

$$Z_c Q_i = 0.3 \cdot 10^5 \hat{a} 3 \cdot 10^5 \text{ N/m}^2 = 0.3 \hat{a} 3 \text{ bar.} \quad (23)$$

As stated before, this value is needed as a reference for pressure pulsations.

One final remark regarding the frequencies of flow and pressure pulsations must be made. Generally, servovalves and servosystems have a bandwidth lower than 200 Hz. For this reason our interest in pulsations is restricted to frequencies lower than 500 Hz, assuming that signals with higher frequencies do not disturb the servosystem.

Before deriving the basic equations in a dimensionless form to analyze pressure fluctuations due to pump flow pulsations, we give some perspective here to clarify our approach.

It is good practice to place a pressure control valve as well as an accumulator close to the pump to reduce the effect of pump flow pulsation. The G-functions given in Fig. 6.16 show that at low frequencies the pressure control valve is dominant over the accumulator. For this reason the pressure control valve is able to control the DC-level of the supply pressure far better than the accumulator. However, at high frequencies (the pump flow pulsation frequency *is* high) the opposite is true. So the accumulator (*with* short connection line) can possibly cancel pump flow pulsations where the pressure control valves fails completely. Yet the accumulator is not a universal remedy to cancel pump flow pulsations. This is particularly true for very long hydraulic lines.

Fortunately, there is another possibility, that of the tuned filters, to eliminate the first harmonic of the flow pulsation completely. The Helmholtz-resonator behaves perfectly as a tuned filter. In theory, this is true for the T-filter as well, but its quality is more sensitive to small errors in tuning.

6.5.2 Strategy and basic equations

Our goal is simple and clear: to suppress pressure fluctuations in the pipeline due to pump flow pulsations. Especially the pressure at the end of the pipeline, near the servovalve, is important, as it is the fluctuation of the supply pressure at the servovalve that could disturb the servosystem. Thus we want to eliminate the pressure fluctuations at the end of the pipeline completely.

The pressure at the other end of the pipeline, near the pump, is less important. Because of the duration of life of the pump, however, pressure fluctuations should not exceed moderate boundaries. By no means is a complete elimination of pressure fluctuations near the pump a strict condition.

In the rest of the pipeline pressures are scarcely interesting as long as there is no risk of bursting. In some German publications [5], [6] it is suggested that the whole length of the pipeline is a major *noise-producer*. There is no firm foundation for this suggestion. On the contrary, we believe that the greater part of the noise is produced by the pump itself. To reduce noise production in special cases, we designed a pumpset with the electric motor and the pump submerged in the oil. Some additional acoustic measures reduced the noise to less than that of an oscilloscope ventilator, in spite of normal fluctuations all over the high pressure line. For this reason we restrict our attention to the ends of the hydraulic lines alone.

To analyze the effect of various components in the pipeline we use the same basic equations each time. The starting point is a long hydraulic line with a fixed displacement pump at its origin. The flow pulsation Q_i of this pump is regarded as an independent variable, the pressure P_i is the dependent variable. At the other end of the pipeline we assume a laminar flow resistance R_o , relating pressure and flow directly: $P_o = R_o Q_o$.

Boundary conditions at the ends being fixed, we place one of the components at a junction in the main line. For this pipeline we have, according to Fig. 6.9

$$A_L^* Q_i + B_L^* P_i = Q_o$$

$$C_L^* Q_i + D_L^* P_i = P_o$$

With $A_L^* D_L^* - B_L^* C_L^* = 1$ and $P_o = R_o Q_o$ it follows directly

$$\frac{P_o}{Q_i} = \frac{R_o}{D_L^* - B_L^* R_o} \rightarrow \frac{P_o}{Z_c Q_i} = \frac{1}{\frac{Z_c}{R_o} D_L^* - Z_c B_L^*} \quad (24a)$$

$$\frac{P_i}{Q_i} = \frac{A_L^* R_o - C_L^*}{D_L^* - B_L^* R_o} \rightarrow \frac{P_i}{Z_c Q_i} = \frac{A_L^* - C_L^*/R_o}{\frac{Z_c}{R_o} D_L^* - Z_c B_L^*} \quad (24b)$$

These dimensionless transfer functions enable us to find the dependent variables P_o and P_i as functions of $Z_c Q_i$. Dividing eqn. (24a) by eqn. (24b) gives an "indirect" relation between the two dependent variables P_o and P_i .

$$\frac{P_o}{P_i} = \frac{R_o}{A_L^* R_o - C_L^*} = \frac{1}{A_L^* - C_L^*/R_o} \quad (24c)$$

Again using Fig. 6.9 we have the following dimensionless parameters

$$\left. \begin{aligned} A_L^* &= \cosh T_L s + Z_c G \sinh T_1 s \cosh T_2 s \\ Z_c B_L^* &= -\sinh T_L s - Z_c G \cosh T_1 s \cosh T_2 s \\ C_L^*/Z_c &= -\sinh T_L s - Z_c G \sinh T_1 s \sinh T_2 s \\ D_L^* &= \cosh T_L s + Z_c G \cosh T_1 s \sinh T_2 s \end{aligned} \right\} \quad (25)$$

with $T_L = T_1 + T_2$

and

$$Z_{c1} = Z_{c2} = Z_c.$$

The G-functions of the various components are taken from Fig. 6.16 and are brought into a dimensionless form also.

$$\text{T-filter: } Z_c G = \frac{Z_c}{Z_{c3}} \tanh T_3 s \quad (26a)$$

$$\text{H-filter: } Z_c G = \frac{Z_c C_a s}{\left(\frac{s}{\omega_H}\right)^2 + 1}, \quad \omega_H = a \sqrt{\frac{A_3}{L_3 V_4}} \quad (26b)$$

$$\text{Pressure control valve: } Z_c G = \frac{Z_c K_p}{\tau_p s + 1} \quad (26c)$$

$$\text{Gas-accumulator + short supply line: } Z_c G = \frac{Z_c C_a s}{Z_{c3} T_3 C_a s^2 + 1}, \quad C_a = \frac{V_a}{n P_a} \quad (26d)$$

Thus the transfer functions (24) can be found by substitution of eqns. (25) and (26). In doing so, we can take advantage of non-dimensional notations. At sinusoidal signals ($s = j\omega$) the dimensionless transfer functions

$$\frac{P_o}{Z_c Q_i}, \frac{P_i}{Z_c Q_i}, \frac{P_o}{P_i}$$

are functions of dimensionless numbers, each consisting of a group of parameters

$$\frac{T_1}{T_L}, \frac{Z_c}{R_o}, \omega T_L$$

Applying the following components, two dimensionless numbers must be added

$$\text{T-filter: } \frac{Z_c}{Z_{c3}}, \frac{T_3}{T_L},$$

$$\text{H-filter: } \frac{Z_c C_a}{T_L}, \frac{\omega}{\omega_H},$$

$$\text{Pressure control valve: } Z_c K_p, \frac{\tau_p}{T_L},$$

$$\text{Gas-accumulator + short connection line: } \frac{Z_c C_a}{T_L}, \frac{T_3}{T_L}.$$

Thus, in each case of a hydraulic line with one branch-away component, we have five independent dimensionless numbers (or any combination of them).

When we bring the viscosity $\nu = \mu/\rho$ into account also, a sixth number appears (see Fig. 6.6)

$$\alpha T_L = \frac{32\nu}{D^2} \cdot T_L \quad (27)$$

Thus, analyzing our transfer functions (24), we have to consider no less than six dimensionless numbers that are relevant to the dynamic behavior. On the one hand, one might say that thanks to this dimensional analysis the total number of relevant parameters has been reduced to its essential minimum, while on the other hand there is still the danger of not seeing the wood for the trees. For this reason we try to simplify the transfer functions by giving three (!) of the six dimensionless numbers fixed (extreme) values in order to allow them to be disregarded. Supposedly, this scarcely influences the characteristics of the transfer functions! Mostly, we disregard αT_L , Z_c/R_o and one of the quantities related to the branch-away component. On this basis we analyze our transfer functions. If necessary, the influence of the quantities disregarded in the initial analysis can be checked one by one to test our assumption. This strategy proved to be very effective, as is shown in the subsequent sections.

6.5.3 Selective filters

Selective filters branched away from the main line between the pump and the servovalve show an amplitude ratio $|G(j\omega)| = \infty$ at a frequency that can be chosen freely. By "tuning" the filter on the pump pulsation frequency ω_p the influence of the pump pulsation can be eliminated completely. To show this we first take the T-filter, which is a simple line closed at the end, branched away from the main line. Its G-function is

$$Z_c G = \frac{Z_c}{Z_{c3}} \cdot \tanh T_3 s$$

Assuming $R_o = \infty$, the combination of eqns. (24), (25), (26a) gives

$$\frac{P_o}{Z_c Q_i} = \frac{1}{j \left[\sin \omega T_L + \frac{Z_c}{Z_{c3}} \tan \omega T_3 \cos \omega T_1 \cos \omega T_2 \right]} \quad (28)$$

$$\frac{P_i}{Z_c Q_i} = \frac{\cos \omega T_L - \frac{Z_c}{Z_{c3}} \tan \omega T_3 \sin \omega T_1 \cos \omega T_2}{j \left[\sin \omega T_L + \frac{Z_c}{Z_{c3}} \tan \omega T_3 \cos \omega T_1 \cos \omega T_2 \right]} \quad (29)$$

$$\frac{P_o}{P_i} = \frac{1}{\cos \omega T_L - \frac{Z_c}{Z_{c3}} \tan \omega T_3 \sin \omega T_1 \cos \omega T_2} \quad (30)$$

It is perfectly clear that $P_o = 0$ if

$$\omega_p T_3 = \frac{\pi}{2} (2k + 1) \quad k = 0, 1, 2, 3, \dots \quad (31a)$$

or

$$L_3 = \frac{2\pi a}{\omega_p} \cdot \frac{2k + 1}{4} = \frac{\lambda_p}{4} (2k + 1) \quad (31b)$$

Thus, making the length L_3 of the side line a quarter of the wavelength λ_p (or and odd multiple of it) of the pump pulsation frequency ω_p , the $P_o = 0$ provided the location of the filter guarantees $\cos \omega_p T_1 \cos \omega_p T_2 \neq 0$. Later on it will be shown that it is always possible to choose a location where $0.5 < |\cos \omega_p T_1 \cos \omega_p T_2| < 1$. For reasons of simplicity we assume for the time being $|\cos \omega_p T_1 \cos \omega_p T_2| \approx 1$.

Tuning the filter according to eqn. (31), the pressure P_i at the entrance of the main pipeline becomes (see eqn. (29))

$$\frac{P_i}{Z_{cQ_i}} = j \tan \omega_p T_1$$

To limit P_i , the location of the T-filter should be such that $|\tan \omega_p T_1| < 1$. This can be effected easily by making L_1 very short.

The above derivations seem to suggest that the T-filter is absolutely perfect, implying that the somewhat more complicated H-filter is no longer of interest. Yet, in our opinion, the H-filter should be preferred, for the following reasons.

The tuning of a filter is never perfect. In the first place, some parameters (particularly E) are never known precisely beforehand. On the other hand, some quantities may change because of change in the nominal pressure. For instance, the pump pulsation frequency itself could depend on the nominal pressure.

Thus, our question is: how effective is the filter if the tuning is not perfect? Assuming a tuning on $\omega = \omega_F \neq \omega_p$, we have

$$\omega_F T_3 = \frac{\pi}{2}$$

$$\omega_p = \omega_F + \Delta\omega, \quad |\Delta\omega| \ll \omega_F$$

Now the G-function at $\omega = \omega_p$ becomes

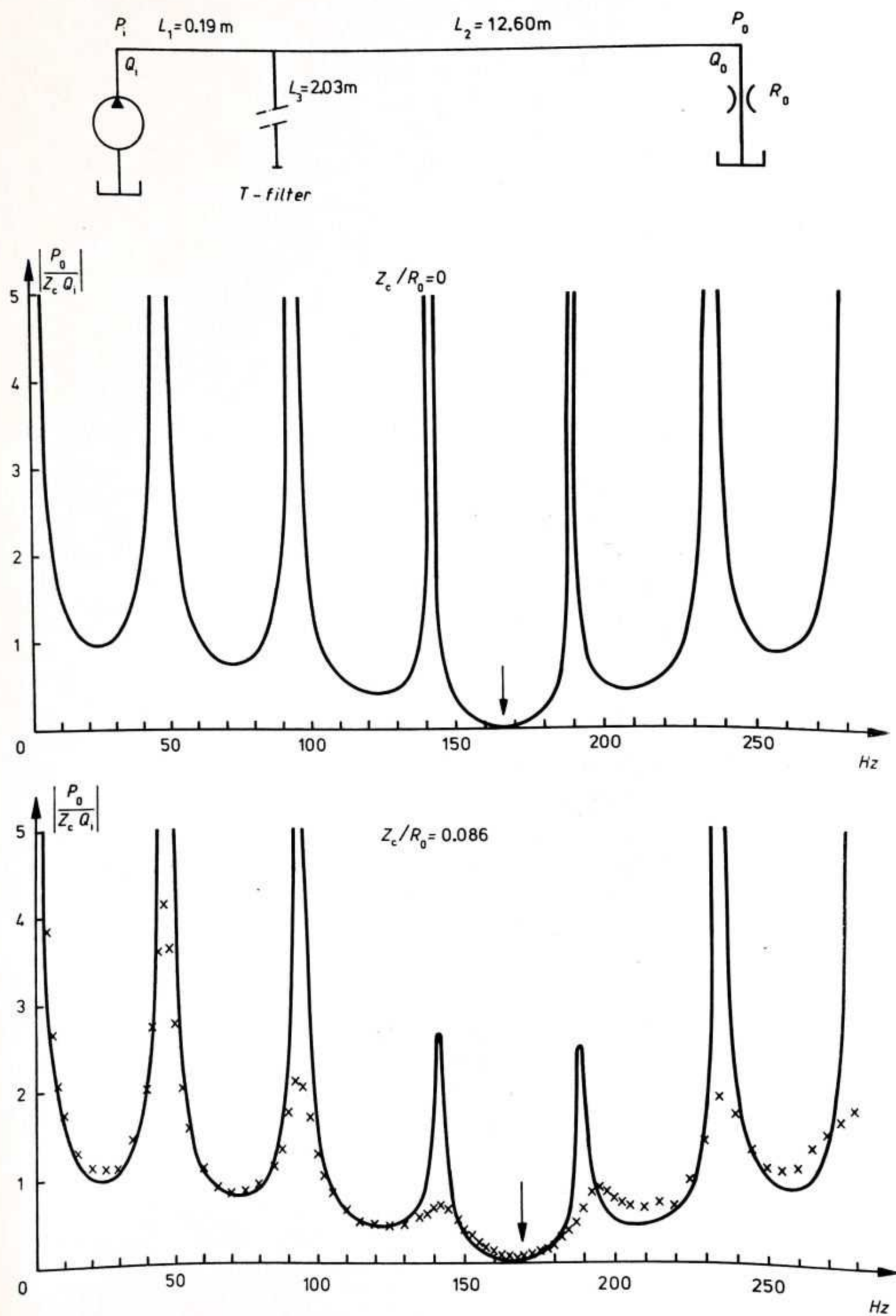


Fig. 6.17 Hydraulic line with T-filter, tuned for 168 Hz.

$$Z_c G(\omega_p) = j \frac{Z_c}{Z_{c3}} \tan \omega_p T_3 = j \frac{A_3}{A} \tan \left(\omega_F T_3 + \frac{\Delta\omega}{\omega_F} \cdot \omega_F T_3 \right) = -j \frac{A_3/A}{\tan \frac{\pi}{2} \frac{\Delta\omega}{\omega_F}} \approx -j \frac{A_3/A}{\frac{\pi}{2} \cdot \frac{\Delta\omega}{\omega_F}}$$

and finally

$$\left| \frac{P_o}{Z_c Q_i} \right| \approx \frac{A}{A_3} \cdot \frac{\pi}{2} \cdot \frac{|\Delta\omega|}{\omega_F}$$

In order to limit P_o as much as possible, we choose $A_3 = A$ (a larger value is unrealistic) Thus

$$\frac{P_o}{Z_c Q_i} \approx 1.6 \frac{|\Delta\omega|}{\omega_F} \quad (32)$$

Assuming $Z_c Q_i \approx 1$ bar (see eqn. (23)) and $|\Delta\omega/\omega_F| \approx 0.1$, then $P_o \approx 0.16$ bar results. This result should be compared with the corresponding result of the H-filter.

Before going to analyze the H-filter, we take a look at some experiments and measurements on hydraulic lines with T-filters. In Fig. 6.17 the experiment is indicated schematically: via an electro-hydraulic servovalve sinusoidal flow pulsations are fed into a pipeline with a T-filter. In fact, the servovalve spool is actuated sinusoidally, measuring Q_i , P_i and P_o related to it. First, the theoretical curve is drawn for a pipeline with

$$\begin{array}{lll} R_o = \infty & Z_c = Z_{c3} = 0.868 \cdot 10^{10} \text{ Ns/m}^5 & a = 1280 \text{ m/s} \\ L_1 = 0.19 \text{ m} & L_2 = 11.57 \text{ m} & L_3 = 2.03 \text{ m} \end{array}$$

The filter is tuned for a frequency of

$$\omega = \omega_F = 1056 \text{ rad/sec (168 Hz), measured instead of 158 Hz.}$$

To provide a DC-component of the flow Q_i , a resistance $R_o \neq \infty$ is introduced for practical experiments. In Fig. 6.17 the theoretical curve (assuming the viscosity $\mu = 0$) as well as measurements for $R_o/Z_c = 11.6$, are presented. The agreement, especially around the tuned frequency ω_F (indicated by an arrow), is sufficient to support the theory developed on tuned filters.

The location of the filter, directly related to the product $\cos \omega_p T_1 \cos \omega_p T_2$ will be discussed in connection with H-filters. There the problem is similar.

The Helmholtz-resonator or H-filter has a transfer function

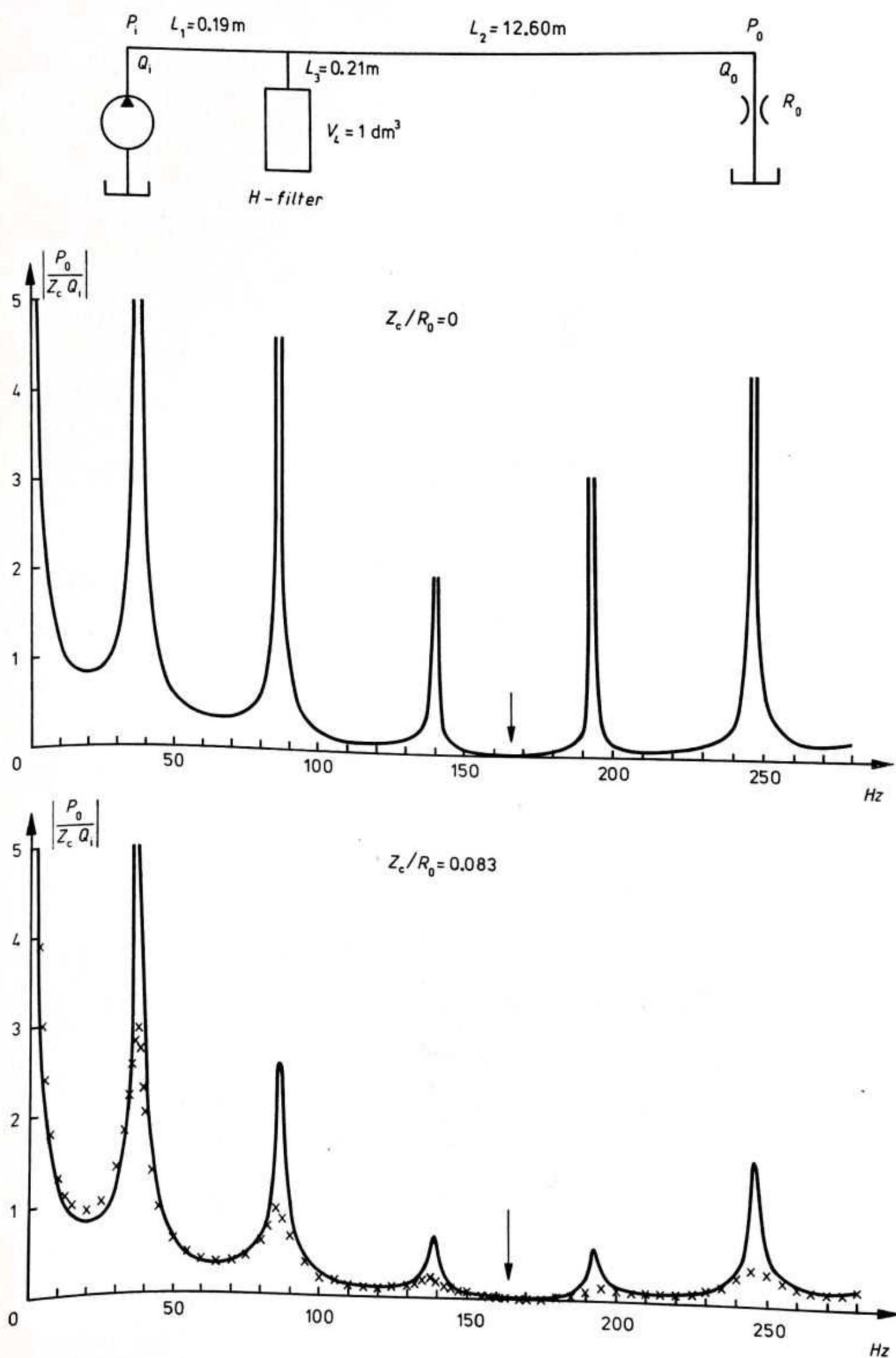


Fig. 6.18 Hydraulic line with H-filter, tuned for 166 Hz.

$$Z_c G = \frac{Z_c C a s}{\left(\frac{s}{\omega_H}\right)^2 + 1}$$

where

$$C_a = \frac{V_4}{E} \quad \text{and} \quad \omega_H = a \sqrt{\frac{A_3}{L_3 V_4}}$$

Again assuming $R_o = \infty$, the combination of eqns. (24), (25), (26b) gives

$$\frac{P_o}{Z_c Q_i} = \frac{1}{j \left[\sin \omega T_L + \frac{Z_c C a \omega}{1 - (\omega/\omega_H)^2} \cos \omega T_1 \cos \omega T_2 \right]} \quad (33)$$

$$\frac{P_i}{Z_c Q_i} = \frac{\cos \omega T_L - \frac{Z_c C a \omega}{1 - (\omega/\omega_H)^2} \sin \omega T_1 \cos \omega T_2}{j \left[\sin \omega T_L + \frac{Z_c C a \omega}{1 - (\omega/\omega_H)^2} \cos \omega T_1 \cos \omega T_2 \right]} \quad (34)$$

$$\frac{P_o}{P_i} = \frac{1}{\cos \omega T_L - \frac{Z_c C a \omega}{1 - (\omega/\omega_H)^2} \sin \omega T_1 \cos \omega T_2} \quad (35)$$

At this filter we find $P_o = 0$ at $\omega = \omega_p = \omega_H$, provided $\cos \omega_p T_1 \cos \omega_p T_2 \neq 0$.
As in the case of the T-filter, we find for P_i at $\omega = \omega_p = \omega_H$

$$\frac{P_i}{Z_c Q_i} = j \tan \omega_p T_1$$

which could be made small by choosing L_1 very short.

The effectiveness, in case this filter is not tuned perfectly, can be found by assuming ω_p to be slightly different from $\omega_H = \omega_F$

$$\omega_p = \omega_F + \Delta\omega, \quad |\Delta\omega| \ll \omega_F = \omega_H$$

gives

$$Z_c G(\omega_p) \approx -j \frac{Z_c C a \omega_F}{2\Delta\omega/\omega_F}$$

and

$$\frac{P_o}{Z_c Q_i} \approx \frac{2}{Z_c C a \omega_F} \cdot \frac{\Delta\omega}{\omega_F}$$

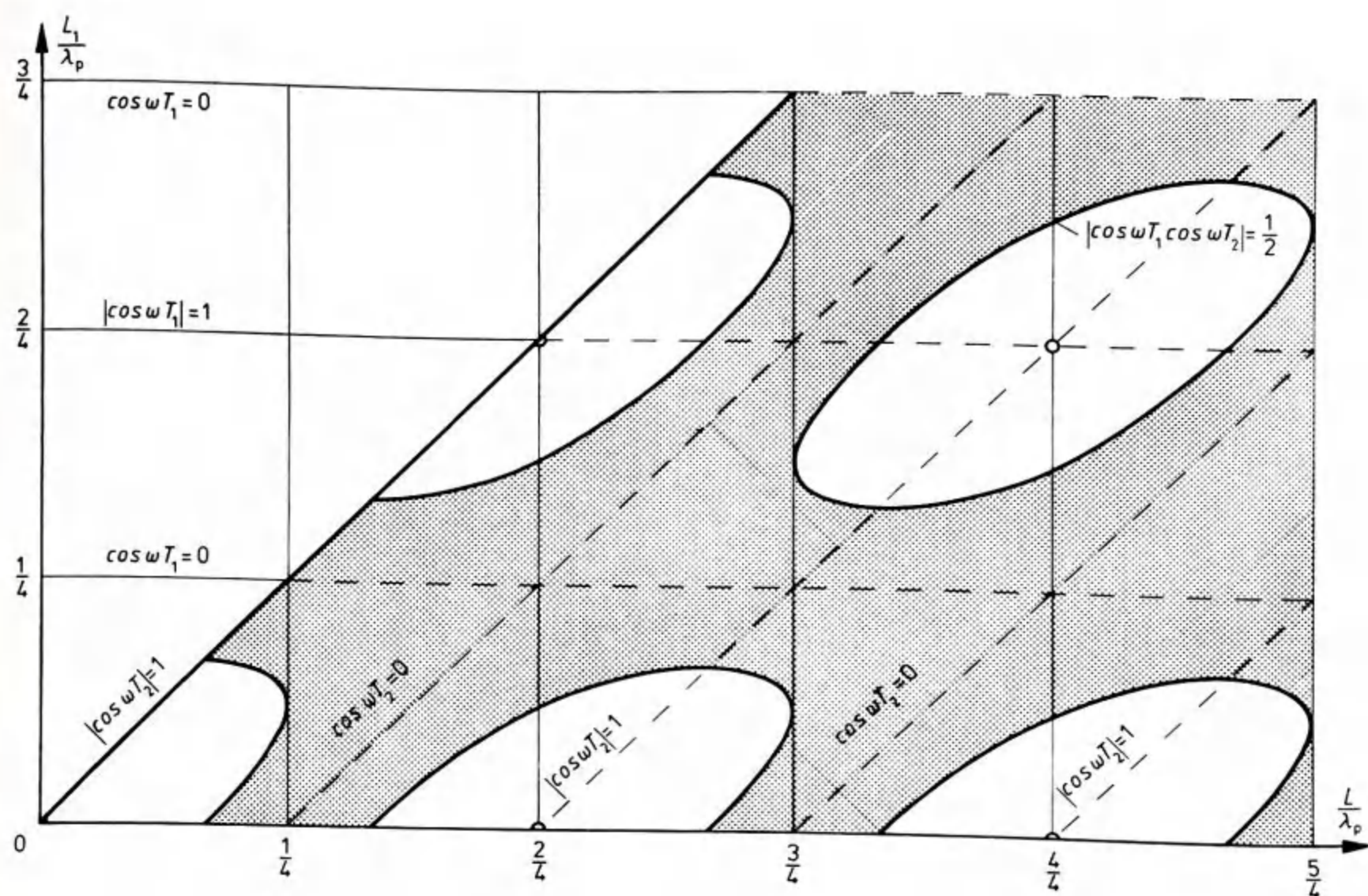


Fig. 6.19 Influence of L_1 and $L = L_1 + L_2$ on the product $\cos \omega_p T_1 \cos \omega_p T_2$.

Here, in fact, only $C_a = V_4/E$ can be chosen freely. However, in order to keep P_o small, V_4 should not be chosen too small. A ping-pong ball size filter might look very attractive, but a larger size is recommended, as the following example explains.

At $E = 1.3 \cdot 10^9 \text{ N/m}^2$, $Z_c = 10^{10} \text{ Nsec/m}^5$, $V_4 = 10^{-3} \text{ m}^3 (= 1 \text{ dm}^3!)$ and $\omega_F = 1500 \text{ rad/sec}$ ($\approx 240 \text{ Hz}$), we find

$$\left| \frac{P_o}{Z_c Q_i} \right| \approx 0.2 \frac{|\Delta\omega|}{\omega_F}$$

This result is so much better than the corresponding result of the T-filter (see eqn. (32)) that for this reason we prefer the H-filter. The H-filter is less selective than the T-filter, meaning that the H-filter is still very effective even when it is not very precisely tuned.

Experiments carried out on an H-filter, tuned on the same frequency as the T-filter of Fig. 6.17, confirm these conclusions. In Fig. 6.18 the theoretical curve is drawn for a pipeline and H-filter with

$$\begin{array}{lll} R_o = \infty & Z_c = 0.868 \cdot 10^{10} \text{ Ns/m}^5 & a = 1280 \text{ m/s} \\ L_1 = 0.19 \text{ m} & L_2 = 11.57 \text{ m} & L_3 = 0.20 \text{ m} \\ V_4 = 1 \text{ dm}^3 & A_3 = 1.33 \text{ cm}^2 & \omega_H = 1043 \text{ rad/s (166 Hz)} \end{array}$$

In addition, a curve has been plotted for $R_o/Z_c = 12.0$. Experimental measurements are indicated too. Comparison with Fig. 6.17 shows a minimum (see arrow) at the tuning frequency, which is considerably more flat.

One final question regarding the location of the filter must be dealt with. As can be seen from eqns. (28) and (33), tuned filters are most effective when at the tuning frequency $\omega_F = \omega_p$ the product $|\cos \omega_p T_1 \cos \omega_p T_2|$ equals 1. Expressing L_1 as well as $L = L_1 + L_2$ in relation to the wavelength $\lambda_p = 2\pi a/\omega_p$ of the pump flow pulsation frequency ω_p , in Fig. 6.19 configurations and locations are indicated where the product is zero and "islands" where

$$0.5 \leq |\cos \omega_p T_1 \cos \omega_p T_2| \leq 1$$

In the centers of the "islands" the product equals 1. So the ideal configuration and location is given by

$$\begin{aligned} L_1 &= k \cdot \lambda_p / 2, & k &= 0, 1, 2, 3, \dots \\ L_1 + L_2 &= m \cdot \lambda_p / 2, & m &= 1, 2, 3, \dots \end{aligned}$$

Even if the total length of the pipeline $L = L_1 + L_2$ cannot be chosen freely, a proper location with $|\cos \omega_p T_1 \cos \omega_p T_2| \geq 0.5$ can easily be found.

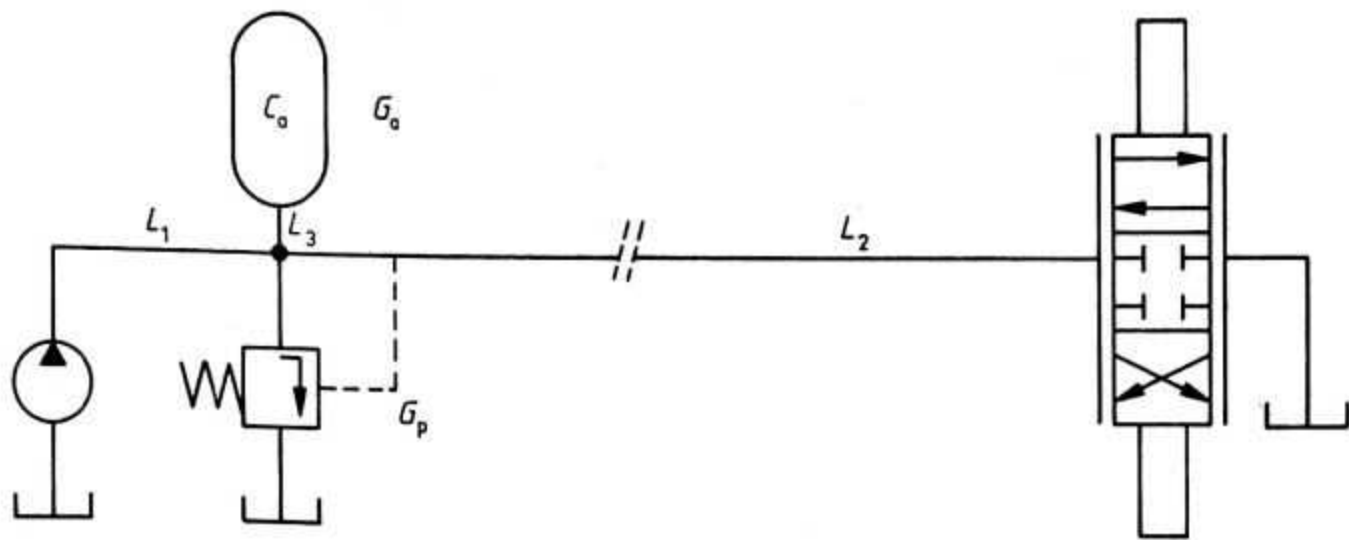


Fig. 6.20 Pressure control valve and gas-accumulator in the hydraulic line between pump and servovalve.

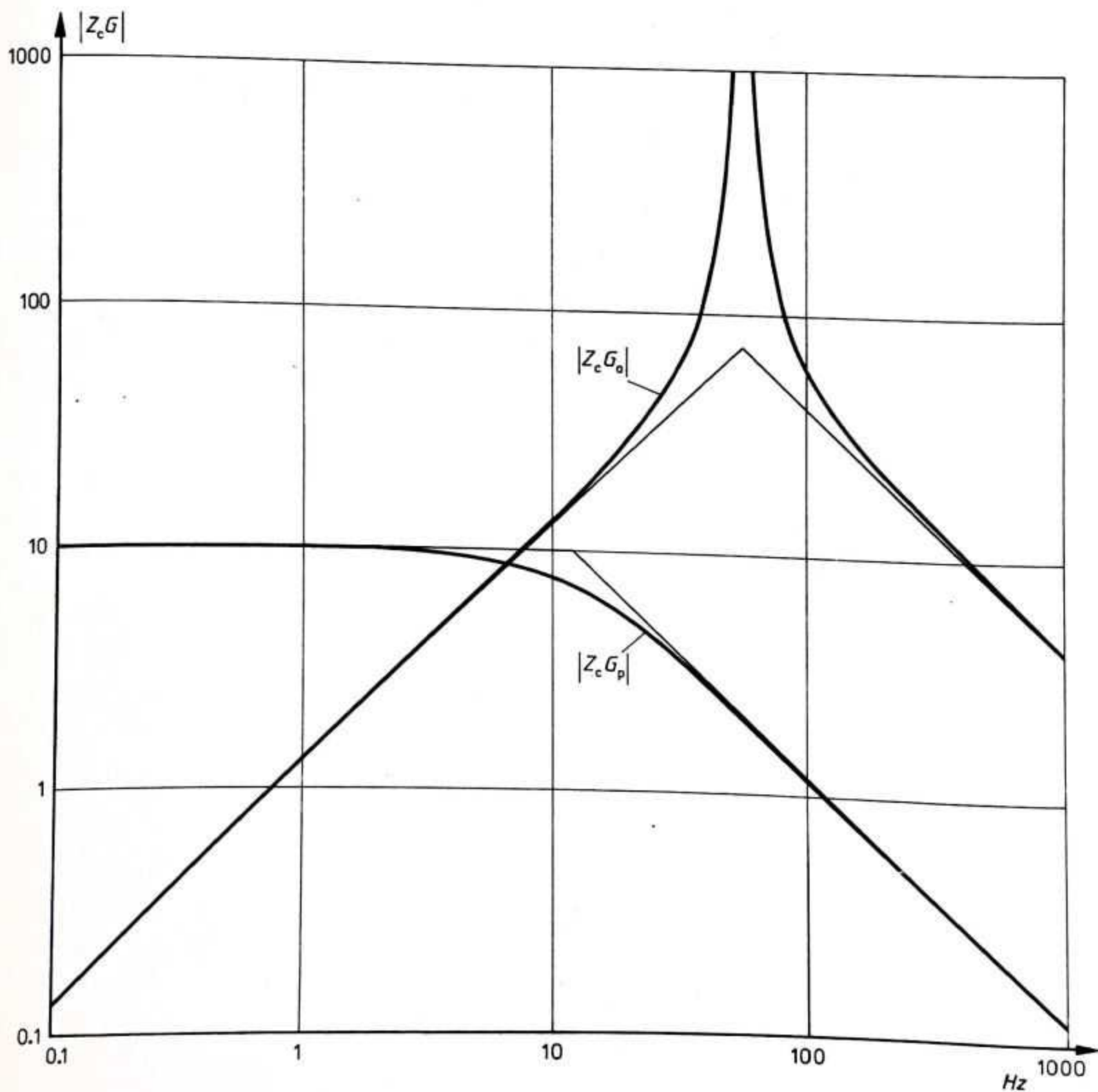


Fig. 6.21 Characteristics of a two-stage pressure control valve (see Fig. 6.14) and a gas-accumulator.

6.5.4 Pressure control valve and gas-accumulator near the pump

It is a common practice to install a pressure control valve and a gas-accumulator near the pump to stabilize the supply pressure and to suppress the pump flow pulsation (see Fig. 6.20). To analyze its effects the G-functions of the two components including their data are shown in Fig. 6.21.

The pressure control valve is representative for a wide range of valves (see Fig. 6.14). The same is true for the gas-accumulator, provided its supply line $L_3 = T_3 a$ is very short.

As the pump flow pulsation frequency is high (mostly over 100 Hz), only the asymptotes for the high frequencies are relevant. So we approximate the G-functions for the pressure control valve

$$Z_{c p} G_p = \frac{Z_c K_p}{\tau_p s + 1} \approx \frac{Z_c K_p}{\tau_p s}$$

and for the gas-accumulator with a short supply line

$$Z_{c a} G_a = \frac{Z_c C_a s}{Z_c C_a T_3 s^2 + 1} \approx \frac{1}{T_3 s}$$

Because $T_3 = L_3/a \ll \tau_p/Z_c K_p$, the effect of the pressure control valve could be neglected compared with that of the accumulator. Thus in the following analysis we restrict our attention to

$$Z_c G = \frac{1}{T_3 s} \quad (36)$$

It should be noted that this G-function by no means represents a selective filter like the T- or H-filter. At high frequencies its effect is the same as that of an open side-line to tank ($P = 0$). As scarcely anything more effective can be thought of than a short, open side-line to atmospheric pressure, some optimism seems to be justified. To verify this, we combine eqns. (24a), (25) and (36) and introduce $s = j\omega$

$$\frac{P_o}{Z_c Q_i} = \frac{1}{j \left[\sin \omega T_L - \frac{\cos \omega T_1 \cos \omega T_2}{\omega T_3} \right]} \quad (37)$$

Since $L_3 = T_3 a$ is short enough to ensure $\omega T_3 \ll 1$ in the relevant frequency range up to 500 Hz (that might disturb the servosystem) the factor $\cos \omega T_1 \cos \omega T_2 / \omega T_3$ is so dominant over $\sin \omega T_L$ that we may neglect the latter. Thus

$$\frac{P_o}{Z_c Q_i} \approx \frac{j\omega T_3}{\cos \omega T_1 \cos \omega T_2} \quad (38)$$

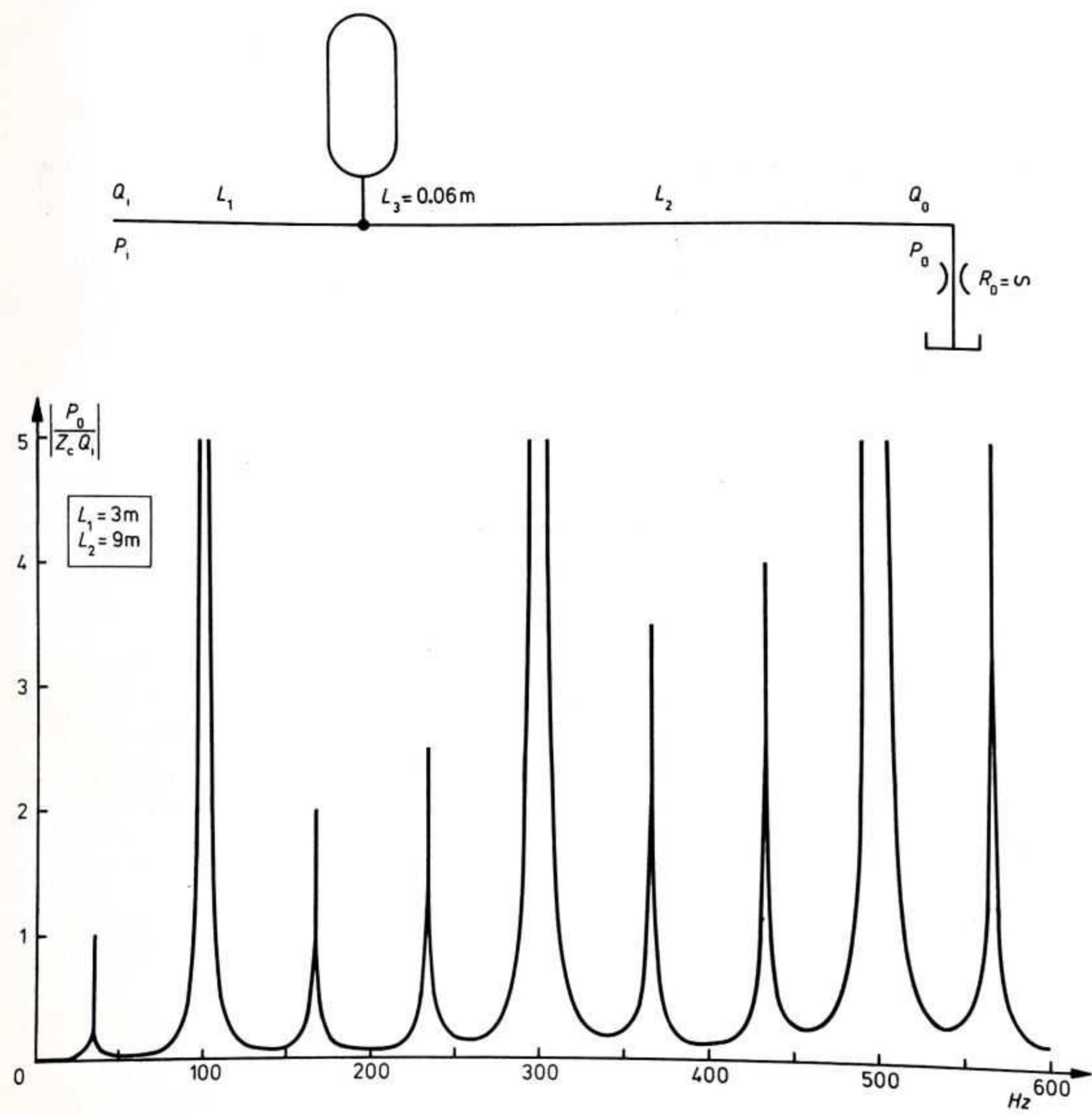


Fig. 6.22 Characteristic of a hydraulic line with an accumulator located far from the pump.

From this transfer function the importance of the location becomes apparent. If L_1 (or L_2) is *not* chosen very short then there are possibly frequencies where both $\cos \omega T_1$ and $\cos \omega T_2$ become zero. Around these frequencies *wide peaks* with $P_o = \infty$ will occur, as is shown in Fig. 6.22. As illustrative data $L_1 = 3$ m, $L_2 = 9$ m, $L_3 = 0.06$ m and $a = 1200$ m/sec have been chosen. In this configuration wide peaks occur at

$$\omega \frac{L_1}{a} = \omega T_1 = \frac{\pi}{2} (2k + 1) \quad k = 0, 1, 2, 3, \dots$$

Thus, at $\omega = 200\pi, 600\pi, 1000\pi, \dots$ rad/sec (100, 300, 500 Hz), the chance that the pump flow pulsation frequency will fall within the frequency areas with a very large amplitude ratio is considerable and must be avoided at all costs. Therefore L_1 should be chosen short enough to guarantee $\omega T_1 < \pi/4$ as long as $\omega < 3150$ rad/sec (500 Hz).

$$\text{Thus } 3150 L_1 / 1200 < \pi/4 \text{ or}$$

(39)

$$\underline{\underline{L_1 < 0.3 \text{ m}}}$$

With this result we approximate $\cos \omega T_1 \approx 1$, so

$$\frac{P_o}{Z_c Q_i} \approx \frac{j\omega T_3}{\cos \omega T_2}$$

(40)

Previous derivations with numerous simplifications and approximations hardly seem worthwhile since the pressure could have infinitely large values at

$$\omega \frac{L_2}{a} = \omega T_2 = \frac{\pi}{2} (2k + 1)$$

$$k = 0, 1, 2, 3, \dots$$

(41)

Assuming, for instance, $L_1 = 0$, $L_2 = 12$ m, $L_3 = 0.06$ m and $a = 1200$ m/sec we have $T_2 = 0.01$ sec. Thus $P_o = \infty$ at $\omega = 50\pi, 150\pi, 250\pi, \dots$ rad/sec (25, 75, 125, ... Hz).

However, the frequency areas in which $|P_o / Z_c Q_i| > 1$ are extremely narrow. Analysis of eqn. (40) yields $|P_o / Z_c Q_i| > 1$ at

$$\frac{\pi}{2} (2k + 1) \left(1 - \frac{T_3}{T_2}\right) < \omega T_2 < \frac{\pi}{2} (2k + 1) \left(1 + \frac{T_3}{T_2}\right) \quad (42)$$

In our case, where $T_3/T_2 = 1/200$, the location of the critical frequency areas is $25 \pm 1/8, 75 \pm 3/8, 125 \pm 5/8, 175 \pm 7/8$ Hz, etc. Figure 6.23, depicting eqn. (40), clearly shows that outside the frequency areas of eqn. (42) the amplitude ratio is readily approaching $\omega T_3 \ll 1$.

The interpretation of Fig. 6.23 is that apart from the narrow peaks the charac-

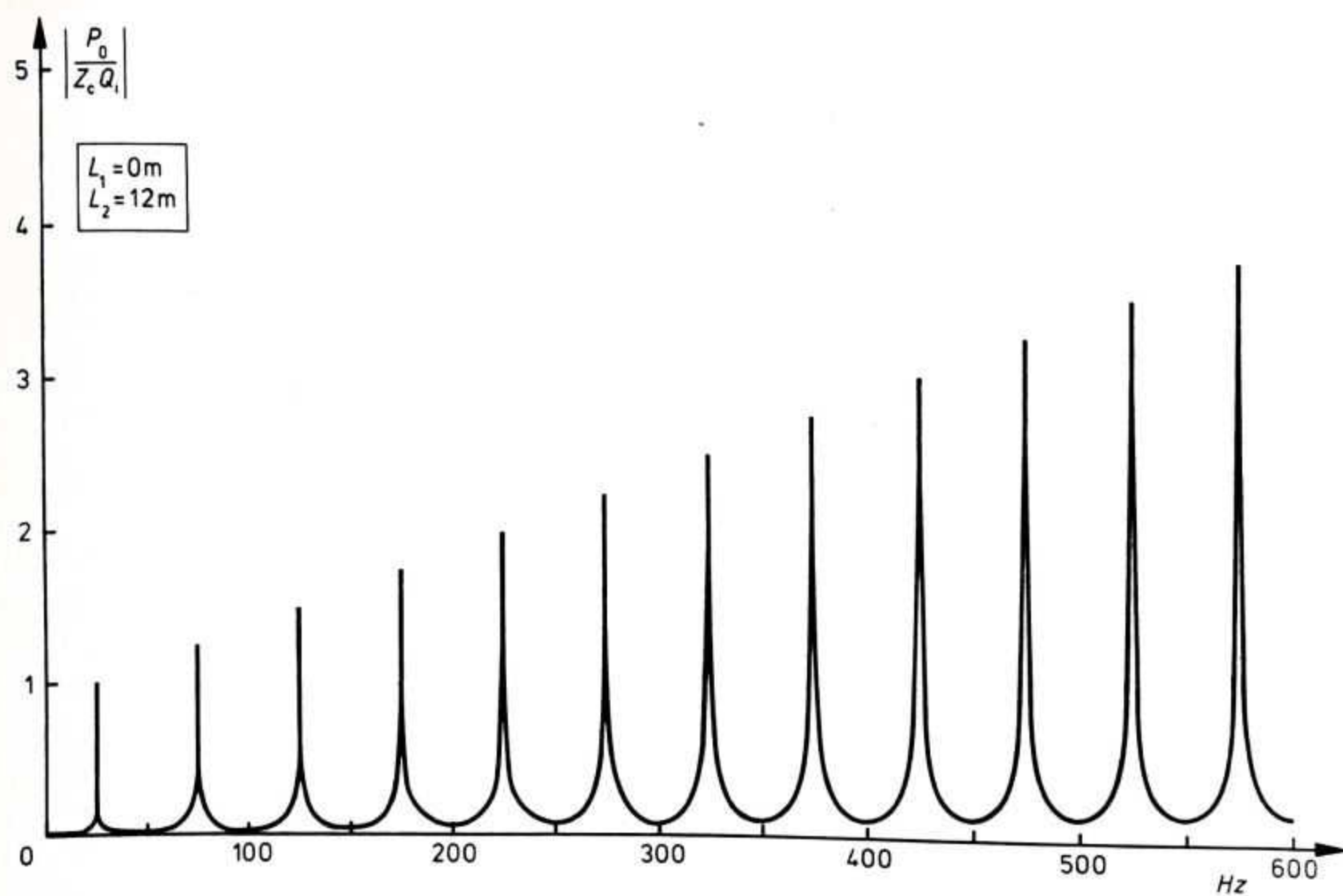


Fig. 6.23 Characteristic of a hydraulic line with an accumulator located very close to the pump.

teristic is quite satisfactory. Generally speaking, a gas-accumulator with a very short supply line installed close to the pump acts like an effective wide band filter. The chance that the pump flow pulsation frequency will fall within the narrow frequency areas is small. Nevertheless, theoretically that chance is present.

However, the situation in practice is considerably better than that depicted in Fig. 6.23. Since the pressure pulsations can disturb the servosystem only if the servovalve is more or less open, a resistance R_o at the end of the pipeline could represent this situation. Again combining eqns. (24a), (25) and (36), and following exactly the same lines as before, we find

$$\frac{P_o}{Z_c Q_i} = \frac{1}{j \left[\sin \omega T_L - \frac{\cos \omega T_1 \cos \omega T_2}{\omega T_3} \right] + \frac{Z_c}{R_o} \left[\cos \omega T_L + \frac{\cos \omega T_1 \sin \omega T_2}{\omega T_3} \right]}$$

$$\frac{P_o}{Z_c Q_i} \approx \frac{\omega T_3}{\cos \omega T_1 \left[-j \cos \omega T_2 + \frac{Z_c}{R_o} \sin \omega T_2 \right]} \approx \frac{j \omega T_3}{\cos \omega T_2 - j \frac{Z_c}{R_o} \sin \omega T_2}$$

At $R_o \gg Z_c$ we observe successive maxima

$$\left| \frac{P_o}{Z_c Q_i} \right|_{\max} \approx \frac{R_o}{Z_o} \cdot \omega T_3 \quad (43)$$

at frequencies very near to $\cos \omega T_2 = 0$.

In spite of reducing the maxima by reducing the length L_3 of the supply line to the accumulator as much as possible we cannot eliminate (moderate) successive peaks, increasing proportionally with frequency.

In view of experiments and measurements to be discussed subsequently, we consider the following data

$L_1 = 0.207 \text{ m}$	$L_2 = 5.108 \text{ m}$	$L_3 = 0.05 \text{ m}$
$\rho = 895 \text{ kg/m}^3$	$E = 1470 \cdot 10^6 \text{ N/m}^2$	$a = 1280 \text{ m/sec}$
$R_o = 135 \cdot 10^9 \text{ Nsec/m}^5$	$Z_c = 863 \cdot 10^7 \text{ Nsec/m}^5$	$R_o/Z_c = 15.6$

The resistance R_o at the end of the pipeline is so large that at a nominal pressure of 90 bar in the steady-state, only a flow of $4 \text{ dm}^3/\text{min}$ ($1/15000 \text{ m}^3/\text{sec}$) occurs. The simplifications and approximations go so far that we need only eqns. (41) and (43) to estimate the location and height of the successive pressure peaks!

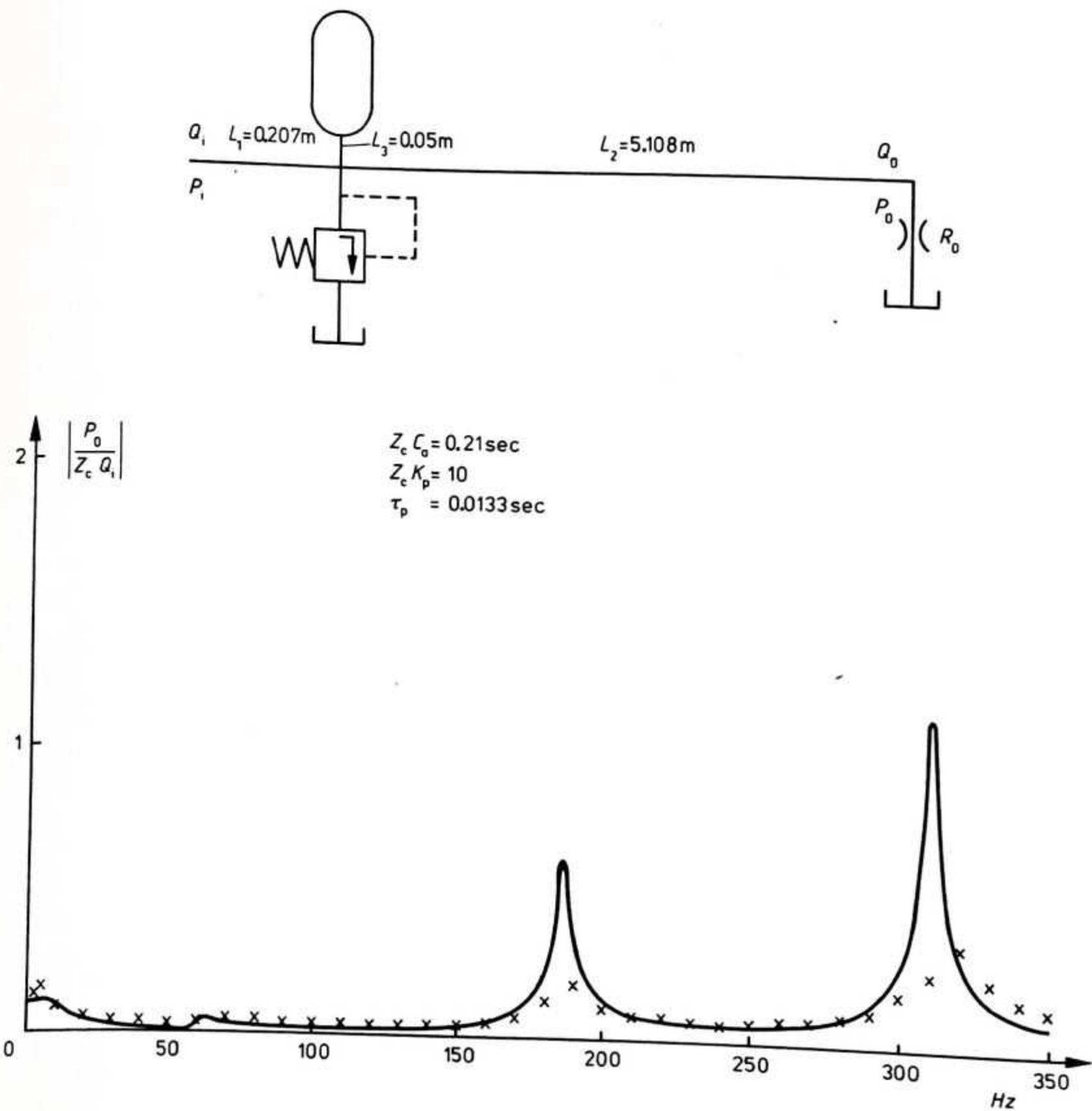


Fig. 6.24 Characteristic of a hydraulic line with a pressure control valve and a gas-accumulator. The peaks in the theoretical curve (viscosity excluded) can be estimated by eqns. (41) and (43). Experimental results are indicated as well, showing moderated peaks, mainly due to the viscosity.

$$\begin{array}{lll} |P_o/Z_c Q_i|_{\max} = 0.24 & \text{at} & 63 \text{ Hz} \\ & 0.72 & \text{at} & 188 \text{ Hz} \\ & 1.20 & \text{at} & 313 \text{ Hz} \\ & 1.68 & \text{at} & 439 \text{ Hz} \end{array}$$

These peaks can easily be recognized in Fig. 6.24, which is based on experiments carried out by Ham [4]. In the experiments a pressure control valve and a gas-accumulator were installed in the pipeline with G-functions according to Fig. 6.21. Without any of the previous approximations the transfer function was calculated by combining eqns. (24a), (25), (26c) and (26d) and the above data. The resulting characteristic (drawn line in Fig. 6.24) justifies remarkably well the process of the numerous approximations leading to the very simple eqns. (41) and (43).

The experiments of Ham were carried out for frequencies up to 350 Hz, owing to the limited bandwidth of the servovalve. The results, indicated by dashes in Fig. 6.24, show perfectly located peaks that are, however, moderated considerably. This can be explained by the viscosity, whose moderating influence was shown earlier in Fig. 6.8.

One final problem regarding the pressure P_i at the entrance of pipeline has been left open so far. For the sake of the duration of life of the pump, extreme pressures are undesirable, so we analyze the transfer function $P_i/Z_c Q_i$ by combining eqns. (24b), (25) and (36), introducing $R_o = \infty$ and $s = j\omega$.

$$\frac{P_i}{Z_c Q_i} = j \frac{\cos \omega T_L + \frac{\sin \omega T_1 \cos \omega T_2}{\omega T_3}}{-\sin \omega T_L + \frac{\cos \omega T_1 \cos \omega T_2}{\omega T_3}} \quad (44)$$

The denominator equals that of $P_o/Z_c Q_i$ (see eqn. (37)). Therefore most of the statements regarding $P_o/Z_c Q_i$ are relevant for $P_i/Z_c Q_i$ also. Thus there are comparable narrow peaks with $P_i = \infty$ around $\omega T_2 = 0.5\pi (2k + 1)$. The numerator of eqn. (44), however, shows several zeros as well, indicating narrow "dips" with $P_i = 0$. These dips are not interesting, since they are harmless.

Neglecting the narrow peaks and dips by omitting $\cos \omega T_L$ and $\sin \omega T_L$ in eqn. (44), we get

$$\frac{P_i}{Z_c Q_i} \approx j \tan \omega T_1 \quad (45)$$

Equation (45) directly indicates the desirability of a very short L_1 (not L_2 !), because otherwise several wide peaks with $P_i = \infty$ will occur and this is intolerable. Thus, choosing $L_1 < 0.3$ m in accordance with an earlier condition (39), we may

approximate eqn. (45) to

$$\left| \frac{P_i}{Z_c Q_i} \right| \approx \omega T_1 < \frac{\pi}{4}$$

This result is completely acceptable (see eqn. (23)).

Finally, we summarize the results obtained before.

- (1) Pressure control valve and accumulator, branched away from the main line between the pump and the servovalve, are quite capable of maintaining the nominal pressure and suppressing pump flow pulsations, provided some strict conditions are met.
- (2) In normal configurations, specified later, gas-accumulators are completely dominant over pressure control valves in suppressing pump flow pulsations.
- (3) Assuming that pulsation frequencies over 500 Hz do not disturb hydraulic servo-systems (because of their limited bandwidth), the accumulator should be placed within 0.3 m of the pump.
- (4) The supply line between main line and accumulator should be as short as possible, preferably between 5 and 10 cm.
- (5) The volume of the gas-accumulator should be sufficiently large to guarantee that the resonance frequency of accumulator with its supply line does not exceed the pump flow pulsation frequency. Normally, 1 dm³ is more than enough.
- (6) Fulfilling the above condition the combination of valve and accumulator acts as a wide band filter.
- (7) If the compound pipeline is (almost) closed at the end, narrow frequency areas with $|P_o/Z_c Q_i| > 1$ may occur. These peaks become wider and higher at increasing frequencies, especially in very long lines ($L > 10$ m).
- (8) In some cases minor adjustments in the total length L of the pipeline works out beneficially.
- (9) To guarantee minimal fluctuations in P_o , the H-filter should be preferred over gas-accumulators. This is even a condition in applications with pressure controlled pumps, where an accumulator near the pump will make the whole supply line completely unstable (see the following sections).

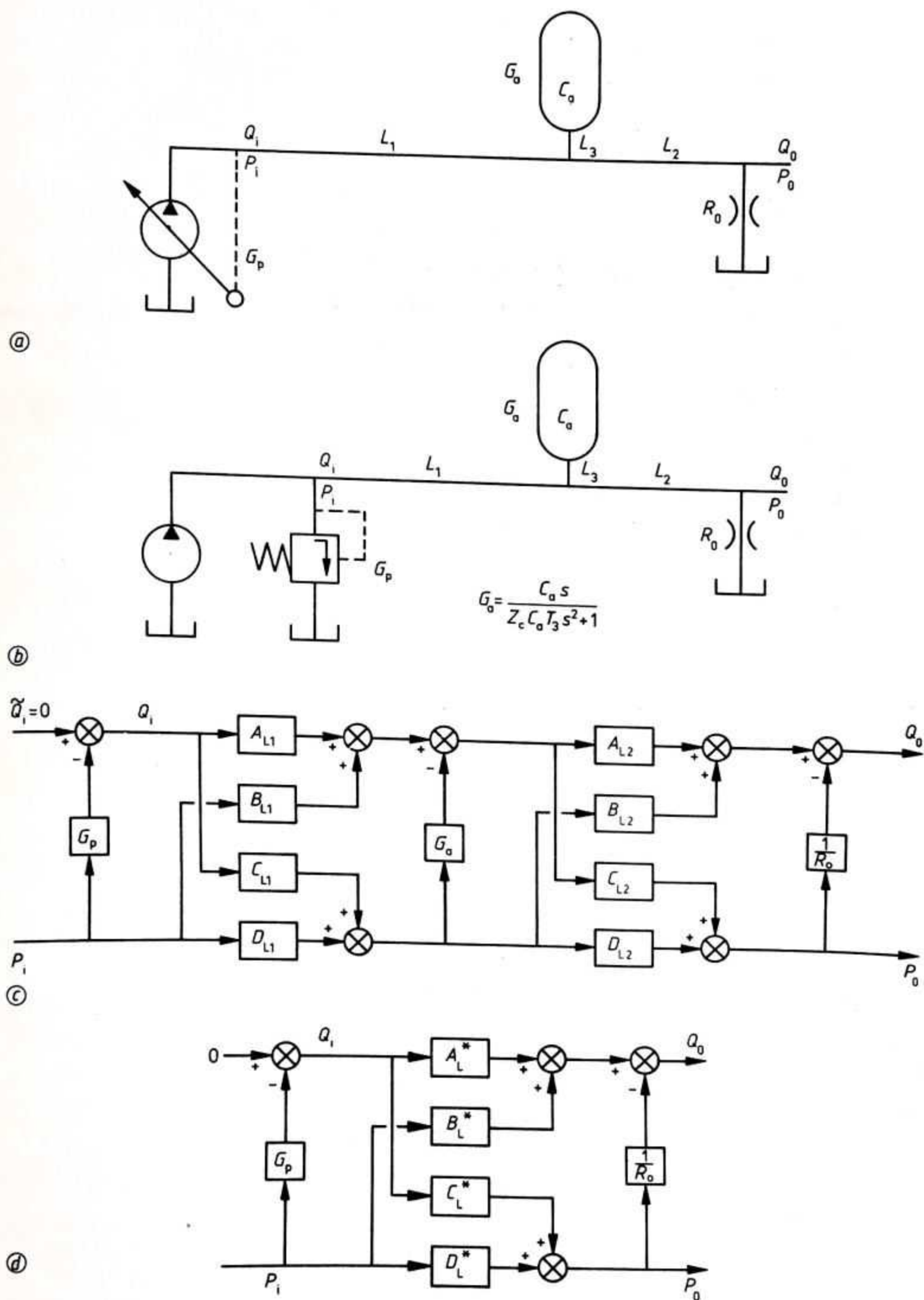


Fig. 6.25 Calculation of outlet pressure fluctuations due to servovalve outlet flow

$$\frac{P_o}{Z_c Q_o} = \frac{1}{z_c \frac{A_L^* - B_L^* G_p}{C_L^* - D_L^* G_p} - \frac{z_c}{R_o}}$$

6.6 Reducing supply pressure variations due to consumption by the servovalve

The influence of disturbances at the entrance of the pipeline having been treated, we start the analysis of the influence of disturbances at the end of the pipeline: variable flows via the servovalve. These disturbances are a direct consequence of displacements performed by the servomotor which could be expressed directly in servovalve flow variations. The servovalve flows are withdrawn from the hydraulic pipeline supplying the servosystem with oil. If we neglect the interaction between pipeline and servomotor, the flow Q_o could be regarded as an independent variable and the pressure P_o at the end of the pipeline as a dependent variable. (In Chapters 1, 2, 3 and 4 the supply pressure has been wrongly assumed to be constant; at best this is an approximation.)

Just as in the case of pump flow pulsations, it makes sense to suppress the effects of the servovalve flow variations by installing an accumulator near the source of the disturbance: the servovalve.

In Fig. 6.25 a compound pipeline is depicted in the case of a pressure controlled pump (a) and in the case of a fixed displacement pump with a pressure control valve (b). As was shown in Sections 6.4.7 and 6.4.8, the G-functions of a pressure control valve and of a pressure controlled pump have the same character (see Fig. 6.16). Neglecting the influence of the very short pipeline between the fixed displacement pump and the pressure control valve, the difference between the pumps is reduced to differences in the parameters K_p and τ_p of their G-functions. Consequently, both systems can be summarized in one block diagram (c), which can be reduced to the simple block diagram (d). The resistance R_o is included in the model to account for a possible interaction between Q_o and P_o , especially in the case of experiments with an hydraulic pipeline having at its end a servovalve only.

Regarding the location of the (large!) accumulator, in analogy with the previous section, it is installed close to the disturbance source: the servovalve. Consequently, the pipeline L_2 is short and the pipeline L_1 is considered to be (relatively) long. Naturally, the supply line between the main line and the accumulator is short also, thus

$$Z_c G_a = \frac{Z_c C_a s}{Z_{c3} C_a T_3 s^2 + 1} \quad (46)$$

Since the accumulator capacity $C_a = V_a / n P_a$ should be large enough to follow Q_o fluctuations reasonably well, the volume V_a should be very much larger than the volume of the accumulator suppressing the pump flow pulsations. For this reason the resonance frequency of the accumulator and its connection line is relatively low. Usually the supply line to the accumulator and the main line have the same diameter, thus $Z_{c3} = Z_c$.

Using Fig. 6.25 (d), the transfer function $P_o/Z_c Q_i$ can be derived directly from

$$\left. \begin{aligned} Q_o &= (A_L^* - C_L^*/R_o) Q_i + (B_L^* - D_L/R_o) P_i \\ P_o &= C_L^* Q_i + D_L^* P_i \\ Q_i &= -G_p P_i \end{aligned} \right\}$$

by elimination of P_i .

$$\frac{P_o}{Z_c Q_o} = \frac{1}{Z_c \frac{A_L^* - B_L^*/G_p}{C_L^* - D_L^*/G_p} - \frac{Z_c}{R_o}} \quad (47)$$

To achieve the best possible surveyable analysis of various influencing factors, we derive $P_o/Z_c Q_o$ further in the most general form, using Fig. 6.9.

$$A_L^* = A_{L1} A_{L2} + C_{L1} B_{L2} - G_a C_{L1} A_{L2} = C_{L1} A_{L2} \left[\frac{A_{L1}}{C_{L1}} + \frac{B_{L2}}{A_{L2}} - G_a \right]$$

$$B_L^* = B_{L1} A_{L2} + D_{L1} B_{L2} - G_a D_{L1} A_{L2} = D_{L1} A_{L2} \left[\frac{B_{L1}}{D_{L1}} + \frac{B_{L2}}{A_{L2}} - G_a \right]$$

$$C_L^* = A_{L1} C_{L2} + C_{L1} D_{L2} - G_a C_{L1} C_{L2} = C_{L1} C_{L2} \left[\frac{A_{L1}}{C_{L1}} + \frac{D_{L2}}{C_{L2}} - G_a \right]$$

$$D_L^* = D_{L1} D_{L2} + B_{L1} C_{L2} - G_a D_{L1} C_{L2} = D_{L1} C_{L2} \left[\frac{B_{L1}}{D_{L1}} + \frac{D_{L2}}{C_{L2}} - G_a \right]$$

Substitution in eqn. (47) yields

$$\frac{P_o}{Z_c Q_o} = \frac{1}{Z_c \frac{A_{L2}}{C_{L2}} \cdot \frac{C_{L1}}{D_{L1}} \left[\frac{A_{L1}}{C_{L1}} + \frac{B_{L2}}{A_{L2}} - G_a \right] - \left[\frac{B_{L1}}{D_{L1}} + \frac{B_{L2}}{A_{L2}} - G_a \right] / G_p} - \frac{Z_c}{R_o}$$

Finally, substituting parameters of the two elements of the pipeline (see Fig. 6.6) we get

$$\frac{P_o}{Z_c Q_o} = \frac{-1}{\frac{1}{\tanh T_2 s} \cdot \frac{\tanh T_1 s \left[\frac{1}{\tanh T_1 s} + \tanh T_2 s + Z_c G_a \right] + \frac{\tanh T_1 s + \tanh T_2 s + Z_c G_a}{Z_c G_p} + \frac{Z_c}{R_o}}{\tanh T_1 s \left[\frac{1}{\tanh T_1 s} + \frac{1}{\tanh T_2 s} + Z_c G_a \right] + \frac{\tanh T_1 s + \frac{1}{\tanh T_2 s} + Z_c G_a}{Z_c G_p}} \quad (48)$$

The results have been summarized in Fig. 6.25. The large number of parameters (G_a and G_p both have two parameters) makes it very difficult to analyze the various factors. To get a grip on the analyzing process we introduce a series of simplified cases:

Simplifications	$Z_c G_a$	$Z_c G_p$	R_o/Z_c	$\alpha = 32 v/D^2$	relevant frequency
case a	$\frac{1}{T_3 s}$	$0 (Q_i=0)$	∞	0	high
case b	$\frac{1}{T_3 s}$	$\infty (P_i=0)$	∞	0	high
case c	$Z_c C_a s$	$\frac{Z_c K_p}{\tau_p s + 1}$	$\neq \infty$	$\neq 0$	low

At first glance these cases seem to be completely arbitrary. Numerous experiments and calculations, however, made it possible to distinguish relevant influences. A very important point concerns the frequencies being involved. From eqn. (6.26d) we see that at higher frequencies $Z_c G_a$ can be approximated by $1/T_3 s$ and at low frequencies by $Z_c C_a s$ (cases a and b versus case c). So it makes sense to distinguish here. It is remarkable, however, that at $Z_c G_a = 1/T_3 s$ the other quantities ($Z_c G_p$, R_o/Z_c and α) have scarcely any influence. They simply don't matter!

At low frequencies, however, where $Z_c G_a = Z_c C_a s$, it will become clear that the exact transfer function $Z_c G_p$ as well as the magnitudes of R_o/Z_c and α are extremely important. So here no approximation is permitted. But in this case the relevant frequencies where instability, etc. might occur are so low that simplifications like $\sin \omega T \approx \omega T$, $\cos \omega T \approx 1$ and $N(s) \approx 1 + \alpha/s$ are permitted, permitting a straightforward analysis.

Thus in case (a) we assume $Z_c G_a = 1/T_3 s$, $G_p = 0$, $R_o = \infty$. The physical implication of $G_p = 0$ is $Q_i = 0$, as can be seen in Fig. 6.25(b). The accumulator is placed close to the servovalve, thus $\tan \omega T_2 \approx \omega T_2 < \pi/4$. Substitution in eqn. (48) gives

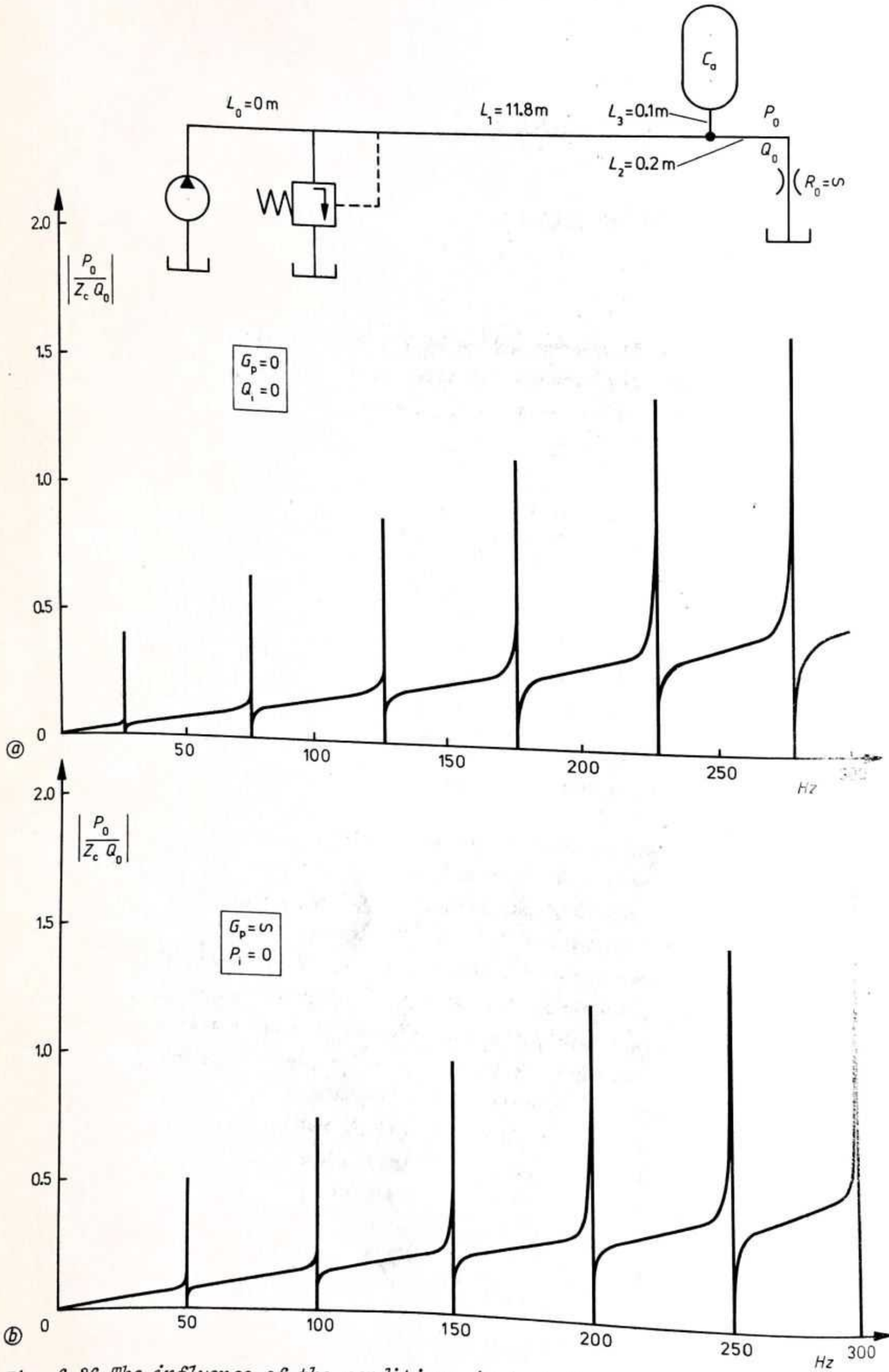


Fig. 6.26 The influence of the conditions (G_p) at the entrance of a pipeline on the outlet pressure fluctuations in a pipeline with a gas-accumulator near the servovalve.

$$\frac{P_o}{Z_c Q_o} = -j \tan \omega T_2 \cdot \frac{\tan \omega T_1 - \frac{1}{\tan \omega T_2} - \frac{1}{\omega T_3}}{\tan \omega T_1 + \tan \omega T_2 - \frac{1}{\omega T_3}} \approx -j \omega T_2 \cdot \frac{\tan \omega T_1 - \frac{T_2 + T_3}{\omega T_2 T_3}}{\tan \omega T_1 - \frac{1}{\omega T_3}} \quad (49)$$

Equation (49) is depicted in Fig. 6.26(a) for $L_1 = 11.8$ m, $L_2 = 0.2$ m and $L_3 = 0.1$ m. We observe narrow peaks and dips. In between there is a monotonously increasing curve. Equation (49) yields

$$P_o = \infty \quad \text{at} \quad \omega T_1 \approx \frac{\pi}{2} (2k + 1) \left(1 - \frac{T_3}{T_1}\right)$$

$$P_o = 0 \quad \text{at} \quad \omega T_1 \approx \frac{\pi}{2} (2k + 1) \left(1 - \frac{T_3}{T_1} \cdot \frac{T_2}{T_2 + T_3}\right)$$

$$\frac{P_o}{Z_c Q_o} \approx -j \omega (T_2 + T_3) \quad \text{at} \quad \omega T_1 \approx \pi k, \text{ where } k = 0, 1, 2, 3, \dots$$

Thus the general curve is satisfactory. Thanks to the short lengths L_2 and L_3 the peaks and dips are very narrow. As we saw in earlier instances (see Figures 6.23 and 6.24), these are "filtered" out by some opening of the servovalve ($R_o \neq \infty$) and by the viscosity ($\alpha \neq 0$).

In case (b) we assume $Z_c G_a = 1/T_3 s$, $G_p = \infty$ (i.e. $P_i = 0$) $R_o = \infty$ and $\tan \omega T_2 = \omega T_2$. Substitution in eqn. (48) gives

$$\frac{P_o}{Z_c Q_o} = -j \tan \omega T_2 \cdot \frac{\frac{1}{\tan \omega T_1} + \frac{1}{\tan \omega T_2} + \frac{1}{\omega T_3}}{\frac{1}{\tan \omega T_1} - \tan \omega T_2 + \frac{1}{\omega T_3}} \approx -j \omega T_2 \cdot \frac{\frac{1}{\tan \omega T_1} + \frac{T_2 + T_3}{\omega T_2 T_3}}{\frac{1}{\tan \omega T_1} + \frac{1}{\omega T_3}} \quad (50)$$

Equation (50) is depicted in Fig. 6.26(b), again for $L_1 = 11.8$ m, $L_2 = 0.2$ m and $L_3 = 0.1$ m. Again we observe narrow peaks and dips as deviations on a monotonously increasing curve. We find

$$P_o = \infty \quad \text{at} \quad \omega T_1 \approx \pi k \left(1 - \frac{T_3}{T_1}\right)$$

$$P_o = 0 \quad \text{at} \quad \omega T_1 \approx \pi k \left(1 - \frac{T_3}{T_1} \cdot \frac{T_2}{T_2 + T_3}\right)$$

$$\frac{P_o}{Z_c Q_o} \approx -j \omega (T_2 + T_3) \quad \text{at} \quad \omega T_1 \approx \frac{\pi}{2} (2k + 1), \text{ where } k = 0, 1, 2, 3, \dots$$

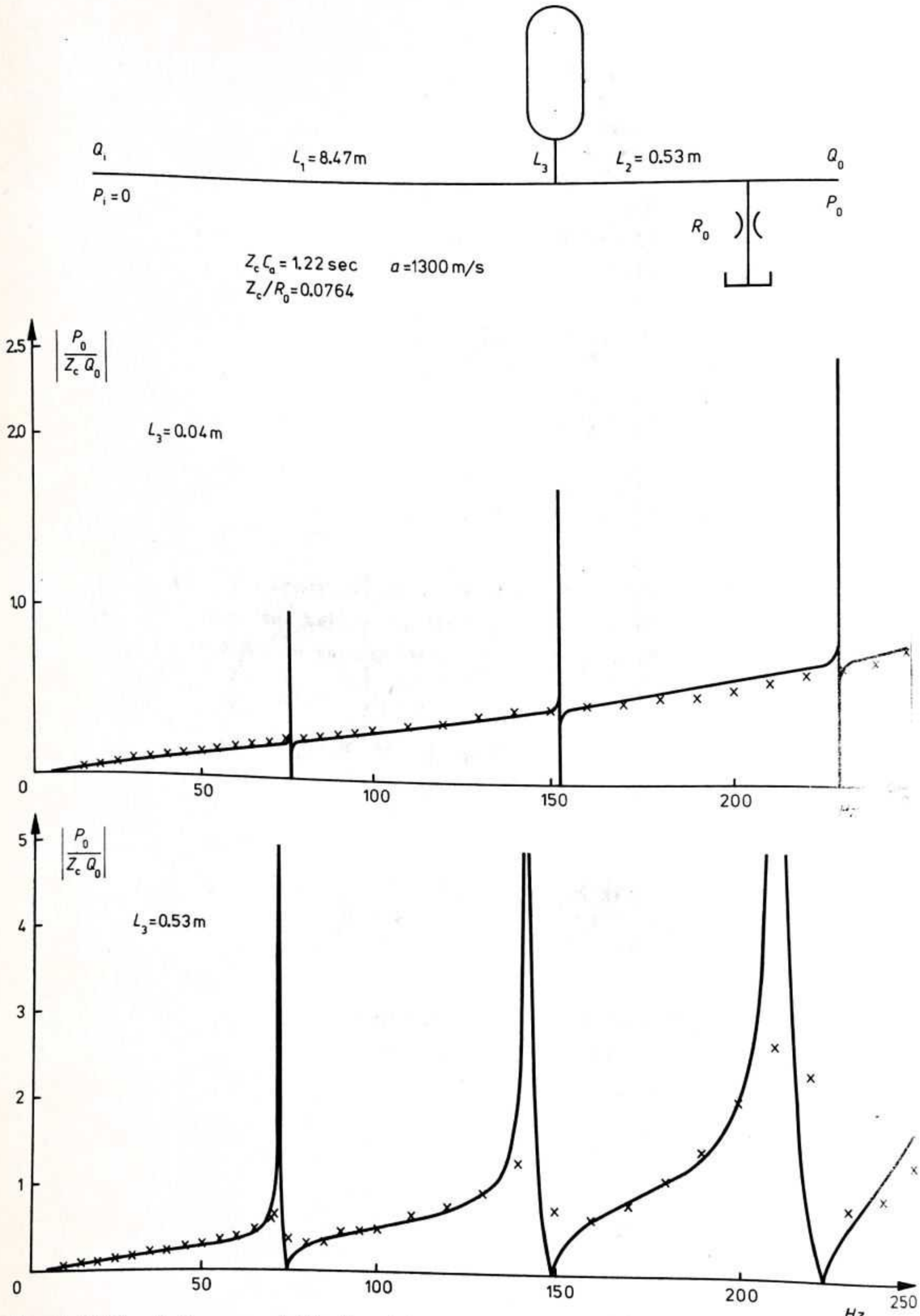


Fig. 6.27 The influence of the location of a gas-accumulator near the servovalve on the outlet pressure fluctuations.

The only difference from case (b) is that peaks and dips occur near $\tan \omega T_1 \approx 0$ instead of $\tan \omega T_1 \approx \infty$. How sharp the curve changes in the areas of peaks and dips are can be shown by calculation of $\Delta\omega = \omega(P_o = 0) - \omega(P_o = \infty)$ in relation to the relevant frequency ω_r . In both the cases (a) and (b) we find

$$\frac{\Delta\omega}{\omega_r} \approx \frac{T_3}{T_1} \cdot \frac{T_3}{T_2 + T_3}$$

This equation once more shows the need to make L_3 as short as possible. Analysis of eqns. (49) and (50) also stresses the need for a short L_2 in order to avoid wide peaks.

To illustrate the previous analysis in Fig. 6.27(a) and (b), the results of some experiments are shown, indicating the importance of L_2 and L_3 . Indeed, when L_2 and L_3 are made short, peaks and dips can no longer be measured (mainly due to the viscosity).

Summarizing so far, we conclude that at higher frequencies the function G_p is not important, so the pump control or the pressure control valve does not influence the dynamic behavior in practice. But we have to install the accumulator very close to the servovalve. In doing so, heavy pressure fluctuations due to "servovalve consumptions" will be damped out almost completely by the accumulator.

According to eqn. (46), at frequencies lower than

$$\omega_H = \frac{1}{\sqrt{Z_{c3} T_3 C_a}} = \sqrt{\frac{n A_3 P_a}{\rho L_3 V_a}}$$

$Z_c G_a$ can be approximated by $Z_c C_a s$, case (c). V_a usually being larger than 10^{-3} m^3 , the frequency ω_H will be in the region 20 - 75 rad/sec (3 - 12 Hz). Assuming $L_1 < 15 \text{ m}$, we have $\omega T_1 < \pi/4$ also. Therefore, we may approximate

$$\tanh T_1 s \approx T_1 s, \quad \tanh T_2 s \approx T_2 s$$

Substituting these results in eqn. (48) yields

$$\frac{P_o}{Z_c Q_o} = \frac{-1}{1 + (T_2 + Z_c C_a) T_1 s^2 + (T_1 + T_2 + Z_c C_a) s / Z_c G_p} + \frac{Z_c}{R_o} \frac{1}{T_1 s (1 + \frac{T_2}{T_1} + Z_c C_a T_2 s^2) + [1 + (T_1 + Z_c C_a) T_2 s^2] / Z_c G_p} \quad (51)$$

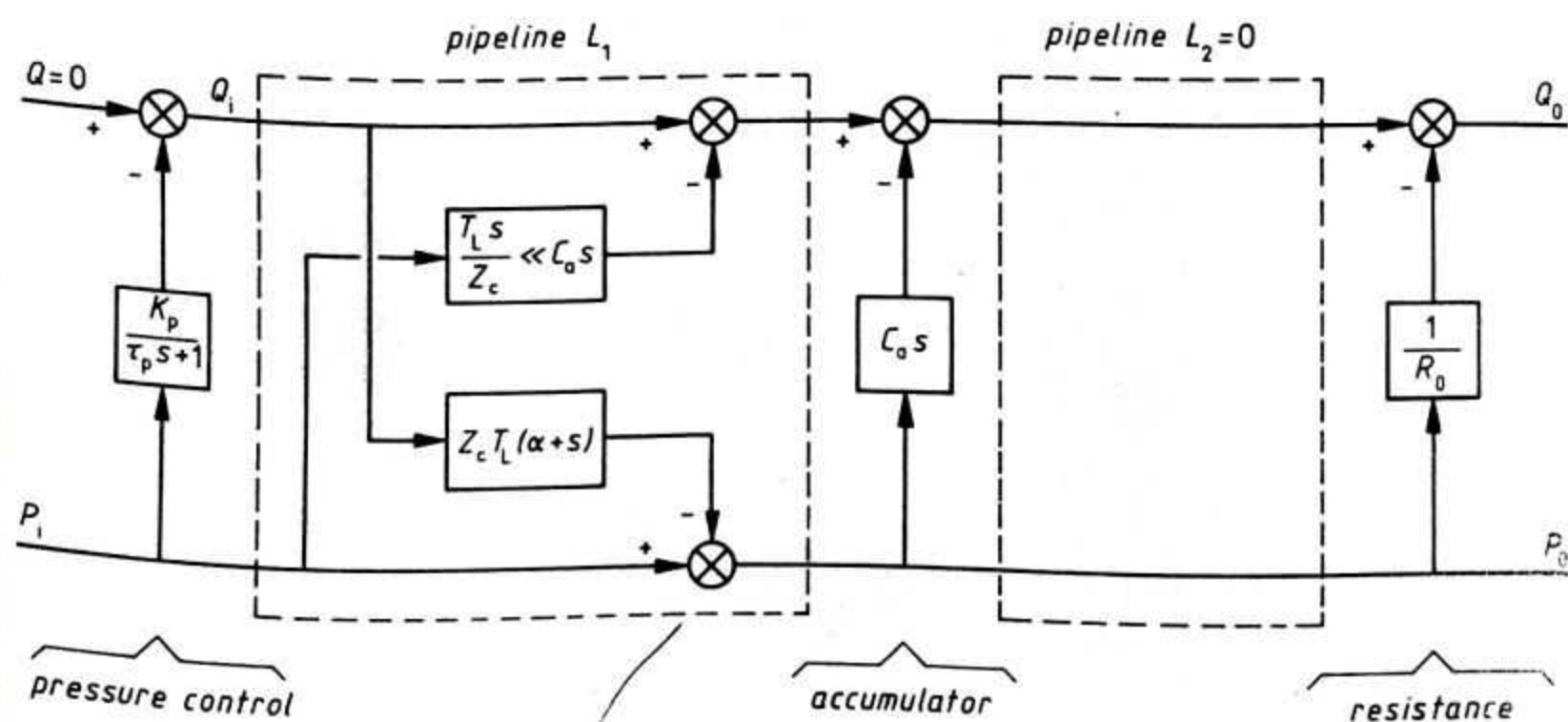
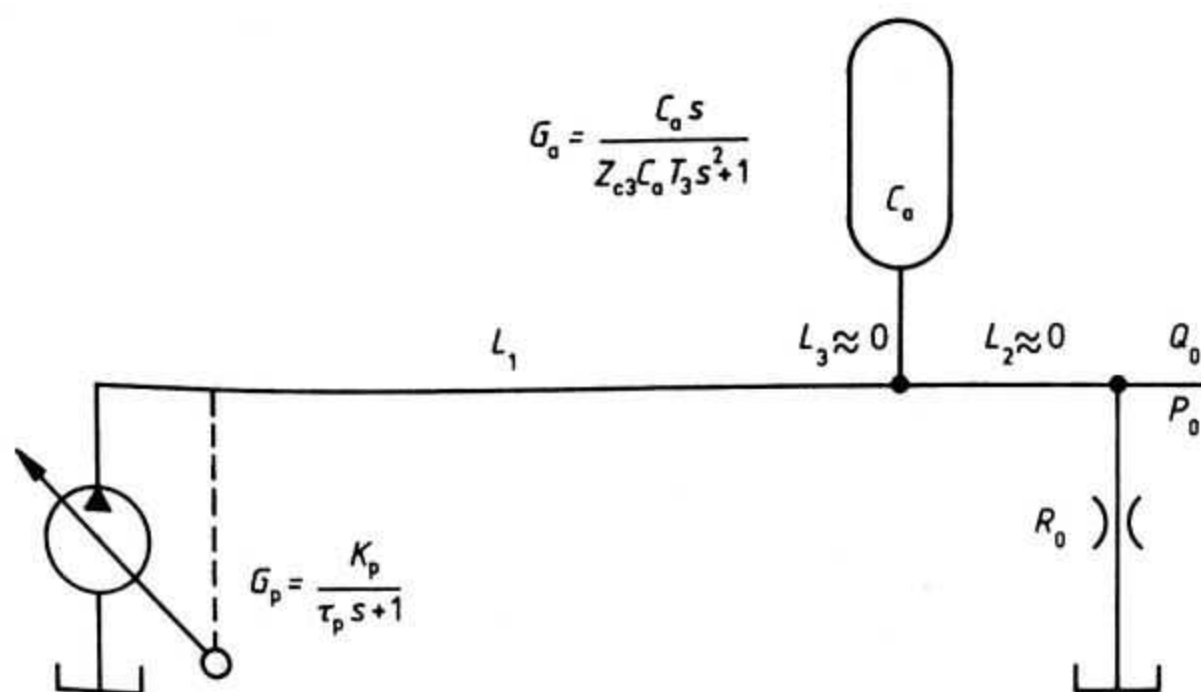


Fig. 6.28 Simplified model to calculate at low frequencies the outlet pressure fluctuations in a long pipeline with

- pressure-controlled pump, or fixed displacement pump with pressure control valve,
- large gas-accumulator,
- servovalve.

At low frequencies the block diagram is reduced to a lumped parameter system.

In this equation both T_1 and T_2 can be neglected in comparison to $Z_c C_a$ since

$$\frac{T}{Z_c C_a} = \frac{L/a}{\frac{\rho a}{A} \cdot \frac{V_a}{nP_a}} = \frac{LA}{V_a} \cdot \frac{nP_a}{\rho a^2} = \frac{LA}{V_a} \cdot \frac{nP_a}{E} \ll 1$$

Choosing L_2 very short, on the grounds already discussed, all factors with T_2 in eqn. (51) are neglected. Thus

$$\frac{P_o}{Z_c Q_o} = \frac{-1}{1 + Z_c C_a T_L s^2 + \frac{Z_c C_a s / Z_c G_p}{T_L s + 1/Z_c G_p} + \frac{Z_c}{R_o}} \quad (52)$$

We now have to introduce

$$Z_c G_p = \frac{Z_c K_p}{\tau_p s + 1}$$

The viscosity should be taken into account by replacing (see Fig. 6.6)

$$Z_c \quad \text{by} \quad Z_c \sqrt{N} \approx Z_c \sqrt{1 + \alpha/s}$$

$$T \quad \text{by} \quad T \sqrt{N} \approx T \sqrt{1 + \alpha/s}$$

The approximation $N(s) \approx 1 + \alpha/s$ is justified by the low frequencies involved (see eqn. (6e)).

The resulting model is summarized in Fig. 6.28, showing a simplified lumped parameter system. Elaborating Fig. 6.28 or substituting $Z_c G_p$, Z_c and T in eqn. (52) yields, after working it out completely,

$$\frac{P_o}{Z_c Q_o} = - \frac{K_L + \frac{Z_c C_a s}{M}}{1 + K_L \frac{Z_c}{R_o} + (K_L + \frac{Z_c}{MR_o}) Z_c C_a s + \frac{(Z_c C_a s)^2}{M}} \quad (53)$$

In this

$$K_L = T_L \alpha + \frac{1}{Z_c K_p} \quad (54)$$

$$M = \frac{\frac{Z_c C_a}{T_L} \cdot \frac{T_L}{\tau_p}}{\frac{T_L}{\tau_p} + \frac{1}{Z_c K_p}} \quad (55)$$

The introduction of the new dimensionless parameters K_L and M permitted the reduction of the total number of dimensionless parameters, characterizing the transfer function (53). In this second order transfer function K_L represents the steady-state pressure drop due to the offset of the pressure control (pressure control valve or pressure-controlled pump) and the pressure drop across the line itself. At $R_o = \infty$

$$\left(\frac{P_o}{Q_o}\right)_{s=0} = -K_L Z_c = -Z_c T_L \alpha - \frac{1}{K_p}$$

$$= -\frac{\rho a}{A} \cdot \frac{L}{a} \cdot \frac{32\nu}{D^2} - \frac{1}{K_p} = -\frac{128\mu L}{\pi D^4} - \frac{1}{K_p}$$

Thus a small viscosity and a large pressure gain K_p implies a small K_L . Equation (53) directly reveals the risk of a small K_L , especially at $R_o = \infty$, namely a low damping.

In the parameter M the effectiveness of the accumulator capacity C_a is heavily influenced by the type of pressure control. Applying a pressure control valve, values of $\tau_p \approx 1/60$ sec are normal. In a pressure-controlled pump a value of $\tau_p \approx 10$ sec has been observed (see Fig. 6.14). Assuming, for instance, $Z_c K_p \approx 20$ and $T_L \approx 0.01$ sec, we get at a

$$\begin{aligned} \text{o pressure control valve} \quad M &\approx Z_c C_a / T_L \approx 100 Z_c C_a \\ \text{o pressure controlled pump} \quad M &\approx Z_c C_a \cdot Z_c K_p / \tau_p \approx 2 Z_c C_a \end{aligned}$$

Thus to obtain the same value of M a pressure-controlled pump requires in this case an accumulator 50 times larger than a pressure control valve.

Analyzing eqn. (53), we bring it in a general form

$$\frac{P_o}{Z_c Q_o} = K \frac{\tau s + 1}{\left(\frac{s}{\omega_o}\right)^2 + 2\beta \frac{s}{\omega_o} + 1}$$

in order to calculate the maximum amplitude ratio $\sigma = |P_o / Z_c Q_o|_{\max}$ as a function of β and $\omega_o \tau$. Laborious but straightforward calculations show at $\omega = \omega_r$ according to

$$\frac{\omega_r^2}{\omega_o^2} = \frac{\sqrt{(\omega_o^2 \tau^2 + 1)^2 - 4\beta^2 \omega_o^2 \tau^2} - 1}{\omega_o^2 \tau^2}$$

a maximum

$$\sigma = \left| \frac{P_o}{Z_c Q_o} \right|_{\max} = \frac{K}{\sqrt{1 - \left(\frac{\omega_r}{\omega_o} \right)^4}}$$

if the following condition is met:

$$2\beta < \sqrt{2 + \omega_o^2 \tau^2}$$

If this condition is not met, then the largest value of $|P_o/Z_c Q_o|$ occurs at $\omega = 0$.

$$\sigma = \left| \frac{P_o}{Z_c Q_o} \right|_{\max} = K$$

Translating the parameters of these results into the corresponding parameters of eqn. (53) finally yields

$$\sigma = \frac{1}{\sqrt{\left[K_L M + \frac{Z_c}{R_o} \right]^2 + M \left[\sqrt{1 + 2K_L \left(K_L M + \frac{Z_c}{R_o} \right)} - 1 \right]^2}} \quad (56)$$

occurring at

$$\omega = \omega_r = \frac{\sqrt{M}}{Z_c C_a} \cdot \sqrt{\sqrt{1 + 2K_L^2 M + 2K_L \frac{Z_c}{R_o}} - K_L^2 M} \quad (57)$$

if the following condition is met:

$$(K_L^2 M - 1)^2 < 2 \left(1 + K_L \frac{Z_c}{R_o} \right) \quad (58)$$

if condition (58) is not met, then

$$\sigma = \frac{K_L}{1 + K_L \frac{Z_c}{R_o}} \quad (59)$$

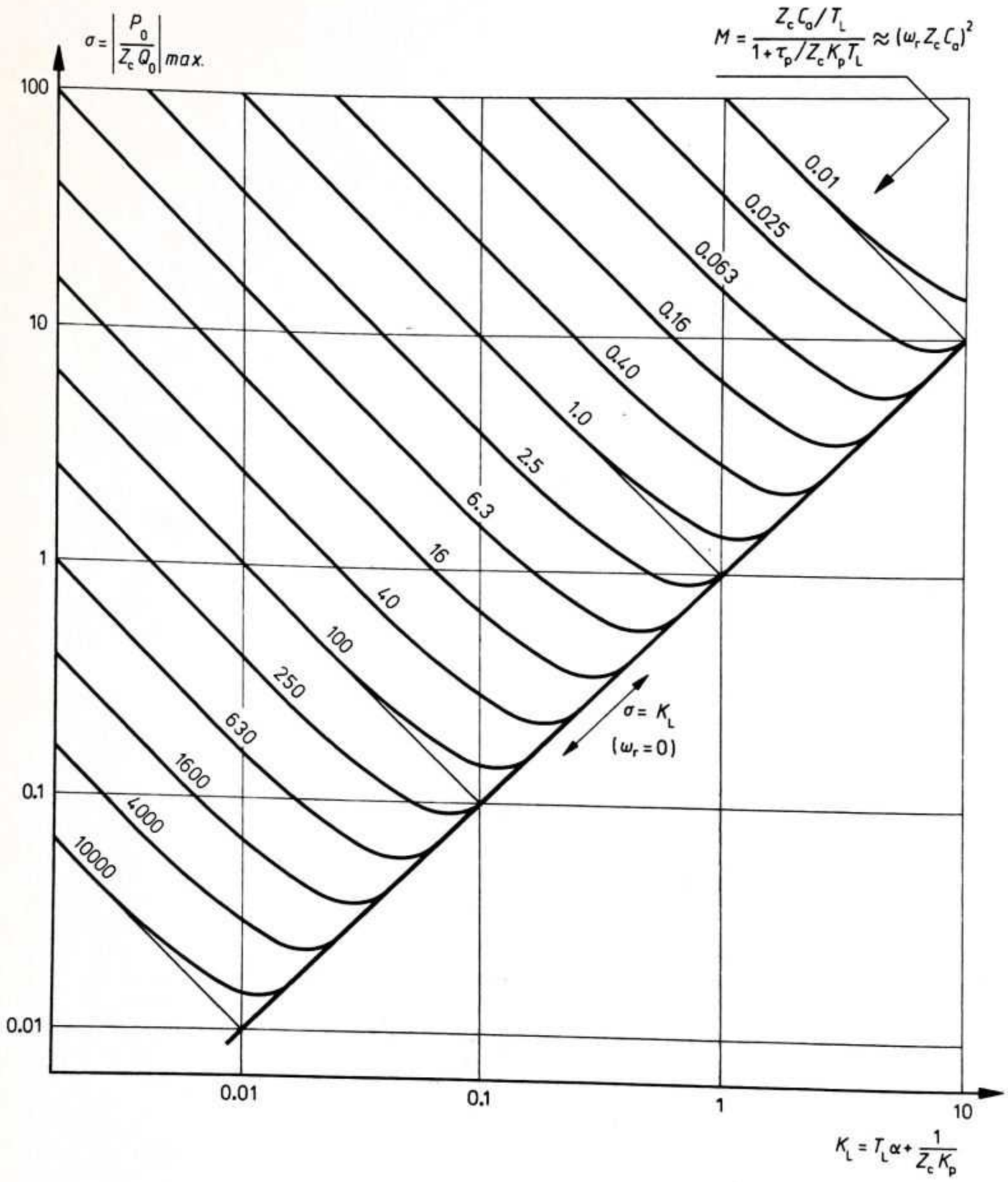


Fig. 6.29 Diagram to calculate σ as a function of K_L and M .

occurring at

$$\omega = \omega_r = 0 \quad (60)$$

Observing in eqns. (56) and (59) a moderating influence of R_o on σ , we assume first $R_o = \infty$ and plot in Fig. 6.29 σ as a function of K_L with M as a parameter. Figure 6.29 clearly shows by its almost straight lines that, accepting an error of 50% in the transitional region, the characteristic could be summarized extremely simply.

$$\text{If } K_L^2 M > 1 \quad \text{then} \quad \omega_r = 0 \quad \text{and} \quad \sigma \approx K_L \quad (61)$$

$$\text{If } K_L^2 M < 1 \quad \text{then} \quad \omega_r \approx \frac{\sqrt{M}}{Z_c C_a} \quad \text{and} \quad \sigma \approx \frac{1}{K_L M} \quad (62)$$

The line $\sigma = K_L$ is the boundary of the characteristic, representing the pressure offset in the steady state for all values of M .

To minimize σ at a given K_L , a design rule

$$M \geq \frac{1}{K_L^2} \quad (63)$$

guarantees a system with dynamic pressure deviations not exceeding the steady-state offset of the pressure. Thus the outline for the design seems obvious. There are two stages. First, the pipeline should not show too large pressure losses in the steady-state, implying a readily obtainable value of

$$K_L = T_L \alpha + \frac{1}{Z_c K_p} \quad (54)$$

In practice, the pipeline diameter D has a major influence since $\alpha = 32\nu/D^2$.

In the second stage, we choose

$$M = \frac{\frac{Z_c C_a}{T_L} \cdot \frac{T_L}{\tau_p}}{\frac{T_L}{\tau_p} + \frac{1}{Z_c K_p}} \geq \frac{1}{K_L^2} \quad (64)$$

At given (normally inflexible) values of $Z_c K_p$, T_L , α and τ_p , this means that the accumulator (C_a) should be chosen large enough to meet condition (64).

This procedure is perfect as long as the resulting volume of the accumulator is acceptable. Normally, this is the case with pressure control valves, having a small τ_p . In case of a pressure-controlled pump with a large τ_p , however, the

procedure could lead to a far too large accumulator.

Then a procedure of trial and error could be carried out, where subsequently τ_p , K_p , T_L , α and Z_c are changed, trying to minimize

$$\frac{P_o}{Q_o} = \frac{Z_c}{K_L M} = \frac{\tau_p}{C_a} \cdot \frac{\frac{T_L}{\tau_p} + \frac{1}{Z_c K_p}}{T_L \alpha + \frac{1}{Z_c K_p}} \quad (65)$$

It may occur in this procedure that (by decreasing the pipeline diameter) *the steady-state pressure drop must be increased deliberately in order to reduce the dynamic deviations of the pressure.*

In view of experiments to be illustrated later on, we give an example using initially a pipeline with the following data:

$L = 12$ m, $D = 12.5 \cdot 10^{-3}$ m, $a = 1200$ m/sec, $T_L = 0.01$ sec.

The oil Shell Tellus 27 at 30°C has:

$\nu = 49 \cdot 10^{-6}$ m²/sec, so $\alpha = 32\nu/D^2 = 10$ sec⁻¹ and $\alpha T_L = 0.1$.

In the experiments we could use:

a pressure control valve with $\tau_p = 0.016$ sec and $Z_c K_p = 20$,

a pressure controlled pump with $Z_c K_p = 20$ and

$\tau_p = 10$ sec (configuration with mechanical feedback),

$\tau_p = 1$ sec (configuration with electro-hydraulic servoloop).

In these configurations three different accumulators could be used:

$Z_c C_a / T_L = 20, 180$ and 300 , respectively.

Summarizing so far, there are $3 \times 3 = 9$ combinations.

In Table 6.3 (upper left) the exact results, using eqns. (56) - (60), are presented.

We observe that

- at $\tau_p = 10$ sec, even the largest accumulator ($Z_c C_a = 3$) fails to reach an acceptable σ ($\sigma = 1.21$),
- at $\tau_p = 1$ sec, the largest accumulator does the job ($\sigma = 0.20$),
- at $\tau_p = 0.016$ sec, a middle-size accumulator is already too large ($\sigma = 0.15$ at $Z_c C_a = 1.8$ and $\sigma = 0.54$ at $Z_c C_a = 0.2$).

To illustrate how a *short pipeline can sometimes upset the behavior*, the pipeline has been halved to $L = 6$ m. As the lower part of Table 6.3 shows, the σ -values of the pressure-controlled pump are worse, and the σ 's of the pressure control valve are better, owing to the lower value K_L .

In all this, viscosity has a vital influence. To show how the results deteriorate, in the upper right part of Table 6.3 the viscosity has been neglected. In seven of the nine possible combinations the σ 's have increased sharply, giving a completely wrong picture.

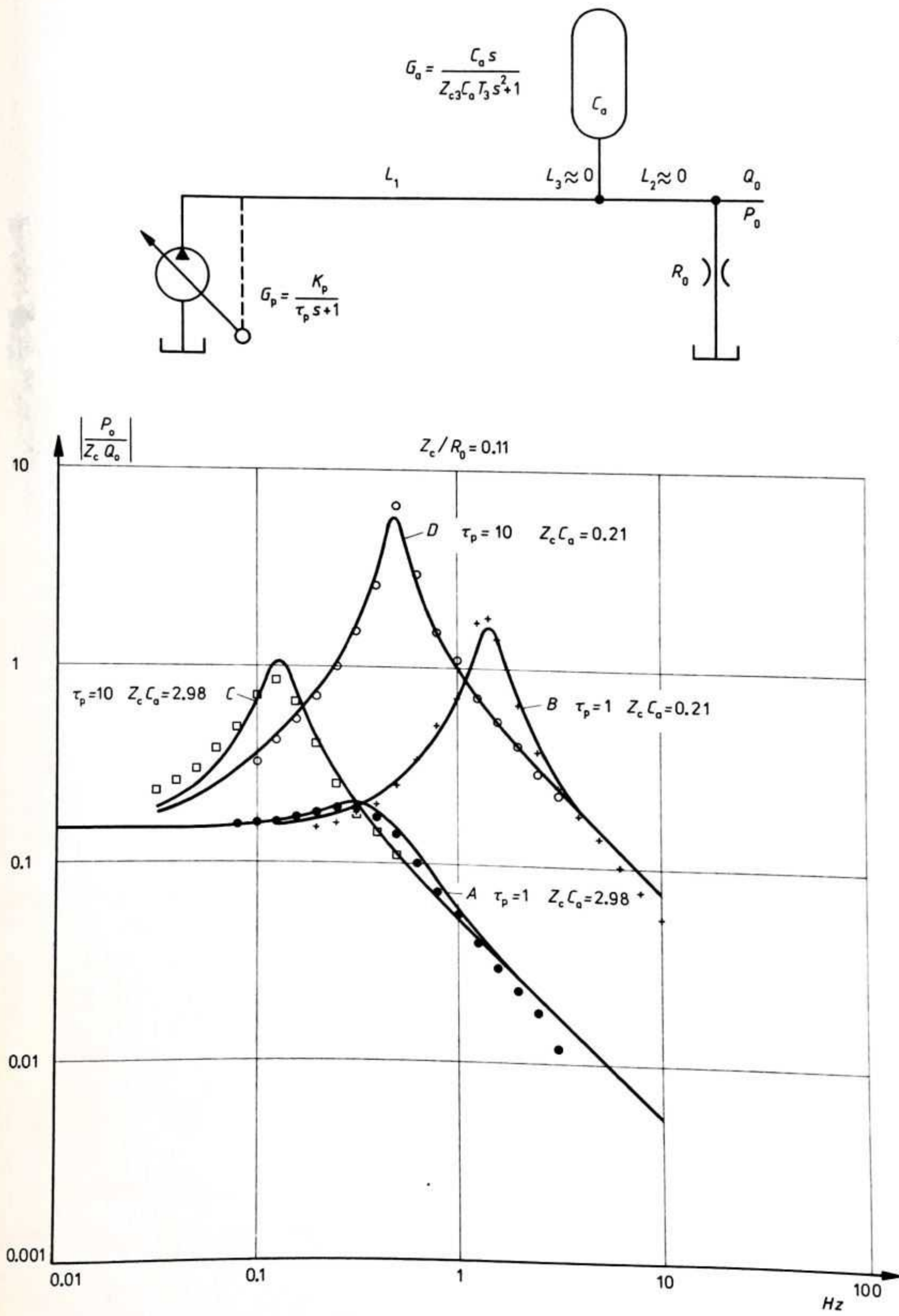


Fig. 6.30 Theoretical results and measurements on long hydraulic pipelines with a pressure-controlled pump at low frequencies.

This becomes even worse if we assume the possibility of a pressure control (valve or pump) without a feedback loop (implying $G_p = K_p/\tau_p s$). In that case, $K_L = 0$ if the viscosity is neglected and, consequently, the pipeline system has no damping at all.

In the Mechanical Control Engineering Laboratory of Delft University of Technology measurements were carried out on a pressure-controlled pump (Abex-Denison PV06-00751R04). The mechanical feedback loop was replaced by an electro-hydraulic servomotor allowing the reduction of the original $\tau_p = 10$ sec to $\tau_p = 1$ sec. Accumulators with $Z_c C_a = 0.21$ and $Z_c C_a = 2.98$ were used, corresponding to the data in Table 6.3.

Other data are: $Z_c K_p = 20$, $Z_c = 87 \cdot 10^8 \text{ Ns/m}^5$, $D = 12.5 \text{ mm}$, $\rho = 900 \text{ kg/m}^3$, $a = 1200 \text{ m/s}$ and $\nu = 49 \cdot 10^{-6} \text{ m}^2/\text{s}$.

At the end of the pipeline a Moog servovalve was installed, actuated sinusoidally around a fixed opening. In Fig. 6.30 the leakage-flow due to the fixed opening is represented by R_o . In the experiments, $Z_c/R_o = 0.11$ was chosen.

In Fig. 6.30 theoretical curves according to eqn. (53) are drawn up in the following cases

A - $\tau_p = 1 \text{ sec}$	$Z_c C_a = 2.98 \text{ sec}$,
B - $\tau_p = 1 \text{ sec}$	$Z_c C_a = 0.21 \text{ sec}$,
C - $\tau_p = 10 \text{ sec}$	$Z_c C_a = 2.98 \text{ sec}$,
D - $\tau_p = 10 \text{ sec}$	$Z_c C_a = 0.21 \text{ sec}$.

Measuring results in the corresponding cases are plotted also, showing a good agreement with theory.

However, in industrial practice the agreement between theory and measurements is insufficient to solve the design problem once and for all, since industrial manufacturers never show values of τ_p and K_p in their catalogs. Even when a special request is made, the answers are completely negative, since the data are unknown (or secret). As measurements, especially of τ_p , require quite an effort, the gap between the theory developed here and practical applications has not yet been bridged sufficiently.

A final comment on dimension analysis could give a clarifying view on the strategy developed in this section. Dimensional analysis of a compound pipeline could bring us directly to the insight that the (dimensionless) amplitude ratio $|P_o/Z_c Q_o|$ is a function of no less than eight dimensionless quantities.

$$\left| \frac{P_o}{Z_c Q_o} \right| = f_1 \left[\frac{Z_c C_a}{T_L}, \frac{T_2}{T_L}, \frac{T_3}{T_L}, Z_c K_P, \frac{\tau_P}{T_L}, \alpha T_L, \frac{Z_c}{R_o}, \omega T_L \right] \quad (66)$$

It seems reasonable that, if T_2/T_L and T_3/T_L are short enough, their influence could be neglected, especially at low frequencies. Thus

$$\left| \frac{P_o}{Z_c Q_o} \right| = f_2 \left[\frac{Z_c C_a}{T_L}, Z_c K_P, \frac{\tau_P}{T_L}, \alpha T_L, \frac{Z_c}{R_o}, \omega T_L \right]$$

The introduction of K_L and M in eqns. (54) and (55) (grouping four dimensionless quantities), provided a major breakthrough.

$$\left| \frac{P_o}{Z_c Q_o} \right| = f_3 \left[K_L, M; \frac{Z_c}{R_o}, \omega T_L \right]$$

If we use $\sigma = |P_o/Z_c Q_o|_{\max}$ as a design criterion the result is fundamentally reduced to

$$\sigma = \left| \frac{P_o}{Z_c Q_o} \right|_{\max} = f_4 \left[K_L, M, \frac{Z_c}{R_o} \right]$$

Since $R_o \neq \infty$ logically will moderate σ , the assumption $R_o = \infty$ is on the very safe side. Thus

$$\sigma = \left| \frac{P_o}{Z_c Q_o} \right|_{\max} < f_5 [K_L, M] \quad (67)$$

The strategy in this section could be characterized by the way in which the nine dimensionless quantities of eqn. (66) have been reduced to the three dimensionless quantities of eqn. (67).

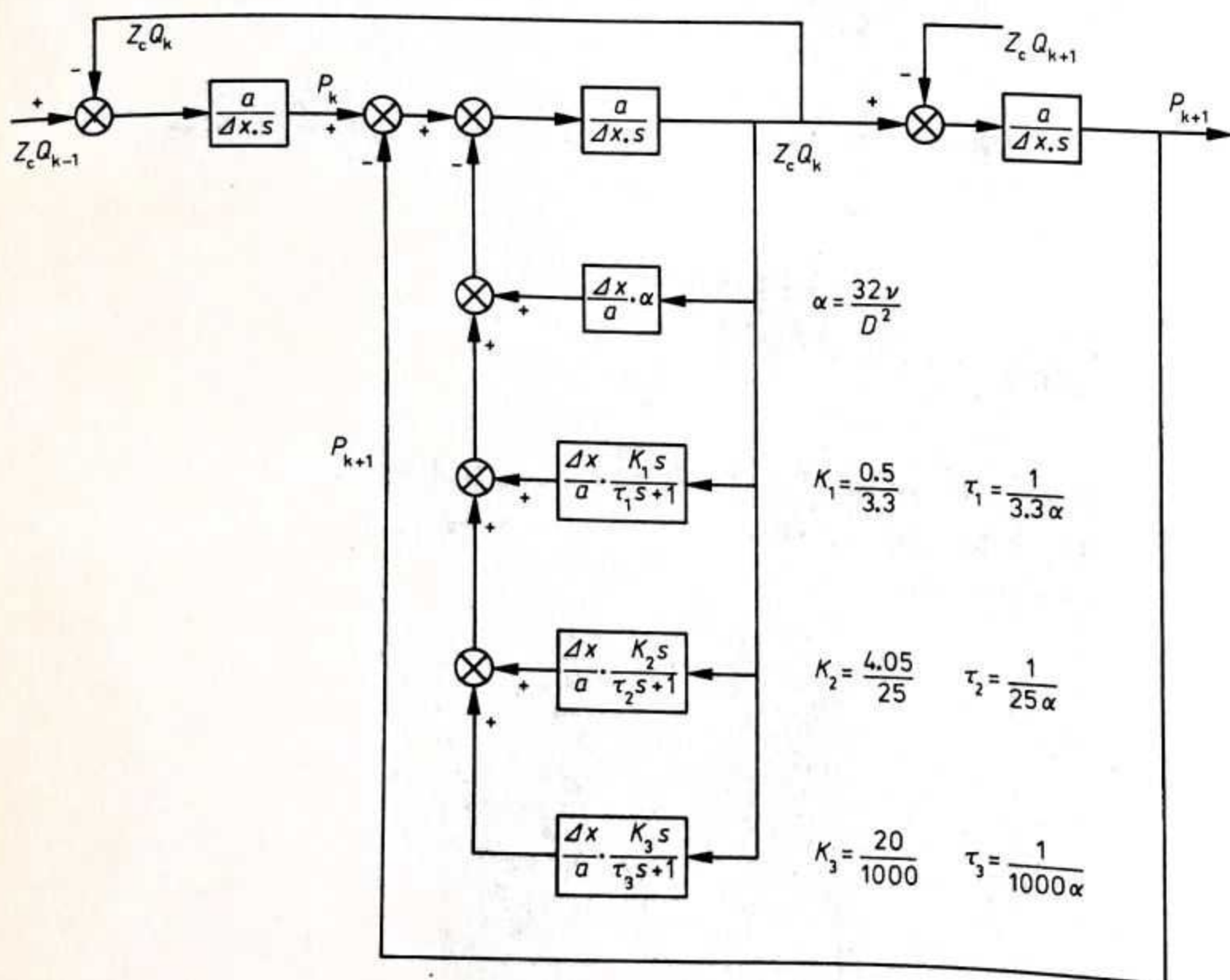
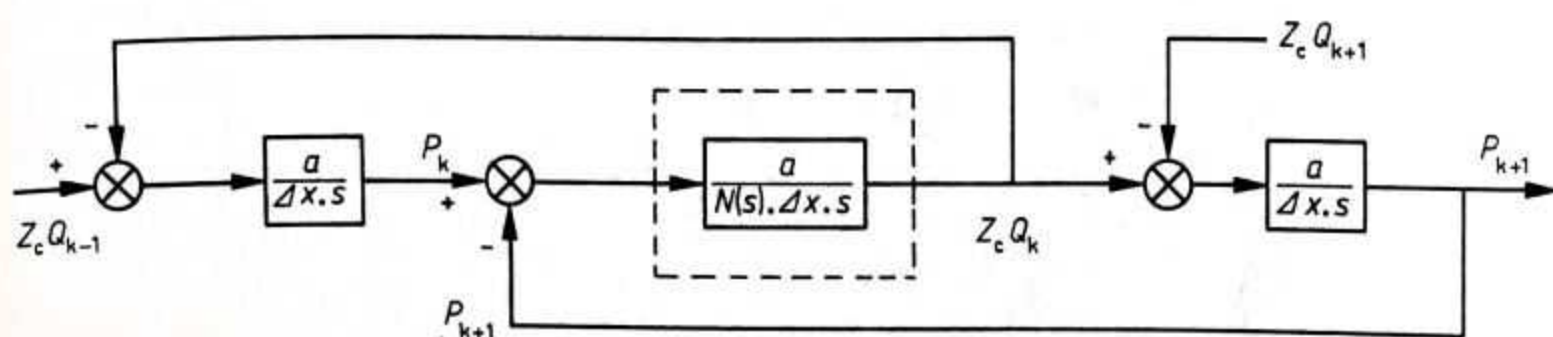
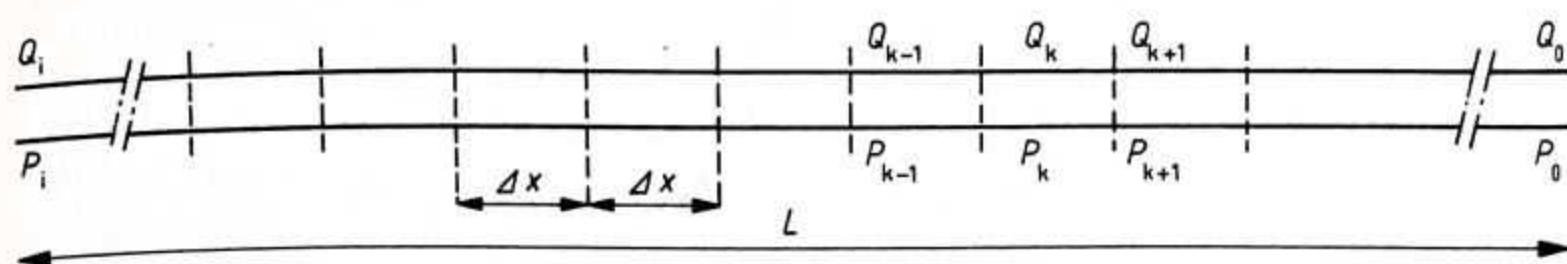


Fig. 6.31 Simulation.

6.7 Simulation

To determine pressures at both ends of hydraulic pipelines, the four-pole equations (10) proved to be very useful for analysis, synthesis and design. By applying the four-pole equations to an arbitrary part of the pipeline (including one end), pressures and flows can be found at any location in the pipeline.

One limitation is that the equations give solutions only in the frequency domain. This is justified by the fact that many disturbances (pump flow pulsations) have a periodic nature. Furthermore, there is the philosophy of control engineers that a "smooth" frequency characteristic implies a "decent" characteristic in the time domain.

Solutions in the time domain (for instance, sudden closing of servovalves leading to sharp pressure peaks) can be found theoretically by transforming solutions found in the frequency domain into the time domain. However, this procedure turns out to be extremely complicated. In particular, singularities (such as infinitely large pressures at very high frequencies) provide insurmountable complications.

A direct approach in the time domain starting with the Navier-Stokes equation and the continuity equation also provides serious problems, mainly related to the viscosity. Naturally, computers are indispensable in this approach, but very lengthy and very expensive programs have caused many scientists serious problems.

For this reason it is worthwhile to consider the use of the four-pole equations for (analogue) simulation. Our previous results allow considerable reduction of costs and programming. Therefore, we cut the pipeline into segments of length Δx and apply the four-pole equations to each of the segments (see Fig. 6.31(a)). The continuity equation (1) and the Navier-Stokes equation (3) can be presented in the form

$$\frac{s}{a} \cdot P(s) + Z_c \frac{dQ(s)}{dx} = 0$$

$$\frac{dP(s)}{dx} + \frac{s}{a} \cdot Z_c Q(s)N(s) = 0$$

Supposing for the elements $k - 1$, k and $k + 1$

$$\frac{dQ}{dx} = \frac{Q_k - Q_{k-1}}{\Delta x}$$

$$\frac{dP}{dx} = \frac{P_{k+1} - P_k}{\Delta x}$$

leads directly to

$$\left. \begin{aligned} P_k &= \frac{a}{\Delta x} \cdot \frac{Z_c (Q_{k-1} - Q_k)}{s} \\ Z_c Q_k &= \frac{a}{N(s)\Delta x} \cdot \frac{P_k - P_{k+1}}{s} \end{aligned} \right\}$$

Figure 6.31(b) shows the block diagram used to simulate one segment. When there are more segments, a series of the configurations depicted must be effected. A line split up into ten segments will require $10 \times 2 = 20$ integrators. A major problem of the simulation is the function $N(s)$. In Section 6.2.3 an optimal approximation (6e) was given, providing reliable responses up to $\omega \approx 300\alpha$.

$$N(s) = \frac{\alpha}{s} + 1 + \frac{K_1}{\tau_1 s + 1} + \frac{K_2}{\tau_2 s + 1} + \frac{K_3}{\tau_3 s + 1}$$

$$K_1 = \frac{0.5}{3.3} \quad K_2 = \frac{4.05}{25} \quad K_3 = \frac{20}{1000}$$

$$\tau_1 = \frac{1}{3.3\alpha} \quad \tau_2 = \frac{1}{25\alpha} \quad \tau_3 = \frac{1}{1000\alpha}$$

By transformation of the block indicated by dashes in Fig. 6.31 (b), a final scheme (c) has been obtained that is very well suited to simulation.

About this final block diagram (c) we make the following remarks:

- The length Δx of each segment should be sufficiently small to guarantee good simulations at higher frequencies. A rule of thumb is $\Delta x = \lambda/10$, where λ is the wave-length of the highest frequency to be relevant.
- The time scale could be adjusted by changing the sonic velocity a and the viscosity α proportional to the time scale factor.
- Hydraulic components like accumulators, tuned filters and pressure control valves can be implemented in the scheme by using their G-functions (see Fig. 6.16).
- An implication of the simulation is that at any moment and at any place pressures and flows can be found.

Scheme (c) can be used as a basis for analogue simulation, which is a very simple, direct and fast method, provided the capacity of the analogue computer is sufficient. But also hybrid simulation (computing digitally the blocks including the viscosity factor α) or even digital simulation is possible, based on scheme (c). The choice of the preferred method depends completely on the length and the complexity of the compound pipeline, the results desired and the computer facilities.

In the Mechanical Control Engineering Laboratory of Delft University of Technology only analogue and hybrid simulations have been carried out, until now mainly to check measurements in the frequency domain. But the simulations seem to be most attractive

for carrying out optimisation in the time domain. Another major advantage could be the investigation of the interaction between the hydraulic pipeline and the hydraulic servomotor.

REFERENCES

- [1] A.F. D'Souza and R. Oldenburger, Dynamic response of fluid lines, J. Basic Eng., Trans. ASME, Ser. D, 86(1964)589.
- [2] A.K. Trikha, An efficient method for simulating frequency-dependent friction in transient liquid flow, J. Fluid Eng., Ser. I, 97(1975)97.
- [3] J.F. Blackburn, G. Reethof and J.L. Shearer, Fluid Power Control, Wiley, New York, 1960, Chapters 3 and 5.
- [4] A.A. Ham, Influence of a pressure control valve on the dynamic behavior of hydraulic pipelines (in Dutch), Laboratory Report A 203 (1975), Delft University of Technology.
- [5] W. Herzog, Geräuschminderung in der Hydraulik, Oelhydraulik und Pneumatik, 20, Nr. 8(1976)515.
- [6] R. Gösele, Mechanismus der Geräuscentstehung in Hydrosystemen durch Förderstromschwankungen, Oelhydraulik und Pneumatik, 22, Nr. 8(1978)441.

Nomenclature

c = radial clearance (at $e = 0$ and $x = 0$)

d = nominal diameter of bearing

e = eccentricity

f = radial bearing force

h = film thickness

l = axial bearing length

p = pressure

$\Delta p = p_2 - p_1$ = pressure drop

q = leakage flow

$t = 0.5|d_2 - d_1|$ = taper

v = axial speed

x = axial coordinate

y = tangential coordinate

μ = dynamic viscosity

Dimensionless quantities

$E = e/c$

$F = f/0.25\pi d^2 (p_2 - p_1)$ refers to single bearings

$F^* = f/0.25\pi d^2 (\Delta p_I + \Delta p_{II})$ refers to double bearings

$\bar{F}$ = value of F at $E = 0.75$ and $V = 0$

$L = l/d$

$T = t/c$

$V = \mu dv/c^2 (p_2 - p_1)$ refers to single bearings

$V^* = \mu dv/c^2 (p_2 + p_1)$ refers to double bearings

V_c = value of V^* where cavitation might occur

$\bar{V}$ = value of V at $E = 0.75$ and $F = 0$

Subscript 1 refers to the narrow end (2 to the wide end) of bearing gap.

CHAPTER 7

THE DESIGN OF CONICAL HYDROSTATIC BEARINGS FOR HYDRAULIC CYLINDERS WITH CONSIDERABLE SPEED

Conical hydrostatic bearings in hydraulic cylinders simply consist of tapered pistons in cylindrical bores and tapered bushings around cylindrical piston rods. The required energy can be supplied by the hydraulic pressures already present.

Performance characteristics are given which show the influence of the axial piston speed. Rules for optimum design are presented and finally industrial applications with unequalled control accuracies are discussed.

Sections 1 to 8 of this chapter summarize very compactly the thesis of Blok [6].

The text was prepared in consultation with Dr. Blok.

Section 9 gives a design example. In the final sections, elaborate derivations of all the equations in Sections 2 and 4 have been given, as well as the approach for numerical solutions.

Chapter 7 can be read and studied separately from the previous chapters.

7.1 Introduction

Closed loop controlled electro-hydraulic actuators or cylinders are frequently used for accurate control of displacements, forces or continuous varying speeds (as applied in machine tools, hydraulic presses and flight simulators). An important factor, affecting the control accuracy, is Coulomb friction of piston and rod in its cylinder and bearing. Coulomb friction in hydraulic actuators easily reaches values between 5% and 15% of the maximum available load capacity. Hydrostatic bearings offer a major opportunity to improve control accuracy as they eliminate Coulomb friction almost completely. Many authors [1], [2], [3] recommend the so-called stepped bearings, which in their opinion have better characteristics and manufacturing qualities than conical bearings.

The arguments are doubtful. First of all, stepped bearings are much more sensitive to clearance variations than conical bearings. In industrial applications one has to consider that due to manufacturing tolerances the occurring clearance may differ considerably from a calculated value. More serious is the fact that the calculated performance of a stepped bearing, usually based on the assumption of one-dimensional flow, gives a far too optimistic idea of the maximum bearing force [4]. Also, optimizing hydrostatic bearings with respect to the ratio nondimensional stiffness to nondimensional leakage [1] is misleading in many cases. Implicitly, it suggests that

the real stiffness could be increased by increasing the leakage. Remarkably, industrial experience has never been mentioned to support the numerous statements that stepped bearings are more easily manufactured than conical bearings. In the Netherlands, more than 100 hydraulic servo cylinders provided with conical bearings have been manufactured without major problems. The majority of these cylinders is operating daily (in machine tools as well as in flight simulators), showing extremely low values of Coulomb friction.

7.2 Single bearings

Conical hydrostatic bearings provide centering forces if the greater of the two pressures, adjoining the bearing gap, prevails at the wide side of the gap. Figure 7.1 shows a very simple application to a hydraulic cylinder with two hydrostatic bearings. As long as $p_2 > p_1 > p_0$ both bearings provide centering forces and a permanent oil film can be maintained, eliminating metal to metal contact and Coulomb friction.

The Reynolds' equation for a two-dimensional laminar flow around a tapered piston in a perfect cylindrical bore (see Fig. 7.2) gives

$$\frac{\partial}{\partial x} \left[\frac{h^3}{12\mu} \frac{\partial p}{\partial x} - \frac{vh}{2} \right] + \frac{\partial}{\partial y} \left[\frac{h^3}{12\mu} \frac{\partial p}{\partial y} \right] = 0 \quad (1)$$

For the analysis we first use a simple axial flow model by neglecting $\partial p / \partial y$. Now eqn. (1) can be solved analytically. With $h = h_1 + tx/l$ and $h_2 = h_1 + t$, the pressure distribution along the piston becomes

$$p = p_1 + (p_2 - p_1) \frac{\frac{1}{h^2} - \frac{1}{h_1^2}}{\frac{1}{h_2^2} - \frac{1}{h_1^2}} + \frac{6\mu lv}{t} \frac{\left(\frac{1}{h} - \frac{1}{h_1}\right)\left(\frac{1}{h} - \frac{1}{h_2}\right)}{\frac{1}{h_2} + \frac{1}{h_1}} \quad (2)$$

Proper integration around the piston gives the lateral force f [5]. In a dimensionless form the result is

$$\frac{F}{L} = \frac{T}{E} \left[\frac{2 + T}{\sqrt{(2 + T)^2 - 4E^2}} - 1 \right] - \frac{24VL}{TE} \left[\frac{\sqrt{(1 + T)^2 - E^2} - \sqrt{1 - E^2}}{T} - \frac{2 + T}{\sqrt{(2 + T)^2 - 4E^2}} \right] \quad (3)$$

with $T = t/c$, $E = e/c$, $L = 1/d$, $F = f/0.25\pi d^2 (p_2 - p_1)$ and $V = \mu dv/c^2 (p_2 - p_1)$.

In Fig. 7.3(a) the characteristics for $T = 2$ are given. For $V = 0$ it shows a bearing force F continuously increasing with the eccentricity E . According to

Table 7.1 Summary of results for $E = 0.75$ and $p_1 = 0$

Bearing parameter		one-dimensional laminar flow			two-dimensional laminar flow			approximation	
								$\bar{V} \approx$ 0.15T/L	$V_c \approx$ 0.1T/L
L	T	$\bar{F}$	$\bar{V}$	V_c	$\bar{F}$	$\bar{V}$	V_c		
0.5	1	0.103	0.340	0.260	0.090	0.352	0.276	0.30	0.20
	2	0.105	0.611	0.422	0.095	0.622	0.435	0.60	0.40
	3	0.097	0.885	0.587	0.089	0.900	0.610	0.90	0.60
	4	0.087	1.160	0.752	0.082	1.180	0.797	1.20	0.80
	5	0.079	1.435	0.919	0.075	1.460	0.999	1.50	1.00
1.0	1	0.206	0.170	0.130	0.129	0.182	0.147	0.15	0.10
	2	0.210	0.306	0.211	0.149	0.317	0.224	0.30	0.20
	3	0.193	0.443	0.293	0.146	0.454	0.310	0.45	0.30
	4	0.175	0.580	0.376	0.138	0.600	0.404	0.60	0.40
	5	0.159	0.718	0.459	0.128	0.730	0.506	0.75	0.50
1.5	1	0.309	0.113	0.087	0.133	0.120	0.103	0.10	0.07
	2	0.315	0.204	0.141	0.165	0.210	0.152	0.20	0.13
	3	0.290	0.295	0.196	0.170	0.300	0.209	0.30	0.20
	4	0.262	0.387	0.251	0.165	0.393	0.271	0.40	0.27
	5	0.238	0.478	0.306	0.158	0.492	0.339	0.50	0.33

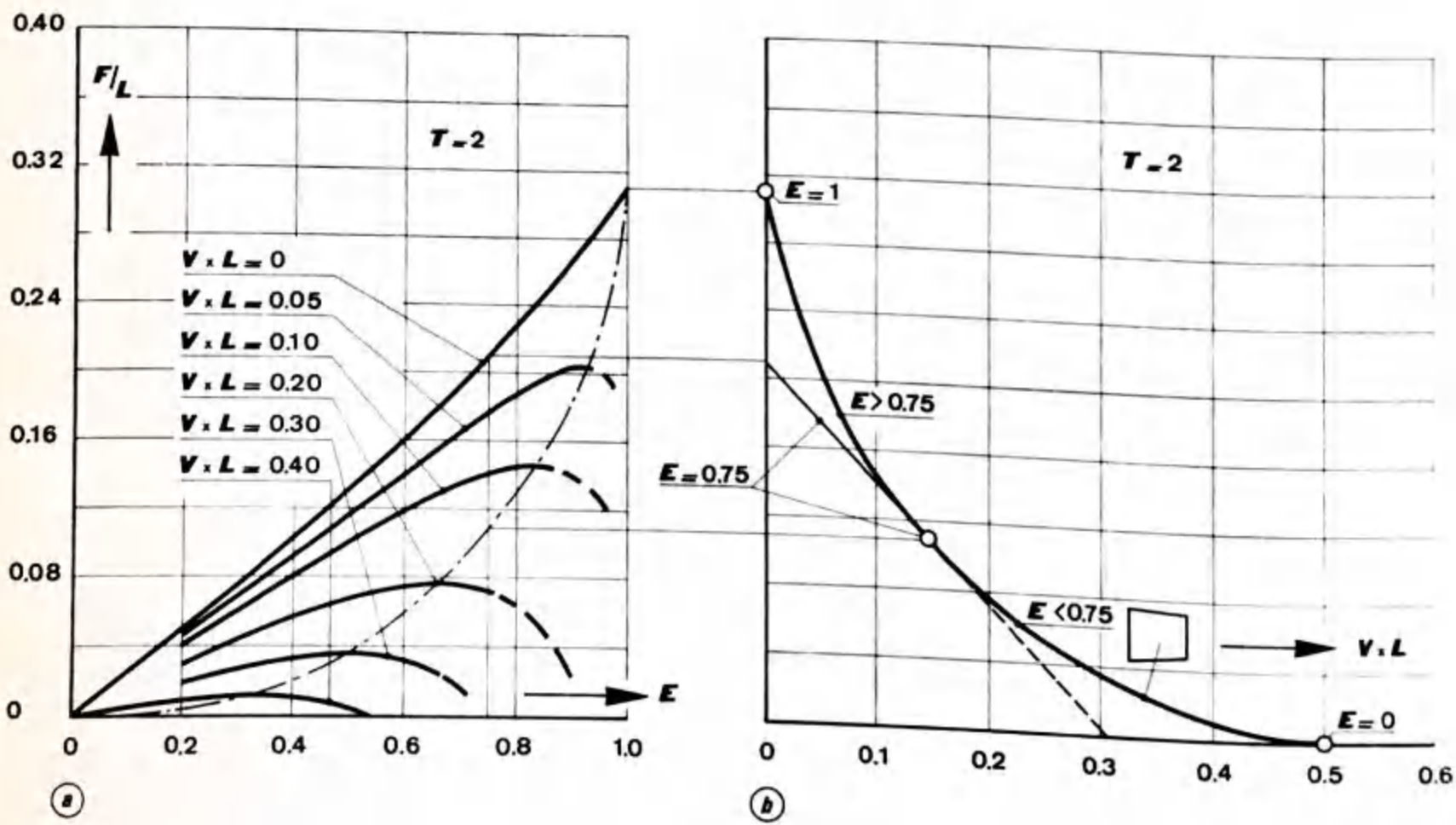


Fig. 7.3 Characteristics of single bearing for axial flow.
(a) Influence of eccentricity and speed on bearing force.
(b) Combination of maximum bearing force and speed.

eqn. (3), the bearing force F is reduced proportionally to the speed V (the positive sign of V is indicated in Figs. 7.1 and 7.2). For each speed V the corresponding curve has a maximum F at a certain value of the eccentricity E . For greater values of E the stiffness dF/dE becomes negative. For single bearings these right parts of the characteristics are meaningless as the corresponding bearing performance is unstable. The locus of the maxima (where $dF/dE = 0$) includes all the possible combinations of maximum bearing forces F and speed V .

The result is shown in Fig. 7.3(b). Equation (3) and Fig. 7.3(b) show that a conical bearing with $T = 2$ and $V = 0$ gives $F/L = 0.31$ at $E = 1$, while $T = 2$ and $F = 0$ gives $VL = 0.5$ at $E = 0$.

If for operational and practical reasons a maximum value of $E = \bar{E} = 0.75$ is allowed, the upper part of the curve in Fig. 7.3(b) has to be replaced by a straight line, tangent to the curve. The straight line follows directly from eqn. (3) for $E = 0.75$. The lower part of the curve is then still valid. But the straight line derived from eqn. (3) for $E = 0.75$, extended to the horizontal axis, gives a very simple and safe approximation of the permitted combinations of maximum F and V . For practical use, in designing conical bearings, the whole range can be covered satisfactorily by

$$\frac{F}{\bar{F}} + \frac{V}{\bar{V}} = 1 \quad (4)$$

$\bar{F}$ and $\bar{V}$ can be derived directly from eqn. (3) with $E = 0.75$. Also, it can be made clear that the length l should not be incorporated in the dimensionless bearing force F (as is done in most publications) nor in the dimensionless speed V .

The ratio $L = l/d$ is a free design parameter, which can be optimized. In Fig. 7.4 the influence of L on the F - V -characteristic is shown for $T = 2$ and $E = 0.75$. An optimal choice of L is possible here, because L is not incorporated in F and V . The above analysis for single bearings with axial laminar flow can be repeated for the more accurate model with two-dimensional flow, also starting from the Reynolds equation (1). A summary of the method is given in the appendix.

The numerical outcomes [6] qualitatively confirm the above results. Analysis of eqn. (1) yields a linear relation between F and V for $E = 0.75$, thus eqn. (4) can be used without hesitation.

The numerical results, summarized in Table 7.1, show that the values of $\bar{F}$ differ considerably from the axial flow model, especially at larger values of L . However, the values of $\bar{V}$ are hardly influenced and can be approximated remarkably well by

$$\bar{V} = 0.15 \frac{T}{L} \quad (5)$$

The procedure for optimizing L is not changed at all, as is shown in Fig. 7.4.

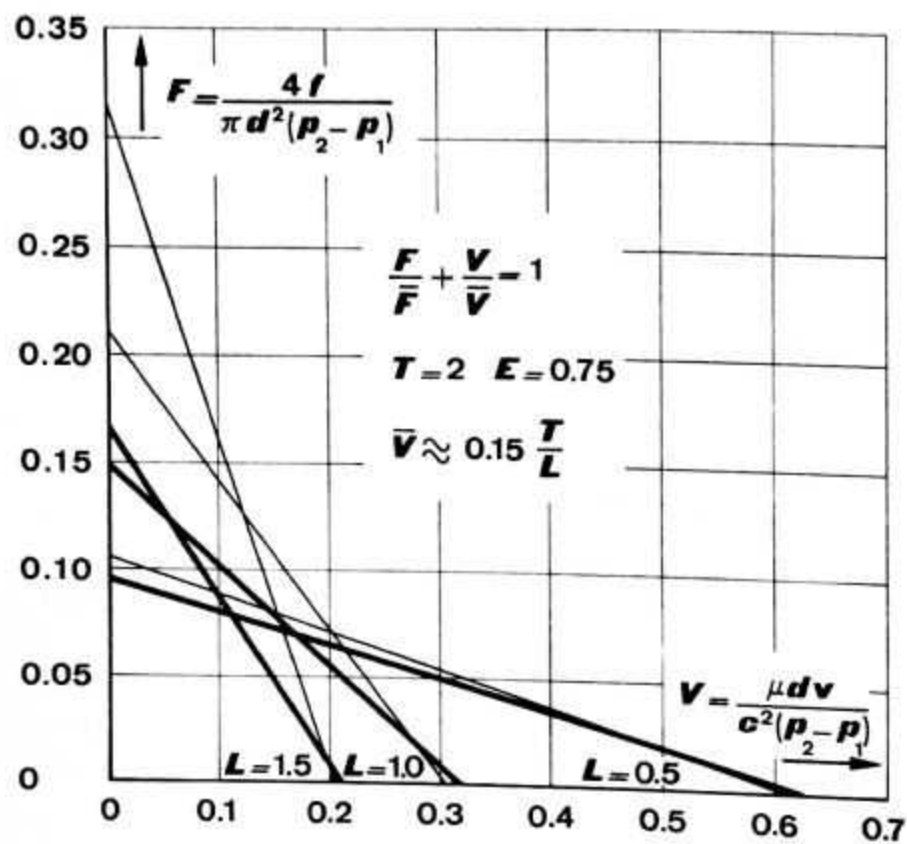


Fig. 7.4 Optimizing the length-to-diameter ratio $L = l/d$ with respect to the F - V -characteristic for $E = 0.75$. The influence of circumferential flow, especially on $\bar{F}$, can be seen by replacing the thin lines (one-dimensional axial flow) by the thick lines (two-dimensional flow). All data are taken from Table 7.1.

Applying hydrostatic bearings in a cylinder, a certain leakage flow must be accepted, otherwise the bearings will not work. Knowing the pressure distribution of eqn. (2) around the piston, the starting point for the calculation of the leakage flow is

$$dq = \frac{h^3}{12\mu} \frac{\partial p}{\partial x} dy.$$

In the axial flow model straightforward integration at a constant x yields

$$q = \frac{\pi c^3 \Delta p}{12\mu L} \left[\frac{2(1+T)^2}{2+T} + \frac{3}{4} (2+T)E^2 + \frac{T^4}{8(2+T)} \left\{ \frac{2+T}{\sqrt{(2+T)^2 - 4E^2}} - 1 \right\} \right]$$

which can be approximated by

$$q = \frac{\pi c^3 \Delta p}{12\mu} \frac{1 + 1.75T + 1.5(1 + 0.5T)E^2}{L} \quad (6)$$

As the leakage flow is always small in comparison with the displacement flow, the influence of the speed is omitted. Numerical calculations for the case of a two-dimensional flow show no significant differences [6].

When considering oil consumption and energy supply, it was found that in careful designs the leakage seldom exceeds a few percentages of the oil flow at maximum speed. In several cases the leakage flow was even smaller than the flow required for the pilot stage of a two-stage electro-hydraulic servovalve.

With regard to the piston position itself, the influence of the leakage must be considered, and compensated for if necessary. In closed loop servomechanisms every offset due to leakage flow is compensated for automatically by feedback and control action.

7.3 New designs as a breakthrough to applications

Applications of hydrostatic bearings in hydraulic cylinders based on the above design principles have strong limitations. Most important is that the bearings provide centering forces only when the greater of the two adjoining pressures permanently prevails at the wide side of the gap. Otherwise, a decentering force will occur. For this reason it will be difficult to apply a simple hydrostatic bearing at the piston of a symmetric cylinder with reversing loads. A symmetric piston (in a symmetric cylinder) with one single bearing provides a bearing force only for a one-directional load.

Another limitation is the speed. It has been shown that the speed directly influences the bearing force. Requiring a positive bearing force therefore limits the maximum speed. Both limitations can be overcome by double bearings which consist of two identical single bearings with opposite tapers, installed side by side.

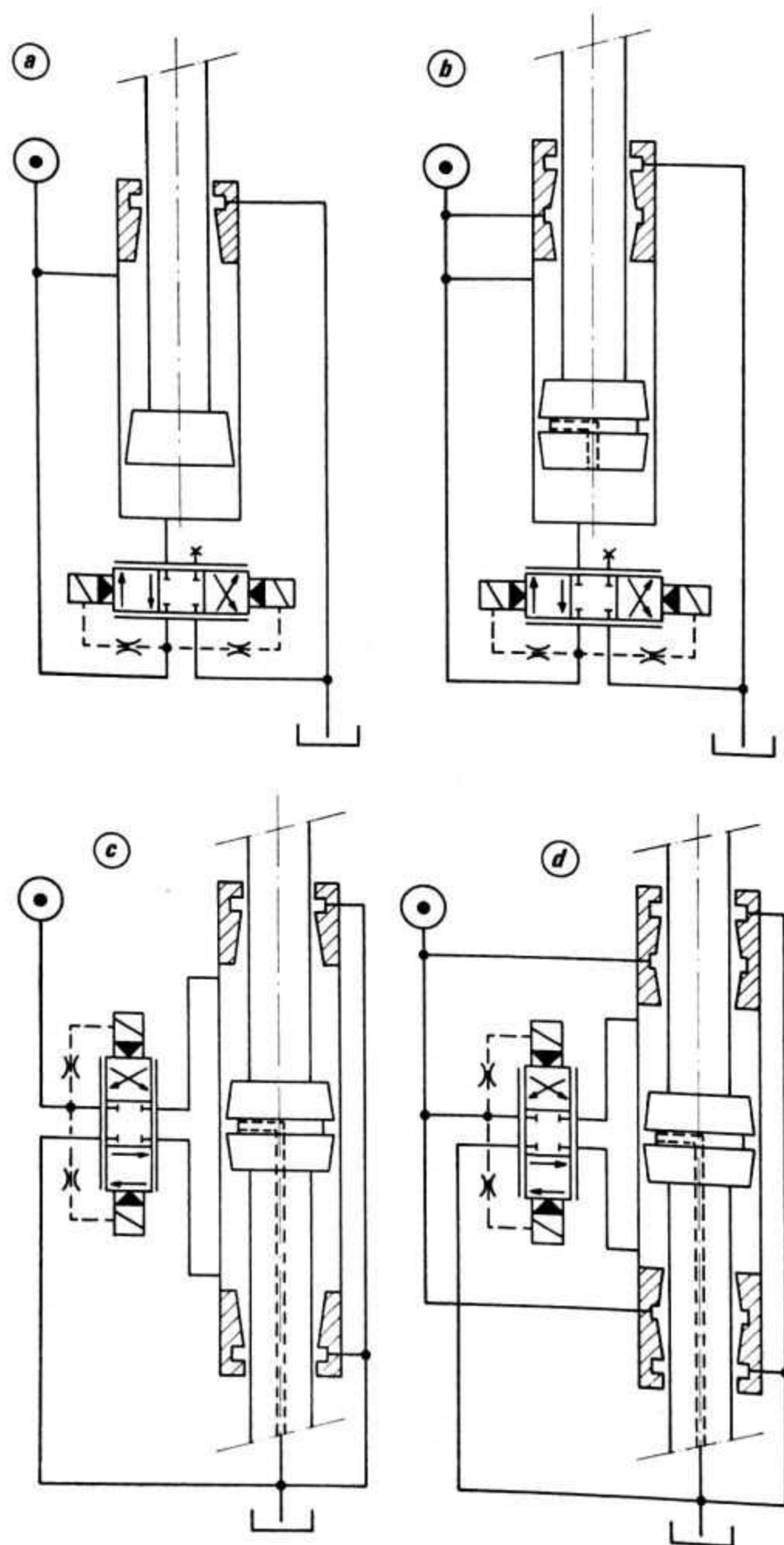


Fig. 7.5 New designs of hydraulic cylinders with conical hydrostatic bearings.
 Designs (b) and (d) are fully provided with double bearings.

Double bearings allowed a breakthrough in application of hydrostatic bearings in hydraulic cylinders [5], [6].

A series of basic design set-ups is shown in Fig. 7.5. Figure 7.5(a) shows the asymmetrical cylinder with the limitations mentioned above. Figure 7.5(b) also shows an asymmetrical cylinder, but now provided with double bearings. Each bearing has been designed symmetrically, the halves having opposite tapers. In this arrangement it is ascertained first of all that no "negative" pressure drops can occur. Thus a positive bearing force is guaranteed. Furthermore, the influence of the speed is eliminated, as the negative influence of the speed on the bearing force of one bearing half is cancelled by the positive influence of the same speed on the other bearing half.

Figure 7.5(c) shows a symmetrical cylinder with a double bearing at the piston. An essential feature is the connection to tank pressure from the slot between the two bearing halves to guarantee positive pressure drop across both bearing halves. In this original but simple way the piston of a symmetrical cylinder with reversing loads can effectively be provided with hydrostatic bearings.

In Fig. 7.5(c) the piston rods still have single bearings. In Fig. 7.5(d) double bearings have been introduced, increasing the maximum speed with positive bearing forces considerably. As can be expected, double bearings also have limitations. Therefore, further analysis is necessary.

7.4 Double bearings

In the axial flow model the total bearing force of a symmetric bearing can be found directly from eqn. (3)

$$f = \frac{TL}{E} \left[\frac{2 + T}{\sqrt{(2 + T)^2 - 4E^2}} - 1 \right] \frac{\pi}{4} d^2 (\Delta p_I + \Delta p_{II}) = F^* \frac{\pi}{4} d^2 (\Delta p_I + \Delta p_{II}) \quad (7)$$

Here Δp_I and Δp_{II} are the pressure drops across both bearing halves. Δp_I is not necessarily equal to Δp_{II} , as is shown in Fig. 7.5(b) where one of the pressure drops equals zero.

The speed V does not directly influence the bearing as the negative influence of the speed on the bearing force of one bearing half is cancelled by the positive influence of the speed on the other bearing half (see eqn. (3)). However, in an indirect way the speed does influence the maximum bearing force. Since negative pressures are physically impossible, it has tacitly been assumed in the calculations made so far that the pressure at any location is always positive. To verify the assumption of positive pressures, the minimum pressure $p_{\min}$ in a bearing with a pressure distribution conforming to eqn. (2) must be examined.

Putting $p_{\min} = 0$ yields, after laborious calculations

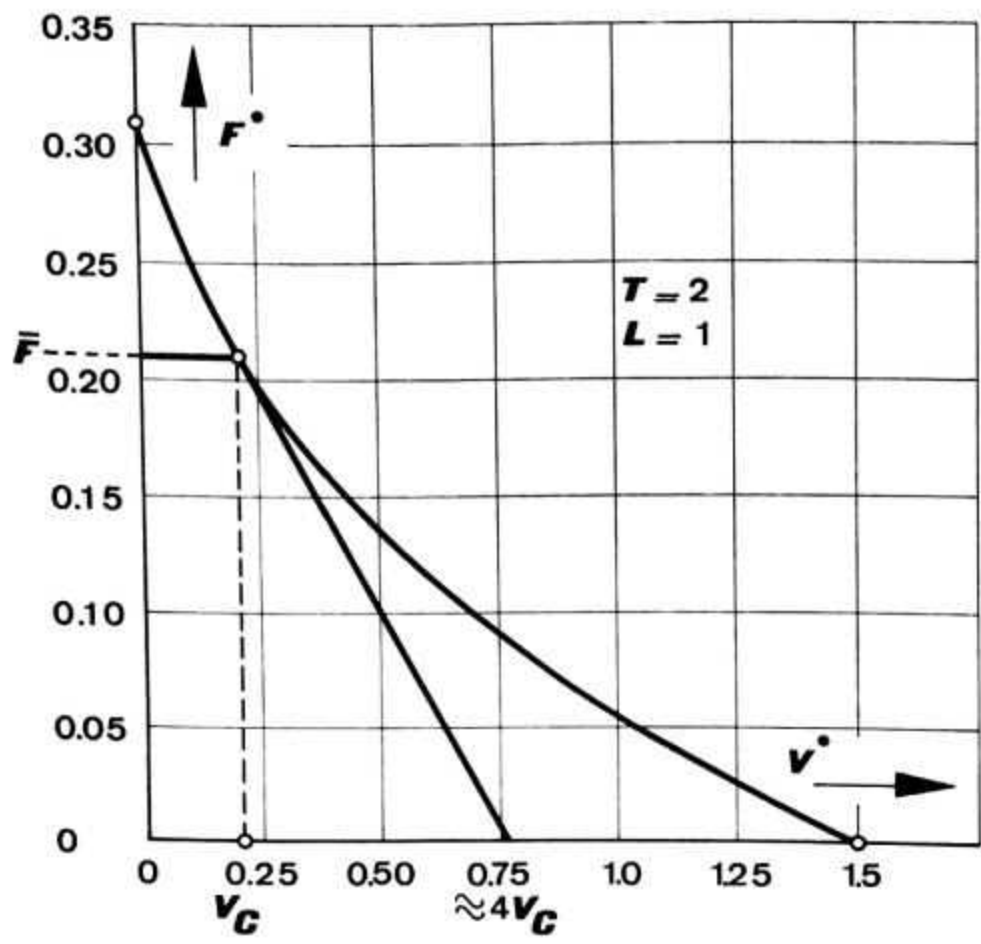


Fig. 7.6 F - V -characteristic of double bearing with one-dimensional flow.

Table 7.2 Values of the speed V^* where cavitation might occur

$E = 0.75,$		$L = 1$	$T = 1$	$T = 2$	$T = 3$	$T = 4$
$\frac{\mu dv}{c^2(p_2 + p_1)}$	eqn. (8)	$p_1 = 0$	0.13	0.21	0.29	0.38
	eqn. (8)	$p_1 = p_2$	0.16	0.23	0.32	0.40
$0.1 T/L$	eqn. (9)		0.10	0.20	0.30	0.40

$$\frac{\mu dv}{c^2} = \frac{(1 - E)(1 + T - E)^2}{3TL} \left[\Delta p + \frac{2 + T - 2E}{1 + T - E} \{p_1 + \sqrt{p_1^2 + p_1 \Delta p}\} \right]. \quad (8)$$

To guarantee the validity of the results - and to avoid any risk of cavitation - the speed of both bearing halves should be smaller than calculated here. Equation (8) shows that the risk of cavitation increases with E . Again allowing $E = 0.75$, one can approximate eqn. (8) by

$$V_c = \frac{\mu dv}{c^2(p_2 + p_1)} \approx 0.1 \frac{T}{L} \quad (9)$$

as is shown in Table 7.2, where in two extreme cases ($p_1 = 0$ and $p_1 = p_2$) the calculated results are compared with the approximation.

Obviously, for double bearings it is very practical to introduce a dimensionless speed $V^* = \mu dv / c^2(p_2 + p_1)$. Otherwise, $V = \mu dv / c^2(p_2 - p_1)$ as well as p_2/p_1 should be used, or a combination of both.

Summarizing so far, it can be seen that eqn. (7) holds for $E = 0.75$ as long as $V^* < V_c = 0.1 T/L$. For $V^* > V_c$ the bearing force as a function of the speed can be found by eliminating E ($E < 0.75$) from eqns. (7) and (8). For $L = 1$, $T = 2$ and $p_1 = 0$ the result is shown in Fig. 7.6, One can simplify the complete characteristic to

$$f = \frac{\pi}{4} d^2 (\Delta p_I + \Delta p_{II}) F^* \quad (10)$$

$$V^* = \frac{\mu dv}{c^2(p_2 + p_1)} < 0.1 \frac{T}{L}, \quad \text{then} \quad F^* = \bar{F} \quad (11)$$

$$V^* = \frac{\mu dv}{c^2(p_2 + p_1)} > 0.1 \frac{T}{L}, \quad \text{then} \quad F^* = \frac{\bar{F}}{3} \left(4 - 10 \frac{LV^*}{T} \right) \quad (12)$$

$\bar{F}$ can be found in Table 7.1, in which the values of V^* where cavitation might occur have been calculated for the two-dimensional flow model also. The differences from the one-dimensional flow model are relatively small. However, it should be realized that in designing a double bearing in practice, $\mu dv / c^2(p_2 + p_1)$ cannot be estimated beforehand with an accuracy much better than 50%.

7.5 Manufacturing

The tapers on pistons and in bearing bushings have been manufactured only by grinding on universal grinding machines. Try-outs on turning lathes failed owing to irregularities in the surface (stick-slip-motion!). The workpieces are first

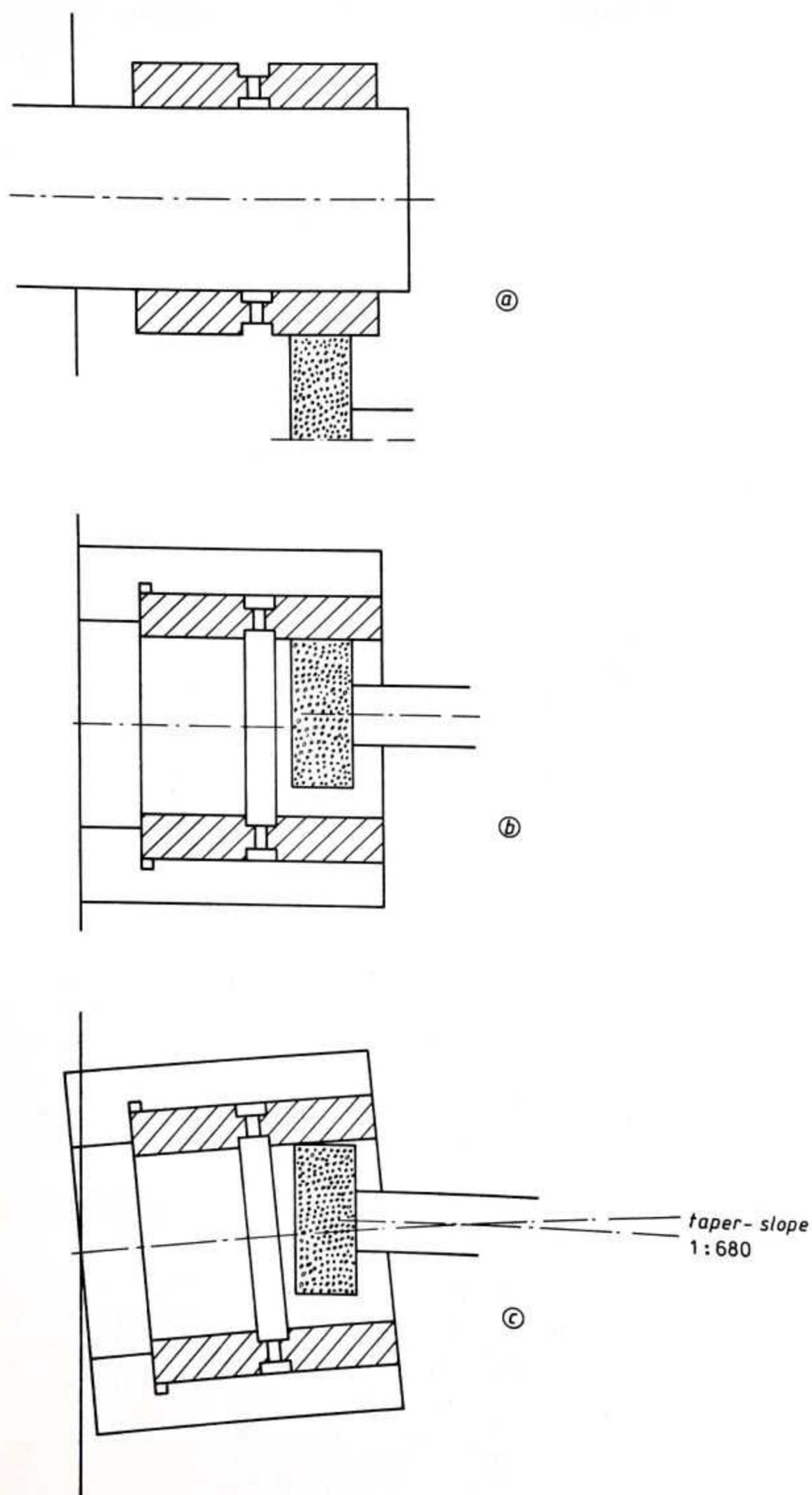


Fig. 7.7 The three most important stages (schematically) of the grinding of a piston rod double bearing. Finish: (a) outer diameter; (b) smallest inner diameter; (c) one cone.

ground cylindrically until the diameter corresponding to the narrowest place in the gap is correct. By adjusting the axis of the workpiece to the desired angle with the guidance of the grinding wheel, the correct taper dimensions can easily be obtained.

To ensure that the original diameter at one side of the bearing is maintained during grinding, the workpiece has to be dyed beforehand. Grinding (under the adjusted angle) must be continued until a coloured ring of about 1 mm is left on the surface [6]. Then the workpiece is finished. This simple and safe manufacturing procedure results in a bearing which is not conically shaped over its whole length, but is cylindrical over a very small area at the narrowest side of the gap.

By way of illustration, Fig. 7.7 shows some important stages of the manufacturing of a double bearing.

7.6 Applications

Conical hydrostatic bearings in electro-hydraulic servo cylinders have been applied in the Netherlands since 1968. In the hydraulic research group of the Mechanical Control Engineering Laboratory of Delft University of Technology work has been done systematically on challenging projects to prove the validity of theories, of experiments and of developments in the laboratory. The projects carried out professionally can be divided roughly into the following groups:

- (a) machine tool slide actuators,
- (b) hydraulic presses and force actuators,
- (c) flight simulators,
- (d) special medical equipment.

Most projects have been discussed by Blok [6], giving design considerations and detailed drawings.

For a profile shaper and a lathe, copying systems have been developed; for numerical control, a linear electro-hydraulic stepping motor has been designed [7]. Owing to Coulomb friction, conventional hydraulic follow-up systems show a considerable dead zone (quasi-stationary behaving like backlash), which affects the control accuracy. A combination of feedforward load compensation [8] and friction reduction by hydrostatic bearings reduced the dead zone from usual values of 10-30 μm to less than 1 μm . Control errors during metal cutting were less than 10 μm , which is quite exceptional.

The most spectacular hydraulic press developed in the laboratory is a three-axis 3,5 MN (350 tons) press to examine mechanical properties of rocks for mining purposes. Each axis can be controlled independently. A load sweep up and down over the full range can be carried out within 2 seconds. Very remarkable in this press is the absence of spurious force transducers (load cells). As there is scarcely any Coulomb friction (less than 21 N), the load can be measured by small pressure transducers. During operation the deviations from linearity of the pressure-to-force characteristic

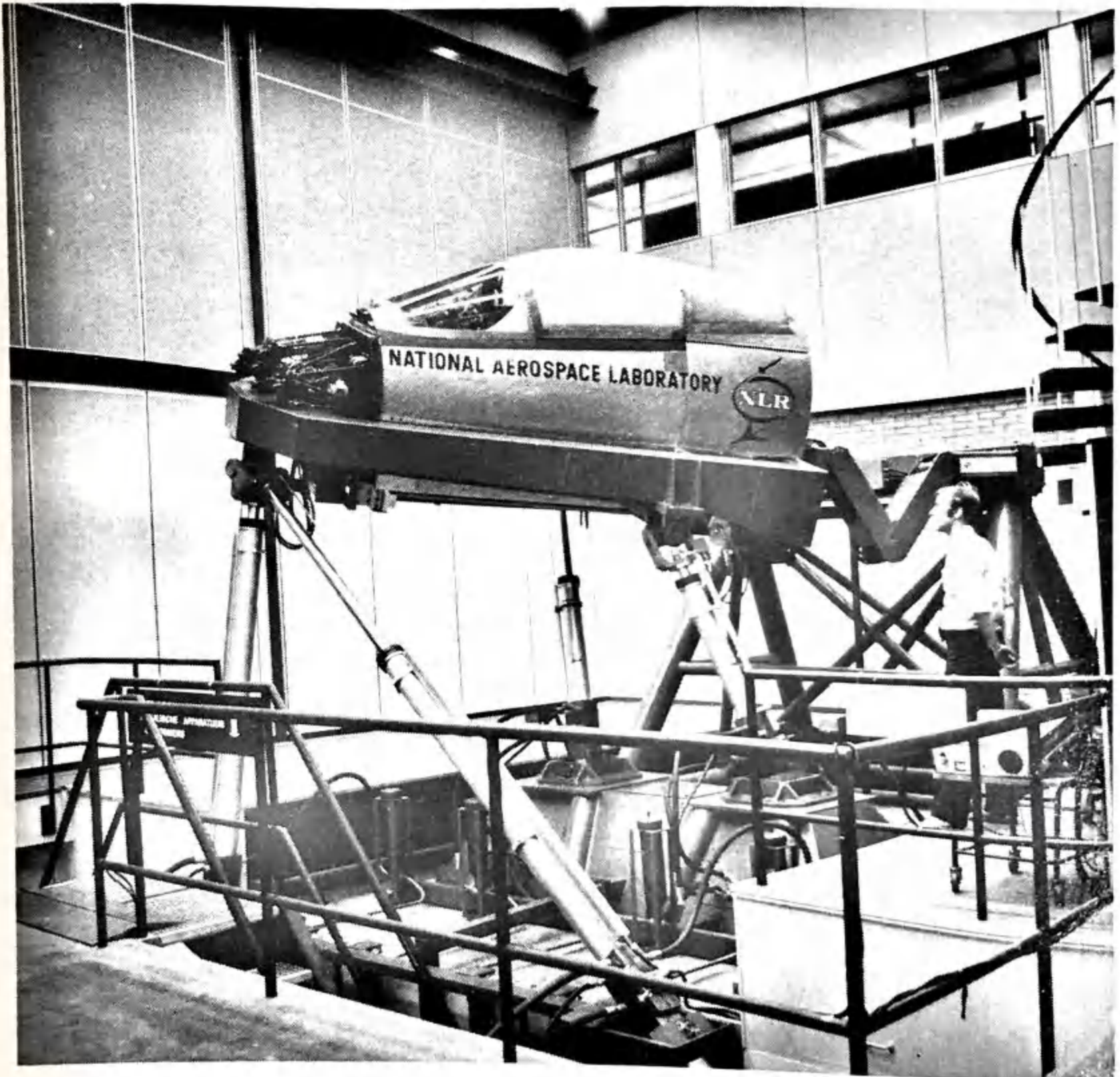


Fig. 7.8 Flight simulator of National Aerospace Laboratory in the Netherlands. The development of special servo cylinders with conical hydrostatic bearings resulted in a simulator performance with acceleration disturbance lower than 0.1 m/s^2 (0.01 g). Copyright: N.L.R. - Amsterdam.

are less than 0.25%.

For the Department of Aeronautical Engineering of Delft University of Technology and for the National Aerospace Laboratory in Amsterdam two flight simulators have been developed, with three and four axes of freedom respectively (see Fig. 7.8). An important complaint of "consumers" of conventional flight simulators is the acceleration disturbance level during smooth simulator movements, reaching values up to 1 m/s^2 (0.1 g). An important source of these disturbances is the Coulomb friction. The development of special servo cylinders with hydrostatic bearings resulted in a simulator performance with disturbances less than 0.1 m/s^2 (0.01 g) during low-frequency oscillations [9]. This is below the human perception level. Thus a practising pilot cannot notice smooth simulator movements up and down. In conventional flight simulators the disturbing accelerations, especially around the extreme positions, always mark the ups and downs in an unnatural way.

7.7 Conclusions and summary

The mathematical model of conical hydrostatic bearings has been developed and condensed so far that it has been possible to derive simple, safe and practical design rules in which the effect of occurring piston speeds is considered.

For values of the length-to-diameter ratio $L = l/d$ ranging from 0.5 to 1.0 a specific taper $T = t/c \approx 2$ can be recommended. Together with the data of Table 7.1 the eqns. (4), (5) and (6) for single bearings, and eqns. (10), (11) and (12) for double bearings provide the information necessary for an optimal design. Bearing forces up to 2000 N can be readily obtained. Since conical bearings can be manufactured with sufficient accuracy, there is an ever-increasing number of successful applications under industrial conditions.

7.8 Appendix

Assuming two-dimensional laminar flow and parallel axes of the bearing components, the Reynolds' equation has been transformed to a difference equation and solved by an iterative relaxation with 80% overrelaxation. The bearing surface therefore has been split up into small rectangles: 40 in both axial and circumferential direction. For reasons of symmetry, only half the bearing surface has been covered. In the symmetry plane itself, $\partial p / \partial y = 0$ is a boundary condition. In the axial direction, $p = p_1$ and $p = p_2$ provide the remaining boundary conditions.

Similar calculations (with $V = 0$) were carried out earlier by Mannam [10], [11], supported by very good corresponding measurements. Since these results differ from each other considerably, at least calculations and measurements of one of these publications must be wrong. Particularly the results of the latter publication seem very doubtful, since the results do not satisfy the requirement that the

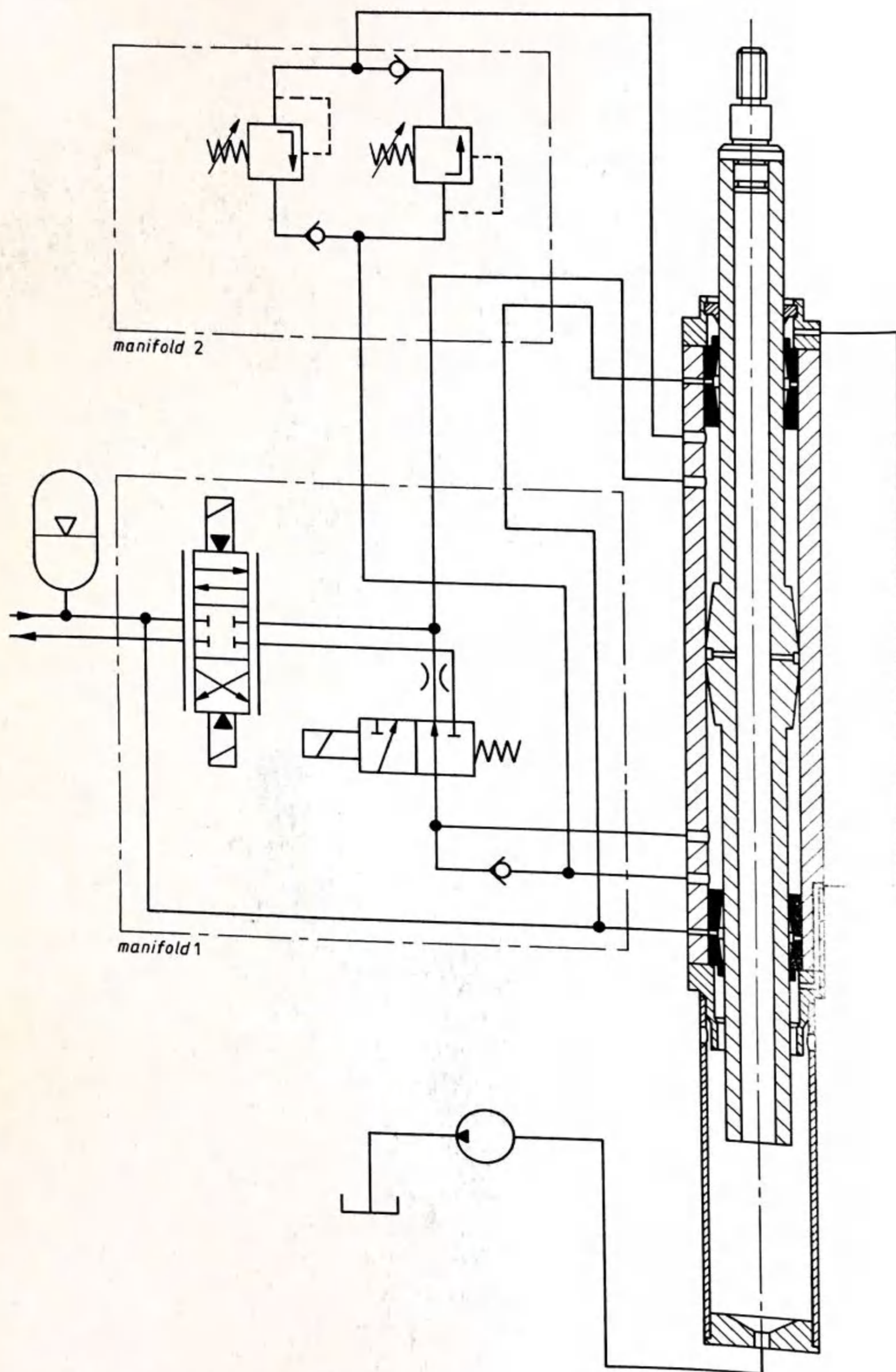


Fig. 7.9 Hydraulic diagram of a servomotor for a flight simulator.

solutions of the two-dimensional flow model for $L = 1/d$ approaching zero must approach the solutions of the one-dimensional flow model. For these reasons alone it is not surprising that Mannam's results differ considerably from the results presented above.

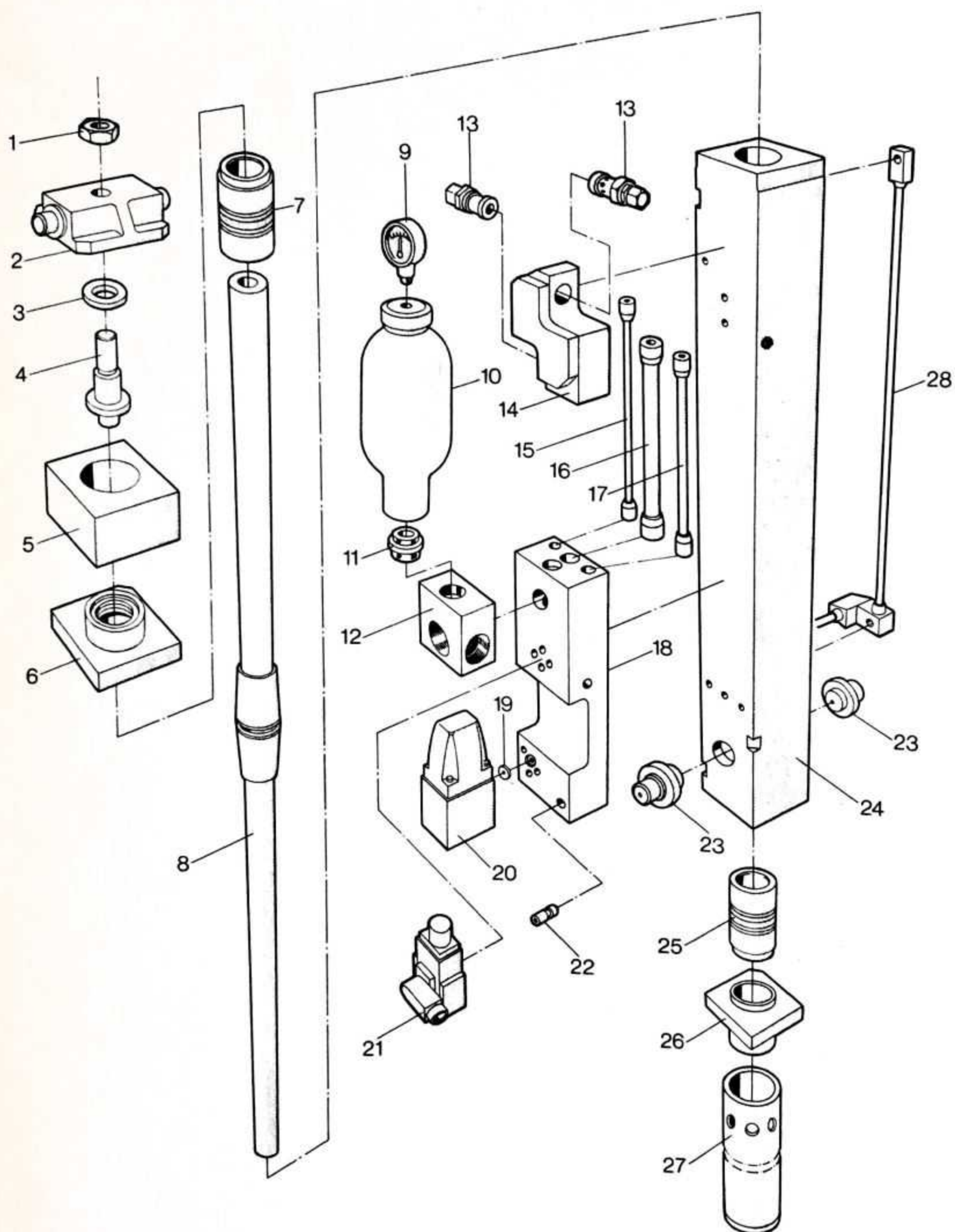


Fig. 7.10 Exploded view of the servomotor of Fig. 7.9.

7.9 Design example

In order to incorporate hydrostatic bearings in hydraulic servomotors a very careful design is necessary, and some traditional conventions are inevitably upset. It should be stressed that a consensus concerning the need for improved motion quality, and consequently for hydrostatic bearings is very important. Otherwise, any obstacle could disturb the network of cooperation that is needed for the introduction.

In some interesting cases the *users* of hydraulic servosystems forced hesitating or reluctant *manufacturers* to introduce hydrostatic bearings in order to get a better system performance. For instance, some major European airlines were so interested in improved motion quality of flight simulators, that one after another the manufacturers yielded to the pressure, mainly to strengthen their position in relation to the competitors. This process was started at Delft University of Technology, where a very simple three-axes-of-freedom flight simulator was developed that showed a very remarkable smoothness of motion of the system without the so-called reversal bump. After the introduction of a flight simulator with hydrostatic bearings, at the National Aerospace Laboratory in Amsterdam (see Fig. 7.8), the attention of European airlines was attracted more and more, leading to the demand described above.

In the first applications, no double bearings were applied, since the need for it was not yet recognized. Recently the "old" simulator in Delft was given a new (and heavier) cockpit, and consequently new servomotors were needed. Naturally, in the new design double bearings were introduced.

In Fig. 7.9 the hydraulic diagram for each of the three (symmetric) servo-systems is depicted. At both ends of the stroke a buffer zone is present to prevent excessive accelerations in order to protect both the pilot and the construction.

In Fig. 7.10 an exploded view of one motor is depicted, showing manifolds, valves and important details of this particular design. The design of the servosystem itself was based on the rules outlined in Chapter 5. The result was a supply pressure $p_s = 210$ bar (3000 psi) and a piston area $A = 11.5 \text{ cm}^2$, obtained by diameters of 67 and 55 mm of the cylinder and the piston rod, respectively. To avoid geometrical imperfections in the bearings, the piston and the piston rod are made in one piece, while the bore of the cylinder is without diameter changes, allowing the end bearings (around the piston rod) to be installed perfectly in line. To ensure correct geometry of the end bearings, the manufacturer should follow strictly the indications of Fig. 7.7.

For the calculation and design of the bearings the following data are important:

- o piston: pressure drop $p = p_c = \frac{1}{2}p_s = 105$ bar across both bearing halves,
- o piston rod: pressure drops $p = p_2 - p_c = 105$ bar and $p = p_s = 210$ bar,
- o maximum speed: $v_{\max} = 1 \text{ m/s}$,
- o oil: Tellus 27 with $\eta = 30 \text{ cSt} = 3 \cdot 10^{-5} \text{ m}^2/\text{s}$, $\mu = 0.027 \text{ Ns/m}^2$.

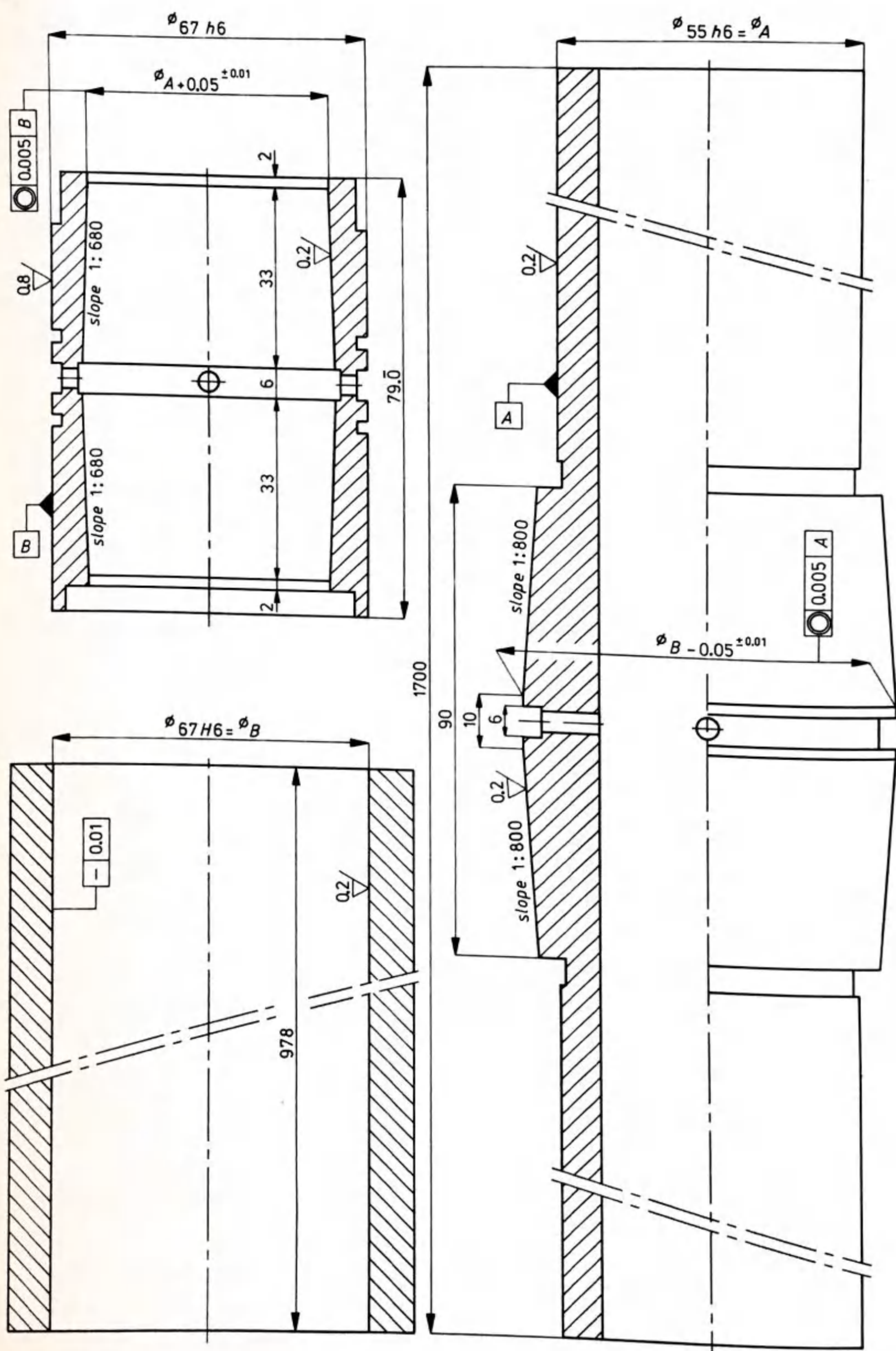


Fig. 7.11 Measures of the bearings of the servomotor of Fig. 7.9.

Since the design of the moving base is such that no large side-loads will occur, the length ratio chosen can be as small as $L = 1/d = 0.6$. A taper ratio $T = 2$ is always the first choice, unless the speed makes a larger value desirable. The clearance c chosen should be as small as normal tolerance field combinations permit, in order to avoid excessive flows. Choosing one diameter as a reference for the other - as is done in Fig. 7.11 - gives optimal results. However, manufacturing facilities and experiences as well as the resulting costs, might change the above recommendation.

The design and the calculations are summarized in Fig. 7.11 and Table 7.3. Together they permit a quick survey and a critical examination.

Table 7.3 Dimensions of bearings. Calculation of performances

Bearing data	piston	piston rod	units
Diameter d_1 , to be measured	67 H 6 = ϕ B	55 h 6 = ϕ A	mm
Length l	40	33	mm
$L = 1/d$	0.6	0.6	
Diameter d_2 , giving c	ϕ B - 0.05 ± 0.01	ϕ A + 0.05 ± 0.01	mm
$c = 0.5 d_1 - d_2 $	0.025 ± 0.005	0.025 ± 0.005	mm
Taper-slope t/l	1 : 800	1 : 680	
Taper t	$40/800 = 0.05$	$34/680 = 0.05$	mm
$T = t/c$	$2 \pm 20\%$	$2 \pm 20\%$	
$\bar{F}$ (see Table 7.1)	0.11	0.11	bar
$\Delta p_I + \Delta p_{II}$	105 + 105	105 + 210	N
$0.25\pi d^2 (\Delta p_I + \Delta p_{II})$	74000	74800	N
f (at $V^* < V_c$)	8100	8200	
$V_c = 0.1 T/L$	0.33	0.33	m/s
$v_{\max}$	1	1	Ns/m ²
μ (Tellus 27)	0.027	0.027	
$(p_2 + p_1)_{\min}$	105	210	bar
$V^* = \mu dv/c^2 (p_1 + p_2)$	0.28	0.11	
$q = \frac{\pi c^3 (\Delta p_I + \Delta p_{II})}{12\mu} \cdot \frac{1 + 1.75T}{L}$	12 + 12	12 + 24	cm ³ /s

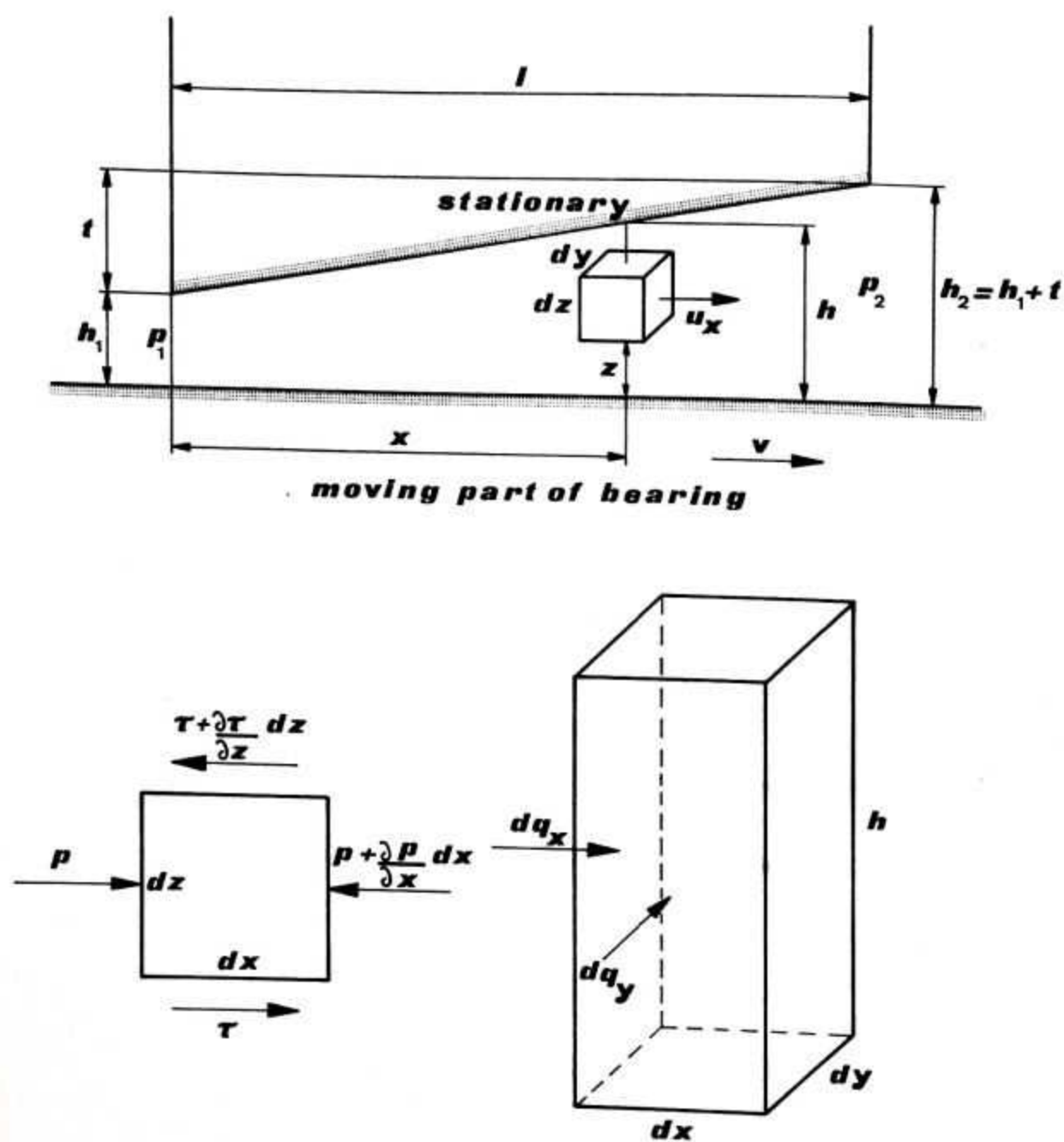


Fig. 7.12 Part of the gap.

*7.10 Derivation of Reynolds' equation (1)

Assuming incompressible laminar flow in a gap, equilibrium of forces on a fluid particle in Fig. 7.12 gives

$$\frac{\partial p}{\partial x} dx \cdot dy \cdot dz = - \frac{\partial \tau}{\partial z} dz \cdot dx \cdot dy, \quad \text{so} \quad \frac{\partial p}{\partial x} = - \frac{\partial \tau}{\partial z}.$$

Laminar flow implies $\tau = -\mu \partial u_x / \partial z$. Combination leads to the *Navier-Stokes* equation

$$\mu \frac{\partial^2 u_x}{\partial z^2} = \frac{\partial p}{\partial x}.$$

Integrating twice gives

$$\mu u_x = 0.5 \frac{\partial p}{\partial x} z^2 + c_1 z + c_2.$$

Using the boundary conditions $u_x = v$ at $z = 0$ and $u_x = 0$ at $z = h$, the constants c_1 and c_2 can be solved.

$$u_x = \frac{1}{2\mu} \cdot \frac{\partial p}{\partial x} (z^2 - zh) + v \left(1 - \frac{z}{h}\right).$$

The axial flow through rectangle $h dy$ is

$$dq_x = \int_0^h u_x dz dy = dy \int_0^h u_x dz = \left[-\frac{h^3}{12\mu} \cdot \frac{\partial p}{\partial x} + \frac{vh}{2} \right] dy.$$

Similarly:

$$dq_y = \left[-\frac{h^3}{12\mu} \cdot \frac{\partial p}{\partial y} \right] dx.$$

Substitution into the continuity-equation for incompressible flow

$$\frac{\partial dq_x}{\partial x} \cdot dx + \frac{\partial dq_y}{\partial y} \cdot dy = 0$$

results in the *Reynolds' equation*

$$\frac{\partial}{\partial x} \left[\frac{h^3}{12\mu} \cdot \frac{\partial p}{\partial x} - \frac{vh}{2} \right] + \frac{\partial}{\partial y} \left[\frac{h^3}{12\mu} \cdot \frac{\partial p}{\partial y} \right] = 0. \quad (1)$$

*7.11 Derivation of pressure distribution, eqn. (2)

Assuming two-dimensional flow, eqn. (1) can be transformed into a *difference-equation*, which can be solved by *iterative relaxation* (see Section 7.15).

In order to gain insight, we first approximate by assuming one-dimensional flow, where $\partial p / \partial y = 0$, $dq_y = 0$, $\partial p / \partial x = dp/dx$ and

$$\frac{dq_x}{dy} = -\frac{h^3}{12\mu} \cdot \frac{dp}{dx} + \frac{vh}{2} = \text{constant.}$$

Thus

$$dp = -12\mu \cdot \frac{dq_x}{dy} \cdot \frac{dx}{h^3} + 6\mu v \cdot \frac{dx}{h^2}.$$

In the gap according to Fig. 7.12, we have

$$h = h_1 + \frac{x}{l} \cdot t, \quad \text{so} \quad dx = \frac{l}{t} \cdot dh.$$

Substitution yields

$$dp = -\frac{12\mu l}{t} \cdot \frac{dq_x}{dy} \cdot \frac{dh}{h^3} + \frac{6\mu vl}{t} \cdot \frac{dh}{h^2}.$$

Integration of dp between h_1 and h yields

$$p - p_1 = \frac{6\mu l}{t} \cdot \frac{dq_x}{dy} \left[\frac{1}{h^2} - \frac{1}{h_1^2} \right] - \frac{6\mu vl}{t} \left[\frac{1}{h} - \frac{1}{h_1} \right]. \quad (13)$$

From this dq_x/dy can be solved as $p = p_2$ at $h = h_1$.

$$\frac{dq_x}{dy} = \frac{p_2 - p_1 + \frac{6\mu vl}{t} \left[\frac{1}{h_2} - \frac{1}{h_1} \right]}{\frac{6\mu l}{t} \left[\frac{1}{h_2^2} - \frac{1}{h_1^2} \right]}. \quad (14)$$

Substitution of eqn. (14) into eqn. (13) yields the pressure distribution

$$p = p_1 + (p_2 - p_1) \cdot \frac{\frac{1}{h^2} - \frac{1}{h_1^2}}{\frac{1}{h_2^2} - \frac{1}{h_1^2}} + \frac{6\mu vl}{t} \cdot \frac{\left[\frac{1}{h} - \frac{1}{h_1} \right] \left[\frac{1}{h} - \frac{1}{h_2} \right]}{\frac{1}{h_2} + \frac{1}{h_1}}. \quad (2)$$

*7.12 Derivation of bearing force, eqn. (3)

The lateral force df in z -direction for a gap with length l and width dy can be found by integration

$$df = \int_0^l p dx dy = dy \int_0^l p dx = \frac{l}{t} dy \int_{h_1}^{h_2} p dh$$

$$\frac{df}{l dy} = \frac{1}{t} \int_{h_1}^{h_2} p dh$$

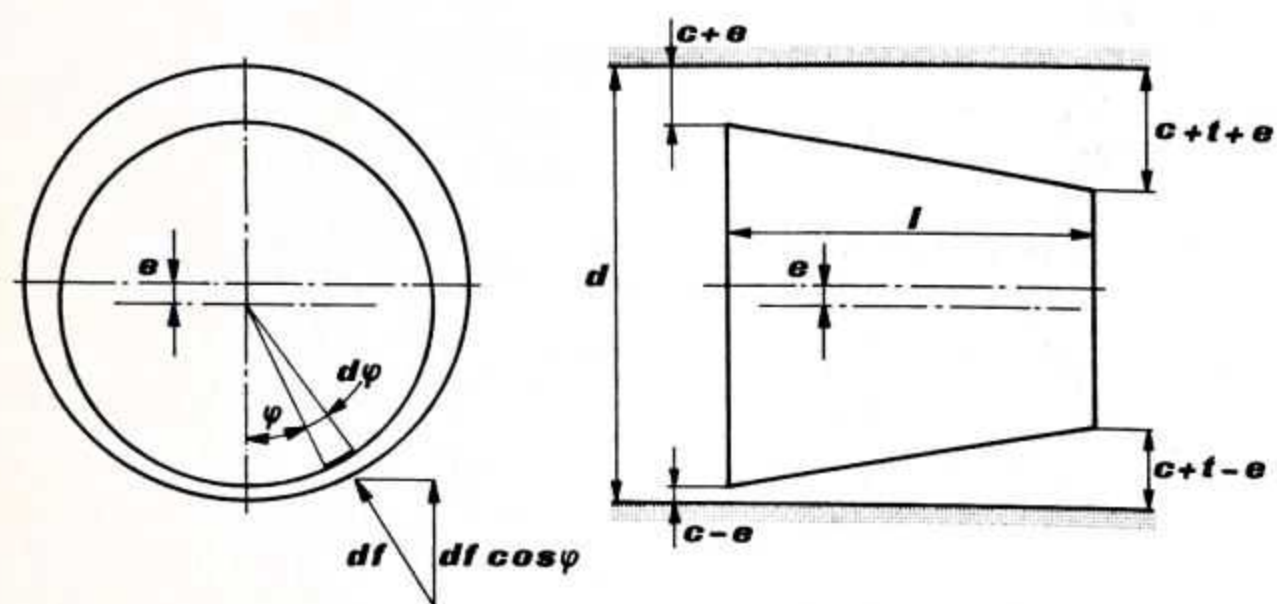


Fig. 7.13 Geometry of the bearing.

Substitution of eqn. (2) and working out gives

$$\frac{df}{dy} = p_1 + \frac{p_2 - p_1}{t \left[\frac{1}{h_2^2} - \frac{1}{h_1^2} \right]} h_1 \int_{h_1}^{h_2} \left[\frac{1}{h^2} - \frac{1}{h_1^2} \right] dh + \frac{6\mu l v}{t^2 \left[\frac{1}{h_2} + \frac{1}{h_1} \right]} h_1 \int_{h_1}^{h_2} \left[\frac{1}{h} - \frac{1}{h_1} \right] \left[\frac{1}{h} - \frac{1}{h_2} \right] dh$$

$$\frac{df}{dy} = p_1 + \frac{p_2 - p_1}{t \left[\frac{1}{h_2^2} - \frac{1}{h_1^2} \right]} \cdot \left[\frac{-1}{h_2} + \frac{2}{h_1} - \frac{h_2}{h_1^2} \right] + \frac{6\mu l v}{t^2 \left[\frac{1}{h_2} + \frac{1}{h_1} \right]} \cdot \left[\frac{2}{h_1} - \frac{2}{h_2} + \frac{h_1 + h_2}{h_1 h_2} \ln \frac{h_1}{h_2} \right]$$

$$\frac{df}{dy} = p_1 + (p_2 - p_1) \frac{h_1 + t}{2h_1 + t} + \frac{6\mu l v}{t^2} \left[\frac{2t}{2h_1 + t} + \ln \frac{h_1}{h_1 + t} \right]$$

$$\frac{df}{dy} = \frac{p_1 + p_2}{2} + \left[\frac{p_2 - p_1}{2} \cdot t + \frac{12\mu l v}{t} \right] \cdot \frac{1}{2h_1 + t} + \frac{6\mu l v}{t^2} \ln \frac{h_1}{h_1 + t}$$

According to Fig. 7.13,

$$h_1 = c - e \cos \phi, \quad h_2 = c + t - e \cos \phi, \quad dy = \frac{d}{2} \cdot d\phi.$$

Integration around the circle in Fig. 7.13 gives the bearing force f .

$$f = \int df \cos \phi = \int \frac{df}{dy} \cos \phi \cdot dy = \int_0^{2\pi} \frac{df}{dy} \cos \phi \cdot \frac{d}{2} \cdot d\phi = d \int_0^{\pi} \frac{df}{dy} \cos \phi \, d\phi.$$

Substitution of df/dy , considering $\int_0^{\pi} \cos \phi \, d\phi = 0$, yields

$$\frac{f}{ld} = \left[\frac{p_2 - p_1}{2} \cdot t + \frac{12\mu l v}{t} \right] \int_0^{\pi} \frac{\cos \phi \, d\phi}{2c + t - 2e \cos \phi} + \frac{6\mu l v}{t^2} \int_0^{\pi} \left[\ln \frac{c - e \cos \phi}{c + t - e \cos \phi} \right] \cos \phi \, d\phi. \quad (15)$$

The above integrals are extremely complicated.

Tables of indefinite integrals give

$$\int_0^{\pi} \frac{dx}{a + b \cos x} = \frac{2}{\sqrt{a^2 - b^2}} \arctan \frac{\sqrt{a^2 - b^2} \cdot \tan \frac{1}{2}x}{a + b} \Big|_0^{\pi} = \frac{\pi}{\sqrt{a^2 - b^2}}.$$

So

$$\int_0^{\pi} \frac{\cos \phi \, d\phi}{2c + t - e \cos \phi} = -\frac{1}{2e} \int_0^{\pi} \frac{2c + t - 2e \cos \phi - (2c + t)}{2c + t - e \cos \phi} d\phi = \frac{-1}{2e} \left[\pi - \frac{\pi(2c + t)}{\sqrt{(2c + t)^2 - 4e^2}} \right]$$

The other definite integral is of the type

$$\int_0^{\pi} \ln(a + b \cos x) \cos x \, dx = \int_0^{\pi} \ln(a + b \cos x) \, d \sin x =$$

$$\sin x \cdot \ln(a + b \cos x) \Big|_0^{\pi} + \int_0^{\pi} \frac{b \sin^2 x \, dx}{a + b \cos x} = \int_0^{\pi} \frac{b - b \cos^2 x}{a + b \cos x} \cdot dx =$$

$$\int_0^{\pi} \left[-\cos x + \frac{a}{b} + \frac{b - a^2/b}{a + b \cos x} \right] dx = \frac{\pi}{b} \left[a + \sqrt{a^2 - b^2} \right].$$

Thus

$$\int_0^{\pi} \ln \frac{c - e \cos \phi}{c + t - e \cos \phi} \cos \phi \, d\phi = -\frac{\pi}{e} \left[c - \sqrt{c^2 - e^2} - (c + t) + \sqrt{(c + t)^2 - e^2} \right].$$

Implementing these integrals into eqn. (15) yields

$$\frac{f}{ld} = \left[\frac{1}{4} (p_2 - p_1)t + \frac{6\mu l v}{t} \right] \cdot \frac{\pi}{e} \cdot \left[\frac{2c + t}{\sqrt{(2c + t)^2 - 4e^2}} - 1 \right] +$$

$$- \frac{6\mu l v}{t} \cdot \frac{\pi}{e} \left[\frac{\sqrt{(c + t)^2 - e^2} - \sqrt{c^2 - e^2}}{t} - 1 \right].$$

Elaborating and introducing

$$F = \frac{f}{\frac{\pi}{4} d^2 (p_2 - p_1)}, \quad v = \frac{\mu dv}{c^2 (p_2 - p_1)}, \quad L = \frac{1}{d}, \quad T = \frac{t}{c}, \quad E = \frac{e}{c}$$

finally gives

$$\frac{F}{L} = \frac{T}{E} \left[\frac{2 + T}{\sqrt{(2 + T)^2 - 4E^2}} - 1 \right] - \frac{24VL}{TE} \left[\frac{\sqrt{(1 + T)^2 - E^2} - \sqrt{1 - E^2}}{T} - \frac{2 + T}{\sqrt{(2 + T)^2 - 4E^2}} \right].$$

(3)

Equation (3) permits some remarkable conclusions.

$$S_o = \lim_{E \rightarrow 0} \frac{DF}{dE} = \frac{2TL}{(2+T)^2} \left[1 - \frac{6VL}{1+T} \right].$$

Thus S_o at $V = 0$ is a maximum for $T = 2$, giving $S_{o\max} = L/4$. $S_o = 0$ at $V = (1+T)/6L = V_{\max}$. At $V > V_{\max}$ no positive bearing force is possible!

It should be noted that *dimensional analysis* of hydrostatic bearings - without any of the above derivations - inevitably leads to the five dimensionless quantities F , V , L , T and E (or a combination of these). Thus the characteristics of hydrostatic bearings must necessarily show a relationship between these five quantities.

When furthermore we assume that the influence of the speed V can be found by superposition, the result becomes

$$F = \phi_1(T, E, L) - V \cdot \phi_2(T, E, L)$$

In the case of one-dimensional flow this result reduces to

$$\underline{\underline{\frac{F}{L} = \phi_3(T, E) - VL \cdot \phi_4(T, E)}}$$

*7.13 Derivation of leakage flow, eqn. (6)

As the leakage flow is small in comparison with the piston displacement flow, the leakage will be derived for $v = 0$. Equation (14) gives

$$\frac{dq}{dy} = \frac{t\Delta p}{6\mu l} \cdot \frac{1}{\frac{1}{h_1^2} - \frac{1}{h_2^2}} = \frac{t\Delta p}{6\mu l} \cdot \frac{h_1^2 h_2^2}{(h_2 - h_1)(h_2 + h_1)} = \frac{\Delta p}{6\mu l} \cdot \frac{h_1^2 (h_1 + t)^2}{2h_1 + t}$$

$$dq + \frac{\Delta p}{12\mu l} \left[h_1^3 + \frac{3}{2} h_1^2 t + \frac{1}{4} h_1 t^2 - \frac{1}{8} t^3 + \frac{1}{8} \frac{t^4}{2h_1 + t} \right] dy.$$

Using $dy = \frac{d}{2} \cdot d\phi$, $h_1 = c - e \cos\phi = c(1 - E \cos\phi)$, $t = Tc$

$$\int_0^{2\pi} \cos\phi \, d\phi = 0, \quad \int_0^{2\pi} \cos^3\phi \, d\phi = 0, \quad \text{we find:}$$

$$q = \frac{c^3 \Delta p}{24\mu l} \int_0^{2\pi} \left[1 + \frac{3}{2} T + \frac{1}{4} T^2 - \frac{1}{8} T^3 + \left(3 + \frac{3}{2} T \right) E^2 \cos^2\phi + \frac{1}{8} \cdot \frac{T^4}{2 + T - E \cos\phi} \right] d\phi.$$

Using the "standard"-integral $\int \frac{dx}{a + b \cos x}$ of Section 7.12 we find

$$q = \frac{\pi d c^3 \Delta p}{12 \mu l} \left[1 + \frac{3}{2} T + \frac{1}{4} T^2 - \frac{1}{8} T^3 + \frac{3}{2} \left(1 + \frac{T}{2}\right) E^2 + \frac{T^4}{8} \cdot \frac{1}{\sqrt{(2+T)^2 - 4E^2}} \right]$$

$$q = \frac{\pi d c^3 \Delta p}{12 \mu l} \left[\frac{2(1+T)^2}{2+T} + \frac{3}{2} \left(1 + \frac{T}{2}\right) E^2 + \frac{T^4}{8(2+T)} \left\{ \frac{2+T}{\sqrt{(2+T)^2 - 4E^2}} - 1 \right\} \right]$$

$$q \approx \frac{\pi d c^3 \Delta p}{12 \mu l} \left[\frac{2(1+T)^2}{2+T} + 1,5 (1 + 0,5T) E^2 \right]$$

$$q \approx \frac{\pi c^3 \Delta p}{12 \mu} \cdot \left[\frac{1 + 1,75T + 1,5(1 + 0,5T) E^2}{L} \right]. \quad (6)$$

The approximation is very good. Calculations on the two-dimensional flow model did not show significant differences.

*7.14 Derivation of cavitation boundary for double bearings, eqn. (8)

The equation for the pressure distribution

$$p = p_1 + \Delta p \cdot \frac{\frac{1}{h^2} - \frac{1}{h_1^2}}{\frac{1}{h_2^2} - \frac{1}{h_1^2}} + \frac{6\mu l v}{h_2 - h_1} \cdot \frac{\left(\frac{1}{h} - \frac{1}{h_1}\right)\left(\frac{1}{h} - \frac{1}{h_2}\right)}{\frac{1}{h_2} + \frac{1}{h_1}} \quad (2)$$

makes clear that the speed may cause "negative" pressures. To avoid cavitation risks, the lowest pressure $p_{\min}$ in the gap must be zero. First the location of $p_{\min}$ must be obtained by letting $dp/dx = 0$ or $dp/dh = 0$. At that location, $p = 0$ sets the cavitation boundary, resulting in the maximum permitted speed.

Calculations easily result in overly complicated formula's. Therefore the essential "stepping stones" for the derivation of the cavitation boundary are summarized. First, eqn. (2) has to be transformed into:

$$\left(h_2^2 - h_1^2\right)\left(\frac{p}{\Delta p} - \frac{p_1}{\Delta p}\right) = h_2^2 + \frac{6\mu l v}{\Delta p} \left[1 - \frac{h_1 + h_2}{h} + \frac{h_1 h_2}{h^2} \left\{ 1 - \frac{h_1 h_2 \Delta p}{6\mu l v} \right\} \right]. \quad (16)$$

Letting

$$\frac{dp}{dh} = 0 \quad \text{yields} \quad h = \frac{2h_1 h_2}{h_1 + h_2} \left\{ 1 - \frac{h_1 h_2 \Delta p}{6\mu l v} \right\} \quad (17)$$

$p = 0$ in eqn. (16), using $1 - \frac{h_1 h_2 \Delta p}{6\mu l v}$ of eqn. (17) yields:

$$0 = \frac{p_1}{\Delta p} (h_2^2 - h_1^2) + h_2^2 + \frac{6\mu l v}{\Delta p} \cdot \left[1 - \frac{h_1 + h_2}{2h} \right]. \quad (18)$$

Substituting h of eqn. (17) into $\{1 - (h_1 + h_2)/2h\}$ yields after considerable working out,

$$1 - \frac{h_1 + h_2}{2h} = - \frac{\frac{6\mu l v}{\Delta p} \cdot \left[\frac{h_2 - h_1}{2h_1 h_2} \right]^2 + 1}{\frac{6\mu l v}{\Delta p} \cdot \frac{1}{h_1 h_2} - 1}. \quad (19)$$

Substituting of eqn. (19) into eqn. (18) yields after multiplying with

$$\frac{6\mu l v}{\Delta p} \cdot \frac{1}{h_1 h_2} - 1 :$$

$$0 = \left[\frac{p_1}{\Delta p} (h_2^2 - h_1^2) + h_2^2 \right] \left[\frac{6\mu l v}{\Delta p} \cdot \frac{1}{h_1 h_2} - 1 \right] - \frac{6\mu l v}{\Delta p} \left[\frac{6\mu l v}{\Delta p} \cdot \left(\frac{h_2 - h_1}{2h_1 h_2} \right)^2 + 1 \right].$$

Rearrangement yields:

$$\left[\frac{6\mu l v}{\Delta p} \cdot \frac{h_2 - h_1}{2h_1 h_2} \right]^2 - 2 \cdot \frac{6\mu l v}{\Delta p} \cdot \frac{h_2 - h_1}{2h_1 h_2} \left[h_2 + \frac{p_1}{\Delta p} (h_2 + h_1) \right] + \left[h_2^2 + \frac{p_1}{\Delta p} (h_2^2 - h_1^2) \right] = 0.$$

The solution is:

$$\begin{aligned} \frac{6\mu l v}{\Delta p} \cdot \frac{h_2 - h_1}{2h_1 h_2} &= h_2 + (h_2 + h_1) \left[\frac{p_1}{\Delta p} \pm \sqrt{\left(\frac{p_1}{\Delta p} \right)^2 + \frac{p_1}{\Delta p}} \right] \\ \mu l v &= \frac{h_1 h_2^2}{3(h_2 - h_1)} \left[\Delta p + \frac{h_2 + h_1}{h_2} \left(p_1 + \sqrt{p_1^2 + p_1 \Delta p} \right) \right]. \end{aligned} \quad (20)$$

As p_1 raises the cavitation boundary, only the positive root is considered. For reasons of symmetry, cavitation will occur first in the (narrowest) gap in the symmetry plane. Thus, according to Fig. 7.13,

$$h_1 = c - e \quad \text{and} \quad h_2 = c + t - e.$$

Using $E = \frac{e}{c}$, $L = \frac{1}{d}$, $T = \frac{t}{c}$, eqn. (20) becomes

$$\frac{\mu d v}{c^2} = \frac{(1 - E)(1 + T - E)^2}{3TL} \left[\Delta p + \frac{2 + T - 2E}{1 + T - E} \left(p_1 + \sqrt{p_1^2 + p_1 \Delta p} \right) \right]. \quad (8)$$

With $p_2 + p_1 = \Delta p + 2p_1$, we find as boundary

$$\frac{\mu dv}{c^2(p_2 + p_1)} = \frac{(1 - E)(1 + T - E)^2}{3TL(\Delta p + 2p_1)} \cdot \left[\Delta p + \frac{2 + T - 2E}{1 + T - E} \left(p_1 + \sqrt{p_1^2 + p_1 \Delta p} \right) \right].$$

Table 7.2 proves that at $E = 0.75$ a safe approximation of the cavitation boundary is given by

$$v^* = \frac{\mu dv}{c^2(p_2 + p_1)} = v_c = 0.1 \frac{T}{L}. \quad (9)$$

Analysis of eqns. (3) and (8) proves that *cavitation can not occur in the "stable" regions of the characteristics of single bearings, represented by the solid lines in Fig. 7.3(a).*

*7.15 Iterative relaxation for conical bearing with two-dimensional flow

Reynolds' equation

$$\frac{\partial}{\partial x} \left[\frac{h^3}{12\mu} \cdot \frac{\partial p}{\partial x} - \frac{vh}{2} \right] + \frac{\partial}{\partial y} \left[\frac{h^3}{12\mu} \cdot \frac{\partial p}{\partial y} \right] = 0$$

After substitution of non-dimensional quantities

$$X = \frac{x}{l}, \quad \Phi = \frac{\phi}{\pi} = \frac{2y}{\pi d}, \quad P = \frac{p - p_1}{\Delta p}, \quad H = \frac{h}{c}, \quad L = \frac{l}{d}$$

$$\frac{\partial}{\partial X} \left[H^3 \cdot \frac{\partial P}{\partial X} - \frac{6\mu lv}{c^2 \Delta p} \cdot H \right] + \left(\frac{L}{\pi/2} \right)^2 \cdot \frac{\partial}{\partial \Phi} \left[H^3 \cdot \frac{\partial P}{\partial \Phi} \right] = 0$$

$$H^3 \left[\frac{\partial^2 P}{\partial X^2} + \left(\frac{L}{\pi/2} \right)^2 \cdot \frac{\partial^2 P}{\partial \Phi^2} \right] + 3H^2 \left[\frac{\partial P}{\partial X} \cdot \frac{\partial H}{\partial X} + \left(\frac{L}{\pi/2} \right)^2 \cdot \frac{\partial P}{\partial \Phi} \cdot \frac{\partial H}{\partial \Phi} \right] - \frac{6\mu lv}{c^2 \Delta p} \cdot \frac{\partial H}{\partial X} = 0.$$

Introduction of difference-network

$$X = i \cdot \Delta X; \quad i = 1, 2, 3 \dots m; \quad m \cdot \Delta X = 1$$

$$\Phi = j \cdot \Delta \Phi; \quad j = 1, 2, 3 \dots n; \quad n \cdot \Delta \Phi = 1$$

$$P(X \pm \Delta X, \Phi) = P_{i \pm 1, j} \quad P(X, \Phi \pm \Delta \Phi) = P_{i, j \pm 1}.$$

Taylor:

$$\left. \begin{aligned} P_{i+1, j} &= P_{i, j} + \Delta X \cdot \frac{\partial P}{\partial X} + \frac{1}{2} (\Delta X)^2 \cdot \frac{\partial^2 P}{\partial X^2} \\ P_{i-1, j} &= P_{i, j} - \Delta X \cdot \frac{\partial P}{\partial X} + \frac{1}{2} (\Delta X)^2 \cdot \frac{\partial^2 P}{\partial X^2} \end{aligned} \right\}$$

$$\frac{\partial P}{\partial X} = \frac{P_{i+1, j} - P_{i-1, j}}{2\Delta X} \quad \frac{\partial^2 P}{\partial X^2} = \frac{P_{i+1, j} - 2P_{i, j} + P_{i-1, j}}{(\Delta X)^2}$$

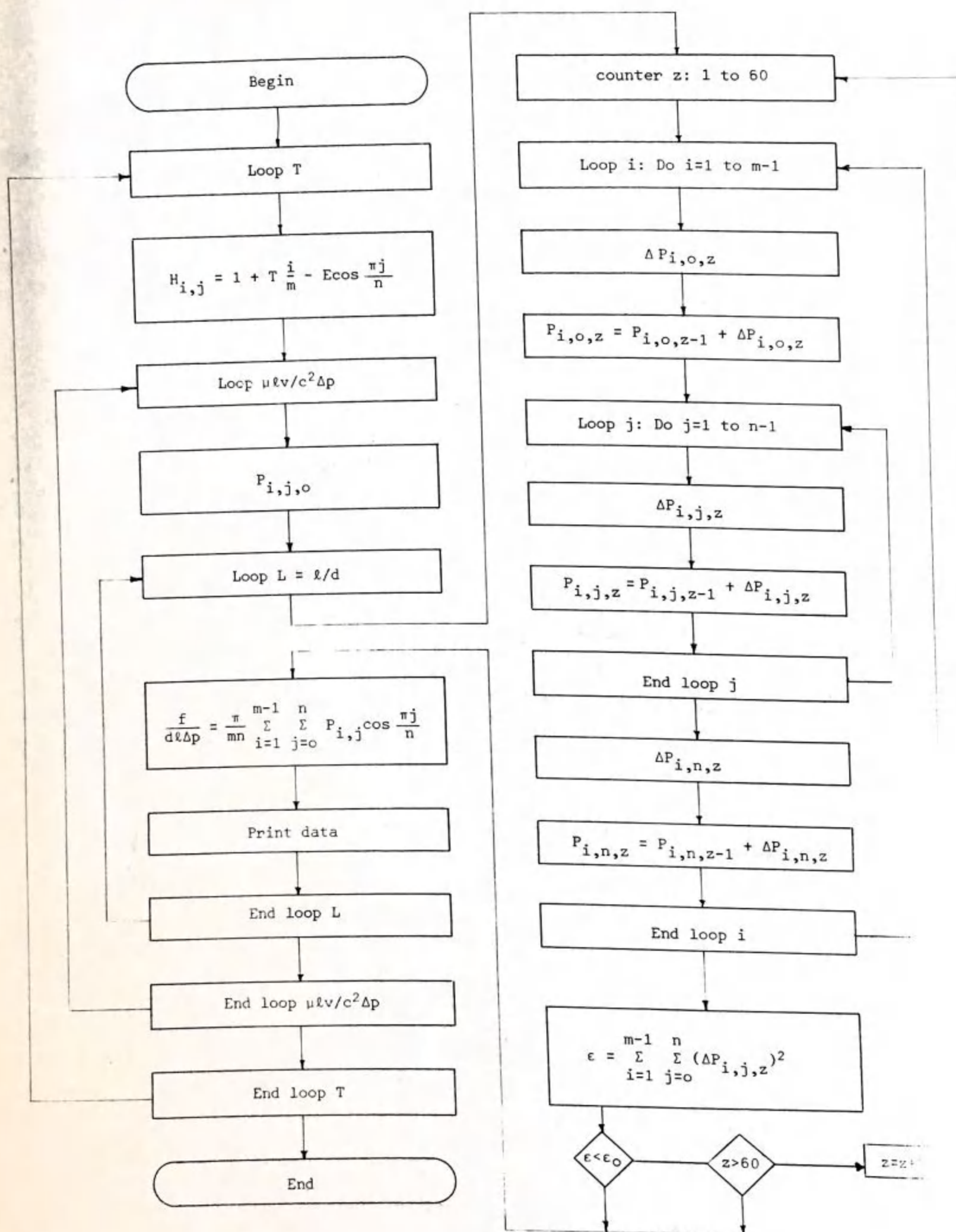


Fig. 7.14 Flow diagram relaxation pressure distribution of conical bearings with two-dimensional flow.

Similarly:

$$\frac{\partial P}{\partial \Phi} = \frac{P_{i,j+1} - P_{i,j-1}}{2\Delta\Phi} \quad \frac{\partial^2 P}{\partial \Phi^2} = \frac{P_{i,j+1} - 2P_{i,j} + P_{i,j-1}}{(\Delta\Phi)^2}$$

$$\frac{\partial H}{\partial X} = \frac{H_{i+1,j} - H_{i-1,j}}{2\Delta X} \quad \frac{\partial H}{\partial \Phi} = \frac{H_{i,j+1} - H_{i,j-1}}{2\Delta\Phi}$$

Substitution of $\partial P/\partial X$, $\partial^2 P/\partial X^2$, etc., using

$$R = \frac{L}{\pi/2}, \quad \Delta X = \frac{1}{m} \quad \text{and} \quad \Delta\Phi = \frac{1}{n}$$

into the differential equation results, after considerable working out, in a difference-equation.

$$P_{i,j} = C_1 P_{i+1,j} + C_2 P_{i-1,j} + C_3 P_{i,j+1} + C_4 P_{i,j-1} + C_5 \frac{\mu l v}{c^2 \Delta p}$$

$$C_1 = \frac{1}{2+2R^2} \left[1 + \frac{3}{4} \frac{H_{i+1,j} - H_{i-1,j}}{H_{i,j}} \right] \quad C_2 = \frac{1}{2+2R^2} \left[1 - \frac{3}{4} \frac{H_{i+1,j} - H_{i-1,j}}{H_{i,j}} \right]$$

$$C_3 = \frac{R^2}{2+2R^2} \left[1 + \frac{3}{4} \frac{H_{i,j+1} - H_{i,j-1}}{H_{i,j}} \right] \quad C_4 = \frac{R^2}{2+2R^2} \left[1 - \frac{3}{4} \frac{H_{i,j+1} - H_{i,j-1}}{H_{i,j}} \right]$$

$$C_5 = \frac{3/m}{2+2R^2} \cdot \frac{H_{i+1,j} - H_{i-1,j}}{H_{i,j}^3}$$

Boundary conditions:

$$P_{i,j} = 0 \quad \text{for } i = 0 \quad \text{and} \quad P_{i,j} = 1 \quad \text{for } i = m$$

$$P_{i,j+1} = P_{i,j-1} \quad \text{for } j = 0 \quad \text{and } j = n$$

According to the flow diagram $P_{i,j,1}$; $P_{i,j,2}$ $P_{i,j,z}$ have been calculated by overrelaxation:

$$P_{i,j,z} = P_{i,j,z-1} + \Delta P_{i,j,z}$$

$$\Delta P_{i,j,z} = \alpha \left[C_1 P_{i+1,j,z-1} + C_2 P_{i-1,j,z-1} + C_3 P_{i,j+1,z-1} + C_4 P_{i,j-1,z-1} + C_5 \frac{\mu l v}{c^2 \Delta p} - P_{i,j,z-1} \right]$$

with $\alpha = 1.7$ for $m \cdot n = 30 \cdot 15$

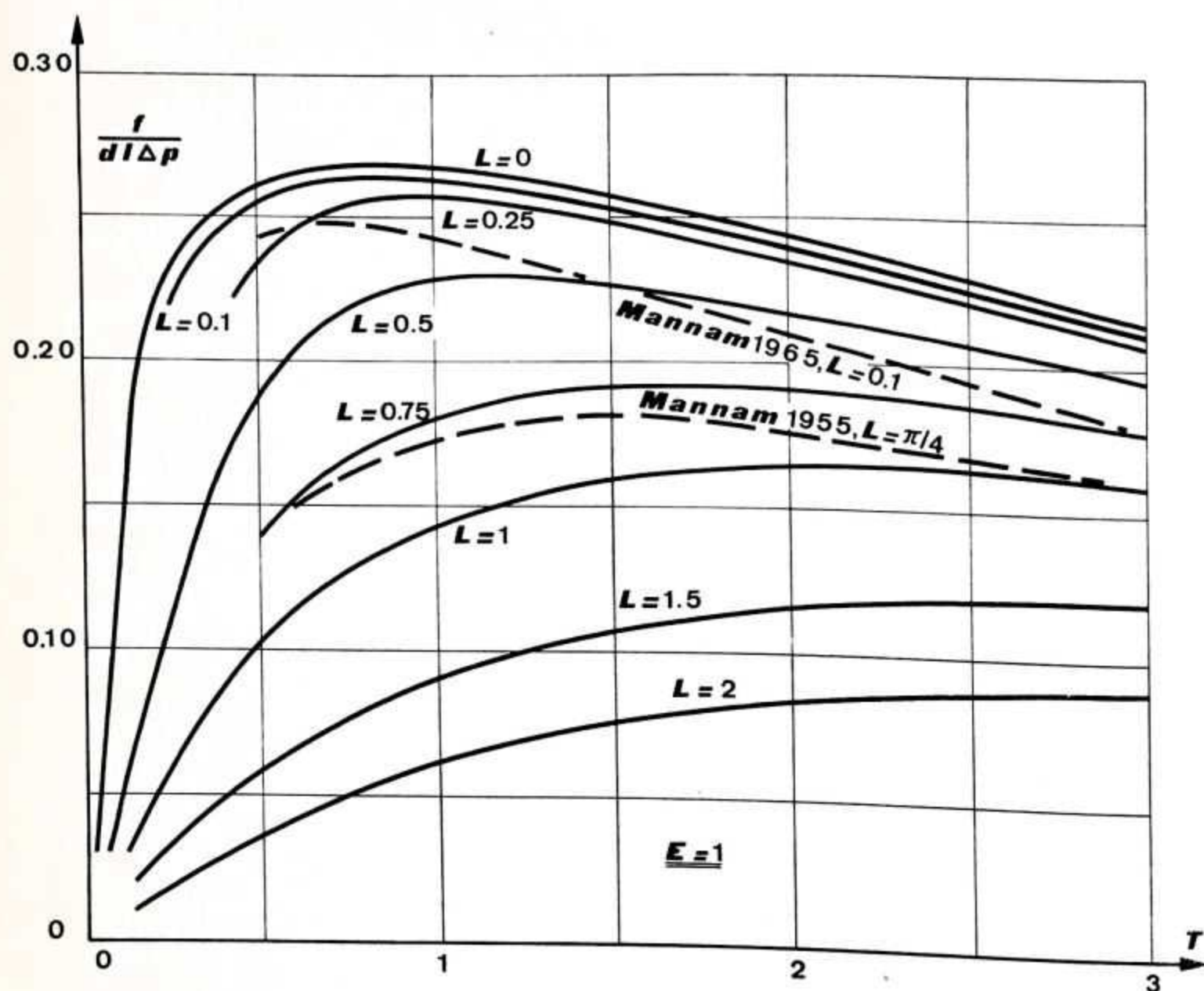


Fig. 7.15 Influence of circumferential flow on bearing force $f/dl\Delta p$ at $E = 1$ (see Table 7.4).

Relaxation is continued until

$$\epsilon = \sum_{i=1}^{m-1} \sum_{j=0}^n (\Delta P_{i,j,z})^2 < \epsilon_0 = 5 \cdot 10^{-7}.$$

As starting condition for the pressure distribution the results of the one-dimensional flow model can be used

$$P_{i,j,0} = \frac{\frac{1}{H_{i,j}^2} - \frac{1}{H_{o,j}^2}}{\frac{1}{H_{m,j}^2} - \frac{1}{H_{o,j}^2}} + \frac{6\mu l v}{c^2 \Delta p} \cdot \frac{\left[\frac{1}{H_{i,j}} - \frac{1}{H_{o,j}} \right] \left[\frac{1}{H_{i,j}} - \frac{1}{H_{m,j}} \right]}{T \left[\frac{1}{H_{o,j}} + \frac{1}{H_{m,j}} \right]}$$

From the pressure distribution, bearing force and leakage can be derived.

$$f = d \int_{\phi=0}^{\pi} \int_{x=0}^l p \cos \phi \, d\phi \, dx = \pi d l \Delta p \int_{\phi=0}^{\pi} \int_{x=0}^l P \cos \pi \phi \cdot d\phi \cdot dX$$

$$\frac{f}{d l \Delta p} = \frac{\pi}{mn} \sum_{i=0}^m \sum_{j=0}^n P_{i,j} \cos \frac{\pi j}{n}, \text{ see Fig. 7.14.}$$

$$q = d \int_0^{\pi} \frac{h^3}{12\mu} \cdot \frac{\partial p}{\partial x} \cdot d\phi = \frac{\pi d c^3 \Delta p}{12\mu} \int_0^{\pi} H^3 \cdot \frac{\partial P}{\partial X} \cdot d\phi = \frac{\pi d c^3 \Delta p}{12\mu l} \cdot \frac{m}{2n} \sum_{j=0}^n H_{i,j}^3 (P_{i+1,j} - P_{i-1,j})$$

*7.16 Results of calculations on two-dimensional flow

The essentials for the design of hydrostatic bearings have been summarized before (see Table 7.1). Other data are given in the following tables, especially on bearings with $T = 2$ and $T = 3$. It is always important to test the calculated values. Since for $L \rightarrow 0$ the results of the two-dimensional model must coincide with those of the one-dimensional model, it is practical (for control of the results only) to use the non-dimensional bearing force $f/dl\Delta p$ instead of $F = f/0.25\pi d^2\Delta p$ and the non-dimensional speed $\mu l v/c^2\Delta p$ instead of $\mu d v/c^2\Delta p$. Plotting the results, the continuity for $L \rightarrow 0$ can then be controlled easily. The Figs. 7.15 and 7.16 corresponding to Tables 7.4 and 7.5 confirm this.

Two more tests are possible.

- o For single bearings, $F = 0$ at $\mu l v/c^2\Delta p = (1 + T)/6$ at $E = 0$
 - o For double bearings, $F = 0$ at $\mu l v/c^2\Delta p = (1 + T)^2/3T$ at $E = 0$ and $P_1 = 0$.
- This is true for one-dimensional as well as for two-dimensional flow. Furthermore, some results of Mannam have been plotted in Fig. 7.15 to "prove" they are incorrect.

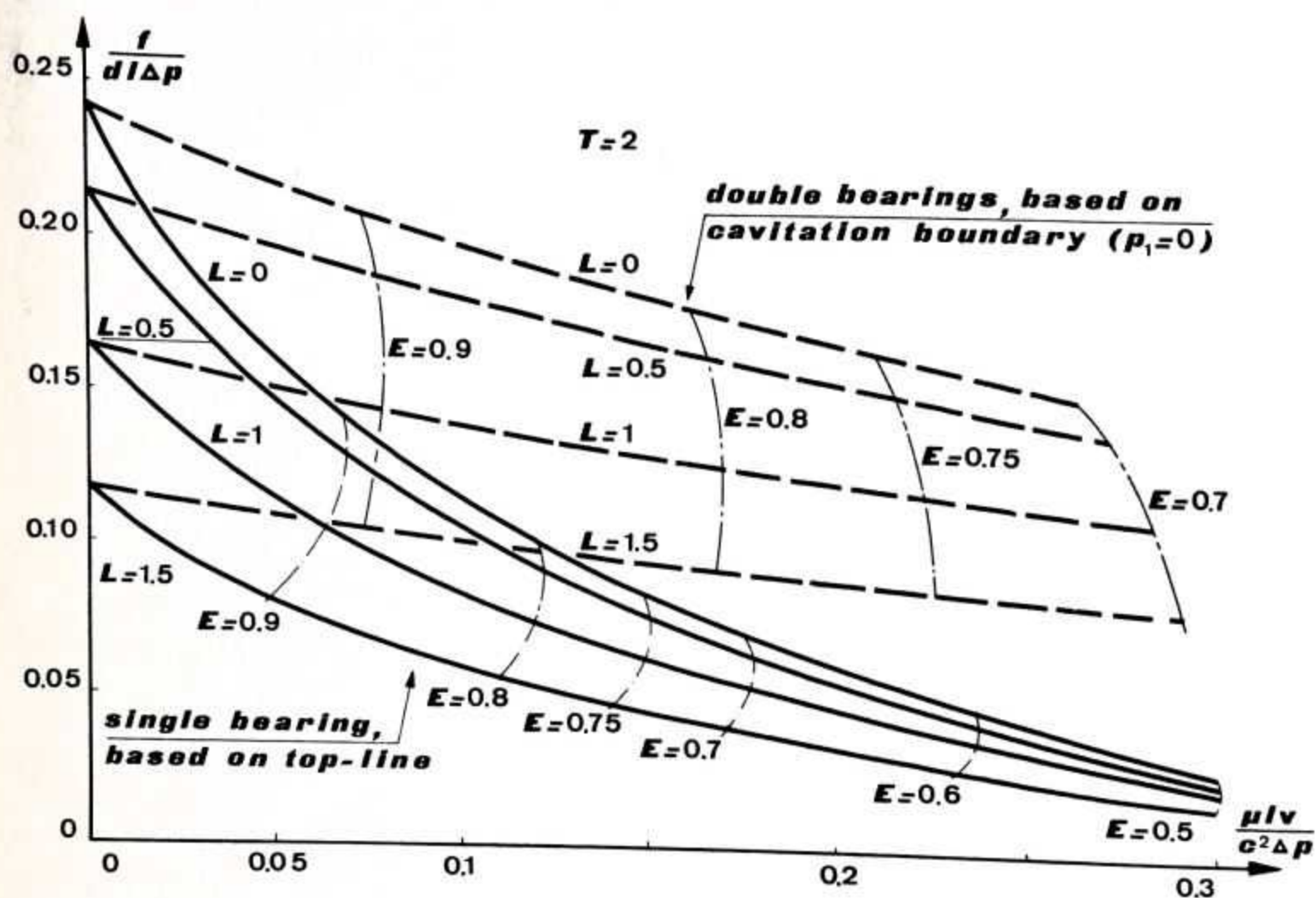


Fig. 7.16 Influence of circumferential flow, considering the axial speed at $T = 2$ (see Table 7.5).

In the right hand part of Table 7.4 the use of the non-dimensional bearing force $F = f/0.25\pi d^2 \Delta p$ can once more be seen to be justified. To choose an optimal $L = 1/d$ (with respect to the bearing force f itself), the presence of 1 in the non-dimensional force must be avoided!

Table 7.4 bearing forces at $E=1, V=0$		Influence circumferential flow					Influence of L on bearing forces				
		$\frac{f}{d\Delta p}$					$F = \frac{4f}{\pi d^2 \Delta p} = \frac{f}{d\Delta p} \cdot \frac{4}{\pi} L$				
		$T=0.5$	$T=1$	$T=1.5$	$T=2$	$T=3$	$T=0.5$	$T=1$	$T=1.5$	$T=2$	$T=3$
single bearings	$L=0$	.262	.268	.257	.243	.215	0	0	0	0	0
	0.1	.256	.264	.253	.239	.210	.033	.034	.032	.030	.027
	0.25	.235	.258	.249	.234	.208	.075	.082	.079	.074	.066
	0.50	.188	.229	.227	.216	.196	.120	.146	.145	.138	.125
	0.75	.140	.180	.191	.190	.178	.134	.172	.182	.181	.170
	1.0	.103	.143	.160	.165	.160	.131	.182	.204	.210	.204
	1.5	.060	.091	.106	.117	.120	.115	.174	.202	.222	.229
	2	.037	.061	.075	.084	.090	.094	.155	.191	.214	.229

Table 7.5			L = 0		L = 0.5		L = 1		L = 1.5		
			$\frac{f}{d\Delta P}$	$\frac{\mu l v}{c^2 \Delta p}$	$\frac{f}{d\Delta p}$	$\frac{\mu l v}{c^2 \Delta p}$	$\frac{f}{d\Delta p}$	$\frac{\mu l v}{c^2 \Delta p}$	$\frac{f}{d\Delta p}$	$\frac{\mu l v}{c^2 \Delta p}$	
Permitted combinations of bearing forces and speeds	Single bearings	E=1.0	.243	0	.2155	0	.1650	0	.1180	0	
		.9	.144	.066	.1260	.073	.1040	.064	.0810	.049	
		.8	.101	.120	.0915	.124	.0740	.120	.0570	.110	
		.75	.084	.149	.0770	.150	.0620	.150	.0490	.137	
		T=2	.7	.069	.177	.064	.178	.0520	.179	.0410	.170
		.6	.046	.237	.043	.238	.0352	.239	.0275	.231	
		.5	.028	.300	.0255	.302	.0215	.303	.0170	.299	
		.4	.0155	.360	.0144	.362	.0115	.366	.009	.368	
	Double bearings	E=1	.2430	0	.2155	0	.1650	0	.1180	0	
		.9	.2090	.0735	.1899	.0803	.1455	.0787	.1048	.0747	
		.8	.1788	.1613	.1625	.1675	.1268	.1703	.0927	.1700	
		.75	.1649	.2109	.1502	.2176	.1180	.2238	.0869	.2275	
		T=2	.7	.1515	.2645	.1383	.2733	.1094	.2826	.0811	.2907
		.6	.1264	.3840	.1159	.3955	.0927	.4132	.0696	.4316	
		(p ₁ = 0)	.5	.1030	.5208	.0950					
		.4	.0810	.6760	.0745						
	Single bearings	E=1.0	.2146	0	.1970	0	.1599	0	.1220	0	
		.9	.1270	.1060	.1136	.117	.0965	.107	.0775	.094	
		.8	.0880	.1880	.0805	.196	.0678	.190	.0545	.179	
		.75	.0738	.2280	.0668	.235	.0565	.231	.0460	.218	
		T=3	.7	.0605	.2680	.0555	.274	.0465	.273	.0385	.259
		.6	.0398	.350	.0367	.354	.030	.359	.0248	.348	
		.5	.0240	.430	.0218	.435	.018	.438	.0145	.436	
		.4	.0130	.510	.0115	.512	.010	.508	.0082	.520	
	Double bearings	E=1.0	.2146	0	.1970	0	.1599	0	.1220	0	
		.9	.1886	.1067	.1741	.1040	.1417	.0901	.1085	.0803	
		.8	.1635	.2275	.1515	.2407	.1240	.2407	.0955	.2295	
		.75	.1517	.2934	.1407	.3049	.1155	.3104	.0893	.3131	
		T=3	.7	.1402	.3630	.1302	.3744	.1072	.3826	.0832	.3898
		.6	.1182	.5137	.1099	.5264	.0910	.5416	.0712	.5571	
		(p ₁ = 0)	.5	.0972	.6806						
		.4	.0769	.8640							

REFERENCES

- [1] I. Etsion, An optimum step design for centering of pistons, J. Fluids Eng., Trans. ASME, Dec. 1975, 621-624.
- [2] W. Backé, Hydraulisches Klemmen von Steuerschiebern, Industrieanzeiger, 20(1961)580-582.
- [3] H.E. Merrit, Hydraulic Control Systems, Wiley, New York, 1967, 110.
- [4] H.J.W.M. Volman, Berekeningen en constructie van het radiaal staplager, Polytechn. Tijdschr., 27(1972)46-52.
- [5] T.J. Viersma, Reibungsfreie Hydromotoren, Oelhydraulik und Pneumatik, 18(1974)562-569.
- [6] P. Blok, Das hydrostatische Keilspaltlager, Thesis, Delft University of Technology, 1976.
- [7] P. Blok, Linear electro-hydraulic stepping motor for NC machines, Hydraulics and Pneumatics, 21, 10(1968)117-123.
- [8] T.J. Viersma, Designing load-compensated fast response hydraulic servos, Control Eng., May(1962).
- [9] W.P. Koeversmans, C.J. Jansen, Design and performance of the four-degrees-of-freedom motion system of the NLR research flight simulator, Proc. Flight Simulation/Guidance Systems Simulation, AGARD Symposium, The Hague, 1975.
- [10] J. Manhajm and D.C. Sweeney, An investigation of hydraulic lock, Proc. Inst. Mech. Engrs., 169(1955)865-880.
- [11] J. Mannam, J.H. Fowler and A.L. Carpenter, Tapered land hydrostatic journal bearings. Proc. Lubrication and Wear, Third Convention, London, 1965, Report 2, 78-84.

A compact and comprehensive textbook for graduate-level study, this book gives a complete account of the analysis, synthesis and design of hydraulic servosystems. It is based on the author's 20 years of experience in research, development and lecturing in industry and university, and aims to fill the gaps between theory, efficient education and engineering practice.

Chapters 1 to 5 deal with the fundamental behavior (motor speed vs. valve displacement and load) related to the geometry and dimensions of the components. Introducing basic control theory, a comprehensive survey is presented of the dynamic behavior, including stability, of all damping methods, control concepts, servovalve characteristics, influence of compliant structures and Coulomb friction. This section of the book ends with a design chapter based on performance diagrams and control concepts. The abundant illustrations provide the reader with an almost complete pictorial summary.

The sixth chapter presents fundamentally new material on the design of hydraulic supply lines for hydraulic servosystems (including accumulators, pressure control valves and tuned filters) and shows how to suppress pressure fluctuations at the servovalve. The last chapter is devoted to the design of newly developed hydrostatic bearings for hydraulic cylinders, eliminating Coulomb friction and slip-stick motions.

The book has been written not only for lecturers and students but also for scientists and qualified engineers in advanced hydraulic industries, aircraft and space industries and defence forces.